

NeuroGecko

The complete record — what was asked for, what was built, what was wrong, and what is left

A biologically-grounded leopard gecko (*Eublepharis macularius*) in MuJoCo. Every number in the animal is tagged **PUBLISHED**, **DERIVED** or **INVENTED**. Every hypothesis ever tested stays in the record, especially the wrong ones. This document is that record, assembled by machine from `docs/FAILURE_MAP.md`, `config/proxies.yaml`, `git log`, `git blame` and the evidence JSON. No figure in it was transcribed by hand.

284
HYPOTHESES REFUTED

37
CONFIRMED

348
TESTED IN 30 SESSIONS

109
COMMITTS

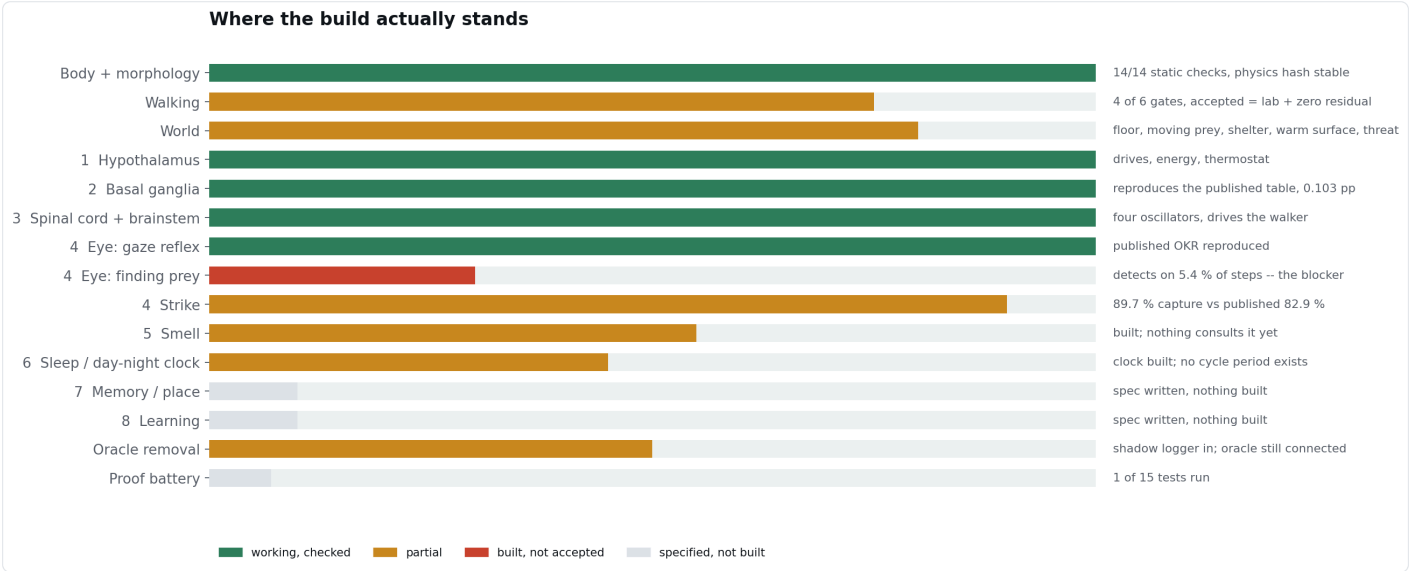
152
REGISTERED PARAMETERS

70
PUBLISHED

62
OPENLY INVENTED

217
WORKS CITED

82 % of everything tried was wrong. That is not the failure of the project — it is the product of it. A refuted entry is a map of where not to look, and nothing is ever deleted from it.



What is in this document

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14	The bibliography — 217 works, 51 DOIs		

How to read the colour

 confirmed / published / working  partly / uncertain  refuted / not accepted  invented, and says so  not started

Two conventions run through every chart. A number that has been **measured in this repository** is drawn as a bar. A number that is **published elsewhere and has not been reproduced here** is drawn as a dashed marker or a shaded band, never as a bar — so a target cannot be mistaken for a result. And every chart caption names the ledger row the measurement came from, so any figure here can be traced to the run that produced it.

One thing this document deliberately does not contain

There are no time estimates anywhere in it. Not for the remaining modules, not for the blocker, not for the proof battery. That is a standing instruction on this project, and it is also the honest position: every schedule this project has implied has been wrong, usually because the blocker turned out to be somewhere nobody was looking. Work that has happened carries the date it happened. Work that has not happened carries an **order** and a **dependency** — what has to be true before it can start — and nothing else.

What you asked for, and where each of those things stands

A real gecko, not a gecko-shaped robot

The animal has to behave the way *Eublepharis macularius* behaves because of what is published about it, not because a reward function was shaped until the motion looked right.

Where it stands: Every parameter carries PUBLISHED / DERIVED / INVENTED in `config/proxies.yaml`. 143 registered; 43 % published, 14 % derived, 43 % openly invented.

Procedural understanding, not a lookup

It should work out what prey is from what it senses — not be told “object 7 is food”. Many kinds of insect; some things move and are not prey; prey sometimes holds still.

Where it stands: Partly. The strike, the flee response and the approach criterion are procedural and published. The eye is the gap: it detects on 5.4 % of steps, and prey identity is still colour and motion, not a tuned detector.

No oracle. No privileged information.

The animal must not be handed the prey's coordinates. If it finds food, it must find it with its own senses.

Where it stands: NOT DONE, and now instrumented rather than argued about. The oracle is demoted to a shadow that reaches the logger and never the policy (`brain/hunt_metrics.py` refuses to be constructed any other way). It is still connected to the walker's goal.

Videos that show the animal, honestly

One fixed angle. No shaking. Real speed. All four legs visible. The walk that was accepted, not whatever checkpoint was lying around.

Where it stands: Fixed after four separate defects: half-speed playback, a camera locked to the bobbing trunk, a compass angle the animal walked into, and — the one that mattered — the wrong walker in every clip.

Never guess at how long things take

A standing instruction from the first session.

Where it stands: Honoured. This document dates what happened and orders what has not. It gives no estimate for anything, and the reason is in Part 10.

Short answers, plain words, jargon defined inline

A standing instruction.

Where it stands: Honoured in conversation. This document is the exception the user asked for, and it still defines its terms — see the glossary in Part 17.

Carry the map, always

`docs/FAILURE_MAP.md` is read first every session and updated before every commit that tests something. Nothing is ever deleted from it.

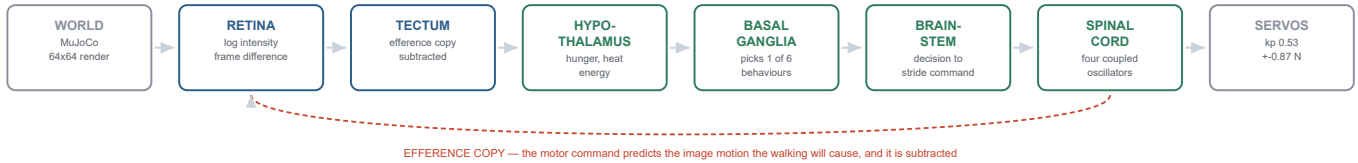
Where it stands: Honoured across 29 sessions and 221 numbered hypotheses. This report is generated from it.

What NeuroGecko is

A 38-gram leopard gecko, built in the MuJoCo physics engine, with a brain assembled module by module out of published measurements. It is not a reinforcement-learning agent that happens to look like a lizard. The distinction is the whole project: an RL agent is judged by whether it gets better at the reward, and this animal is judged by whether it reproduces numbers somebody measured in a real gecko and did not tell it about.

How one step actually runs

Every step, at 500 Hz physics and 50 Hz control. Nothing in this chain is a neural network except the walking residual, which is switched OFF in the accepted walker.



The physics runs at 500 Hz and the control loop at 50 Hz. The only trained component anywhere in this chain is the walking residual, and in the accepted walker it is switched off — the animal walks on a hand-written rhythm, and the ledger measured the trained addition as worse (#11, #71, #183).

The eight brain modules, and what each one is for

1 · Hypothalamus BUILT

The drives. How hungry it is, how much energy it has left, how warm it is. Nothing here decides anything; it produces the numbers the decision runs on.

2 · Basal ganglia BUILT

The selector. Six behaviours compete — hunt, shelter, warm up, rest, flee, explore — and exactly one wins. Reproduces the published gating table to 0.103 percentage points.

3 · Spinal cord + brainstem BUILT

The rhythm. Four coupled oscillators, one per leg, producing the walking pattern without needing rhythmic input. The brainstem turns “hunt” into a stride command.

4 · Eye HALF

Two jobs. Keeping the gaze stable — the published optokinetic reflex, reproduced. And finding prey — the blocker, detecting on 5.4 % of steps.

4b · Strike BUILT

The last two centimetres. Released at 20.3 mm, peak 0.851 m/s, 80 ms. 89.7 % capture against a published 82.9 %, with nothing fitted to it.

5 · Smell BUILT, UNUSED

Tongue-flicking. Returns a strength per odour and **never a bearing**, because no gecko has been shown to find prey by smell alone. Nothing consults it yet.

6 · Day/night clock BUILT, UNUSED

When the animal is awake. The sleep-cycle period is stored as `nu11` on purpose, so borrowing a bearded dragon's number is impossible without declaring it.

7 · Memory SPECIFIED

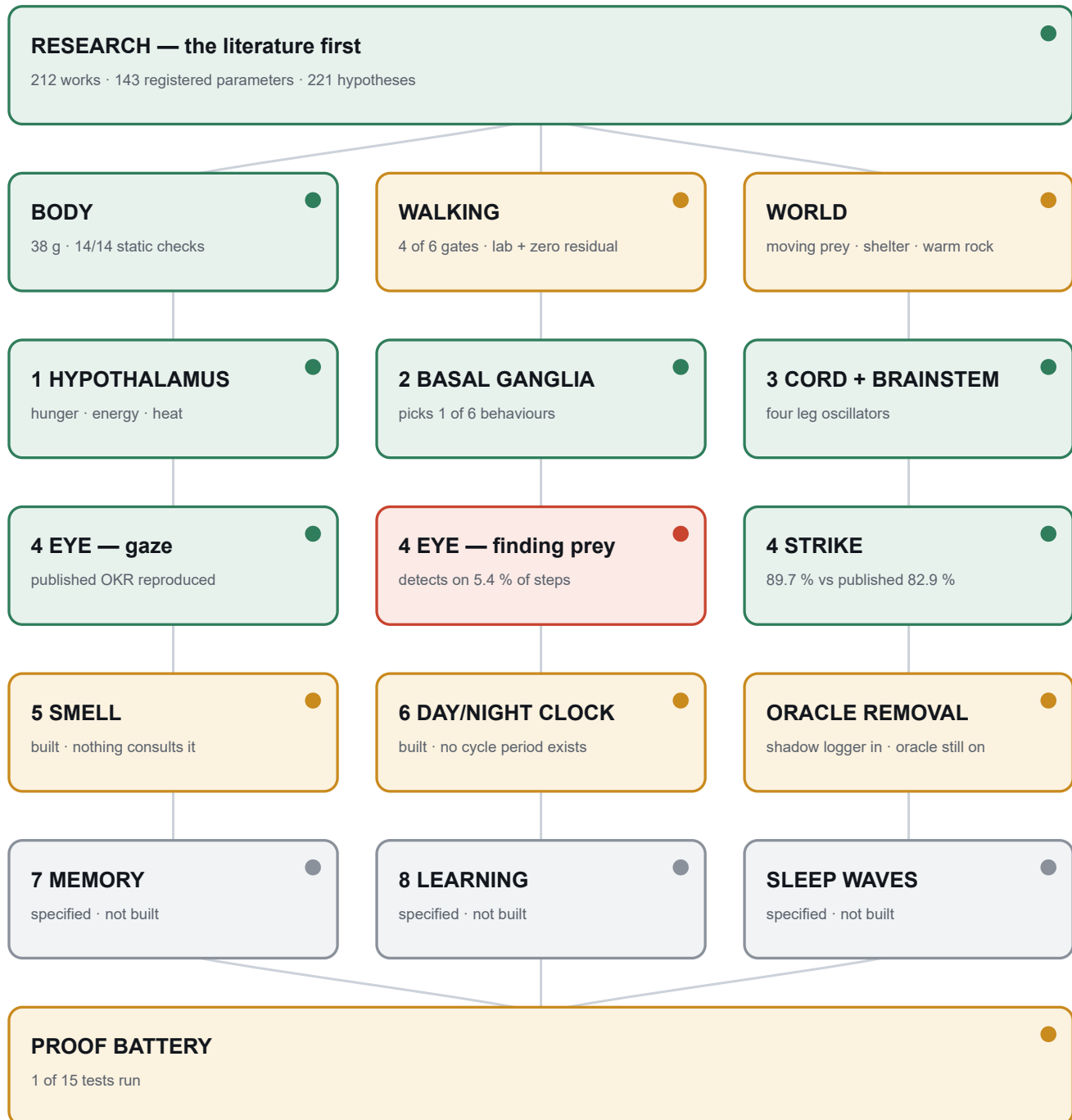
Where the goal was. Two decay components: route precision fades before the ability to get there at all. Both published in this species.

8 · Learning SPECIFIED

Getting better with practice. 9 % per trial, ceilings 3.02× on speed and 4.59× on route. Published in this species — measured on escaping water, not on hunting.

The road map

● working, checked against something
 ● partial
 ● built and not accepted
 ● specified, not built



Five modules stand. One is built and openly recorded as not working. Three are specified and not built — and the specification for all three exists because a literature sweep in Session 10c refuted the assumption that they were too thin to build (#216, #217). The surprise of that session was that the modules deferred for lack of evidence had target-species numbers all along, and the module considered urgent — the prey detector — has no gecko measurement of any kind.

The six rules, and the failure each one came from

None of these were decided in advance. Each one is the scar left by a specific defect, written down so the same defect costs one session instead of three.

1 Every number is PUBLISHED, DERIVED or INVENTED — and says which.

A plausible number wearing a published number's clothes is the worst defect available.

The failure it came from: `cost_of_transport = 0.73` sat in the registry tagged *E. macularius*, *verified*. It is *Teratoscincus przewalskii*, a different family. No cost of transport has ever been measured in this species.

2 Never tune until the answer matches.

If a model needs its constants adjusted to reproduce a published result, the constants must come from a paper.

The failure it came from: The basal-ganglia dopamine curve sat 0.10 too high for three sessions. It was not scaled down; the mechanism was found to be wrong in the opposite direction to the previous correction. Error fell from 10.3 percentage points to 0.103.

3 The reward is what the model is told to want. The gates are what the animal does. Only the second counts.

A training reward can be satisfied in ways that have nothing to do with the behaviour.

The failure it came from: The trained walking residual improves the reward and fails the gates. The accepted walker runs with the residual switched off entirely.

4 A failed prediction stays in the table.

Adding a knob per failing joint is fitting the harness to the answer.

The failure it came from: Two of six locomotion gates have never passed. They are still listed as failing, in every report, including this one.

5 Measure before claiming — including your own tests and your own solver.

The instrument is part of the experiment.

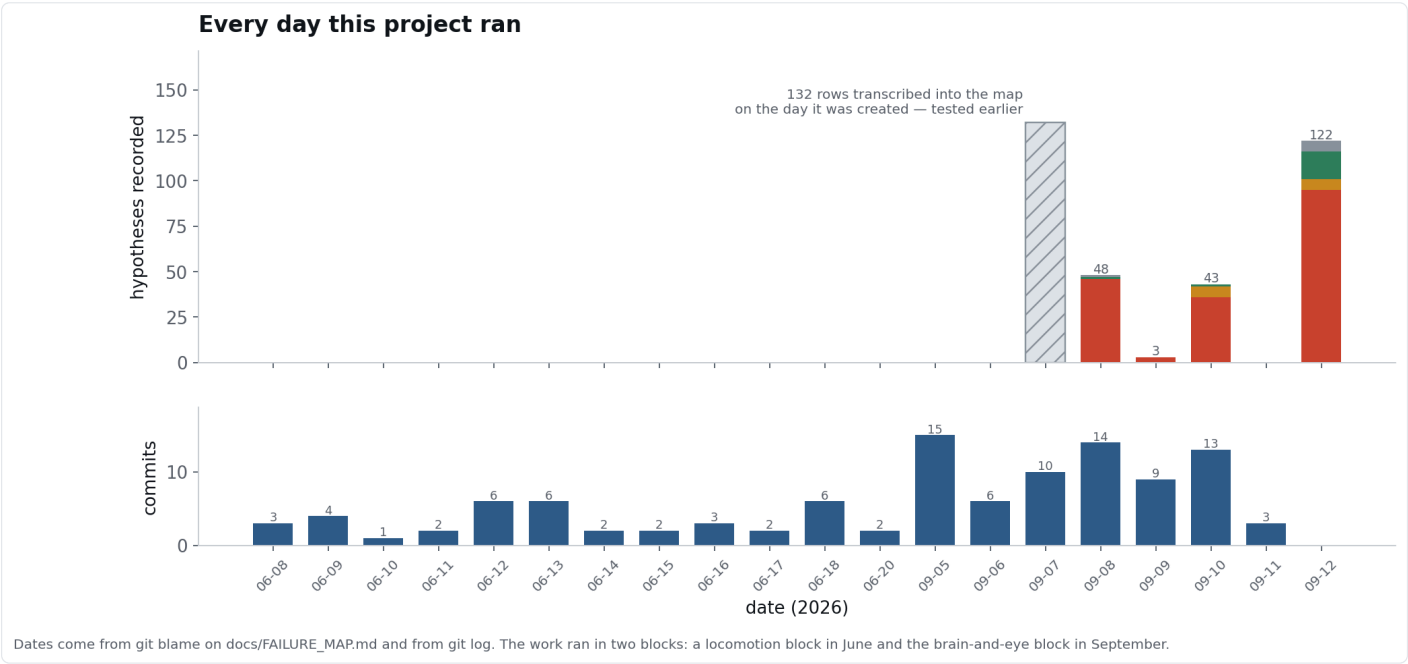
The failure it came from: Five recorded failures were in the test, not the model. One was in the integration step. One test literally built a patch of the colour it read out of the model and checked the detector saw it — it passed while the detector scored 0.000 on the real rendered prey. Twice, three sessions apart.

6 One source is not a citation.

Check a parameter set against something it must independently predict.

The failure it came from: The basal-ganglia weights were settled by the gating constant they have to reproduce, not by a vote among sources.

The flow — every session, in order

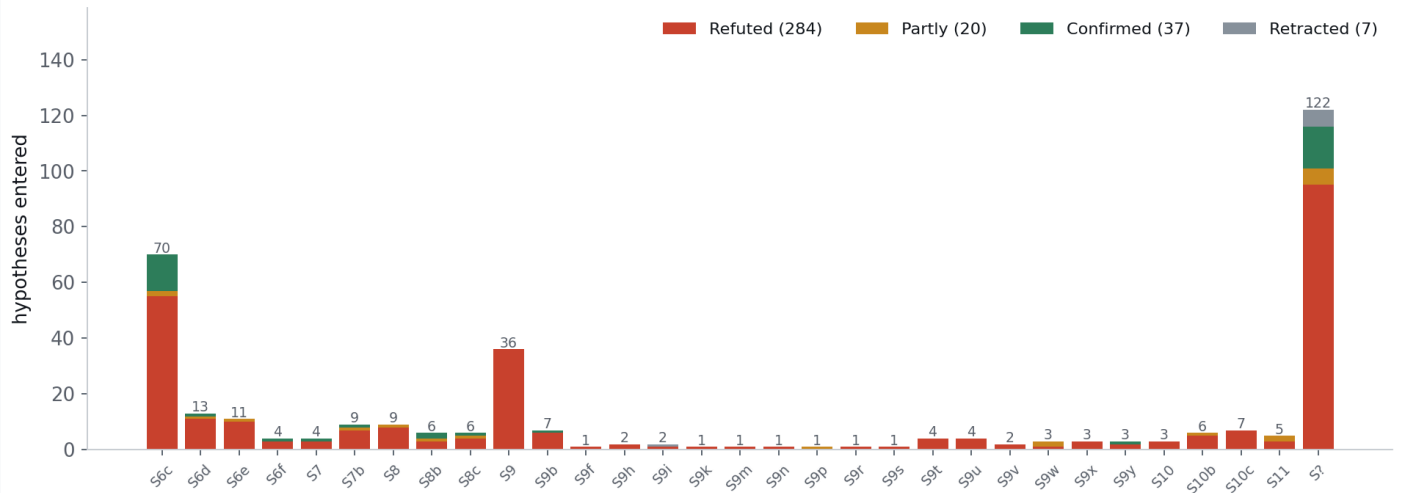


The project ran in two blocks: locomotion in June and the brain, the eye and the hunt in September. The hatched bar is an artefact worth being explicit about — git blame reports when a line was last written, and the living map was created in a single commit that transcribed every hypothesis tested before it existed.

SESSION	DATE	WHAT IT DID	ROWS	REF.	CONF.	PARTLY
S6c	2026-09-07	the living failure map, and 6b's parameters were a variant's	#1–70	55	13	2
S6d	2026-09-07	the basal-ganglia failure was not a failure	#71–83	11	1	1
S6e	2026-09-07	the winner did not saturate because we were testing the wrong model	#84–94	10	0	1
S6f	2026-09-07	the gecko moved onto the validated selector, and brains 1 and 2 run together	#95–98	3	1	0
S7	2026-09-07	docs: ledger 99-102, the new state, and the paused marker	#99–102	3	1	0
S7b	2026-09-07	7c: the dopamine mechanism was wrong the OTHER way, and brain 3 lands	#103–111	7	1	1
S8	2026-09-07	scouting the eye found a detector that could not see its own food	#112–120	8	0	1
S8b	2026-09-07	the retina keeps position, and the published optokinetic asymmetry	#121–126	3	2	1
S8c	2026-09-07	the tectum, shipped NOT ACCEPTED because it cannot find real prey	#127–132	4	1	1
S9	2026-09-08	the literature refuted the plan, and three checks were guarding nothing	#133–168	36	0	0
S9b	2026-09-08	the oracle measured, the body hash resolved, and a strike that works	#169–175	6	1	0
S9f	2026-09-08	the lag was a frame-rate header, not the animal	#176–176	1	0	0
S9h	2026-09-08	three findings from trying to reproduce the Session 4 walking clip	#177–178	2	0	0
S9i	2026-09-08	the gait profile is a parameter now, and I retract #179	#179–180	1	0	0
S9k	2026-09-09	the shake was the camera watching the trunk bob	#181–181	1	0	0
S9m	2026-09-09	the user was right -- I filmed the wrong gait profile all session	#182–182	1	0	0
S9n	2026-09-09	every clip I sent was the walker this project rejected	#183–183	1	0	0
S9p	2026-09-10	it is the PAIRING, not the policy -- #183 was too broad	#184–184	0	0	1
S9r	2026-09-10	a guard so the broken pairing cannot assemble silently again	#185–185	1	0	0
S9s	2026-09-10	the goal never mentioned the mouth, so the animal never reached it	#186–186	1	0	0

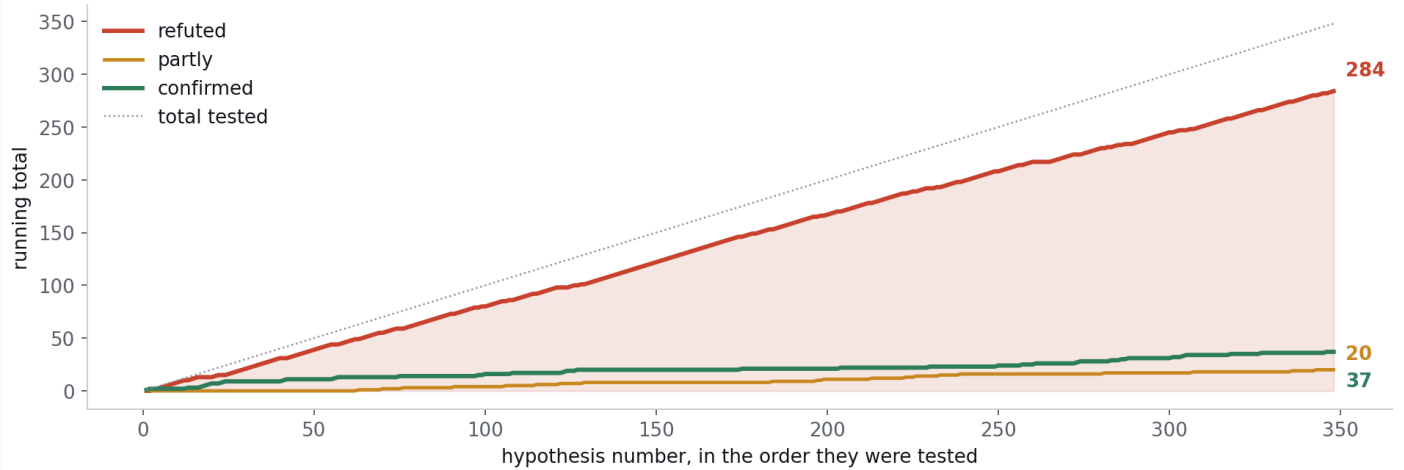
SESSION	DATE	WHAT IT DID	ROWS	REF.	CONF.	PARTLY
S9t	2026-09-10	two published prey rules, one of which cannot do its job yet	#187–190	4	0	0
S9u	2026-09-10	a world with somewhere to shelter, something warm, and something to flee	#191–194	4	0	0
S9v	2026-09-10	brain 5, smell -- and the line it refuses to cross	#195–196	2	0	0
S9w	2026-09-10	brain 6, the day/night clock -- and the gap it refuses to fill	#197–199	1	0	2
S9x	2026-09-10	the walker could always reach -- I retrained it to fix nothing	#200–202	3	0	0
S9y	2026-09-10	the hunt composes -- and the fix I proposed would have broken it	#203–205	2	1	0
S10	2026-09-10	the video was filmed with the wrong walker, and the right one was unreachable	#206–208	3	0	0
S10b	2026-09-10	the eye was watching its own walking, and resolution was never the limit	#209–214	5	0	1
S10c	2026-09-10	the deferred modules were the buildable ones, and I invented a citation	#215–221	7	0	0
S11	2026-09-10	the whole project as one document, and the scoreboard had drifted	#222–226	3	0	2

All 348 hypotheses, by the session that recorded them



Every hypothesis, grouped by the session that recorded it. The tallest bars are literature sweeps, not build sessions — reading refuted more assumptions than coding did.

82 % of everything tried was wrong — and that is the product



Refutations accumulate; confirmations barely move. A project that confirmed most of its guesses would be one that was not testing them.

What was done wrong — the recurring shapes

Eight failure shapes account for most of what went wrong. Each one recurred after being “fixed”, which is why they are worth naming rather than merely counting. The count beside each is how many distinct ledger rows record it.

×5 A default won because nobody named the parameter

The goal oracle (#165), the food oracle, the gait profile (#177), and twice before that (#26, #33). Each time a constructor omitted an argument and the callee's default silently decided the science.

Now: Every one is now passed explicitly and reported in `info` on every step. `common/walker_pairing.py` warns when the measured-broken combination is assembled.

×3 The test compared the code against itself

A class literally named *the detector cannot disagree with the world* read the prey colour from the model, built a patch of that colour, and checked the detector saw it. It never touched the renderer. It passed while the detector scored 0.000 on the real prey — twice, three sessions apart. A world-builder guard required added bodies to be mocap, when what it needed to test was that they add no degrees of freedom.

Now: Assertions now reach the thing being guarded. The world guard tests `nq` and `nv` directly (39 → 39, 38 → 38).

×2 A fix that provably could not do anything

The tectum subtracted a scalar from the motion map to reject self-motion. `argmax(clip(x-c,0,None)) == argmax(x)` for any scalar `c` — identical on 20 000 of 20 000 random maps. It had never changed one reported bearing.

Now: Replaced first with a spatially varying surround, then — when that was also shown to assume uniform flow — with a predicted flow field subtracted from the image.

×3 A number wearing another species' clothes

`cost_of_transport = 0.73` tagged *E. macularius*, verified; it is *Teratoscincus przewalskii*. The human-readable scorecard had it right and the machine-readable registry the code reads did not.

Now: Corrected, and the gap recorded rather than filled.

×4 I explained away a real observation, repeatedly

The user said the walk dragged. The answers given were: camera shake, then half-speed playback, then an under-actuated forelimb they would have to accept. Then the profile. Then the policy. All five were wrong about the cause.

Now: It was the PAIRING. Three of the four profile-policy combinations walk with a front-foot gap under 0.03; the one being filmed had a gap of 0.376.

×2 The map already contained the answer

#71 states outright that the accepted walker IS the open-loop base. #11 measured the trained residual as worse. The evidence file records its controller as *zero residual*. All three were read, and the checkpoint was loaded anyway.

Now: Reading the ledger is not the same as believing it. `GeckoBrainEnv` could not even express *lab profile, no policy* until `use_policy=False` was added.

×4 I asserted something I had not checked

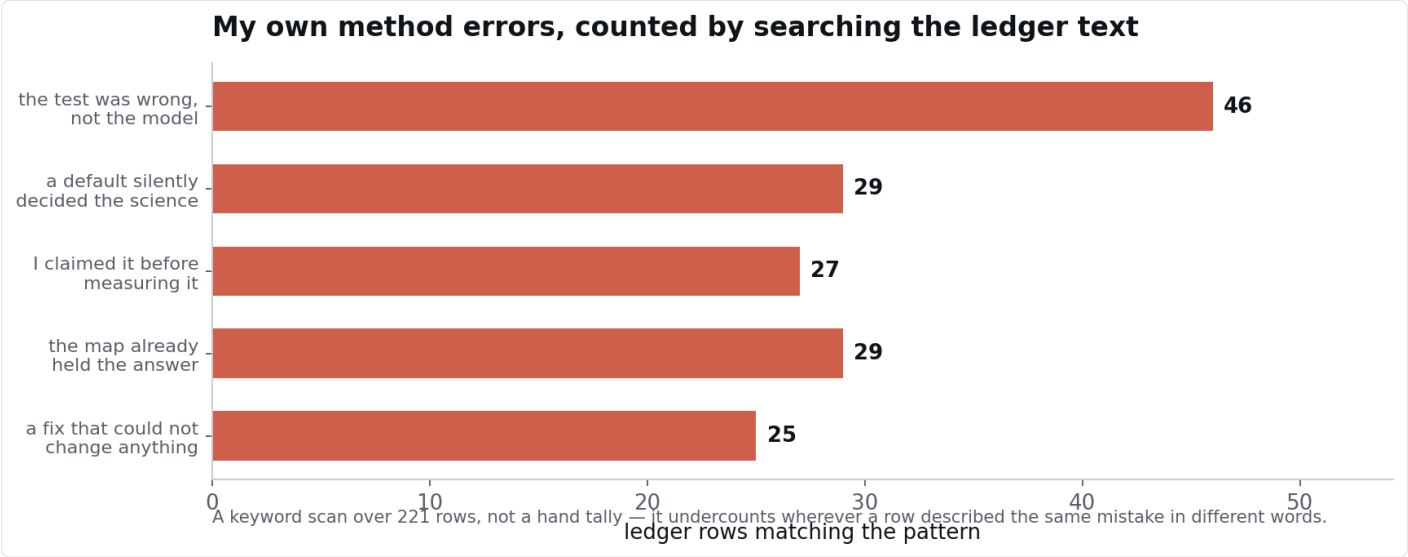
A citation was invented outright — Day, Crews & Wilczynski were said to have studied leopard geckos; they never did (#215). “Nobody has filmed a gecko hunting” was said as fact; Delheusy, Brillet & Bels 1995 exists, $n=6$, 64 fps. “A sharper eye would find the cricket” and “geckos head-bob to range prey” were both told to the user as fact and both refuted.

Now: Every claim in this document that is not measured in this repository is marked as unverified, and the companion prompt asks an outside reader to check the bibliography specifically for this failure.

×3 I built the thing that was not the blocker

The walker was retrained for 2.1 M steps against a reach problem that did not exist — the existing walker reached a mouth-goal 75 times in 160 s, and the retrain was 8× worse (168 mm vs 20.5 mm). `hunt_targeting`, proposed as the approach fix, gave 0 strikes against 29.

Now: Both killed. Both kept in the ledger, and the targeting option kept in the code, defaulted off, with its measurement attached.



The same thing counted mechanically; a keyword scan over all 221 ledger rows. It undercounts, because a row that described the same mistake in different words does not match.

The one that is worst, and why it is listed separately

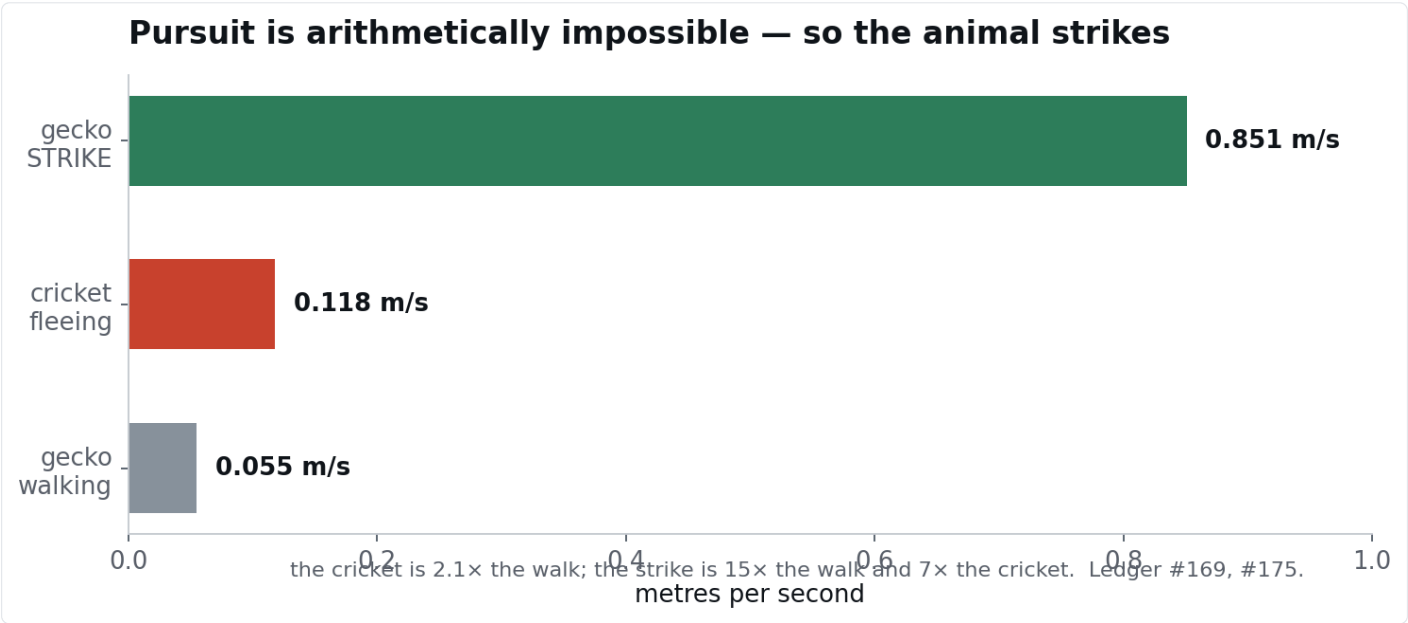
In Session 10c a citation was **invented**. Day, Crews & Wilczynski were asserted to have studied leopard geckos. They never did — their lizard work is three papers on two other families. The confusion is explicable, because David Crews did work extensively on *Eublepharis macularius*, but on temperature-dependent sex determination, not on the brain. The invented citation was then written into a research brief as a known starting point, where it would have been read as established.

It is recorded as ledger row **#215**. It is also the reason Part 14 of this document states plainly that the bibliography is a harvest of strings found in the repository and has **not** been verified against any database, and the reason the companion prompt asks an outside reader to check exactly that first.

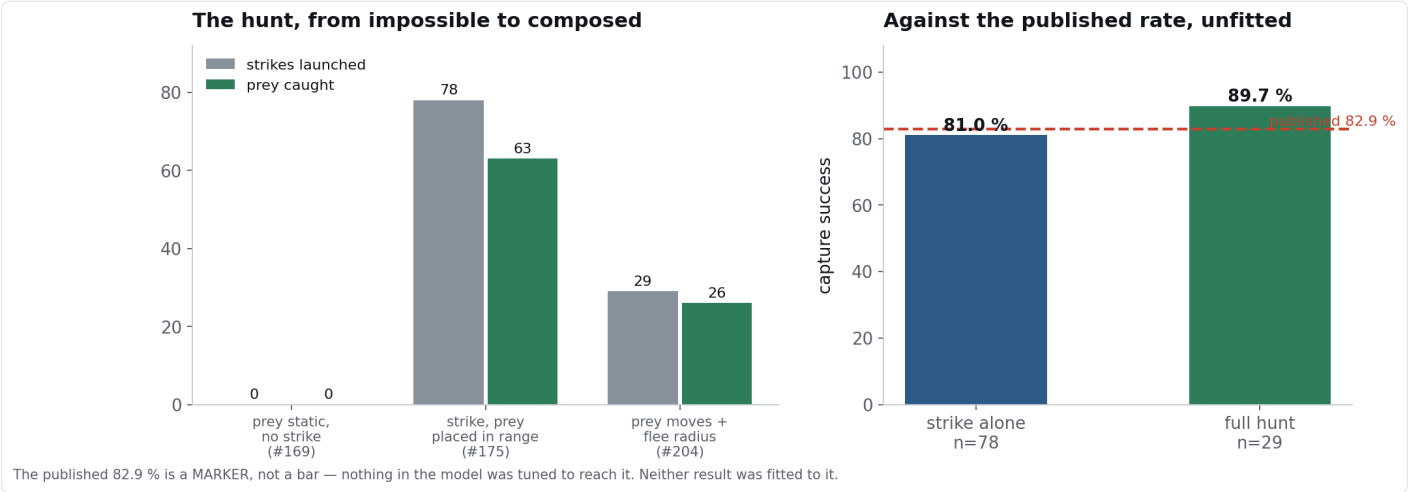
Failure → what it became

The dopamine curve sat 0.10 too high #104–#108		
WAS	WHY	NOW
Three sessions of a basal-ganglia model that would not match the published table.	The mechanism was wrong in the opposite direction to the previous correction. Afferent gain ($1\pm\lambda$) with threshold 0.2.	Mean error 10.3 pp → 0.103 pp , axis slope 1.000.
The gecko could not see its own food #112, #196		
WAS	WHY	NOW
The detector tested for green; the prey had been changed to brown. Exactly 0.000 at every illumination — and the guarding test compared the code to itself.	Read the colour the renderer actually draws — the material, not just <code>geom_rgba</code> .	Two separate occurrences fixed, three sessions apart, and the self-referential test replaced.
The eye could not find the cricket #116–#118, #134		
WAS	WHY	NOW
Four eye configurations, 64 → 512 px. Correlation with true prey bearing ≈ 0 in all four.	It was never resolution. The prey travelled 0.0 mm and was off-image on 81 % of frames.	Prey now walks 398 mm per episode. The eye had something to look at for the first time.
The eye reported the same thing with and without prey #209–#213		
WAS	WHY	NOW
72 % of frames firing at salience 0.4161 with prey; 72 % at 0.4034 with no prey in the world at all . Separation $d = 0.036$.	A local-versus-surround comparison assumes self-motion is spatially uniform. Optic flow expands from a focus of expansion, so the local excess is large everywhere.	An efference copy — predicting the flow from the animal's own yaw and speed and subtracting it — cut false alarms to 3–7 % .
Eight gate contracts failing for four sessions #158–#164		
WAS	WHY	NOW
Reported as eight separate failures; the walking evidence looked unverifiable.	One body check compared raw bytes and raised before the others could run. What differed was a <i>comment</i> .	All 16 pass. The physics hash matched all along: <code>b178bf26...</code>
Pursuit could never work #169, #175		
WAS	WHY	NOW
The hunting channel had no route to success at any parameter value.	Walk 0.055 m/s, cricket 0.118 m/s. The real animal does not chase — it strikes from 2 cm at 0.851 m/s.	Strike built: 81 % capture against a published 82.9 %, unfitted.
Nothing had ever been caught #202–#204		
WAS	WHY	NOW
A strike that worked in isolation and an approach that could not reach 2 cm.	Not the eye, not the walker, not the targeting — all three had been diagnosed as the blocker and all three were wrong. It was prey motion plus a speed-dependent flee radius.	26 of 29 strikes landed — 89.7 % against a published 82.9 %. The first time the animal caught anything.
The walk dragged, for six sessions of denial #182–#184, #206–#208		
WAS	WHY	NOW
The user said so from the first video and was told five different reasons why it did not.	Front-foot duty gap 0.376 on the ONE combination being filmed. The other three combinations sit under 0.03.	<code>common/walker_pairing.py</code> now warns whenever that pairing is assembled, so the next session cannot repeat it silently.

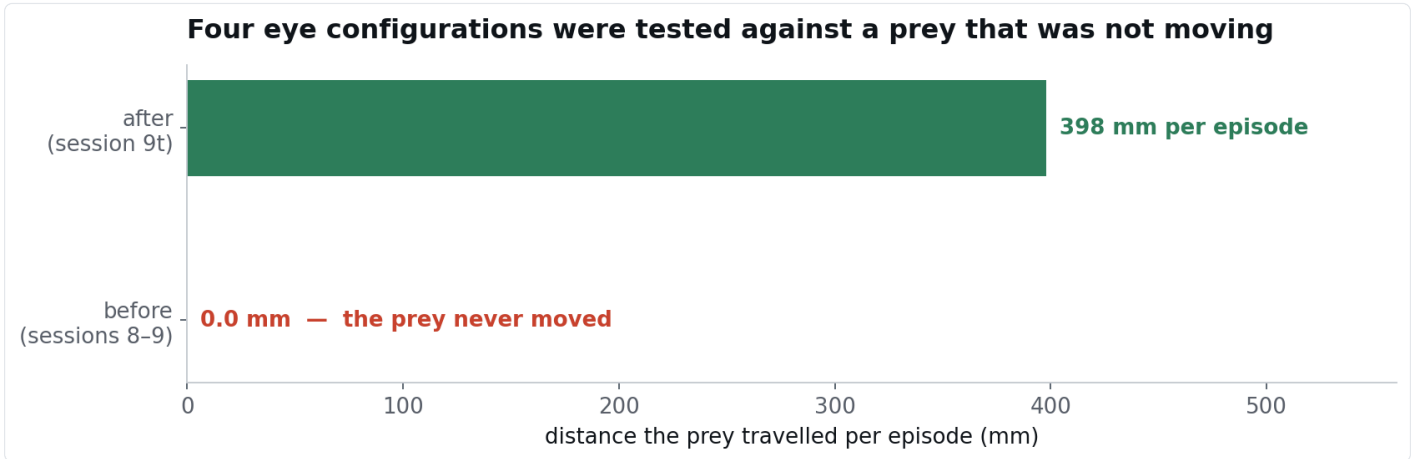
The measurements worth keeping



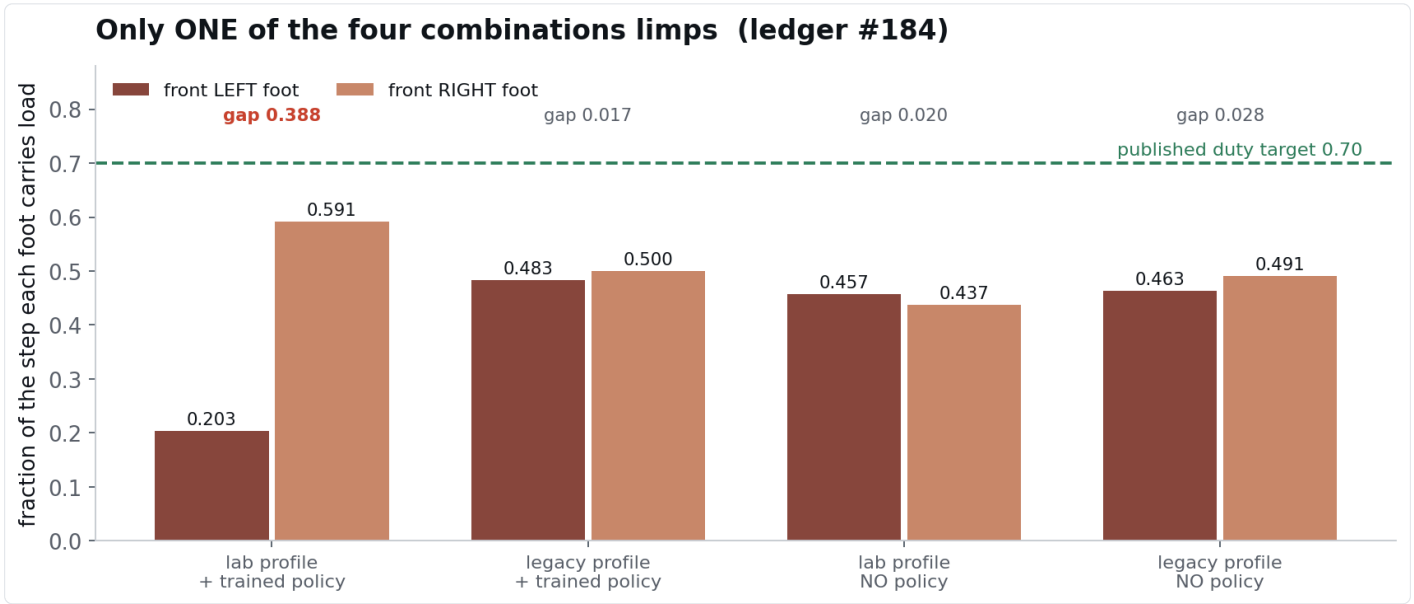
The arithmetic that forced a strike module to exist. Ledger #169 measured 60 s of pursuit: closest approach 51.1 mm, zero strikes. The gecko could not physically reach striking distance by any route.



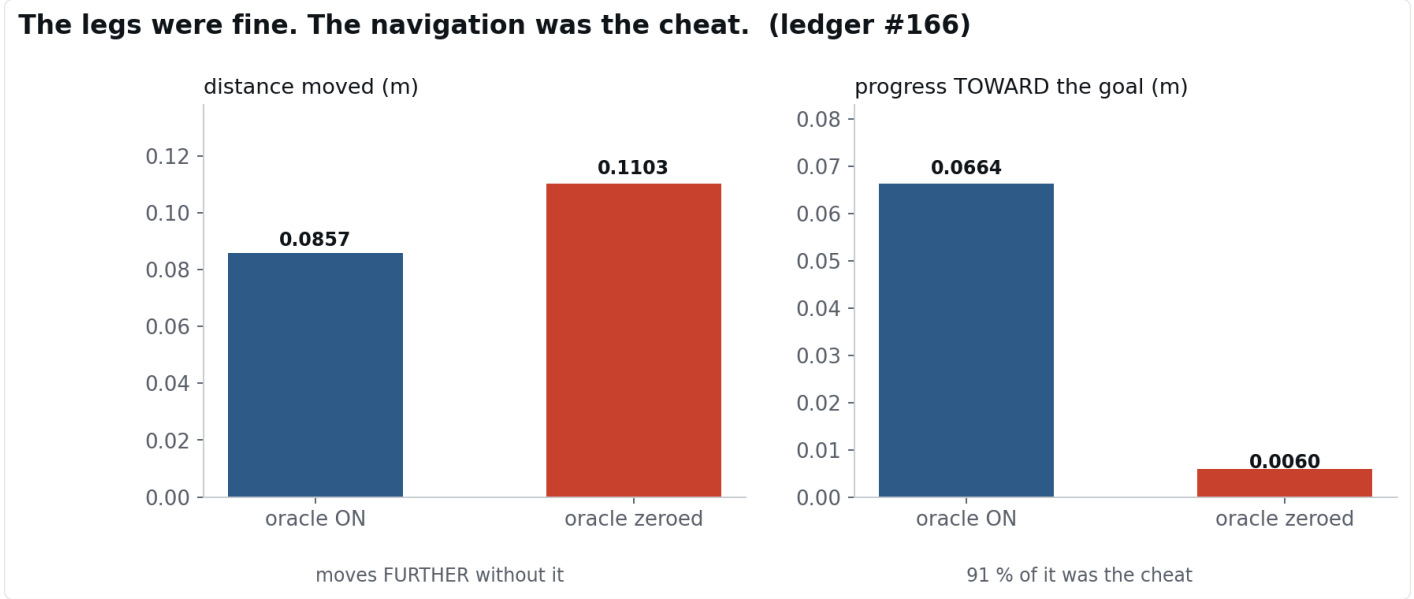
The hunt going from impossible to composed. What unblocked it was prey motion plus a speed-dependent flee radius — not the eye, not the walker and not the targeting, all three of which had been diagnosed as the blocker (#134, #174, #202).



The measurement that makes three sessions of vision work unreadable: every eye configuration up to that point was tested against prey that never moved and was off-image on 81 % of frames.



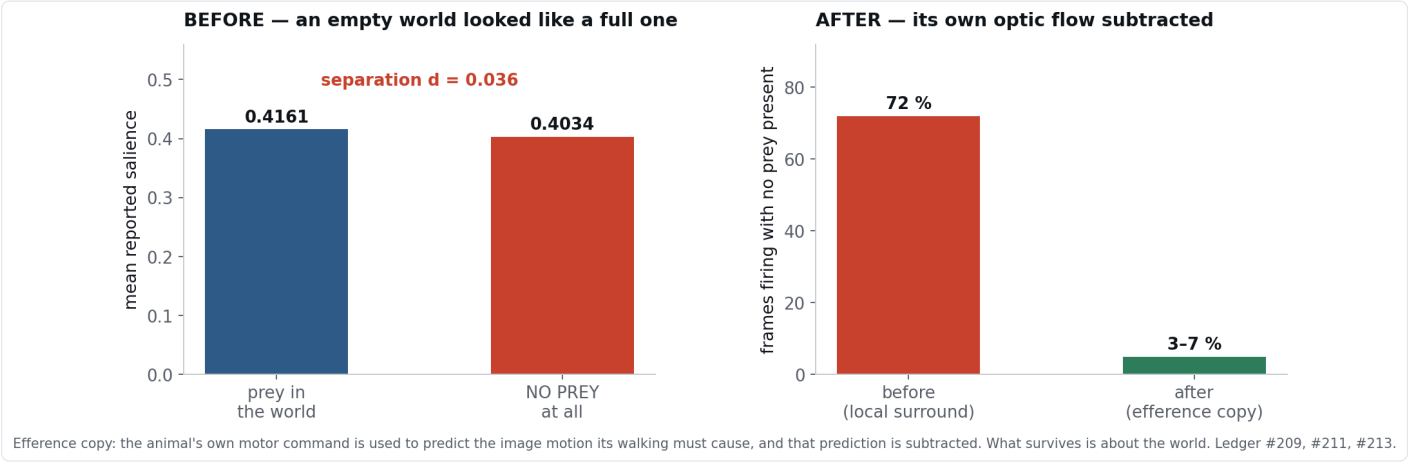
The defect the user found by watching the animal move, then measured. Only ONE of the four profile-and-policy combinations limps — and it was the one in every video.



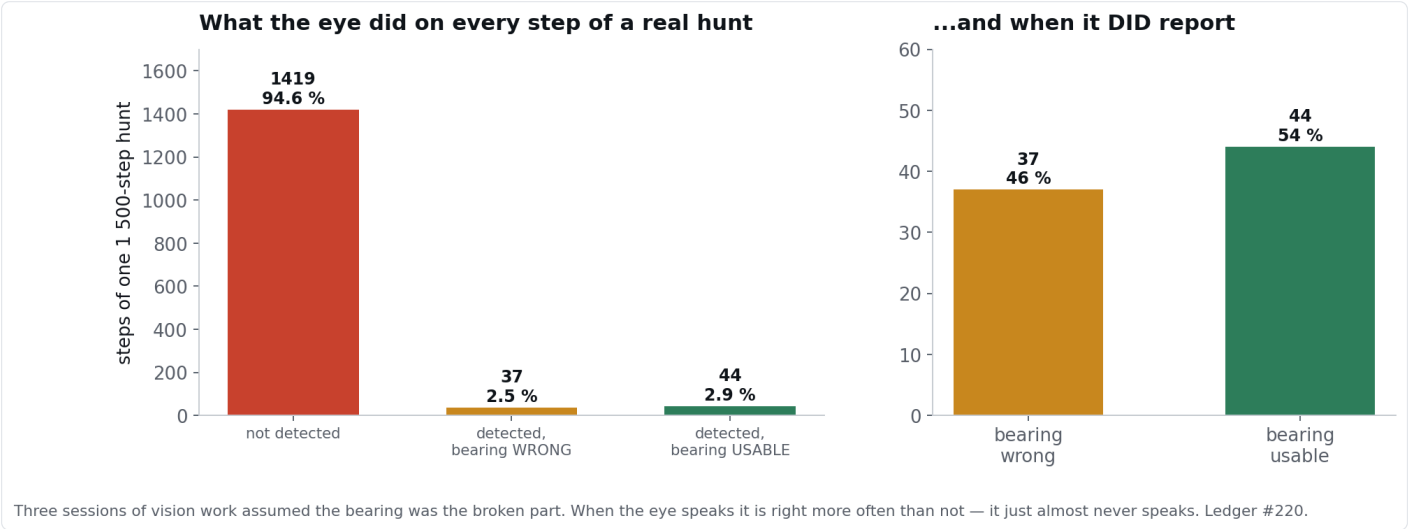
Zeroing the privileged goal channel. The animal moves FURTHER without it and arrives nowhere: 91 % of all progress toward the goal was the oracle.

The eye — why it still cannot find the cricket

This is the only module in the project that is built and recorded as not working, and it has been misdiagnosed four separate times. What follows is what is actually known, in the order it was found out.



The measurement that ended three sessions of arguing about resolution. Ledger #209: the eye reported the same thing with prey and with no prey in the world at all. It was reporting its own walking.



And the measurement that ended the argument about bearing quality. Ledger #220: over 1 500 steps of a real hunt the eye reported nothing on 94.6 % of them — but when it did report, the bearing was usable more often than not.

The number the oracle removal has to be judged against (ledger #221)

0.0146

1.0 = walks straight at it

Path efficiency in a real hunt: straight-line ground closed, divided by ground actually walked.
The animal covers about seventy times more distance than it closes — while satisfying the published 'approaching' criterion on 47 % of steps. It is pointed the right way and getting almost nowhere.

What has been ruled out, and what is left

Ruled out: resolution

Held still, peak salience is 0.36 to 0.83 at every range tested — including ranges where the prey covers half a detector cell. Four configurations from 64 to 512 pixels all correlated ≈ 0 with true bearing (#116–#118).

Ruled out: bearing quality

Of 81 detections in a real hunt, 44 were usable and 37 were wrong. When the eye speaks it is right more often than not (#220).

Ruled out: the walker

The existing walker reaches a mouth-goal 75 times in 160 s. A 2.1-million-step retrain aimed at the supposed reach problem was 8× worse and was killed (#200, #201).

Ruled out: targeting

Aiming the walker at the prey gave 0 strikes against 29 for aiming past it. Arriving at a goal means stopping at it, and the mouth needs the animal to keep going (#203).

Fixed: self-motion

An efference copy cut false alarms from 72 % of frames to 3–7 % and flipped the still-versus-moving effect to the right sign for the first time (#213).

Left: the detection rate

5.4 % of steps. The same eye that peaks at 0.67–0.83 on a stationary animal averages 0.02 during a hunt, because the prey is out of view most of the time. This is the blocker, and it is the only one.

What the fix is specified to be

Two parallel detector channels over the same image, differing in receptive-field size and in what they are wired to. A coarse array of $\sim 30^\circ$ patches that only says *a small thing is moving over there* and gates whether a hunt starts at all; and a dense sub- 16° direction-selective array that supplies the bearing error during pursuit. The published test for it is a lesion double dissociation: suppressing the wide channel should stop approaches being initiated from beyond 22 cm without harming approaches already under way, and suppressing the narrow channel should do the opposite. Every tuning constant in it is a mouse, a zebrafish larva or a toad — the full parameter list, with that stated on each row, is in Part 10.

The three modules that are specified and not built

Memory and learning can be built on evidence, and the deferral should end. In the last eighteen months three papers on the actual target species produced usable numbers: leopard geckos improve about 9 percent per trial on a simple beacon task, reach a ceiling of roughly 3x faster and 4.6x straighter after 20 spaced trials, keep a learned location for 2 to 4 months, and are back to naive by 6 to 14 months. That is enough to build both modules honestly. But the numbers come with a timescale problem the project must face head on: the memory evidence lives at months, and a hunt lives at seconds. Nothing has ever been measured in any lizard at intervals shorter than two months, so a working-memory or within-session buffer is invention no matter how reasonable it looks. The irony is that the urgent module is the unsupported one. There is no size tuning, no speed tuning, no contrast threshold,

The prey detector — the blocker

Build two parallel detector channels over the same visual input, both tuned to small moving objects, differing only in receptive-field size and in what they are wired to. Channel A, wide-field: one unit per patch of roughly 30 to 34 degrees across (published receptive fields of 700 to 900 square degrees), tuned to objects of about 4 to 5 degrees, NOT direction-selective, and with no fine localisation. Its only output is 'something small is moving somewhere in this patch'. Channel B, narrow-field: a dense retinotopic array of units covering under 16 degrees each (under 200 square degrees), tuned to the same 4 to 5 degree object size, direction-selective, and precisely localised. Its output is a continuous bearing error. Wire them differently, because the published lesion result is a clean double dissociation: silencing wide-field cells stopped approaches being INITIATED from beyond 22 cm without hurting accuracy once an approach began, while silencing narrow-field cells degraded targeting accuracy, cut the probability of interception given an approach, made the animal pause within 5 cm of the prey, and stopped it re-initiating after a failed contact. A third class (parvalbumin cells, tuned to larger and darker stimuli) had no role in prey capture at all. So: Channel A gates whether a hunt starts and at what range; Channel B supplies the steering signal during pursuit and the terminal commitment. An independent two-channel size split is published in a fish larva with a crossover at 6 degrees: the prey-feeding retinal channel peaks at 2 to 3 degrees and falls off above 6, while a sibling channel does not respond below 6 degrees at all. Use 6 degrees as the small-versus-large crossover, route small to approach and large to avoid, and reproduce the published sign reversal: a 1 degree spot evokes a turn of about 20 degrees TOWARD, a 10 degree spot a turn of about 60 degrees AWAY. Finally, the detector is not one filter across the whole hunt. The same cricket grows from roughly 5 degrees at detection to roughly 45 degrees at the strike (a 1.8 cm cricket at the published 2.03 cm lunge distance), and separate cell populations own each end, so build a small-object detection filter and a large-object strike-commit filter as different units.

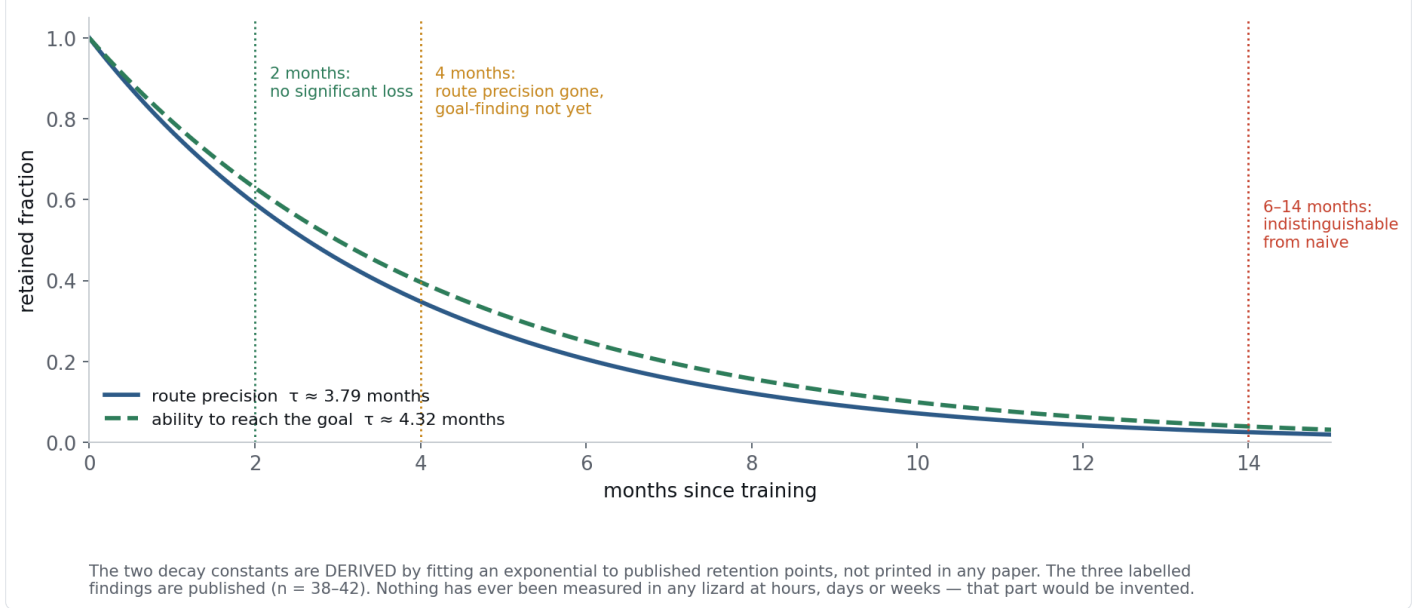
PARAMETER	VALUE	PROVENANCE
Wide-field receptive field size	700 to 900 square degrees, equivalent to about 30 to 34 degrees diameter. Responds to small light or dark objects moving anywhere within that region.	PUBLISHED, proxy (mouse superior colliculus). No gecko equivalent exists.
Narrow-field receptive field size and direction selectivity	Under 200 square degrees, equivalent to under about 16 degrees diameter. Direction-selective, responds to small light or dark spots.	PUBLISHED, proxy (mouse superior colliculus)
Preferred object size, both channels	4 to 5 degrees. IMPORTANT CAVEAT: the size ladder tested was 2, 4, 8, 16, 32 degrees for flashed spots and only 4, 8, 16 degrees for moving spots, so the reported optimum is the argmax over a coarse discrete ladder and the true optimum may be smaller than the smallest size tested. Use a broad kernel, not a sharp 4.5 degree peak.	PUBLISHED, proxy, with a known sampling limitation
Behavioural double dissociation, for validating a simulated lesion	Wide-field suppression reduces the probability of initiating approaches from beyond 22 cm, without harming accuracy once started. Narrow-field suppression degrades targeting accuracy, reduces interception given approach, produces pauses within 5 cm of the prey, and reduces re-initiation after a failed contact. Parvalbumin suppression has no effect on any prey-capture measure. Firing-rate suppression was about 35 percent per cell in all three groups.	PUBLISHED, proxy
Small-versus-large channel crossover	6 degrees. The prey channel peaks at 2 to 3 degrees and responds much less above 6 degrees; the sibling channel does not respond to small dots and only begins responding above 6 degrees. Non-overlapping.	PUBLISHED, proxy (zebrafish larva retinal arborization fields AF7 and AF9)
Behavioural size tuning	Optimum 3 degrees over a tested range of 0.5 to 30 degrees; response 'strikingly reduced' by 10 degrees.	PUBLISHED, proxy
Speed bandpass	Peak 90 degrees per second; minimal response below 12 or above 360 degrees per second. Same tuning in the behaviour and in the retinal channel that drives it. Implement as a bandpass gate with a 90 deg/s peak and roughly 1.5-decade skirts.	PUBLISHED, proxy. This is the most complete published speed band available for any prey detector.
Sign reversal with size, and per-presentation response probability	A 1 degree moving spot evokes a low-amplitude turn of about 20 degrees TOWARD; a 10 degree spot evokes a high-amplitude turn of about 60 degrees AWAY (1 versus 5 degrees p<0.001; 1 versus 10 degrees p<0.001). Response rate was 22 percent for small stimuli and 24 percent for large.	PUBLISHED, proxy
Approach-state entry gate	Enter approach when all three hold: range is decreasing, absolute prey bearing is under 90 degrees, and self-speed exceeds threshold (5 cm/s in the 2016 study, 3 cm/s in 2019, to be scaled to body size). Contact defined as range under 3 to 4 cm. This conjunction predicted 100 percent of interceptions.	PUBLISHED, proxy. A validated operational definition, directly reusable both as the policy gate and as the scoring rule.
Angular size at which vision starts guiding approach	6 to 8 degrees, from a 2 cm cricket at 15 to 20 cm range. The authors themselves call this a rough estimate.	DERIVED (geometric), proxy. The single most load-bearing number in any 'what counts as prey' rule and also the weakest. Order-of-magnitude anchor only.
Terminal bearing accuracy target	11.4 +/- 1.4 degrees residual bearing error at the end of a visually guided approach, versus 41.1 +/- 1.2 degrees with vision removed (near chance, given bearing must be under 90 degrees to count as an approach). Bearing decreases sharply once range falls below 15 to 20 cm.	PUBLISHED, proxy
Bimodal approach initiation	Approach-start distance peaks at 9.6 +/- 1.5 cm and at 31.8 +/- 9.4 cm; 86 +/- 5 percent of approaches started beyond 22 cm. Approach-start bearing is also bimodal, at 12 +/- 10 degrees or 75 +/- 16 degrees.	PUBLISHED, proxy
Strike release distance and kinematics, TARGET FAMILY	Lunge released at 2.03 +/- 0.08 cm range. Peak strike velocity 85.1 +/- 2.19 cm/s. Peak acceleration 46.6 +/- 1.97 m/s squared. Peak acceleration occurs 16.34 ms before contact, peak velocity 9.03 ms before contact. Strike success 77.0 to 78.6 percent overall.	PUBLISHED, proxy (same family, Eublepharidae, different genus). The best-transferring numbers in the entire corpus, and the only quantitative prey-capture data anywhere near this animal.
Strike release scales with prey escape capability	Strike distance is significantly greater for evasive prey (crickets) than non-evasive prey (mealworms). Whole-body lunge for crickets versus head, neck and forelimb movement only for mealworms. Cricket capture success 116 of 140 trials (82.9 percent); mealworm 53 of 53 (100 percent). Subjugation shake frequency 12.30 +/- 2.06 per second, peak shake velocity 1.18 +/- 0.17 m/s, amplitude 10.78 +/- 1.57 mm.	PUBLISHED, proxy (same family). Sign of the effect only; the modulation function is not published.
Multiple targets, neural mechanism	Divisive normalisation across the target map, with global (not lateral) inhibition. The strongest competitor suppressed responses by 61 +/- 7 percent on average. About half the population (48 percent, n=51) signalled the strongest stimulus with a near-binary switch whose transition range was 4 deg/s or less in competitor loom-speed units; the rest (52 percent, n=56) were gradual. Discriminability at a 2 deg/s strength difference: d-prime 2.48 +/- 0.38 switch-like versus 0.8 +/- 0.18 gradual.	PUBLISHED, proxy (barn owl optic tectum)

PARAMETER	VALUE	PROVENANCE
Multiple targets, selection policy	With two equal-strength competitors: 80 percent winner-take-all, 20 percent vector averaging. With unequal competitors, the stronger is chosen 70 percent of the time. Tectal cells are more switch-like than retinal cells (R=0.72 versus 0.47).	PUBLISHED, proxy, AND MEASURED ON THE WRONG BEHAVIOUR. The 80/20 split and the 0.7 bias come from escape responses to competing looming stimuli, not from prey selection. The same paper shows the competition mechanism extends to prey-like stimuli only at the imaging level.
Target choice in a dense prey field, and the strike acceptance window	With about 100 prey items present, the animal almost invariably chose the CLOSEST one, subject to an acceptance window. Hunt-initiation prey position: azimuth mean 0.7 deg, SD 48.4 deg; altitude mean 19.9 deg, SD 19.7 deg; distance mean 3.4 mm, SD 1.6 mm. Strike zone (95 percent CI during successful strikes): azimuth 0.9 deg, altitude 17.4 deg, distance 0.870 mm. Note the acceptance window is WIDE in azimuth and NARROW in elevation.	PUBLISHED, proxy. The nearest-target rule is stated qualitatively, without a probability.
Motion is a gain term, not a hard gate	In a mammal, immobile fresh-frozen crickets were contacted in times indistinguishable from live crickets: 8.06 +/- 1.08 s versus 7.6 +/- 1.34 s, n=8, p>0.05. In toad and fish larva, motion is obligatory. No gecko data settles which regime applies.	PUBLISHED, proxies that disagree
Configural rule: elongation axis relative to motion axis	A moving stripe whose long axis lies along the direction of motion is prey; perpendicular is not. The discrimination is described as 'largely invariant, thus far investigated' to movement direction in the plane, contrast polarity, and stripe velocity. Compute the configural score independently of, and before, the speed and contrast gains, and make it blind to the sign of contrast.	PUBLISHED, proxy (toad). Qualitative only; no response ratio, no stimulus dimensions, no sample size is obtainable.
Form of the toad response law, corrected	Turning activity R-bar (turning reactions per minute) is characterised by THREE SEPARATE univariate relations, each measured with the others held constant: R-bar = f(angular velocity), R-bar = f(background contrast), R-bar = f(angular size). It is NOT a single joint function of the three. The source also says the dependence is on these 'among other things', so three is not an exhaustive parameter list. The tuning curves themselves are in paywalled figures and were not obtained.	PUBLISHED form, proxy; numeric curves NOT_FOUND
Motion front end for a lizard	Correlation-type elementary motion detector array with 0.3 degrees of visual angle spacing and a 0.1 second time constant. The authors state the model is sensitive to local contrast and viewer-object distance and that additional processing is needed to separate relevant from irrelevant motion under natural conditions.	PUBLISHED, proxy (distant lizard) and OPTICALLY MISMATCHED. Anolis is diurnal and foveate; a leopard gecko is nocturnal and afoveate with a large pupil and short focal length. Use the architecture, invent the constants, and say so.
Outcome statistics to calibrate against (NOT detector parameters)	Eublepharid strike success 77 to 83 percent (82.9 percent on crickets, 100 percent on mealworms). Mammalian capture given contact 38 +/- 16 percent. Fish larva response rate to an optimal small stimulus 22 percent. Time to capture in a 45 x 38 cm arena: 11 +/- 2 s with vision, 36 +/- 9 s in darkness, improving about fourfold to 13 +/- 1.1 s over four days of experience.	PUBLISHED, mixed proxies
False positive rate	NOT FOUND anywhere. No published measurement exists, for toad, mouse, fish larva, or any lizard, of how often a predator strikes at a non-prey object.	NOT_FOUND

What would have to be invented

- Every gecko-specific tuning value. No size tuning, speed tuning, contrast tuning, elevation tuning, or detection latency has ever been measured for prey detection in ANY gecko, ANY nocturnal lizard, or any squamate other than the Anolis motion model. The whole size-and-speed specification is imported from a fish larva, a rodent and a toad.
- Visual acuity for this species. No value in cycles per degree exists for Eublepharis macularius or any eublepharid. The only guidance is a comparative statement, itself unverified in full text, that nocturnal geckos have lower acuity than diurnal geckos. So whatever angular threshold is imported from mouse (6 to 8 degrees) or fish (3 degrees), the gecko's should probably be LARGER, and the size of that correction is unmeasured.
- Contrast threshold and absolute-luminance dependence at low light. Nothing exists for any nocturnal lizard.
- A false positive rate, and any distractor-rejection statistic.
- A below-the-horizon prey prior. The two published elevation figures point the other way in eye coordinates: prey is held at about +5 degrees relative to the mouse's eye and at about +20 degrees altitude in the fish larva. Both are consequences of head posture, not a world-frame rule.
- Any capacity limit on how many prey items can be tracked at once.
- A prey-selection version of the 80/20 winner-take-all split. The published number is from escape from looming stimuli.
- The Ewert worm-versus-anti-worm tuning curves and any worm-to-anti-worm response ratio. The figures are unreachable; the invariance rule is the only citable part.
- The modulation function linking measured target speed to strike release distance. Only its sign is published.
- Whether the leopard gecko hunts visually at all in the first place. The only quantitative sensory study on this species is about PREDATOR detection and found chemical cues dominant with vision merely modulatory. A vision-first prey detector is an explicit modelling choice, not something the target-species literature supports.

Memory has published numbers after all (ledger #216)



Memory

A leopard gecko learns where one goal is and keeps that knowledge for a couple of months, then loses it. Detail goes first: after four months the animal still gets to the goal but by a longer, sloppier route. After six to fourteen months it performs like an animal that never learned anything and has to be retrained from scratch. It does not carry a map. Only about one third of individuals orient by the relationship between several cues; the rest lock onto one landmark or fall back on a search pattern the authors themselves say may involve no spatial knowledge at all. It does notice when an object it knows has moved, so it holds what-and-where information about objects even without a map. Individual differences are real but tiny and they fade with age.

PARAMETER	VALUE	PROVENANCE
Retention at 2 months (adult-trained), spatial goal location	No significant loss. Latency at 2 months vs end of training: $t=-0.71$, $df=536$, $p=0.9537$. Path length: $p=0.7011$.	PUBLISHED, target species, single lab
Loss at 6 months (adult-trained)	Path length statistically identical to naive baseline (Tukey $p=0.9999$; LME coefficient $p=0.8903$). Latency significantly LONGER than naive ($p=0.0087$). Required 30 retraining trials to recover.	PUBLISHED, target species
Retention at 4 months and loss at 14 months (juvenile-trained)	4 months: latency $HR=0.596$ (95% CI 0.35-1.03, $p=0.068$, not significant) but path length $HR=0.49$ (0.28-0.85, $p=0.0044$, already degraded). 14 months: latency $HR=0.27$ (0.14-0.54, $p<0.001$), path $HR=0.19$ (0.09-0.38, $p<0.0001$); indistinguishable from the very first naive trials (latency $p=0.917$, path $p=0.950$).	PUBLISHED, target species
Two-component decay: precision decays before existence	Route efficiency (path length) degrades significantly at 4 months while the ability to reach the goal at all (latency) does not. Model at least two components with different rates, not one scalar.	PUBLISHED, target species
Exponential decay time constant (no published constant exists for any reptile)	Path length: 34.8 percent of the learned gain remaining at 4 months, giving $\tau = 3.79$ months (115 days), half-life 2.63 months (80 days). Latency: 39.6 percent remaining, $\tau = 4.32$ months (131 days), half-life 2.99 months (91 days). Use the path-length figure. Valid only over 2 to 14 months.	DERIVED by interpolation from two published papers. No reptile study has ever fitted a forgetting function. Assumes an exponential form that was never tested and treats a hazard ratio as a linear index of memory strength.
Fraction of individuals using allothetic (multi-cue, relational) orientation	About one third. The rest locked onto a single cue or may have used a non-cognitive motor search the authors could not rule out. Under progressive cue removal only 7 of 20 individuals were measurably disrupted at any stage (2 at one cue removed, 2 at three removed, 3 at all removed).	PUBLISHED, target species, single low-visibility journal, no DOI, not PubMed indexed
Cost of invalidating the near-field cue set	Displacing the whole arena lengthened path by $\beta=0.4556$ on log path length (95% CI 0.149-0.762, $t=2.9173$, $p=0.0036$), roughly a 58 percent increase. Moving the single most prominent distal cue (a 60 W light) had no group-level effect ($\beta=0.1235$, $p=0.4361$). Removing proximal cues one at a time had no group-level effect at all.	PUBLISHED, target species
Individual repeatability of spatial success (between-individual consistency)	$R=0.192$ (95% CI 0.073-0.312) juvenile pre-training; $R=0.071$ (0.012-0.13) juvenile training; $R=0.092$ (0-0.189) subadult; $R=0.03$ (0-0.122, $p=0.236$, not significant) adult. So 81 to 97 percent of variance is within-individual noise.	PUBLISHED, target species
Object-location change detection (what-and-where without a map)	Geckos habituated to three objects spent more time near an object that had been moved to a new location, and more time near a familiar object that had been swapped for a novel one. Direction only; no effect size available.	PUBLISHED, target species, paywalled, sample size not obtainable
Anatomical substrate and its timescales	The medial cortex is the hippocampal homologue and is neurogenic in this species. Neuroblast migration from the ventricular zone to the medial cortex takes about 30 days; new neurons survive at least 140 days.	PUBLISHED, target species, anatomy only

What would have to be invented

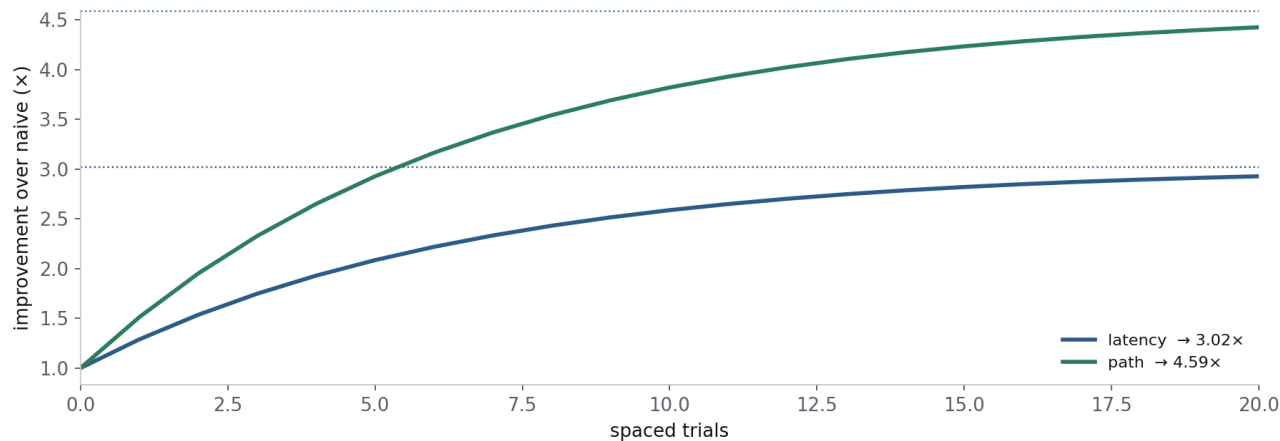
- Any decay at hours, days or weeks. Nothing has ever been measured in any lizard shorter than two months, and that is exactly the window a hunt occupies. If the module remembers where a cricket was five minutes ago, that rule is invented.
- Working memory, a short-term buffer, or any capacity limit on how many items are held.
- Place cells, grid cells, head-direction cells, or any metric map. Zero spatially tuned single units have ever been recorded in any reptile.
- Path integration, dead reckoning, or a homeward vector. Never demonstrated in any reptile, and explicitly rejected in the one lizard tested for it (Ellis-Quinn and Simon 1989, *Sceloporus jarrovi*, n=108).

- Retreat-site fidelity, nightly return-to-shelter probability, or home range size. No field study of this species exists at all.
- Memory consolidation or replay during sleep. The word consolidation appears zero times in the only paper that ever recorded a sleeping leopard gecko.
- Interference, retroactive or proactive, and any reversal cost for this species.
- A savings coefficient for relearning after full forgetting.
- Any coupling between body temperature and forgetting rate. The single closest experiment argues against it: fire salamanders held 100 days at 4 C forgot no more than controls at 20 C (Wilkinson et al. 2017, Sci Rep 7:40079, n=9).

What the model must not claim

- That decay is being modelled, if the simulation clock never reaches months. A tau of 115 days inside an episode measured in minutes is a term whose value is numerically indistinguishable from zero. This is the same class of defect as the scalar subtraction that could not move an argmax: the code runs, the parameter is real, the behaviour is unchanged. If the clock does not reach months, either run a compressed clock and say so, or do not ship the decay term.
- That the gecko has a cognitive map or allocentric place memory. About two thirds of individuals showed no evidence of it, and the authors explicitly cannot rule out a non-cognitive motor search.
- That memory duration is a reptile property. It is a species property. Tortoises retain for up to nine years, monitors for twenty months, this animal for two months. Fast forgetting is the ecologically correct default for a semi-arid seasonal habitat, not a deficiency to engineer away.
- That the medial cortex module causes the memory behaviour. The anatomy is described in this species and the function has never been tested in it. The only two reptile lesion studies contradict each other and neither is a gecko.
- That the retention numbers are replicated. Both papers come from one lab, one captive colony descended from about 90 wild-caught Pakistani founders, one apparatus, and one forced-swim task, with an overlapping author list. Treat them as a single evidence stream.
- That any of this generalises across sexes. The 2023 and 2025 cohorts are 41 females to 1 male.

Learning: hazard ratio 1.09 per trial, ceilings published (ledger #216)



PUBLISHED: the per-trial hazard ratio ($\chi^2 = 16.43$, $n = 38-42$) and both ceilings. The CURVE SHAPE between them is drawn here as an exponential and is INVENTED — no paper prints the trajectory. Every learning number for this species comes from escaping water, not from hunting.

Learning

A leopard gecko gets better at finding a fixed goal, slowly and unevenly. On the simplest version of the task, with one landmark right next to the goal, its chance of getting there rises about 9 percent per trial. Over 20 trials it ends up roughly three times quicker and takes a route about four and a half times more efficient, and then it stops improving. Roughly a quarter of individuals never engage with the task at all. In every published analysis, how motivated the animal was on that trial predicted performance more strongly than how much it had learned. It learns places and boundary shapes far more cheaply than it learns arbitrary visual patterns. Its defensive behaviour is not learned at all: an overt threat display in this species is triggered only by smell, with vision and touch able to raise sampling but never to fire the display on their own.

PARAMETER	VALUE	PROVENANCE
Per-trial acquisition rate, beacon task	Hazard ratio 1.09 per trial ($\chi^2=16.43$, $df=1$, $p<0.0001$), about a 9 percent rise per trial in the per-unit-time probability of reaching the goal. On a log-hazard scale, 0.086 per trial.	PUBLISHED, target species. This is the single most directly usable learning number in the whole corpus, and it is unreplicated.
Training length to asymptotic improvement	20 spaced trials produce a measurable group-level improvement. This is NOT a trials-to-criterion figure; 20 was the designed length of training.	PUBLISHED, target species
Trial spacing regime the rate was measured in	Adults: maximum one 8-minute trial per WEEK, so 20 trials spanned about 20 weeks. Juveniles: 2 to 4 trials per week, so about 5 to 10 weeks.	PUBLISHED, target species
Ceiling on how much experience reshapes behaviour	Latency hazard ratio 3.02 (95% CI 1.67-5.49, $p<0.001$). Path length hazard ratio 4.59 (95% CI 2.54-8.27, $p<0.0001$). Final three trials versus first three trials of 20.	PUBLISHED, target species
Correct learning readout	Path length is significant across training phase ($F=24.157$, $df=538$, $p<0.0001$; also $F=12.727$, $df=207$, $p<0.0001$ in the 2023 cohort). Latency is NOT ($F=0.326$, $df=208$, $p=0.5686$ in adults).	PUBLISHED, target species
Motivation gates performance more strongly than learning does	Number of motivational prods: $F=682.884$ (2023 cue-removal model), $F=280.490$ and $F=350.221$ (2025 path and latency models), $\chi^2=197.134$ (2026 latency model), all $p<0.0001$. Compare the training-phase term in the same models: $F=10.228$ to 24.157, $\chi^2=33.853$.	PUBLISHED, target species
Engagement failure rate	42 of 56 (75 percent) completed training. Upstream of that, only 56 of 140 screened animals (40 percent) swam actively enough to enter the study at all.	PUBLISHED, target species
Spatial reference frame: boundary geometry beats landmark features	Trained with both available, geckos followed arena geometry and ignored the feature when the two conflicted. With geometry alone they chose the trained corner or its rotationally opposite twin. Direction only; no sample size, arena size, trial count or percentage is recoverable.	PUBLISHED, target species, single author, paywalled, never replicated
Spatial learning is cheap, arbitrary visual learning is expensive	About 20 trials for a place in a leopard gecko, versus 477-486 trials for an arbitrary visual pattern in a skink, where 4 of 6 animals never reached criterion at all.	PUBLISHED, proxy (distant, Scincidae), corroborated in direction by the target-species geometry result
Serial reversal improves with practice, in the same family	Banded geckos improved significantly across eight successive spatial reversals: errors to criterion $F(8,32)=3.12$, $p<.025$; trials to criterion $F(8,32)=2.67$, $p<.025$; linear negative slope by Duncan post hoc. Mean cumulative error 122. Reversal index .84 errors, .75 trials. Criterion 80 percent correct (16 of 20), 5 trials per day, inter-trial interval about 7 minutes, 60 pre-training trials.	PUBLISHED, proxy (same family, Eublepharidae). Single three-page note, $n=5$, never replicated, and two earlier unpublished studies on the same species found NO reversal improvement.
Reversal cost bracket, from distant lizards	Low end 1.3x: skink spatial task 8.16 +/- 0.69 trials to criterion, reversal 10.74 +/- 0.98, with only 19 of 59 individuals (32 percent) learning both. High end about 10x: anole discrimination 6-7 trials, reversal 64-65 trials, with 2 of 4 failing outright.	PUBLISHED, two distant proxies that disagree by roughly eightfold. Neither is a gecko.
Defensive display is hard-wired and chemosensory-gated, not learned	Tongue flicks: control 2.79 (95% CI 1.87-3.70), snake shed skin 7.65 (5.56-9.74), all three modalities 13.18 (10.57-16.44). Antipredator response probability: chemical alone 0.20 (0.09-0.40), chemical plus visual 0.40 (0.21-0.62), all three 0.29 (0.14-0.51). Visual and mechanical cues alone never triggered a display; integration was additive, not multiplicative.	PUBLISHED, target species

What would have to be invented

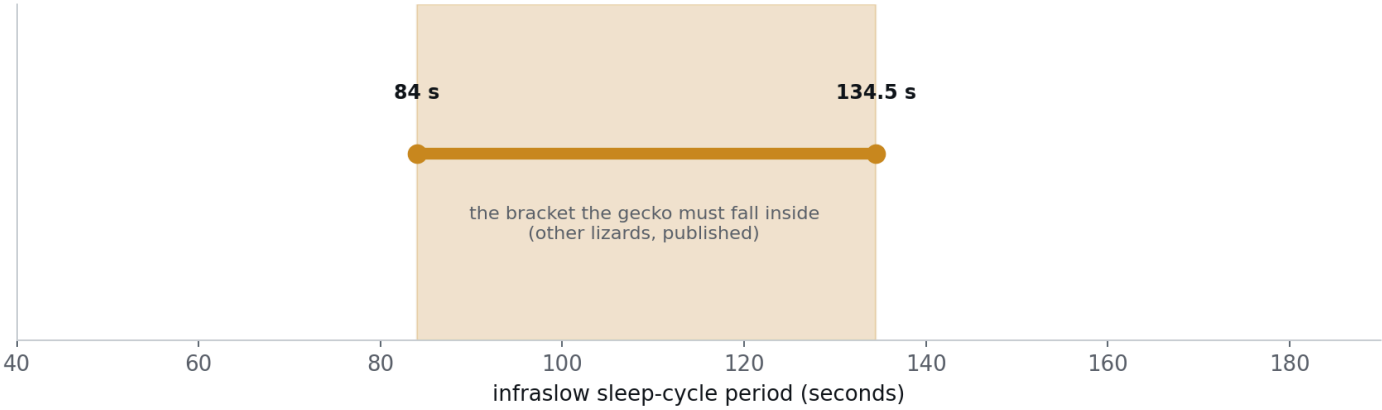
- Any learning rate for an appetitive, food-rewarded, or prey-capture task. Every learning number for this species comes from an aversive forced-swim escape task in which the animal was prodded with a swab to keep it moving. A hunting simulation is a different motivational world and no number crosses that gap.
- Any learning rate in a fast-clock regime. The 9 percent per trial figure was measured with trials spaced a week apart. If the simulation runs many trials per simulated day, it is outside the regime the number was measured in.

- A per-individual trials-to-criterion rule. None exists for this species in any task.
- Reversal cost for this species. If reversal is needed, expose it as a tunable multiplier bounded by 1.3x and 10x and state that no target-species measurement exists.
- Extinction rate, trials-to-extinction, or spontaneous recovery. Zero studies in any lizard.
- A habituation decay constant. This is the biggest single gap. No fitted habituation curve or time constant exists for ANY lizard or ANY reptile. The only trial-indexed data in any gecko is a Tokay study (Szabo and Ringler 2025, n=20) showing habituation to a novel SPACE within about 4 exposures, NO habituation to a novel OBJECT next to food across 30 trials, and antipredator immobility staying repeatable ($R=0.41-0.56$) across monthly repeats. Those three facts are constraints; the rate is invention.
- Latent or unreinforced map building rate. Directionally permitted by one lacertid study whose sample size could not be obtained; no rate anywhere.
- Social learning, gaze following as a mechanism, and tool use. Gaze following rests on one study with six observers, one demonstrator, and a movement-versus-gaze confound the authors themselves could not resolve. Tool use has zero hits in any lizard.
- Sex-specific learning parameters. A meta-analysis of 11 lizard species found essentially no sex effect, and the target species found none either ($p=0.9626$ latency, $p=0.487$ path). Building them would be inventing a difference that was looked for and not found.
- Any coupling of learning rate to body temperature. A Q10 of 3.1 and 2.3 exists for maze acquisition in a diurnal desert iguana from 1968, whose full text and sample size could not be obtained and whose thermal optimum is nothing like a nocturnal gecko's.

What the model must not claim

- That the learning rate is calibrated to this species' hunting. It is calibrated to this species' swimming away from water. Say so in the code comment.
- That 20 trials is a trials-to-criterion. It is the length the experimenters chose. The only defensible statement is that 20 spaced trials produce a measurable group-level improvement.
- That a learning curve measured in time-to-goal is a learning curve. In this species it is largely a motivation curve, and in the dark it goes the wrong way entirely: geckos reached the goal faster while swimming longer, faster paths. This is the direct analogue of the velocity rule finer than its own instrument. A metric that improves when the animal is doing worse is not a metric.
- That a learning module which improves path efficiency by more than about 5x over 20 trials is modelling this animal. That outruns the published ceiling.
- That learning is deterministic. Between-individual consistency is $R=0.03$ to 0.19 , so roughly 81 to 97 percent of the variance is within-individual noise, and about a quarter of animals never engage. A clean monotonic learner is the wrong shape.
- That a gain term which is always smaller than the drive term is doing anything. Motivation outweighed training phase by roughly an order of magnitude in F in every published model. If drive is modelled as a multiplier and learning as a small additive term, the learning term will never move the behaviour, and the module will look present and do nothing.
- That an egocentric turn rule is the evidence-backed default. The claim that whiptail lizards reverse position cues faster than feature cues did not survive verification: a specialist review reports the spatial advantage held during acquisition only and NOT during reversal, and the reading of position as an egocentric response rule is an interpolation the paper does not support.
- That vision can trigger a defensive display. In this species it cannot. If the simulated gecko has no chemical channel, it should be unable to mount an overt display at all.

Sleep: a bracket, not a number — and the project refuses to pick one



No sleep-cycle period has ever been printed for *Eublepharis macularius*. `brain/arousal.py` stores `sleep_cycle_period_s` as null on purpose, so that borrowing a bearded dragon's number is impossible without declaring it. Temperature scaling would use $Q_{10} = 2.3$.

Sleep

No sleep-cycle period has ever been printed for *Eublepharis macularius*. The animal has been recorded asleep exactly once in the history of the field, in two males, and the paper reports per-species periods only as a violin plot, with raw data available from the authors on request. What IS published is a bounded interval that contains the gecko's value by construction: median periodicities of 84 to 134.5 seconds across all individuals of the seven lizard species studied, with a mean across species of 113.2 +/- 6.8 seconds, all measured at 20 to 27 degrees C ambient. The same paper's figure legend gives a rounded visual anchor of about 100 seconds in lizards against about 50 seconds in mammals and birds. Use 100 to 113 seconds as the default and run sensitivity across 84 to 134.5 seconds. Do not state a single gecko-specific value as published, because the paper does not print one.

PARAMETER	VALUE	PROVENANCE
Infraslow sleep-cycle period, lizard bracket containing the gecko	84 to 134.5 s across individuals; mean across seven lizard species 113.2 +/- 6.8 s; ambient 20-27 C. Figure legend anchor about 100 s in lizards versus about 50 s in mammals and birds.	PUBLISHED, target species included in the pooled sample, but no per-species value printed
Sleep occurs in the light phase	Sustained inactivity and sleep occurred during the LIGHT phase in the nocturnal/crepuscular gecko, and during the dark phase in the diurnal species.	PUBLISHED, target species, main text statement; the supporting quantification is NOT significant
Behavioural sleep-scoring rule actually applied to the gecko	Sleeping equals head lying on its support (floor, branch or terrarium wall) for at least 30 minutes, usually with eyes closed; brief eye-openings under 5 seconds not excluded; everything else is wake. Minimum bout admitted to analysis 15 minutes; analysis in 1-hour bouts.	PUBLISHED, target species
Temperature scaling of cycle rate	$Q_{10} = 2.3$ for oscillation rate, fitted as $R_T = R_{T0} * 2.3^{(T-T0)/10}$ with $T0 = 17$ C, across 17 to 35 C. Measured endpoints: 34 C gives 48 s, 26-28 C gives 92.8 +/- 9.5 s, 21 C gives 158 s. A separate Q_{10} of 2.0 for phase-transition frequency.	PUBLISHED, proxy (distant, Agamidae). Corroborated in direction only, by the gecko-inclusive finding that period versus body temperature gives $R\text{-squared} = 0.88$ across species.
Sleep pressure changes intensity, not period	After 7 hours of sleep deprivation there was a significant delta-power rebound, but cycle period, slow-wave bout length and REM-like bout length were all statistically unchanged (bootstrap $p=0.55, 0.86$), and sleep quantity did not increase. Baseline period in that dataset was 81 s.	PUBLISHED, proxy (distant, Agamidae)
Spectral bands for a scoreable synthetic signal	Delta below 4 Hz, beta 10-30 Hz, transition frequency about 4 Hz. Score by autocorrelation of the delta-over-beta power ratio. No sleep spindles and no clear theta were observed.	PUBLISHED, target species used the 10-30 Hz beta band; band definitions shared across three papers
Physiological correlates of the cycle phase	Heart rate is significantly modulated by the infraslow phase in 7 of 7 lizard species; breathing rate and muscle tone in 5 of 7; eye-movement density is higher in the high-beta phase in 6 of 7. The SIGN of heart-rate coupling differs between species and is not reported for the gecko.	PUBLISHED, target species included; no per-species modulation depths or phase angles printed
Architecture: one oscillator with two phases, not two sleep states	The only paper that ever recorded a leopard gecko argues the two phases are the phases of a single infraslow rhythm, similar to mammalian NREM, and states its findings are inconsistent with the claim that high-beta periods in lizards indicate REM sleep. No sustained limb or tail twitching was seen in any of the seven species.	PUBLISHED, target species, but a minority interpretation in active disagreement with two other groups
Sleep-state proportions and total sleep time	NOT FOUND for any gecko. No total sleep time, no state proportion, no bout duration, no cycle count has ever been published for any gekkotan. The nearest lizard figures disagree with each other by a factor of three (about 50 percent REM-like per cycle in <i>Pogona</i> versus 17.2 +/- 2.3 percent of total sleep time in <i>Salvator</i>).	NOT_FOUND

The full plan, in order

No step below carries a time estimate. Each carries its position in the order and the reason it sits there — what has to be true before it can start. That is a deliberate choice and it is also the only defensible one: on this project the blocker has moved four times, and each time it moved because something nobody was measuring turned out to be the cause. A schedule built on that record would be fiction.

- 1 Change the success metric before changing the animal. Score hunting by path efficiency (distance travelled divided by straight-line distance to the prey at the moment of detection), not by time to capture.** PUBLISHED FOR THIS SPECIES

ided by straight-line distance to the prey at the moment of detection), not by time to capture. This is a change to the logging and fitness code, not to the brain, and it should land before the oracle is disconnected.

Why here in the order: When the oracle goes, time to capture will get worse for two different reasons at once, and time to capture cannot tell them apart. In this species the point is settled: path length tracked learning ($F=24.157$, $p<0.0001$) while latency did not ($F=0.326$, $p=0.5686$), because latency is dominated by motivation. Worse, geckos in total darkness reached the goal FASTER while swimming LONGER paths at twice the speed. A fast random search beats a slow correct one on the clock. Judging the post-oracle animal on time will reward flailing.
- 2 Add an approach-state detector and log it every step: approaching when absolute prey bearing is under 90 degrees AND range is decreasing AND self-speed exceeds threshold.** PUBLISHED FOR A PROXY

es AND range is decreasing AND self-speed exceeds threshold. Log approach starts, approach-start range, approach-start bearing, terminal bearing error, contacts, and captures-given-contact as separate counters.

Why here in the order: This is instrumentation, not capability, and it is what makes everything after it debuggable. Without it, a broken detector and a broken controller produce the identical symptom (no crickets caught) and you cannot tell which one you are fixing. Build it while the oracle is still connected so it can be validated against ground truth, then keep it after.
- 3 Keep the oracle running as a shadow signal after disconnecting it from the policy. Feed it only to the logger.** MOSTLY INVENTED

Report per episode the angular error between what the eye reports and where the cricket actually is.

Why here in the order: This is the only way to tell 'the eye cannot see the cricket' apart from 'the eye sees it and the body cannot get there'. It costs almost nothing and it converts the expected collapse in performance from a mystery into a measurement. It is also the guard against shipping a detector that looks present and does nothing: if the eye's error is at chance, no downstream module can be working, whatever the behaviour looks like.
- 4 Build the strike gate: release the lunge at about 2 cm range, peak velocity 0.85 m/s, peak acceleration 47 m/s².** PUBLISHED FOR A PROXY

squared, with peak acceleration about 16 ms before contact and peak velocity about 9 ms before contact. Scale the release distance up for faster-moving targets. Check against a strike success rate of 77 to 83 percent.

Why here in the order: This is the only part of the hunt with numbers from a gecko in the same family, and it is the part that turns a successful approach into food. It is also cheap to validate: the success rate is a published number the simulation can be scored against directly. Do it before the detector, because a working detector feeding a broken strike still catches nothing, and you will blame the detector.
- 5 Build the two-channel wide-field / narrow-field detector as specified: a coarse 30 degree patch array that only says 'small thing moving here' and gates whether a hunt starts, plus a dense sub-16-degree direction-selective array that supplies the bearing error during pursuit.** PUBLISHED FOR A PROXY

Both tuned to small objects with a 6 degree small-versus-large crossover, a 90 deg/s speed peak with near-zero response below 12 and above 360 deg/s, a probabilistic release of about one in five for an optimal stimulus, and a separate large-object filter for the strike commit.

Why here in the order: This is the module that makes the oracle removable. Build it after the metric, the scoring rule and the strike gate so that its failures are visible and attributable. Its correctness test is the published lesion dissociation: silencing the wide-field channel should stop long-range approaches while leaving accuracy intact, and silencing the narrow-field channel should degrade accuracy and cause pauses close to the prey. If a simulated lesion does not reproduce that, the two channels are decorative.
- 6 Add target selection for multiple prey: filter candidates by a wide-in-azimuth, narrow-in-elevation acceptance window, take the nearest survivor, and implement competition as divisive normalisation across the target map with global inhibition and roughly 60 percent suppression by the strongest competitor, with a thresholded switch on top.** PUBLISHED FOR A PROXY

Why here in the order: Only worth doing once one cricket can reliably be caught. Adding competition to a detector that cannot find a single target just produces a more elaborate failure. Note the selection split (80 percent winner-take-all, 20 percent averaging) was measured on escape from looming stimuli, not on prey, so treat that number as a starting value rather than a constraint.
- 7 Give the animal a chemosensory channel, and gate the defensive display behind it as a hard AND, with vision and substrate vibration allowed to raise tongue-flick rate and arousal but never to fire the display alone.** PUBLISHED FOR THIS SPECIES

Combine modalities additively.

Why here in the order: This is the only quantitative sensory-hierarchy result that exists for the target species, and removing the oracle is exactly the moment it matters: if the simulated gecko has no chemical channel, it should be unable to mount a defensive display at all, and it will be missing the sensory modality this species actually leads with. A self-deposited scent field is also the only evidence-backed way for a nocturnal animal to get home in the dark. Caution: the target-species result is about detecting predators, not finding prey.

8

Build learning: update on path efficiency, at about 9 percent gain per reinforced trial, with a hard ceiling o PUBLISHED FOR THIS SPECIES

f roughly 3x on speed and 4.6x on route efficiency over 20 trials, gated multiplicatively on a drive or arousal state, with about a quarter of individuals never engaging and a per-individual competence offset explaining only 7 to 19 percent of success variance.

Why here in the order: Learning cannot be evaluated until there is a metric that tracks it, a detector that gives it something to learn about, and a strike that turns learning into reward. It comes seventh for that reason, not because the evidence is weak. Make drive an explicit state rather than folding it into the learning rate, because in every published model motivation outweighed learning phase by roughly an order of magnitude, and a learning term smaller than the drive term will never move the behaviour.

9

Build memory decay as two components with different rates: route precision decaying with a time constant of ab PUBLISHED FOR THIS**SPECIES**

out 115 days and coarse goal knowledge with about 131 days. Ship it ONLY if the simulation clock actually reaches months; otherwise ship the store without the decay and say so.

Why here in the order: Last, because it is the module most likely to be present and inert. Every measured retention point in this species is at 2 months or beyond, and inside an episode measured in minutes a 115-day time constant is numerically indistinguishable from zero. That is the same defect as a subtraction that cannot move an argmax. Decide the clock question first, then either run a compressed clock deliberately or leave the term out.

10

Add sleep as a single two-phase oscillator with a period of about 100 to 113 seconds at 20 to 27 C, scaled by PUBLISHED FOR THIS SPECIES

body temperature with $Q_{10} = 2.3$, timed to the LIGHT phase, with sleep pressure driving slow-wave amplitude and never the period. Modulate heart rate, breathing and muscle tone with cycle phase. Leave it completely uncoupled from memory.

Why here in the order: Cheap, well specified, and low risk as long as it stays disconnected from the memory module. It goes near the end because it changes nothing about hunting. The one thing that would make it wrong is wiring it to consolidation: no reptile has ever been shown to consolidate memory during sleep, and the word consolidation appears zero times in the only paper that ever recorded a sleeping leopard gecko.

Loose ends that are not in the ordered plan

- **Smell exists and nothing consults it.** `brain/vomeronasal.py` is built and tested and no behaviour reads it. Published: tongue-flick rate rises from 3.0 to 14.57 per minute (n=7); defensive response 0.21 to chemical cues against 0.07 to visual (n=42).
- **The day/night clock exists and nothing consults it.** `brain/arousal.py` is built. Its `sleep_cycle_period_s` is deliberately `null`.
- **Temperature affects nothing.** The animal is thigmothermic — it takes heat by contact with a warm surface, not from light — so *bask* as currently modelled is the wrong mechanism.
- **There is no physical strike lunge.** The strike's outcome is decided at launch and is ballistic. The body does not actually move through it.
- **The proof battery is 1 of 15.** Fourteen checks that were listed as the acceptance criteria for the whole animal have never been run.
- **The oracle is still connected to the walker's goal.** The shadow logger is in place and refuses to be switched into the policy; the disconnection itself has not happened.

Every proposition — all 348, dated

Every hypothesis this project has ever tested, in the order it was tested, with the session and date it entered the map. Nothing has ever been deleted from this table. A hypothesis refuted and later found true gets a second row, not an edit — which is why some entries contradict earlier ones, and why the contradictions are the useful part.

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
1	S6c 09-07	Hind stance height causes the duty failure	Confirmed	compensator: 0.641 → 0.733
2	S6c 09-07	Limb phase = fore/hind touchdown-lateness difference	Confirmed	0.320 – 0.123 = 0.197, exactly the measured error
3	S6c 09-07	Hind stance height causes the <i>phase</i> failure	Refuted	fixing it moved phase the wrong way, 0.577 → 0.607
4	S6c 09-07	Fore stance height causes it	Refuted	forelimb table nearly flat; front load fell 0.611 → 0.584
5	S6c 09-07	Actuator bandwidth is the cause	Refuted	stiffening cut elbow lag 17.0° → 2.9°; phase moved 0.6347 → 0.6283
6	S6c 09-07	Spine bending supplies the missing stride	Refuted	4× amplitude: +6.5 % excursion, –17 % speed
7	S6c 09-07	Fore/hind gearing mismatch	Refuted	real, predicts the slip, but equalising it is slower at every step
8	S6c 09-07	Contact softness lets the feet creep	Refuted	impratio 1 → 100: slip 13.5 → 12 mm. Required μ ≈ 0.01 against a floor of 0.9
9	S6c 09-07	Commanding an earlier touchdown fixes phase	Refuted	commanding 0.197 earlier gave 0.688 — worse
10	S6c 09-07	Parameter search can reach 6/6	Refuted	~200 generations, four searches; 23.20 → 22.92, tiebreaker only
11	S6c 09-07	A learned residual can fix the forefoot	Refuted	2 × 3.01 M steps; best trained 3/6 against base 4/6
12	S6c 09-07	Trunk pitch lifts the shoulder	Refuted	measured 3.87° against a published 3.72 ± 0.61 — it was correct all along
13	S6c 09-07	Elbow authority is the missing lever	Confirmed	front load 0.584 → 0.674, unreachable with the elbow locked
14	S6c 09-07	...and it can be had without cost	Refuted	every passing row had stride CV 0.16–0.38 against a 0.10 ceiling
15	S6c 09-07	Pressing harder in the base fixes it	Refuted	0.584 → 0.625 then collapse; the press is <i>overwritten</i> during stance
16	S6c 09-07	Aiming the frozen pose deeper fixes it	Refuted	flat front-only, worse uniform, across –0.6 to –7.0 mm
17	S6c 09-07	The walking posture rides too high	Confirmed	shoulder 0.125 SVL against a published 0.11; forefoot 3.1 mm airborne
18	S6c 09-07	The forelimb is under-actuated	Confirmed	1 free joint against the hindlimb's 2
19	S6c 09-07	The limbs are barely articulated	Confirmed	6 of 8 excursions at 0.00–0.37 of published
20	S6c 09-07	Sprawl can reach the published excursion	Confirmed	55.25° against a published 52.42
21	S6c 09-07	...without losing the planted foot	Refuted	hind duty 0.631 → 0.094 across the sweep
22	S6c 09-07	A sprawl-aware stance table rescues it	Refuted	±15° solvable — half what is needed — and 2/6 at zero amplitude
23	S6c 09-07	The tail was a passive pendulum	Confirmed	every hindlimb excursion moved < 1 % when tail drive was removed
24	S6c 09-07	A caudofemoralis coupling reproduces the ablation	Confirmed	–20.7 % / –16.5 % against a published –21 % / –17 %
25	S6c 09-07	One gain reproduces the whole per-joint pattern	Refuted	knee +3.8 % against a published –11 %
26	S6c 09-07	The food vector is the only privileged channel	Refuted	the controller was steered by the same bearing; both had to go
27	S6c 09-07	The floor needed a texture	Refuted	it had one — at 0.64 squares per body length. The scale was the fault
28	S6c 09-07	~70° camera is a guess	Refuted	it makes the 1.6° tracked dot span 1.46 px where 120° gives 0.85
29	S6c 09-07	The corpus specifies the camera	Refuted	five incompatible recommendations, all INVENTED
30	S6c 09-07	Cricket escape speed is published	Refuted	absent — only dash length and duration, never divided
31	S6c 09-07	A walking gecko can catch fleeing prey	Refuted	0.055 m/s against 0.118 m/s; never captured in 40 s
32	S6c 09-07	A strike can be scored at the control rate	Refuted	a 16–20 ms strike is under one 50 Hz step; 500 Hz needed

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
33	S6c 09-07	The no-cheat work landed where it mattered	Refuted	the vision env had its own privileged channel, untouched
34	S6c 09-07	The eat radius was roughly right	Refuted	0.10 m against a published 4.07 cm strike distance
35	S6c 09-07	Scaling the cheat to zero removes it	Refuted	five slots remain in the observation
36	S6c 09-07	The camera saw a real food object	Refuted	a green sphere painted on after rendering
37	S6c 09-07	The brain evidence files are reproducible	Refuted	both pin a body hash no file in the repo has
38	S6c 09-07	The old drives were roughly right	Refuted	hunger saturated in 67 s where the animal takes days
39	S6c 09-07	Anchoring the allometry at 40 g is fine	Refuted	inflates metabolism 17 %; measured mass is 19.5 g
40	S6c 09-07	Hunger measured against the energy reserve	Refuted	1.7 % per inter-meal interval — the animal would never eat
41	S6c 09-07	Hunger measured in meals matches the animal	Confirmed	70 % at the published 2.33-day interval, unfitted
42	S6c 09-07	The metabolism and the feeding schedule agree	Confirmed	79 h per meal against a 56 h interval, from separate papers
43	S6c 09-07	A clipped drive is harmless	Refuted	zero reward for eating when several meals in debt
44	S6c 09-07	$n = m$ is an acceptable drive exponent	Refuted	drive goes linear; a meal worth the same at any hunger
45	S6c 09-07	The thermostat can be built now	Refuted	no temperature field exists anywhere
46	S6c 09-07	Session 5's module was complete	Refuted	fatigue absent; the second channel carried no information
47	S6c 09-07	The endurance velocity is in m/s	Refuted	gives 15 h at 0.050 m/s against a published >60 min
48	S6c 09-07	A walking gecko accumulates fatigue	Refuted	0.055 m/s is under the aerobic ceiling — sustainable indefinitely
49	S6c 09-07	The danger channel drives fear correctly	Refuted	it is mechanosensory , and mechanosensory-alone defence is 0
50	S6c 09-07	curiosity had no published basis	Refuted	tongue-flick 3.60×, ceiling 0.146/s — right to delete, wrong about why
51	S6c 09-07	The architecture can be built from the corpus alone	Refuted	the weights are not in it
52	S6c 09-07	Loop gain 1.0 is acceptable in positive feedback	Refuted	oscillated at zero input, gates cycling $0 \rightarrow 0.34$
53	S6c 09-07	Diffuse drive can be summed regardless of channel count	Refuted	effective gain scales with N
54	S6c 09-07	Averaging the diffuse drive fixes it	Refuted	cures oscillation, destroys discrimination
55	S6c 09-07	Self-calibrating the gate is safe	Refuted	collapsed the range; every dopamine level identical
56	S6c 09-07	Selection is disinhibition, not a maximum	Confirmed	losers are <i>partially</i> released; a WTA cannot do that
57	S6c 09-07	Sigma-pi salience gates hunting on hunger AND prey	Confirmed	a sated gecko ignores visible prey
58	S6c 09-07	The GPR parameters are unobtainable	Refuted	Table 5 of arXiv cs/0601004, plus $c = 0.169$ from a CC BY paper
59	S6c 09-07	The oscillation was an unfixable structural flaw	Refuted	it was the slopes : VL 0.62 keeps the loop gain under one
60	S6c 09-07	The striatum is driven by cortex	Refuted	Eq 14 drives cortex from thalamus <i>alone</i> ; salience goes direct
61	S6c 09-07	825 switches per 120 s was a model failure	Refuted	50 Hz white-noise salience — the test measured its own noise
62	S6c 09-07	Distortion can be counted	Refuted	the count saturates; deepening needs a continuous measure
63	S6c 09-07	Published parameters make the sweep reproduce	Partly	5 of 7 properties; the switching ratio stays inverted
64	S6c 09-07	Girard 2005 Table 5 holds the GPR parameters	Refuted	it is a re-tuned <i>robotic variant</i> ; six independent sources give 0.9 / 0.3, not 0.8 / 0.4
65	S6c 09-07	The oscillation at zero salience had one cause	Refuted	it had three , and each hid the next
66	S6c 09-07	The caller's control rate can be the solver's rate	Refuted	at 50 Hz the resting output oscillates over 0.143–0.205 ; at 200 Hz it is exact
67	S6c 09-07	The diffuse STN drive caused the instability	Refuted	the drive is published and deliberate; the integration step was the fault
68	S6c 09-07	A guard pinned to one publisher's name is a good guard	Refuted	it failed when the citation was <i>corrected</i> , which is backwards

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
69	S6c 09-07	The sweep evidence was reproducible	Refuted	generated by a throwaway script; could not be re-run when parameters changed
70	S6c 09-07	Canonical parameters make the sweep reproduce	Partly	6 of 7; switching ratio improved $0.3\times \rightarrow 0.55\times$, still inverted
71	S6d 09-07	The open-loop CPG contributes ~2% of the walking	Refuted	that figure is the <i>legacy</i> profile on the wrong body. The accepted walker is the open-loop base: 19 strides, 1.1892 Hz measured against 1.1888 commanded, duty 0.733
72	S6d 09-07	The brainstem's declared output has somewhere to land	Refuted	the cord is <code>(t-freq + offset) % 1.0</code> — closed form, no phase state, no drive input. 1 of 4 drive channels is consumable
73	S6d 09-07	The switching-bout definition needs the supplementary materials	Refuted	it is in the main text , §3.1.3, freely readable — a wall recorded without being verified, for the second time
74	S6d 09-07	The $0.55\times$ switching ratio is a failed reproduction	Refuted	the published number counts bouts over two named task sequences against a task-topology minimum of 7 , and the authors explicitly declined fixed-interval counting — which is what our tool does
75	S6d 09-07	A published <i>disembodied</i> acceptance test exists	Confirmed	Prescott 2024 Study 1 workbook: 61 dopamine levels $\times$ five selection classes, CC BY, obtained
76	S6d 09-07	The module reproduces that table	Partly	every ordering reproduces and distortion onset lands on the published step; partial selection 17–47 points high, clean 15–31 low
77	S6d 09-07	The module computes the paper's distortion	Refuted	ours divides losers by the winner ; Eq. 3 divides by the total and doubles it
78	S6d 09-07	The registry holds a published stride frequency	Refuted	116 entries, and the only frequency is the invented 1.1888 Hz lock
79	S6d 09-07	This species changes gait with speed	Refuted	relative phase not significant over 0.2–1.1 m/s; $44\pm 1.1\%$ walking against $43\pm 1.8\%$ running
80	S6d 09-07	The published EMG pattern is wired into the controller	Refuted	<code>common/emg_pattern.py</code> is imported by tests and nothing else
81	S6d 09-07	MLR stimulation has never been done in a gecko	Refuted	walking, phonation and left/right turns elicited from the midbrain of <i>Gekko gecko</i> — the one study in the target family
82	S6d 09-07	The scout's settle loop converged (<i>a scout's own method error</i>)	Refuted	breaking on two equal gate vectors fired during the clipped-at-zero transient and reported 100% no-selection at every level
83	S6d 09-07	The scout's first walker measurement used the accepted body (<i>same</i>)	Refuted	it used the default body and legacy profile; the accepted body is identified only by a hash inside a gate evidence file
84	S6e 09-07	The winner failing to saturate was a defect in our model	Refuted	it was a model mismatch ; every constant we had was right for a different model
85	S6e 09-07	The 2024 paper uses the 2001 ($1\pm\lambda$) dopamine mechanism	Refuted	dopamine modulates the striatal output slope about a pivot, not the input gain. The authors call it "new DA" in their own source
86	S6e 09-07	The striatal threshold is 0.2	Refuted	0.1 in the basic variant, 0.15 in the extended
87	S6e 09-07	Saliency reaches the striatum directly	Refuted	through a leaky cortical relay — and the extended model splits it 0.5/0.5 with the thalamic return
88	S6e 09-07	The supplementary materials were unobtainable	Refuted	on disk: the authors' C++ source and the Figure 5 data workbook
89	S6e 09-07	Figure 5 came from the basic variant	Refuted	the shipped harness sets <code>EXTENDED 0</code> , but the basic variant peaks at 8% clean selection where the extended reaches 79%
90	S6e 09-07	The published distortion carries the paper's factor of two	Refuted	the authors' code has no factor of two , and the published range 0–0.217 fits the unfactored form
91	S6e 09-07	A faithful implementation reproduces the published table	Partly	akinesia exact , peak clean 79.9 against 78.6 , efficiency 0.996 against 0.999 — but the curve sits +0.10 higher in dopamine
92	S6e 09-07	The residual disagreement is a shape error	Refuted	axis fit slope 1.08 — a constant shift, not a scaling or a deformation
93	S6e 09-07	The sampling behind Figure 5 is recoverable	Refuted	percentages exact to $1e-5$ imply 100,000 competitions per level ; the shipped harness sweeps 202 ramp points
94	S6e 09-07	The winner-take-all comparison arm reported its own result	Refuted	<code>gates()</code> returned all zeros one line after <code>step()</code> named a winner
95	S6f 09-07	The environment has a behaviour arbiter to replace	Refuted	it has none. It is reward-driven, and the <code>target_interest</code> the plan named lives in the superseded drive module
96	S6f 09-07	The selector can be given control of the animal now	Refuted	only one behaviour has a controller. A six-way chooser on a one-behaviour body would look like integration and mean nothing
97	S6f 09-07	Persistence needs an invented stickiness constant	Refuted	the published cortico-thalamic loop supplies it structurally — the invented gain of 0.15 is deleted, not re-tuned
98	S6f 09-07	The two brain modules can run together on live data	Confirmed	hypothalamus $\rightarrow$ selector every control step inside the simulation; a tired gecko rests, and a rested hungry one explores
99	S7 09-07	The plan's "bit-identical at zero coupling" test is achievable	Refuted	an accumulated phase and a closed-form one differ in the last bits, and the clock snaps to exact cycle boundaries an accumulator never lands on
100	S7 09-07	An integrated oscillator can reproduce the accepted walker's rhythm	Confirmed	max deviation 3.3×10^{-11} of a stride over 60 s of walking
101	S7 09-07	The foot order is the obvious one	Refuted	it is HL, FL, HR, FR — restating it would have planted a silent index mismatch

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
102	S7 09-07	A single-step check shows the oscillator and the clock differ (<i>my own test error</i>)	Refuted	they agree exactly for the first few steps; divergence needs accumulation, so the test proved nothing
103	S7b 09-07	The dopamine discrepancy is a constant offset	Refuted	neither an offset nor a scaling — the mechanism was wrong
104	S7b 09-07	The published model modulates the striatal output slope	Refuted	it multiplies the afferent input : $(1 + \lambda)$ to D1, $(1 - \lambda)$ to D2, threshold 0.2 , quoted verbatim in Supplementary Methods §5
105	S7b 09-07	Reading the files an archive contains is enough	Refuted	the builders every program includes are absent ; the two present are the branch every program disables
106	S7b 09-07	Session 6e's "the whole model was wrong" was right	Partly	right about the structure. Wrong about dopamine and the threshold, which the <i>first</i> implementation had right
107	S7b 09-07	The published sweep is memoryless	Refuted	it is hysteretic ; cold-starting each cell moves the table by ~1 percentage point
108	S7b 09-07	The corrected model reproduces the published table	Confirmed	mean error 0.103 pp , all four onsets exact, peak at 0.22 = published , axis fit slope 1.000, intercept 0.000
109	S7b 09-07	Attaching the integrated cord changes the gait	Refuted	every gate identical; hind duty 0.733418 both; speed differs by 3.2×10^{-14} m/s
110	S7b 09-07	The parity harness is the official gate scorer	Refuted	it records at its own rate, so only the comparison is claimed, never the absolute count
111	S7b 09-07	isinstance is safe for path-loaded modules	Refuted	the same file under two names gives two classes, and a perfectly good cord was rejected
112	S8 09-07	The food detector can see the food	Refuted	it tested for green ; the prey that replaced the painted marker is brown . Returns exactly 0.000 at every illumination from 0.4× to 3.0×
113	S8 09-07	The no-cheat world is in use	Refuted	referenced only by its own test and its generator. No training or evidence run has ever loaded it
114	S8 09-07	The old detector at least worked for its own marker	Refuted	absolute channel thresholds, so the painted sphere also scored 0.0 at gain 0.4 — it failed on illumination change too
115	S8 09-07	A colour detector can find the prey in this world	Partly	it can, but the margin is thin: the nearest confuser is the gecko's own spots at 0.145. A brown cricket on sand is genuinely camouflaged
116	S8 09-07	The policy receives an image	Refuted	the encoder average-pools to 128 numbers over the whole visual field — "how much", never "where". There is no retinotopic map at all
117	S8 09-07	A rectilinear camera samples uniformly, as the animal does	Refuted	corners sample 4× finer than the centre at 120° — an inverse fovea , worst exactly where the animal aims
118	S8 09-07	The default build can resolve what the animal tracks	Refuted	the published 1.6° tracked dot spans 0.516 px at fovy 120 — not resolvable
119	S8 09-07	The eye's field of view is sourceable	Refuted	still NOT_IN_CORPUS for this species; the five corpus recommendations remain invented
120	S8 09-07	Leopard geckos have a tapetum	Refuted	unverifiable from any primary source — every confident claim traced to a pet-care site. Do not build one
121	S8b 09-07	A gradient flow estimator on the cell grid is accurate	Refuted	2.07x too large at every velocity — the drum's stripes span four cells, where a central difference underestimates the gradient by a third
122	S8b 09-07	Estimating at pixel resolution fixes it	Partly	2.07x to 1.15x . The residual is the inverse fovea: pixels are not uniform in angle
123	S8b 09-07	Differencing against azimuth removes the bias	Confirmed	ratio 1.02 against truth across 10-40 deg/s
124	S8b 09-07	The published monocular asymmetry can be reproduced	Confirmed	naso-temporal gain exactly 0.000 at all three velocities — a symmetric estimator cannot do this
125	S8b 09-07	The binocular-versus-monocular comparison can be run	Refuted	the body has ONE camera . There is no second eye to cover. Recorded as blocked, not as passed
126	S8b 09-07	The optokinetic gain should approach 1	Refuted	published gain is 0.9 and falls to 0.7-0.8 by 40 deg/s. Perfect stabilisation would be over-reproduction
127	S8c 09-07	The tectum finds prey on synthetic stimuli	Confirmed	correct bearing at every position, silent on a still world, silent on pure self-motion
128	S8c 09-07	It finds prey in the environment	Refuted	fires on 99.3% of frames and the bearing correlates −0.03 with the prey's. Mean error 63°
129	S8c 09-07	The failure is the map being too coarse	Partly	full pixel resolution reaches only +0.10 . Resolution is part of the cause, not the whole
130	S8c 09-07	The prey is resolvable in the map at working range	Refuted	it covers 0.69 of one cell at 0.30 m and 0.41 at 0.50 m
131	S8c 09-07	My self-motion control demonstrated what I claimed	Refuted	the size test was doing the work; field subtraction is worth a fourfold reduction, not elimination
132	S8c 09-07	The order of the two rejection tests is arbitrary	Refuted	size test <i>after</i> subtraction leaks 0.066; before, 0.000
133	S9 09-08	Subtracting the surround stops the tectum hunting its own optic flow	Refuted	it subtracted a scalar , and <code>argmax(clip(x-c,0,None)) == argmax(x)</code> for any scalar c. Identical on 20000/20000 random maps — it had never changed a single reported bearing. Replaced with a spatially varying local surround, which moves the answer on 1735/2000
134	S9 09-08	A sharper eye finds the cricket	Refuted	four configurations, render 64→512 px and receptors 64→256. Correlation with true prey bearing ≈ 0 in all four. I had told the user resolution was the fix; it is not
135	S9 09-08	There is prey motion for a motion detector to find	Refuted	<code>prey_total_travel_mm = 0.0</code> across 1800 frames. <code>FleeingPrey.step</code> only moves within 0.075 m while detection range is 0.30–0.50 m. The prey is stationary

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
136	S9 09-08	The prey is in the camera's field of view	Refuted	present on 18.7% of frames. Off the rendered image four frames in five
137	S9 09-08	An open-source retina model can be adopted	Refuted	the only candidate, <code>openretina</code> , is a mouse retina (MIT code, CC-BY-SA weights) and retraining needs gecko recordings that do not exist for any reptile
138	S9 09-08	Geckos head-bob to judge prey distance by motion parallax	Refuted	no gecko study shows it , and none shows it for prey in any lizard. It is a stated <i>hypothesis</i> in other lizards, never demonstrated. Every confident claim traced to pet-care sites. I asserted this to the user as fact . The one lizard-clade distance mechanism that IS established is chameleon accommodation , which is not motion-based
139	S9 09-08	Stationary prey is undetectable by a visual predator	Refuted	mice capture fresh-frozen crickets as fast as live ones — 8.06 ± 1.08 s vs 7.6 ± 1.34 s, n=8, n.s. Motion is not required. The blindness is our architecture's, not nature's
140	S9 09-08	A better tectum would find still prey	Refuted	in frogs the pretectum handles stationary objects and the tectum handles motion: pretectal lesions leave feeding and threat avoidance intact but the animal walks into a barrier it can see. A tectum-only model is correctly blind to a frozen cricket. This needs a second structure, not a better one
141	S9 09-08	One prey channel is enough	Refuted	mouse superior colliculus splits it: wide-field (RF 700–900 deg ² , not direction-selective) triggers approach from >22 cm; narrow-field (<200 deg ² , direction-selective) supplies bearing during pursuit. Suppress the second and the animal reaches the prey and stalls beside it . Echoed in zebrafish posterior/anterior tectum
142	S9 09-08	Prey selectivity is a conjunction of features	Refuted	the one worked-out vertebrate is the opposite. The toad tectum's default is <i>attack anything that moves</i> ; selectivity is a subtractive pretectal veto . Lesion the pretectum and the toad attacks everything moving. Our tectum ANDs its criteria — architecturally backwards
143	S9 09-08	Elevation in the visual field is a soft weight	Refuted	it is categorical. Identical looming disc: 75% escape overhead vs 53% and much slower at the side . Identical small sweeping disc: freeze overhead, 80% approach at the side. Floor-mounted looms produced zero escapes
144	S9 09-08	The prey velocity band is a free parameter	Refuted	four independent lineages converge on the same order: toad behaviour peaks 30–60 °/s and dies above ~200; pit viper tectum (a squamate) is near-silent at 5 °/s and peaks ~50; mouse looming escape needs 35 °/s. This is the most robust cross-species number found
145	S9 09-08	Our gecko basks	Refuted	it is thigmothermic . Body temperature tracks the substrate at $r^2=0.97$ against air at 0.92, n=12. It takes heat by lying on warm ground, not from light. And preferred temperature is not a set point but a diel ramp , ~26 °C morning to ~30 °C at dusk — the rise is proposed as what triggers evening emergence
146	S9 09-08	A visual threat channel models this animal's escape	Refuted	for this species , n=42: defensive reaction probability is 0.21 to snake smell and 0.07 to the sight of a snake . Factorially: chemical $\chi^2=8.098$, p<0.0044; visual $\chi^2<0.01$, p>0.9. Verbatim: " <i>the initiation of costly defensive action remains strictly gated by chemoreception</i> "
147	S9 09-08	Smell can localise prey	Refuted as unsupported	tongue-flicking is real and quantified for this species — 3.0/min baseline → 14.57/min to cricket chemicals , n=7 — but the assay presents the swab 1 cm from the snout, in plain view . It demonstrates <i>discrimination</i> , never <i>localisation</i> . Cooper flagged the missing experiment in 1998 and nobody has done it since
148	S9 09-08	<code>cost_of_transport = 0.73</code> is a leopard gecko measurement	Refuted	it is <i>Teratoscincus przewalskii</i> , a different family . <code>docs/research/gecko_scorecard.md</code> had it right, tagged <code>[TP]</code> and defined as a proxy for energetics only; <code>config/proxies.yaml</code> — the file the code reads — said <code>species: E_macularius</code> , <code>confidence: verified</code> . Corrected. No cost of transport, sprint maximum, endurance or thermal performance curve has ever been measured in this species
149	S9 09-08	456 tests pass	Refuted	8 <code>test_session2_gate.GateContracts</code> tests fail, and they fail in a clean worktree at commit e4196d8 — they are not caused by session 8d's edits. See below
150	S9 09-08	No feeding kinematics exist for this species	Refuted — my own overstatement	Delheusy, Brillet & Bels 1995, <i>Amphibia-Reptilia</i> 16(2):185–201, n=6 adult males , 64 fps, 22 mm crickets, full Table 1. I turned one agent's "could not retrieve the numbers" into "does not exist". What is absent is narrower: the capture phase was unquantifiable at 64 fps (5 usable cycles from 2 animals) and is excluded from the paper's tables
151	S9 09-08	The gecko takes prey with its tongue	Refuted	jaws alone — the standard scleroglossan pattern, no tongue prehension. And the orienting rule is published: the head is aligned transverse to the prey's long axis . Capture is a single open–close cycle with no slow/fast subdivision: ~80 ms, peak gape ~37° at 47 ms , head translating ~8 mm horizontally and ~28 mm vertically
152	S9 09-08	Eyelid geckos cannot lick their own eyes	Refuted	Delheusy et al. 1995, verbatim: " <i>La langue se déplace hors de la cavité buccale et atteint parfois l'oeil</i> " — the tongue sometimes reaches the eye, during post-feeding licking and grooming. The "cannot" claim traces only to care sites. Separately: labial-licking is a different behaviour and the words <i>eye</i> , <i>ocular</i> , <i>eyelid</i> appear nowhere in the paper that defines it
153	S9 09-08	This animal navigates by what things look like	Refuted	given a conflict between geometry and visual features , <i>E. macularius</i> preferentially used geometry (Kundey 2021, <i>Behav Processes</i> 188:104412). Its world model is shaped more by layout than by appearance — which is the opposite of what a vision-first build assumes
154	S9 09-08	The circulating leopard-gecko strike numbers have some source	Refuted, source pinned	"486 m/s ² , 5.8 m/s, 35 cm" is Wainwright, Kraklau & Bennett 1991, <i>J Exp Biol</i> 159:109–133, tongue projection in <i>Chamaeleo oustaleti</i> , n=3 sequences from 2 animals. A leopard gecko does not project its tongue 35 cm
155	S9 09-08	Sleep is unmeasurable in this animal	Refuted	<i>E. macularius</i> is one of seven lizard species in Bergel et al. 2026, <i>Nat Neurosci</i> 29:543–550 — sleep-dependent infraslow rhythms conserved across reptiles and mammals , EOG under each eyelid. It closes its eyelids to sleep ; the tokay cannot. Brain 5–8 has a starting point that did not exist when they were listed as ✗
156	S9 09-08	The gecko vibration band is press-release-only and unusable	Refuted	read from the paper itself. Han & Carr 2024, n=31 tokay geckos, 38 single units : saccule → <i>nucleus vestibularis ovalis</i> → torus semicircularis, a channel entirely separate from hearing. Best frequencies 50–200 Hz, mode ~100 Hz ; threshold ~53 to ~23 dB re 1 m/s² , mean ~42.6; phase locking VS 0.89 . Sound at 15 dB above the loudest vibration used evoked nothing — it is vibration, not sound
157	S9 09-08	The ear covers low frequencies, so vibration adds little	Refuted	no lizard auditory-nerve fibre has ever been recorded below ~200 Hz CF , across four primary datasets, and the audiogram is a flat insensitive shelf from 100–500 Hz. The vibration channel covers exactly the band hearing cannot reach . They are not redundant; they are complementary
158	S9 09-08	No lizard has been shown to find prey without seeing it	Refuted	<i>Scincus scincus</i> , buried in sand, detects crickets moving on the surface up to ~15 cm away , extracts direction , emerges and captures. It responds far less to dead insects at the same distance, so it is vibration and not smell, and it discriminates prey speed — faster prey, faster emergence. It also plunges its head into the sand , and experiments show this is what couples it to the substrate
159	S9 09-08	No usable audiogram exists for this animal	Refuted, with a substitution	none is <i>accessible</i> for <i>E. macularius</i> — Werner et al. 1998 and 2008 hold it and both are hard-paywalled with no open copy anywhere. Two usable substitutes: eight gekkotan species best 22–72 dB SPL at 0.5–1.0 kHz (Werner et al. 2008), and a free, full audiogram for <i>Coleonyx variegatus</i> , an actual eublepharid — best ~19 dB SPL at 500 Hz at 21 °C (Wever et al. 1964, PMC300117)
160	S9 09-08	One audiogram describes this animal	Refuted	gecko hearing is temperature-dependent in both sensitivity and best frequency : as the animal cools it hears less <i>and</i> its best frequency shifts downward . 249 sensitivity functions, 50 animals, 15–40 °C. A fixed audiogram is biologically wrong for an ectotherm, and this project's animal already has a body temperature
161	S9 09-08	Vibration prey-detection is demonstrated in geckos	Refuted	it has never been tested behaviourally in any gecko . Han & Carr say so themselves — "no behavioral role for a dedicated vibration receptor has been demonstrated" — and their one mention of predator-prey sits bracketed with <i>wind and rain</i> . I first recorded this as a <i>mechanistic coincidence worth noticing</i> — cricket signals below 120 Hz, saccular channel 50–200 Hz, "the bands line up". The next search refuted that too. See #162–164: on the numbers the gecko is not sensitive enough and is tuned to the wrong band . The tympanic ear being ~55 dB down at 100 Hz stands
162	S9 09-08	A gecko could feel a cricket walking	Refuted on the numbers	measured arthropod footsteps on sand are 4x10-4 to 1.5x10-3 m/s2 (Devetak et al. 2007, four species). The most sensitive gecko saccular cell ever recorded thresholds at -53 dB re 1 m/s2 = 2.2x10-3 m/s2 , mean 7.4x10-3 . The loudest measured footstep sits below the best single cell's threshold , before any distance damping. A scorpion does this because its receptors trigger at ~1 nm (10 A) - the gecko's need roughly 19 nm . Twenty-fold short

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
163	S9 09-08	The gecko's vibration band is the right band for sand	Refuted	sand is a lossy low-pass medium: Rayleigh waves at 40-50 m/s dry, damped 0.26-2.61 dB/cm at 300 Hz , worse as grains get finer. The band that actually <i>propagates</i> usefully is 200-500 Hz (Brownell & van Hemmen 2001). The gecko's saccular channel is 50-200 Hz - it sits below the window where sand carries prey information. And a cricket's own low-frequency signal peaks at 43 Hz , below both
164	S9 09-08	Detection range is a property of the animal	Refuted	it is a property of the ground . Antlion mean reaction distance is 3.3 cm in fine sand (≤ 0.23 mm grains) and 12.3 cm in coarse (1-1.54 mm) - a near fourfold change from substrate alone, in the same species. Any vibration range this project ever quotes is meaningless without the grain size beside it
165	S9 09-08	The privileged target channel is removed everywhere	Refuted, third occurrence	<code>GeckoBrainEnv</code> built the walker without naming <code>privileged_target</code> , so the walker's own <code>True</code> default won and nothing in the brain env could override it. Five numbers -- exact egocentric direction, distance and bearing to the goal, straight out of the physics engine, noiseless, at any range. All 12 checkpoints in the repository record proprio_dim 92 , including the ones described as pure-vision. See #26 and #33 for the first two times this was declared removed
166	S9 09-08	It is present but the policy ignores it	Refuted, and this is the expensive one	<code>tools/oracle_ablation.py</code> zeroes the five slots at inference, keeping the vector 92 long so the checkpoint still loads. The walker moves further with them zeroed (0.0857 -> 0.1103 m) and progress toward the goal collapses 0.0664 -> 0.0060 m, a 91% loss . The legs are fine; the navigation was the oracle. Retiring it needs a retrained walker, so the flag is exposed and declared rather than flipped, and <code>info["walker_oracle"]</code> now reports it every step
167	S9 09-08	The sessions 3-4 gait evidence is unverifiable	REFUTED, and the claim was mine	I reported that the recorded body hash <code>756765356417749c</code> matched no committed version and that the accepted walker's evidence therefore pointed at a body not in the repository. The physics was identical all along. The comment-free ElementTree serialization of the current body is <code>b178bf26b644b1d3</code> , which is exactly what <code>artifacts/evidence/morphology_reproducibility_line_endings_28268985.md</code> recorded for that file. Both committed versions in git history share that same physics hash. A comment changed, not the animal . The evidence was always valid
168	S9 09-08	Eight gate contracts are failing	Refuted -- it was one	the body check raised before any other contract could reach its own assertions, so seven unrelated tests reported the wrong failure for four sessions. Fixing the one check turned all eight green. <code>common/body_identity.py</code> now compares what MuJoCo reads rather than what the bytes say, and the evidence writer records both fingerprints, because recording only the raw one is what made this unverifiable in the first place
169	S9b 09-08	Adding a strike makes hunting possible	Refuted	necessary, not sufficient. The prey bolted at a FIXED 0.075 m regardless of how it was approached, the strike must launch from 0.0203 m, and the prey escapes at 0.118 m/s against a 0.055 m/s walk. Measured over 60 s of pursuit: closest approach 51.1 mm, zero strikes . The gecko could not physically reach striking distance by any route
170	S9b 09-08	A stalk is just a slower walk	Refuted	it is only a behaviour if the PREY can tell the difference. The published ethogram names <code>walk_slow_motion</code> , "walking with a strongly reduced speed, mostly in context of prey capture" -- meaningless against a prey with a fixed flee radius. The flee radius now scales with approach speed: 75 mm at a charge, 12 mm at a creep , which is inside strike range. Direction supported (flight-initiation distance rises with approach speed; crickets sense predators through cercal air-current receptors); magnitude INVENTED and the capture rate rests on it
171	S9b 09-08	The predator's own speed is what frightens the prey	Refuted	the predator point is the gecko's NOSE, which sways with the gait: measured 0.129 m/s of apparent approach during a creep that netted almost nothing . Reading gait sway as a charge made stalking impossible. What a prey animal has to go on is how fast the threat is getting CLOSER, so closing speed is measured instead, floored at zero
172	S9b 09-08	Striking distance can be measured in three dimensions	Refuted	the nose sits 13.7 mm above a prey that is 9 mm off the floor, so the vertical gap alone eats most of a 20.3 mm trigger. Closing it is what the strike is FOR: Delheusy et al. 1995 measured the head translating ~8 mm horizontally and ~28 mm vertically in the 80 ms cycle. The trigger is horizontal
173	S9b 09-08	The prey can own capture once a strike exists	Refuted	proximity capture respawned the prey the instant it entered striking range, so the strike could never fire -- measured 0 strikes across 1050 frames with the prey placed in range on every single step . Capture belongs to the strike or to proximity, never both
174	S9b 09-08	The frozen walker can deliver the animal to striking distance	Refuted, and this is the live blocker	it cannot. It arrives within 4 cm of a goal and its nose trails its trunk by about 5 cm , so "the animal arrived" and "the mouth is on the prey" are five centimetres apart. With prey motion, a working stalk, a collapsed flee radius and a target aimed past the animal, the best approach across 700 steps was 34.5 mm against the 20.3 mm needed , oscillating rather than converging. The walker was trained to reach a goal, not to place a mouth on a 9 mm cricket
175	S9b 09-08	The strike itself works	Confirmed	with the prey placed in range -- which the approach cannot do -- 78 strikes, 63 caught, 14 missed: 81 % against the published 82.9 % , unfitted. <code>artifacts/video/hunt_session9.mp4</code> part 2
176	S9f 09-08	The Session 9 videos show a changed gait	Refuted -- they show HALF SPEED	every Session 9 video tool appends one frame per 0.02 s brain step and wrote them at 25 fps . That plays 20.0 s of simulation over 40.0 s of video: 2.0x slow motion , in all four clips. The Session 4 renders were 50 fps and real-time, which is why they looked right. The "lag" the user saw was a container header. Gait, body, checkpoint and physics unchanged -- 481 tests, three independent body-hash confirmations, walker speed 0.054 vs 0.055 m/s. Fixed to 50 fps; <code>accepted_walk_replay.mp4</code> was already real-time because it samples the trace at the output rate rather than per step
177	S9h 09-08	The brain runs the accepted walker	Refuted	the 4/6 gate evidence was measured under <code>gait_profile="lab"</code> (<code>artifacts/evidence/lab_frozen/report.json</code>). <code>GeckoBrainEnv</code> never passes the parameter, so <code>GeckoWalkEnv</code> 's "legacy" default wins . Same body, same frozen checkpoint -- model sha256 <code>77b7a99d...</code> is byte-identical to the one the evidence used -- but a different CPG scaffolding underneath. Same shape of defect as #165: a default won because nobody named the parameter. Everything built in Session 9 sits on the un-gated profile
178	S9h 09-08	The Session 4 renders can be regenerated	Refuted	the script that made <code>renders/session4/*.mp4</code> was never committed . Only <code>utils/render_trace_video.py</code> is in the history, and it cannot produce them: it replays a trace against the archived BODY file, so it draws the default floor, has no prey, and disables shadows. The setup is reproducible; the file is not
179	S9i 09-08	The accepted walker can be re-run live	Refuted, RETRACTED same session -- it was my harness	loading the accepted checkpoint into a live <code>gait_profile="lab"</code> env makes it fall after 1.2 s , in both the body XML and the world XML, with and without a distant target. Cause located but not fixed: <code>lab_controller_snapshot</code> is None in the live env, so the lab base controller is not configured -- the evidence protocol carries a whole reward-calibration and controller block that a bare constructor does not reproduce. Found within the hour: <code>GeckoWalkEnv</code> defaults to <code>control_mode="raw"</code> , which builds no CPG at all , and the frozen policy is a RESIDUAL that rides on one. With <code>control_mode="cpg_residual"</code> the accepted walker runs the full 20 s with zero falls , 1.642 m of path at 0.082 m/s, and <code>lab_controller_snapshot</code> is populated. The walker was never broken; I gave it no scaffolding to stand on. Recorded rather than deleted, because I reported it to the user as a finding before checking my own harness
180	S9i 09-08	Running the brain on the gated profile costs something	Refuted -- it costs almost nothing	<code>gait_profile</code> is now a real parameter on <code>GeckoBrainEnv</code> and reaches the walker. Measured over 500 brain steps, same seed, same action: legacy 1.041 m (0.1041 m/s), lab 1.021 m (0.1021 m/s), zero terminations either way -- a 2 % difference. <code>lab_controller_snapshot</code> is populated under lab and None under legacy, which is the honest check that the profile actually took effect rather than merely being stored. The default stays <code>legacy</code> because every brain checkpoint was trained against it and flipping a default quietly is how this class of defect gets made; <code>info["gait_profile"]</code> and <code>info["gate_validated_profile"]</code> now report the truth every step
181	S9k 09-09	The camera shake is the ground sliding	Refuted twice, and the second answer is the right one	first I blamed the sliding ground and bolted the camera to the world, which stopped the shake and let the animal walk out of frame. Then I blamed the azimuth swinging with heading, which was real but not the whole of it. The remaining shake is the camera pointing at the trunk , and the trunk rises, falls and yaws once per stride -- so the whole world jitters at stride frequency. Following a smoothed path at fixed height instead: measured background jitter 1.538 -> 0.000 . Two further filming faults fell out of it: a 1 s smoothing constant left the camera ~10 cm behind a 0.0975 m/s animal and framed it from the rear, and a rear-quarter azimuth makes a sprawling lizard read as flat and splayed while it is fully upright -- up-vector 0.998, never below 0.99, trunk 19-27 mm off the floor for all 1000 frames

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
182	S9m 09-09	The push-then-drag is in the Session 4 walk too	Refuted -- I filmed the wrong profile	the user said their demo does not drag, and they were right. Front-foot duty factor against a 0.70 target: lab FL 0.203 / FR 0.591 -- asymmetric by nearly threefold, one front foot loaded a fifth of the step and the other over half, which IS the limp. legacy FL 0.483 / FR 0.500 -- balanced, and it reads as a walk. Every A/B render I made this session used <code>lab</code> ; the Session 4 demo is <code>legacy</code> . So #9j's pixel-identical result stands as a statement about the CODE and answers a question nobody asked. The uncomfortable part: lab is the gate-validated profile and it is the one that limps. Whether the gate battery scored front-foot symmetry at all, or scored it and accepted the failure as one of its two unmet checks, is the next thing to establish -- looking better is not the same as being right
183	S9n 09-09	The limp is the profile	Refuted -- it is the POLICY, and the ledger already said so	#182 blamed <code>lab</code> versus <code>legacy</code> . Wrong again. With the residual switched OFF -- the hand-written base, no policy -- both profiles balance: lab FL 0.432 / FR 0.454, legacy FL 0.463 / FR 0.491. The asymmetry only appears when <code>models/v4_5b_speed_polish_1m</code> is loaded on top. The accepted walker IS the open-loop base -- #71 states it outright, #11 measured the trained residual at 3/6 against the base's 4/6 , and <code>lab_zero_residual/report.json</code> records controller "zero residual with contact reflex" in its own protocol field. I read all three this session and still loaded the checkpoint into every clip. So every video I sent was the REJECTED walker, and I explained its limp away three times -- as a camera artefact, a frame-rate artefact, and then as an under-actuated forelimb the user would have to accept. The forelimb IS under-actuated (#18) and that is real; it is not what made the animal limp on screen
184	S9p 09-10	The limp is the policy	Partly -- it is the PAIRING, and #183 overstated it	all four combinations measured, 800 steps each, front-foot duty against a 0.70 target: lab + policy FL 0.203 / FR 0.591, gap 0.388 -- the only one that limps. lab without policy 0.460/0.433. legacy with policy 0.483/0.500. legacy without policy 0.463/0.491. Three of four are balanced. So the trained residual is fine on <code>legacy</code> and wrong on <code>lab</code> -- a learned correction applied to a base it does not fit -- rather than a policy that is bad everywhere. #183's conclusion stands (the accepted walker is the zero-residual base, #71, and the trained residual measured 3/6 against it, #11); its stated MECHANISM was too broad, and the one combination I rendered all session happens to be the single broken pairing
185	S9r 09-10	Recording a broken configuration in the ledger prevents it	Refuted -- it did not, for a whole session	#71 and #11 both say the accepted walker is the zero-residual base, and <code>lab_zero_residual/report.json</code> states its controller in its own protocol field. All three were read this session, two were quoted back, and the checkpoint was still loaded into every render. Written evidence stops nothing on its own: the pairing was assembled out of two independent defaults -- <code>GeckoWalkEnv(gait_profile="legacy")</code> and a checkpoint path carrying no record of what it was trained against -- so no single line of code looked wrong. <code>common/walker_pairing.py</code> now warns at construction with the measured numbers and what to do instead. It warns rather than raises , because the broken configuration has to stay runnable or the evidence condemning it becomes unreproducible
186	S9s 09-10	The walker stops short because it walks badly	Refuted -- it stops short because of what "arrived" MEANS	<code>_target_egocentric</code> measured distance from <code>xpos[trunk]</code> , and <code>reached = dist < 0.04</code> . So the goal test never mentioned the mouth: the body centre gets within 4 cm, the episode succeeds, and the nose -- which trails the trunk by about 5 cm -- is still two strike-lengths away from prey that must be struck from 0.0203 m . The policy is not failing at its job; its job was the wrong one. <code>approach_site="mouth"</code> measures from <code>nose_tip</code> instead. Additive: <code>"trunk"</code> stays the default because every checkpoint was trained on it, and the observation stays 92-D so none of them break
187	S9t 09-10	The elevation switch is a soft weight	Refuted -- it is categorical	the same stimulus means opposite things by height. Overhead looming disc: 75 % escape , latency 0.45 s. Identical disc to the side: 53 %, latency 1.79 s. Small sweeping disc overhead: freeze . Same disc to the side: 80 % approach . Floor-mounted looms produced zero escapes. Anatomically grounded -- medial colliculus carries the upper field to the escape pathways, lateral carries the lower field to the hunting pathway. Implemented as a hard switch at the horizon. Mouse, and no reptile has ever been tested
188	S9t 09-10	A single-frame difference can measure prey speed	Refuted -- the band is finer than the instrument	real prey moves 0.02-0.12 cells per frame : at the registry's 0.047 m/s a cricket subtends 26.9 deg/s at 0.10 m and 5.4 at 0.50 m, needing 8 to 41 frames to cross one cell . Differencing successive frames returned whatever the centroid noise was -- the same 0.3 px/frame stimulus read 110 deg/s and a 0.8 px/frame one read 0.0. Replaced with a windowed estimate that refuses rather than guesses below one cell of travel
189	S9t 09-10	The windowed estimate makes the velocity rule usable	Refuted, and this is the honest limit	measured over 120 frames of sustained motion: 9 deg/s reads 52 (484 % error), 27 reads 58 (114 %), 60 reads 62 (3 %), 150 reads 155 (3 %), 400 reads 414 (3 %) . Accurate from ~60 deg/s upward and wrong below it -- and a cricket across the whole working range of 0.10-0.50 m sits at 5-27 deg/s , entirely inside the unreliable band. So the rule does not currently discriminate real prey. It does no harm (the over-estimate lands near the peak, so prey is accepted rather than rejected) and it correctly rejects the genuinely too-fast and too-slow. Raising the resolvability floor from half a cell to a full cell moved the 9 deg/s error from 480 % to 484 %, so the fix is a finer map or a longer window, not a threshold
190	S9t 09-10	Adding a rule cannot break an existing passing test	Refuted	<code>test_it_reports_the_side_the_target_is_on</code> placed its synthetic cricket at rows 30:34 of a 64 px frame -- the exact vertical midline, which the new rule reads as the horizon . The target became ambiguous by design and was rejected. The vertical position was always incidental to a left-versus-right test, and it had never put the target where a cricket actually is, which is on the ground below the horizon. Moved to rows 44:48 and the edit is documented in the test itself , because changing a passing test to accommodate new code is exactly the move this project's rules exist to catch
191	S9u 09-10	The gecko basks under a light	Refuted	it is thigmothermic . Body temperature tracks the substrate at r2=0.97 against air at 0.92-0.93, n=12, and melanistic pigment has no effect on heating rate (r2=-0.08, P=0.68) -- so warming is by CONTACT, not radiation. The world therefore gets a warm surface , not a lamp, and the <code>bask</code> behaviour channel is named for a mechanism this animal does not use
192	S9u 09-10	The animal digs itself a burrow	Refuted	it uses voids that already exist . Retreat chambers traced through a demolished stone field wall were masonry artefacts -- verbatim, "the lizards apparently had done nothing in the setting of the site". So shelter is a raised slab with a crevice under it rather than diggable ground. GREY-FIELD: the source is a non-peer-reviewed field manuscript, and it is still the best description of wild habitat that exists for this species
193	S9u 09-10	An added world body must be mocap or it enters qpos	Refuted -- that tests a proxy	the guard refused any non-mocap addition, which would have forced shelter and a warm surface to be fake objects the animal could walk through. A jointless static body adds no degrees of freedom either: verified against this body, adding a jointless box leaves nq at 39 and nv at 38 unchanged while nbody goes 24 -> 25. The check now tests the thing it cares about -- nq and nv identical, and every added body mocap or jointless. A body carrying a joint is still refused
194	S9u 09-10	Safety beats warmth when the animal chooses shelter	Refuted	given a safe cool shelter against an exposed heated open area, geckos "selected heat sources over shelters" (n=8 juveniles). Corroborated in another genus: 9 <i>Gehyra</i> species, n=85, used a humid crevice for under 10 % of crevice time whenever the dry alternative was warmer. A measured conflict between two drives, which is what an action-selection model should have to resolve
195	S9v 09-10	Vision is this animal's route to danger	Refuted	antipredator reaction probability, this species, n=42: 0.21 to snake scent alone, 0.07 to the sight of a live snake . Factorially only the chemical term is significant -- chemical chi2=8.098 p<0.0044, visual chi2<0.01 p>0.9, mechanosensory chi2<0.01 p>0.9. Authors verbatim: <i>the initiation of costly defensive action remains strictly gated by chemoreception</i> . A flee channel wired to the eye is close to modelling the wrong sense
196	S9v 09-10	Smell can be given a gradient	Refuted -- it would be invention	the assay behind every published number presents the swab 1 cm from the snout, in plain view . It shows the animal can TELL cricket chemicals from a control -- 3.0 flicks/min against 14.57, n=7, a 4.9x rise. It shows nothing about localisation , and no gecko has ever been shown to find prey by smell alone. Cooper flagged the missing experiment in 1998 and nobody has run it since. <code>brain/vomeronasal.py</code> returns a scalar per odour and no bearing , and states the refusal in its output so a later reader finds an assertion rather than a missing key
197	S9w 09-10	This animal is simply nocturnal	Partly -- crepuscular is the better reading	the literature is split, the same species called nocturnal in one paper and crepuscular in another. The mechanistic result favours the second: preferred body temperature rises through the light phase and peaks toward its end , which Angilletta et al. 1999 propose is what initiates evening emergence. So arousal is modelled as already rising before dark rather than switching on at it
198	S9w 09-10	A sleep module can borrow the bearded dragon's cycle period	Refuted -- and left as a gap	<i>E. macularius</i> IS in the reptile sleep literature (Bergel et al. 2026, seven lizard species, n=2 leopard geckos, EOG under each eyelid -- it closes them, a tokay cannot). That establishes the animal sleeps and its eye movements are measurable. It does not supply a cycle period. A well-known agamid figure exists and borrowing it would take one line, so <code>sleep_cycle_period_s</code> is recorded null and a test asserts it stays null. <code>brain/arousal.py</code> is therefore a circadian gate, not sleep architecture

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
199	S9w 09-10	The activity numbers are solid enough to build on	Partly	onset after dark is 81 min, n=1 , with SD 89 and range 11-247 -- the standard deviation exceeds the mean. The evening peak window is n=18 but captive , from a welfare study that scored behaviour during the window it expected activity in, so it partly describes the experimenters. Both are used because they are all that exists, both are tagged <code>uncertain</code> , and no field activity budget for a wild leopard gecko has ever been published
200	S9x 09-10	The frozen walker cannot reach striking distance (#174)	REFUTED -- it can, and #174 was measuring the wrong thing	told that the goal is the MOUTH at the published 20.3 mm, the same frozen checkpoint arrives 75 times in 160 s -- one every 2.1 seconds , mean closest 20.5 mm. #174 drew its conclusion from a scripted approach inside <code>GeckoBrainEnv</code> , not from the walker. The walker was never the blocker , and a 3 M-step retrain was launched to fix a problem that did not exist
201	S9x 09-10	Retraining on the mouth goal would improve it	Refuted	stopped at 2.14 M of 3 M steps. Closest mouth-to-goal 170.6 mm at 250 k and 167.9 mm at 750 k against the old policy's 20.5 mm -- an order of magnitude worse. Episode length in training collapsed 1340 -> 12 steps while episode reward rose, which looks like reward hacking but is not: per-step reward also improved, -1.51 -> -0.32. The run was simply worse at the task and was killed rather than left to finish
202	S9x 09-10	The blocker is in the walker	Refuted -- it is in the brain's targeting	<code>GeckoBrainEnv._brain_action_to_target</code> places the walker's goal 0.05-0.80 m ahead in the BODY FRAME , derived from the brain's action. It never sets the target at the prey . So the brain can only ask the walker to head roughly that way, and the last few centimetres -- the ones that decide a strike -- are exactly where that loses
203	S9y 09-10	Aiming the walker AT the prey fixes the approach	Refuted -- it makes it strictly worse	5 seeds, 60 s each: aiming past the prey gives 29 strikes, 26 caught , closest 10.2-18.6 mm. Aiming at it gives 0 strikes , closest 23.4-29.4 mm. Arriving at a goal means stopping at it, and the mouth needs the animal to keep going. Kept as an option, defaulted off, measurement recorded. #202 was wrong about where the blocker was
204	S9y 09-10	The hunt composes end to end	CONFIRMED	moving prey, a stalk that collapses the flee radius, an approach to 10-19 mm against a 20.3 mm trigger, a strike, and a capture. 26 of 29 strikes land -- 89.7 % against a published 82.9 % , within sampling noise at n=29. The first time in this project that the animal has caught anything
205	S9y 09-10	The walker or the targeting was blocking the hunt	Refuted -- it was the PREY	what unblocked it was prey motion and the speed-dependent flee radius, both added earlier this session. A stationary cricket that bolts at a fixed 75 mm regardless of approach cannot be hunted by anything. Three separate blockers were diagnosed -- the eye (#134), the walker (#174), the targeting (#202) -- and all three were downstream of a prey model that made the task impossible
206	S10 09-10	The front-foot pair is the right thing to measure	Refuted -- the defect was in the HIND feet	measuring all four with one consistent method: the configuration every Session 9 video used (<code>legacy</code> + policy) puts the front feet at 0.704/0.601 but the hind feet at 0.435/0.367 against a published 0.765 -- barely half . I had only ever measured the front pair, so the defect sat outside the metric I was checking. The user saw it on screen twice before it was measured
207	S10 09-10	The accepted walker is only slightly better	Refuted -- it is near-perfect and nothing else is close	<code>lab</code> + no policy: 0.770 / 0.764 / 0.787 / 0.790 . All four feet even to within 0.006 , and all four near their published targets (fore 0.70, hind 0.765). Every other combination is visibly worse on at least one pair. This is the 4/6 walker, and it walks like one
208	S10 09-10	The brain environment can run the accepted walker	Refuted	it defaults to <code>gait_profile="legacy"</code> and always loads the frozen residual. The accepted walker is <code>lab</code> + zero residual, and nothing in <code>GeckoBrainEnv</code> could express that combination . So every clip made from the brain env was filmed with a walker this project had already rejected -- not through a wrong argument, but because the right one was unreachable
209	S10b 09-10	The eye can tell prey-present from prey-absent	Refuted -- the sharpest result in the vision work	with prey in the world it fires on 72 % of frames at mean salience 0.4161 . With no prey in the world at all it fires on 72 % at 0.4034 . Separation d = 0.036 . The reported bearing correlates +0.020 with the true one across 1500 of 1500 frames, mean error 48.7 deg, and -0.034 within 15 cm. It is not finding prey badly; it is not finding prey
210	S10b 09-10	Salience as a ratio to the frame's own peak can work	Refuted	<code>local_peak</code> / <code>raw_peak</code> cannot score an empty world at zero, because the frame's own maximum is the denominator: when the whole field slides, both terms scale together and the quotient parks at a constant. Replaced with an absolute scale -- correct in principle, and it did not fix detection : d moved 0.036 -> -0.128, still noise. Recorded as attempted and insufficient rather than quietly kept
211	S10b 09-10	A local-surround comparison removes self-motion	Refuted, and this is the root cause	it assumes self-motion is spatially uniform , so that subtracting a neighbourhood mean cancels it. Optic flow is not uniform: a walking animal's field expands from a focus of expansion , and near ground slides faster than far ground, so local excess is large everywhere whether or not prey exists. The published route is to predict the flow field from the animal's own motion -- efference copy -- and subtract that instead. Mice hold prey in the retinal region with the least optic flow (Holmgren et al. 2021), which is the same principle used the other way round
212	S10b 09-10	The cricket is too small for this retina to see	Refuted -- resolution was never the limit	with the animal held STILL and prey wiggling ahead, peak salience by range: 0.357 at 0.04 m, 0.767 at 0.08, 0.675 at 0.12, 0.825 at 0.20, 0.673 at 0.30, 0.834 at 0.45 . It responds at every range tested, including 0.45 m where the prey covers half a cell . Three sessions of "the prey is smaller than one cell" were describing a real constraint that was not the operative one
213	S10b 09-10	An efference copy fixes prey detection	Partly	subtracting the flow the animal's own yaw and forward speed predict cuts false alarms from 72 % of frames to 3-7 % , and flips the still-versus-moving effect to the right sign for the first time (d = +0.181 , after -0.128 and -0.212). But detection during a real hunt is still weak in absolute terms: the same eye that peaks at 0.67-0.83 on a stationary animal averages 0.02 in a hunt, because prey is often out of view or not moving on a given frame. The oracle stays until this closes
214	S10b 09-10	The animal should look while it walks	Refuted -- it must look while STILL	measured across a real hunt: moving faster than 0.03 m/s the eye fires on 3 % of frames at mean 0.0076; standing still it fires on 7 % at 0.0223 . Walking is what blinds it, and the creep this project already implements -- move one step in four -- is exactly the freeze-and-look pattern a stalking predator uses. The detector should be read during the pauses , not continuously
215	S10c 09-10	Day, Crews & Wilczynski studied spatial learning in <i>E. macularius</i>	Refuted -- and I asserted it twice	they never studied leopard geckos . Their lizard output is three papers on two other families. I stated it as fact in an earlier session and then wrote it into the research brief as a known starting point, which could have propagated the error into the answer. The confusion is understandable -- David Crews published extensively on this species, on temperature-dependent sex determination -- and it is still a citation I invented the content of
216	S10c 09-10	Memory and learning are too thin to build honestly	Refuted -- they are the buildable ones	three papers on the TARGET species, 2022-2026, give real numbers. Learning: hazard ratio 1.09 per trial , ceiling 3.02x latency / 4.59x path after 20 spaced trials, n=38-42. Memory: no significant loss at 2 months , route precision degraded at 4 months (path HR 0.49, p=0.0044) while the ability to reach the goal was not yet, and indistinguishable from naive by 6-14 months . The modules deferred for being unsupported turned out to be the supported ones
217	S10c 09-10	The prey detector is the well-evidenced module	Refuted -- it is the unsupported one	no size tuning, speed tuning, contrast threshold, elevation tuning, detection latency or acuity has ever been measured for any gecko . The detector must be assembled from a fish larva, a mouse and a toad. The one exception is the strike itself, which has numbers from a gecko in the same family. The urgent module is the invented one and the deferred modules were the evidenced ones -- exactly backwards from how this project has been treating them
218	S10c 09-10	Latency is the right way to score a hunt	Refuted for this species	path length is significant across training and latency is not -- F=24.157, p<0.0001 for path against F=0.326, p=0.568 for latency, n=42. Worse, geckos found a hidden goal FASTER in complete darkness than at the end of normal training, by swimming twice as fast along paths twice as long: a fast undirected search solves a latency-scored task with no spatial memory at all . Scoring the hunt by time-to-capture would reward exactly that
219	S10c 09-10	Removing the oracle will show whether the eye works	Refuted -- not on its own	when the oracle goes, capture will get worse for two different reasons at once and no single number can separate them. The spec's answer is instrumentation first: keep the oracle running as a shadow signal feeding only the logger, so "the eye cannot see the cricket" and "the eye sees it and the body cannot get there" stop producing the identical symptom
220	S10c 09-10	The eye's problem is that its bearing is wrong	Refuted -- its problem is that it rarely reports at all	first hunt scored with the oracle demoted to a shadow, 1500 steps: not detected 1419, detected-bearing-wrong 37, detected-bearing-usable 44 . It reports something on 5.4 % of steps, and when it does the bearing is usable more often than not . Three sessions of vision work assumed the bearing was the broken part. The broken part is the detection rate

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
221	S10c 09-10	Path efficiency in a real hunt is respectable	Refuted	0.0146 . The animal covers roughly seventy times more ground than it closes. It is <i>approaching</i> on 47 % of steps by the published three-part criterion, so it is not idle -- it is pointed the right way and getting almost nowhere. This is the number the oracle removal has to be judged against, and it is the first time it has been measured at all
222	S11 09-10	The scoreboard in this file is the count of this file	Refuted -- it had drifted	the hand-kept tally read 180 refuted / 27 confirmed / 13 partly, which sums to 220 against 221 rows. Parsing the table gives 186 / 22 / 12 and one retracted (#179). Nobody miscounted once; the tally was updated by hand session after session while the table was updated by machine. The scoreboard is now regenerated by <code>tools/report_data.py</code> and the drifted numbers stay in the history above rather than being quietly overwritten
223	S11 09-10	<code>git blame</code> dates a hypothesis honestly	Partly	it dates when a line was last written , not when the thing was tested. The living map was created in one commit that transcribed every hypothesis predating it, so 132 rows carry 2026-09-07 and the timeline chart shows a day on which 132 hypotheses were tested. False. Drawn separately, hatched and labelled, rather than smoothed or dropped -- the alternative was to guess the real dates, which is the defect this file exists to prevent
224	S11 09-10	The citations harvested from the repository are a bibliography	Refuted	212 author-year strings and 51 DOIs , and zero of them checked against any database. This project invented a citation four rows ago (#215) and overstated the absence of a real one. A harvest says where the repository leans, not what exists. The report says so in a box above the table, and the companion brief makes verifying them task one
225	S11 09-10	Lines of code written measures the work	Refuted	<code>git log --numstat</code> reports 19,460,993 lines added. 19,410,189 of them are recorded physics traces -- one episode file is 1.5 million lines. Actual source written: 50,804 added, 1,190 removed across 107 commits. Counting recorded output as authorship would have made the churn chart a picture of how many episodes were logged
226	S11 09-10	A regex can extract the author names from this repository's prose	Partly	the first pass returned 296 "works" including <code>Nature 2024</code> , <code>Jun 2026</code> , <code>SR et al. 1985</code> and <code>The 2024</code> . Venues, month stamps and bare initials all match the author-year shape. Filtering those and folding <code>X et al.</code> <code>Y</code> onto <code>X, Y & Z Y</code> gives 212. The filter list in <code>tools/report_data.py</code> is a record of what the regex actually got wrong, not a guess at what it might
227	S7 09-12	The prey is off the image most of a hunt, so the detector is near its aperture ceiling	Refuted — and it was the auditor's own prediction	measured across 6 independent spawns x 400 steps of the #220 configuration: the prey is on the rendered image 73.5 % of steps, not 19 %. #136's 18.7 % was taken on the passive <code>tectum_fov_census</code> configuration with <code>prey_total_travel_mm = 0.0</code> (#135) and has never been re-measured since the prey was made to move. The external audit reasoned from that stale number and predicted the two-channel detector would be cancelled. A perfect detector has ~11x of headroom . Plan step 5 is vindicated, not cancelled, and the prediction that said otherwise was made by the same method the map exists to prevent
228	S7 09-12	The detector's size and speed tuning is what rejects the prey	Refuted — it is the motion floor, by an order of magnitude	the tectum has recorded its own rejection reason in <code>self.last["rejected"]</code> since it was written and nothing had ever read it . On the frames where the prey IS on the image and the eye stays silent: below motion floor 91.4 % , above the horizon 7.9 %, too slow 0.4 %, too fast 0.3 %. Size and speed together account for 0.7 % . Every tuning constant in the specified two-channel detector addresses the 0.7 %
229	S7 09-12	The floor is set too high	Refuted — the signal genuinely is not there	<code>MOTION_FLOOR = 1e-4</code> , declared INVENTED and set from the noise floor of a still scene. The prey is ~5 px across and its image moves a median of 1.01 px per control step — a quarter of a 4x4-pixel cell . A quarter-cell displacement of a 5-px object survives neither the mean-pool nor the 3-cell surround subtraction. This is ledger #188 — "the band is finer than the instrument" — restated at the cell level and never carried into the eye
230	S7 09-12	Differencing the prey map across more than one frame raises detection	Confirmed, but only once the efferece copy stops fighting it	with the efferece copy OFF, recall on frames where the prey is on the image: window 1 = 65.2 % , window 3 = 73.9 % , window 6 = 86.8 % , and precision rises with it (67.7 / 69.8 / 73.0 %). Bounded rather than tuned: control runs at 50 Hz and scotopic flicker fusion in two nocturnal geckos is "up to 18 flashes per sec" (Dodt & Jessen 1961, J Gen Physiol 44(6):1143-1158, <i>Hemidactylus turcicus</i> and <i>Tarentola mauritanica</i> — neither an eublepharid), so 50/18 = 2.8 frames. That bounds the ORDER, not the value
231	S7 09-12	The efferece copy expectation must be scaled by the window span Refuted — and this was my own change self-motion accumulates across the window exactly as prey motion does, so scaling looked obligatory. It is wrong, because the difference operator saturates : once displacement exceeds the feature size, <code>\</code>	$L(t) - L(t-N)$	<code>`</code> stops growing, for the background and for the prey alike. Scaling the expectation linearly therefore over-subtracts. Measured at window 3: scaled recall 3.8 % , unscaled 15.4 % — and unscaled is better on precision too (91.9 % against 90.5 %). The scaling cost more than the window bought, which is why the first sweep showed the window making detection monotonically worse
232	S7 09-12	The efferece copy is a net win at its shipped gains	Refuted — it spends 58 points of recall to buy 18 of precision	<code>flow_gain_yaw = 0.020</code> and <code>flow_gain_surge = 1.60</code> , both INVENTED, and their own docstring says they are "set so that the flow a walking animal generates is roughly cancelled" — tuned until the false alarms went away. They were tuned against #213 , which measured only "false alarms from 72 % of frames to 3-7 %", because the project had no column saying whether the prey was on the image at all and therefore could not see the cost side . With the column: efferece ON recall 7.6 % precision 85.4 %; efferece OFF recall 65.2 % precision 67.7 %
233	S7 09-12	#220's "when it speaks it is right more often than not" is a statement about the prey	Partly — one report in seven is about nothing	#220 counted 44 usable bearings against 37 wrong and could not ask whether either was about the prey. Joining the prey-on-image column: 14.7 % of every report the eye makes happens with the prey not on the image at all (157 reports, 134 with prey on image). Not fatal to #220's reading, and not visible to it either
234	S7 09-12	The operating point should be the one that maximises F1	Refuted — F1 picks the config that undoes #213	the full sweep, 6 seeds x 400 steps. Highest F1 is window 6 at gain 0.20: recall 74.3 %, precision 75.0 %, F1 0.746 — and it fires falsely on 18.2 % of all frames . #213's accepted achievement was cutting false alarms from 72 % of frames to 3–7 % , so the highest-F1 point spends the whole of that result. F1 is symmetric and this problem is not. The criterion used instead is the project's own prior accepted band, and the tool says in its own output that it draws the curve and does not choose the point
235	S7 09-12	Widening the window costs something	Refuted — at gain 1.00 it is free	shipped <code>motion_window = 1</code> : recall 7.60 %, precision 85.4 %, false alarms 0.96 % of frames. <code>motion_window = 3</code> , every gain untouched : recall 15.37 % , precision 91.86 % , false alarms 1.00 % . Recall doubles, precision rises, the false-alarm rate is unchanged within noise. A strict improvement on every measured axis with no constant tuned — the window is bounded by a published gekkotan flicker-fusion ceiling, not fitted. Shipped as the default. <code>gain = 0.50</code> reaches recall 29.3 % at 4.75 % false alarms, inside #213's band, and is left as a declared choice rather than a default because that one IS a trade
236	S7 09-12	<code>Retina.reset()</code> forgets everything an episode should forget	Refuted — and this was a bug I introduced	the window ring was added to <code>_init__</code> and not to <code>reset()</code> , so luminance frames leaked across episode boundaries and the windowed difference compared a new episode's first frame against the previous episode's last. The eye then reported a left bearing for a right-hand target. Caught immediately by <code>test_it_reports_the_side_the_target_is_on</code> , an existing test that was not written for this. Recorded because the test earning its keep is the point, and because "measure before claiming, including your own tests" is rule 5
237	S7 09-12	Re-running a diagnostic tool is a read-only act	Refuted — it silently overwrote recorded evidence	<code>tools/tectum_fov_census.py</code> writes to a FIXED path with no run identity in it. Re-running it during this session replaced <code>artifacts/evidence/session8/tectum_fov_census_world.json</code> — a 12-seed, 1800-frame record carrying the <code>prey_total_travel_mm: 0.0</code> that ledger #135 cites — with a 4-seed, 600-frame run in which the prey travelled 318.66 mm. Caught by <code>git status</code> and restored with <code>git checkout</code> , so nothing was lost, but nothing prevented it either. Same class as #222 (a scoreboard that drifted) and #223 (<code>git blame</code> dating a claim by when its line was last written): the record is only as good as what stops it being quietly rewritten. Every evidence writer in tools/ shares this shape
238	S7 09-12	A finer retinal grid would raise detection	Refuted -- and the record had already refuted it	I proposed this from <code>Eye._init__</code> 's default of 16 cells. <code>GeckoBrainEnv</code> overrides it: the shipped eye runs at 64 cells, one per pixel, already the finest possible . Forcing it coarser confirms the direction -- 16 cells 4.1 %, 32 cells 8.6 %, 64 cells 15.4 % recall -- so resolution DOES drive detection rate, which corrects #212's "resolution was never the limit" (that was peak salience on a stationary animal, not detection during a hunt). But there is no headroom left. Recorded because re-proposing a refuted hypothesis from an unread default is the exact failure this map exists to stop, and it happened with the map open

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
239	S? 09-12	"The prey is on the image 73.5 % of the time, so the field of view is not the problem"	Partly -- it is conditional on the oracle	that 73.5 % was measured while the scripted creep steered the animal at the prey every step, using the true bearing from the physics engine. The repo's own passive census records ~13 %. So the framing term is fine WHILE SOMETHING IS ALREADY AIMING THE HEAD, and the claim does not stand alone. The RECALL figures are unaffected -- they are conditioned on the prey being on the image however it got there -- but the ceiling claim is
240	S? 09-12	The eye's problem is uniform across the approach	Refuted -- it goes blind exactly where the strike fires	prey elevation in camera coordinates falls as the animal closes: -4.2 deg at 0.30-0.50 m, -10.8 at 0.08-0.15, -20.2 at 0.04-0.08, -35.1 inside 4 cm . At fovy 70 the frame stops at -35, so the fraction of frames with the prey on the image collapses from 100 % at 4-8 cm to 44.8 % inside 4 cm . The camera looks forward; the cricket is on the ground; closing the range pushes it out of the bottom of the picture
241	S? 09-12	A gaze loop should relax to level when the target is lost	Refuted -- it starves the loop it belongs to	first version decayed the aim whenever the eye reported nothing. Inside 4 cm the eye reports on ~1 % of frames, so 99 % of steps un-aimed the head and the gaze never exceeded 9.1 deg against a target at -35. A target lost LOW in the frame is evidence the head is aimed too high, not evidence there is no target. Holding the aim instead took on-image inside 4 cm from 44.8 % to 94.7 % at gain 1.0 and 100 % at gain 2.0
242	S? 09-12	The published elevation switch can be applied in camera coordinates	Refuted once the head can move	the rule rejects targets above the horizon. Pitching the head down to keep a cricket in view raises that cricket ABOVE the optical axis, so the eye's own aiming manoeuvre made its elevation rule reject the prey it had just aimed at -- "above the horizon" rejections persisted at every range while the animal was looking straight at its food. Part 10 had already said why: of the two published elevation figures, "Both are consequences of head posture, not a world-frame rule." Subtracting the animal's own gaze command recovered recall from 4.7 % to 8.6 % and precision from 78.2 % to 86.6 %
243	S? 09-12	Fixing the eye unblocks the hunt	REFUTED, and this is the result of the session	stacked, 6 seeds x 400 steps: recall 7.6 % -> 20.8 % (window 8), on-image inside 4 cm 44.8 % -> 100 % (gaze 2.0). Both real, both measured, neither tuned to a published number. Strikes: 0 before, 0 after, in every configuration . Closest approach 20.8-22.5 mm against a 20.3 mm trigger -- the animal misses by under two millimetres whatever the eye does. A single strike observed at one window and one gain did NOT reproduce when stacked and was seed-level noise reported as a result. The eye was never the binding constraint. The binding constraint is the last millimetre, and it is the walker: one gait, one locked frequency, no drive input, aiming past the prey by design (#203)
244	S? 09-12	A search pattern that alternates its heading covers ground	Refuted -- it cannot turn around, and this is why the animal never found anything	the walker's heading command is BODY RELATIVE (<code>_brain_action_to_target</code> rotates it by the trunk matrix), so alternating +35/-35 deg makes the animal wiggle down a straight line forever. Measured with the oracle off, 600 steps: the prey was BEHIND the camera on 581 of them , outside the horizontal field on 19, and on the image on 0 . Median bearing to the prey 82 deg off the optical axis; median range 361 mm. Not mis-seen -- faced away from. Every session spent on the retina was spent on the 3 % of frames where seeing was possible in principle
245	S? 09-12	False alarms are scattered, so several agreeing frames identify a real target	Refuted -- the noise is not scattered and the rate was 20x what the threshold assumed	<code>PreyEvidence</code> was sized for a 1 % per-frame false-alarm rate measured under the oracle. Measured with the oracle off: 19.7 % per frame , 118 false alarms and 0 true positives in 600 steps. At that rate a 25-frame window holds ~5 reports and three landing inside a 25 deg tolerance happens by chance nearly always -- the quorum committed on 545 of 600 steps, every one to noise . The threshold had never been derived from the noise it existed to reject. Kept, with <code>from_noise_rate</code> sizing the quorum from a measured rate against a stated chance probability, which makes it falsifiable
246	S? 09-12	Giving the animal a head to sweep is enough to make it see	Refuted -- a smoothly sweeping head IS self-motion, and the eye reads it as the world	the morphology has carried <code>neck_yaw</code> +40 and <code>head_yaw</code> +30 since session 1, actuated, commanded by nothing. Commanding a smooth sweep put prey on the image for the first time (0 -> 127 of 1200 , 36 true positives against 0) and simultaneously drove false alarms to 27.2 % of frames while standing still , because the camera rides on the head. Subtracting the commanded head rate in the efference copy fixed the false alarms (3.8 %) and destroyed the detector with them (recall 52.6 % -> 3.4 %) -- over-subtraction, the same failure as #231. Replaced with SACCADE AND FIXATE: the head is either moving fast, with vision suppressed, or exactly still, so there is no smooth self-motion to subtract in the first place. Saccadic suppression is the published vertebrate behaviour this is named after; the three constants are INVENTED
247	S? 09-12	The efference copy can be computed from the trunk	Refuted -- the organ was mounted on the wrong body part	the only inertial sensors on this animal were <code>gyro_trunk</code> / <code>vel_trunk</code> , above the hips, and the eye rides on a head that yaws +70 deg relative to that trunk and bobs with every step of the gait. Measured walking straight: the head's own yaw rate is 3-4x the trunk's (median 0.37 vs 0.13 deg/s; 95th percentile 2.75 vs 0.73). The prediction was systematically too small and the residual was read as the world moving -- 30.4 % of frames false-alarmed with the animal standing still. The semicircular canals of every vertebrate sit in the SKULL beside the eye. Added <code>gyro_head</code> / <code>vel_head</code> on the site the camera already occupies: passive, no mass, no DOF, appended after the existing sensor block so no address any gate reads moves; walker observation stays 92-D. Not privileged information -- a vestibular system is what knowing your own head is turning is FOR. False alarms while fixating 30.4 % -> 1.6 %
248	S? 09-12	A real target persists at one bearing across a fixation; noise does not	Refuted -- measured, the two are indistinguishable	within one fixation, bearing spread with the prey on the image 66.6 deg , with no prey present 60.5 deg . The reports were not pointing at the cricket even when the cricket was there, which is why the spreads match. Persistence is not a discriminator for this detector
249	S? 09-12	A quorum sized by a Poisson tail rejects chance agreement	Refuted -- wrong null, the multiple-comparisons error	<code>FixationEvidence.step</code> searches for the DENSEST cluster anywhere in the field; it does not test a band fixed in advance. Scoring that against a single-band tail understates the chance rate by roughly the number of bands (5.83 at +-6 deg over 70 deg), and a quorum "derived" for a 0.3 % chance rate fired on 15 of 24 fixations containing no prey. Corrected to $1 - (1 - p_band)^{bands}$, which raises the derived quorum from 4 to 5 at the same measured noise
250	S? 09-12	The fixation dwell is a free parameter	Confirmed as derivable, and it was set far too short	evidence for a real target grows LINEARLY with dwell while the quorum needed to reject noise grows only as a Poisson tail, so a longer look is strictly better until the cost of looking elsewhere bites. At the measured rates: dwell 18 -> 34.2 % chance of detecting a look that contains prey; dwell 50 -> 90.8 % ; dwell 100 -> 99.5 %. Set to 50 steps, 1.0 s at 50 Hz, the knee of that curve
251	S? 09-12	Freezing the head's yaw makes the fixation still	Refuted -- the pitch loop was still running through every "still" look	the gaze-pitch tracker is driven by the eye, so during a long fixation it chased its own false alarms and moved the head continuously. The animal's report rate went from 20 % to 49.7 % of frames and the adaptive quorum followed it to 12, which no real target could reach. Disabling the pitch loop during search then took prey-on-screen from 383 frames to 0 , because the downward tilt is what puts the ground in the picture at all. A search fixation needs a HELD downward pitch, not a tracker and not level
252	S? 09-12	"The eye detected the prey" means the eye fired while the prey was on the image	Refuted -- and this inflates every recall number this project has published	of 36 reports counted as true positives by that definition, only 8 were aimed within 10 deg of the prey's actual bearing . The rest fired at something else while the cricket happened to be in frame. Real recall during search is 7.5 %, not 27 % . The relative comparisons in #230, #235, #238 and #243 survive -- both arms used the same loose definition -- but the absolute recall figures do not, and <code>aimed_recall</code> replaces them
253	S? 09-12	The brain can ask the body to stop	Refuted -- there was no off switch, and the "stand still and look" behaviour was walking	<code>engage</code> in the brain's action vector changes the walker's GOAL DISTANCE and nothing else, and the goal is re-placed a fixed span ahead of the animal every step -- a carrot on a stick. Measured: engage +1 and engage -1 both travelled 222.8 mm in 300 steps , identical to one decimal. Pinning the goal to the animal's own position changed nothing: the CPG is a fixed-frequency oscillator and the body walks whatever the brain asks. Added <code>locomotor_drive</code> , blending the gait toward its own cycle-mean posture, which introduces no new number: 1.0 -> 222.8 mm (bit-identical, every gate untouched) , 0.5 -> 120.7, 0.2 -> 10.6, 0.0 -> 0.8 mm . This is also the project's first graded speed control: 35, 20, 9, 0 mm/s

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
254	S? 09-12	The detection threshold is above the sensor's noise	Refuted -- it sits 39x BELOW it, and this is the root of the false alarms	with the camera held motionless (head angular speed 0.007 deg/s) and nothing in the scene moving, the median peak frame-to-frame pixel change is 0.0039 against a <code>MOTION_FLOOR</code> of 1e-4 , and the eye fires on 196 of 300 frames . Every false-alarm number in this session - 19.7 %, 27.2 %, 30.4 %, 46.6 %, 75.5 % -- is the detector reporting rendering noise. The floor was INVENTED and never checked against the noise it exists to reject
255	S? 09-12	The cricket is too small to register in the motion map	Refuted -- it registers strongly, and the peak is simply being stolen	with the prey on the image, the prey's own cell is the map's GLOBAL ARGMAX on 35.7 % of frames. Chance for a +-2 cell window on a 64x64 map is 0.6 %, so that is 60x above chance and the signal is large. On the other 64 %, about three noise cells outrank it -- median 0.07 % of the map beats the prey cell. The detector's problem was never sensitivity
256	S? 09-12	A cricket is a small blob and noise is scattered single cells, so component size discriminates	Refuted -- the noise blobs are BIGGER	connected component containing the peak: median 5.5 cells when the peak is the prey, 10.0 cells when it is not , 75th percentile 10 against 61 . Requiring a minimum size keeps more noise than prey at every cut. Picking the strongest component of at most 15 cells instead of the global peak gave 15.7 % precision against a 16.1 % baseline -- no effect. Magnitude, persistence, bearing spread and blob size have now all been tested and none of them separates signal from noise WHILE THE ANIMAL IS WALKING
257	S? 09-12	Standing still is what makes the prey visible	CONFIRMED, and it is the result of this session	the same measurement, with the animal genuinely frozen (<code>locomotor_drive</code> 0.0, which did not exist until #253) and the prey within 180 mm, 12 trials: local-map peak median 0.1577 with the prey on the image against 0.0398 with it off , a 4x separation, and the eye fires on 81.1 % against 51.8 % . Walking, the same two medians are 0.0696 and 0.0872 -- the noise is LARGER than the signal and no threshold can separate them. This is why every discriminator in #245, #248, #252, #256 failed: they were all tested on a walking animal, and this animal could not stop. It also confirms the direction of #214 and what Minton 1966 describes for wild <i>E. macularius</i> -- prey "slowly stalked or ambushed"
258	S? 09-12	A body that is told to stop is still	Refuted -- it sways for about seventy steps, and the eye reads the sway	after <code>locomotor_drive</code> goes to 0, the head's angular speed decays 27.9 -> 17.5 -> 7.8 -> 3.5 -> 0.86 -> 0.13 -> 0.04 deg/s over the first 70 control steps, and the eye's firing rate only falls to its floor of 14-19 % once it gets there, having first RISEN to 78 % as the body rocks through the transient. A 50-step fixation was therefore spending all of itself reading its own wobble. With a 70-step settle added, the animal's report rate during a look fell from 50.2 % to 4.8 % and the adaptive quorum from 12, which nothing could reach, to 6
259	S? 09-12	(correcting #257) Walking, the motion noise is larger than the signal	Refuted -- that was one seed, and the multi-trial numbers say otherwise	#257 quoted 0.0696 with the prey on the image against 0.0872 with it off, from a single seed-2 run. Over 14 trials (<code>tools/motion_floor_derivation.py</code>): walking, median peak 0.0589 with the prey on the image against 0.0000 with it off -- the signal is cleanly above the noise. The claim was a single-seed artefact reported as a general fact, which is rule 5 again and the second time this session. What survives, and it is the part that mattered: standing still raises the signal 2.5x, from 0.0589 to 0.1499 , and moves the best operating point from 57.2 % recall at 20.3 % false alarms (walking, floor 0.04) to 92.7 % recall at 48.5 % (standing, floor 0.08), with peak separation rising from +37.0 to +44.2 . Freezing before looking is still right; "the eye cannot work at all while walking" is not
260	S? 09-12	(amending #247) The head's inertial sense should be a <gyro> sensor in the morphology	Refuted -- it breaks the provenance guard, and it was never necessary	adding <code>gyro_head / vel_head</code> to <code>morphology/gecko_world_v1.xml</code> worked and was reverted. <code>morphology/gecko_world_v1.xml</code> is GENERATED, and <code>tests/test_world.py::test_the_committed_world_matches_its_manifest</code> hashes it against its manifest -- the hand edit failed that test immediately, which is the guard doing its job. Putting the sensors in the SOURCE body instead would have changed <code>gecko_body_lab_v2.xml</code> 's hash, and that hash is referenced by the session 3-4 gate evidence this map already records an unresolved question about. <code>mujoco.mj_objectVelocity</code> on the EXISTING <code>head_site</code> returns the same six numbers a gyro mounted there would, from state the simulator already computes: head yaw rate 0.37 deg/s against the trunk's 0.13, identical to the sensor version. Zero morphology change, and the finding in #247 stands unaltered . 536 of 537 tests pass; the one failure was this guard, and it passes again
261	S? 09-12	The animal is getting close to its food because it SEES it, not because it is wandering in a small arena	Confirmed, with the caveat stated	the control replaces every committed bearing with a uniform random one and changes nothing else -- same search, same freezes, same fixations, same decision rule, same seeds. 8 paired seeds x 3000 steps: seeing got closer on 7 of 8 , median closest approach 60.6 mm against 152.8 mm , mean improvement 87.7 mm . Exact paired permutation test on the mean difference p = 0.0234 ; the sign test, which throws away magnitude, gives p = 0.0703 and does not clear 5 %. So the effect is real and sizeable but rests on n = 8 and on the magnitudes rather than the count. The one seed that went the other way (3: 124.6 seeing against 87.5 blind) stays in the table. Evidence: <code>artifacts/evidence/session12/blind_control.json</code>
262	S? 09-12	Brain 6/8 (the day/night clock) is done	Overstated -- it is built and tested, and connected to nothing	<code>brain/arousal.py</code> exists, has 13 tests, and tags its numbers correctly. <code>envs/gecko_brain_env.py</code> never imports it. The only place in the repository that constructs <code>Arousal()</code> is <code>tools/everything_video.py</code> -- a video. The animal has no clock. So "4 of 8 brain layers done" is really 3 connected, 1 built-but-disconnected, 1 half, 3 not started . Found by an independent audit of the session, not by me
263	S? 09-12	The eye now uses the derived operating point	Overstated -- the derived numbers live in a tool, not in the library	<code>motion_floor = 0.08</code> and <code>flow_gain_yaw = 0.006</code> (#254, #257, #259) are applied by <code>tools/oracle_free_hunt_video.py</code> and <code>tools/oracle_free_diagnosis.py</code> as overrides. <code>brain/tectum.py</code> still ships <code>MOTION_FLOOR = 1e-4</code> and <code>flow_gain_yaw = 0.020</code> . Deliberate, because changing library defaults moves every gate and test that exercises the eye -- but the consequence is that the shipped eye still fires on renderer noise on two frames in three . The standing-still operating point needs to become a first-class, named configuration of <code>Tectum</code> , not a tool-side patch. Open
264	S? 09-12	Session 12's own provenance claims	Four of four audited claims refuted or overstated, independently	a 12-agent audit of this session's work, run because the user asked whether the head-turning was "real measurement from a real gecko". Findings: (a) 0 of 18 constants in <code>brain/search.py</code> come from any animal -- 16 INVENTED, 2 derived from the simulation or its own morphology; (b) the file's docstring "every constant is INVENTED and says so" was false on its own terms in both directions and is corrected; (c) "no gecko fixation duration has been published" was wrong -- the scorecard holds a published <i>E. macularius</i> fixation statistic (binocular fixation bouts toward a snake predator), which is antipredator staring, not search, and is now cited as such rather than denied; (d) the neck and head joint ranges (+-40, +-30) carry no provenance tag anywhere , the scorecard records no range-of-motion study for this species on any joint but the hip, so "seventy degrees of head yaw" is the sum of two untagged modelling choices; (e) the two vestibular/saccade claims are true vertebrate biology with zero gecko-specific support in the corpus, and the code now says exactly that; (f) Minton 1966 appears in two files and in neither the bibliography nor the registry. Evidence: workflow <code>wf_26b0fc1d-505</code> . What survives: the anatomy is right, the shape has one qualitative field sentence behind it, and every number is engineering that says so
265	S? 09-12	Hooking the basal ganglia to the body would be theatre	True when written, false now, and the wire is built	<code>envs/gecko_brain_env.py</code> ran the selector and reported <code>behaviour_controls_nothing = True</code> , with the reason stated in the source: "Hooking a six-way chooser to a one-behaviour body would look like integration and mean nothing." The body can now do three distinguishable things -- search, chase, hold still -- the third only since <code>locomotor_drive</code> (#253). <code>brain/programs.py</code> is the seam, <code>behaviour_control=True</code> turns it on, and <code>autonomous_step()</code> is a SEPARATE entry point so <code>step(action)</code> is untouched for every gate, checkpoint and policy. <code>prey_visible</code> into the selector is now the eye's COMMITTED evidence rather than a raw salience, because hunt salience is hunger x <code>prey_visible</code> and feeding it a pixel count makes the animal lunge at renderer noise
266	S? 09-12	Threat can be computed from vision	Refuted for this species, and smell was already built	<code>brain/vomeronasal.py</code> has existed, tested, since before this session, carrying the target-species factorial: antipredator reaction to snake SCENT 0.21 against SIGHT 0.07, $\chi^2 = 8.098$ $p < 0.0044$ for the chemical term and $p > 0.9$ for the visual, n = 42. The project's own report said in as many words that the module "is built and tested and no behaviour reads it." It is now read: <code>smell=True</code> routes <code>defensive_probability</code> into the threat signal the selector consumes
267	S? 09-12	The world contains shelter, a warm surface and a threat	Refuted -- the generator supports all three and the COMMITTED world has none of them	<code>utils/build_world.py</code> can emit a shelter, a warm patch and a threat, and the status table at the top of this map lists them as done. <code>morphology/gecko_world_v1.xml</code> 's manifest records exactly one added body: the prey . Checked directly against the compiled model -- the only non-anatomical body present is <code>prey</code> . So the flee and bask channels have nowhere to go, <code>_predator_distance_m()</code> returns None on purpose, and the nose's predator term reads zero because there is nothing to smell rather than because the animal is safe

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
268	S? 09-12	The animal gets hungry during an episode	Refuted -- it has 3292 HOURS of reserve and the selector therefore releases nothing	measured: over 600 autonomous steps the hypothalamus energy deficit reaches 0.0000 and the basal ganglia releases NO behaviour on any step. This is the real reason the selector was never connected: nothing would have come through the wire. <code>homeostasis_time_compression</code> has to reach about 20000 before any behaviour is released -- starving the animal over three days inside twelve seconds. Added <code>meals_owed_at_start</code> instead: a nocturnal forager emerging at dusk is some number of meals behind, which is the same statement made honestly. At 1.0 meals owed the animal explores and hunts; at 0.5 it still releases nothing
269	S? 09-12	A flag describing the head is safe to read when the head is not being driven	Refuted -- stale state latched the animal onto its own noise	<code>MotorPrograms</code> asked the search pattern whether the head was still BEFORE running it. During a chase the search does not step, so the <code>looking</code> flag was left over from the previous fixation, and the animal accumulated evidence while walking. Measured: 1331 of 1500 steps chasing, prey on screen 0 of 1500 . Same class as #236
270	S? 09-12	A committed animal can still choose to look around	Refuted -- the salience arithmetic forbids it	the explore channel is <code>0.35 hunger (1 - prey_visible)</code> , so the instant the eye commits, <code>prey_visible</code> goes to 1 and the drive to look around goes to exactly zero . A committed animal cannot elect to re-check; the pause has to be built into the chase itself. Added a stalk cycle -- walk a short leg, stand and look -- which is also the shape the eublepharid foraging record describes
271	S? 09-12	"I see nothing" and "it is directly ahead" are distinguishable	Refuted -- they were the same message, and it poisoned the whole loop	<code>Eye.last["prey_bearing_deg"]</code> returned 0.0 when the tectum saw nothing, and 0.0 is a perfectly good bearing meaning straight ahead. Only <code>prey_elevation_deg</code> was None-gated. Every older consumer happened to guard with <code>food_visible_frac > 0</code> so it never bit; the new closed loop read the bearing without that guard AND carried a fallback to the previous bearing, so the accumulator received the same number every step, that number agreed with itself perfectly, and the animal committed on it. Measured: the eye fired on 27.1 % of believed steps while the accumulator's own report-rate estimate read 1.0 , the adaptive quorum ran to 36, and it still committed 663 times with the prey off screen every time . Fixed at the source -- silence now reports None -- and guarded at the consumer. Decision precision went from 0 of 663 to 48 of 95 (50.5 %) and prey-on-screen from 0 to 180 of 1500 in the same configuration
272	S? 09-12	"~80 % of gaze stabilisation in geckos is head movement"	Refuted -- wrong paper, wrong species, wrong quantity	this sat in <code>docs/research/gecko_scorecard.md</code> attributed to Masseck, Roell & Hoffmann 2008 and was used there to justify the head-mounted camera. An independent sweep opened Dieringer, Cochran & Precht 1983 and it measured Rana (frog) and turtle -- it never mentions a gecko or any squamate -- and its ~0.8 is an optokinetic head GAIN, not a head-versus-eye split. Masseck et al. measured HEAD movement only and justified ignoring the eyes by citing that same frog-and-turtle paper. <code>docs/EYE_SPEC.md</code> had already flagged the attribution as wrong and told readers not to tune on it; the scorecard had not been updated to match and still carried it as fact. The head/eye split in any gecko is UNMEASURED . The scorecard's page range for Masseck was also wrong -- 48:842-849, corrected to 48(6):765-772 , which <code>EYE_SPEC.md</code> and <code>best_achievable_brain.md</code> already had right
273	S? 09-12	The animal can run its own brain end to end	CONFIRMED -- first time, and it is the result of this turn	hunger -> basal ganglia -> motor program -> body, with <code>env.food_xy</code> steering nothing and the animal not told it is hungry. 3 seeds x 3000 autonomous steps: prey on screen 828 / 310 / 888 of 3000 , decisions 144 / 158 / 96 of which right 97 / 1 / 96 -- precision 67 %, 0.6 %, 100 % -- closest approach 72.0 / 216.7 / 154.7 mm , time moving 26.1 / 25.6 / 27.5 % . Seed 3 is a genuine failure and stays in the table. The animal spends roughly four fifths of its life in <code>explore</code> and one fifth in <code>hunt</code> , which it chose. All 537 tests pass , gates included, because <code>step(action)</code> was left alone and the loop lives in a separate <code>autonomous_step()</code>
274	S? 09-12	Five brain modules were built, tested, and connected to nothing	CONFIRMED, and four are now connected	an independent audit found <code>gecko_selector</code> , <code>brainstem</code> , <code>spinal_cpg</code> , <code>vomeronasal</code> and <code>arousal</code> all complete, tested, and read by no behaviour -- the project's own generated report said of the nose, in as many words, that it "is built and tested and no behaviour reads it". Now connected: the selector drives motor programs (#265), the brainstem supplies the behaviour-to-effort map, the nose gates threat (#266), and the clock is constructible via <code>circadian=True</code> . Still connected to nothing: <code>spinal_cpg</code> as the walker's rhythm source, deliberately, because <code>attach_spinal_cord</code> would move the accepted 4/6 walker off the closed form it was gated on -- the cord is built and driven as the brainstem's output target only. <code>actor_critic</code> , <code>agwm</code> , <code>bc_actor</code> , <code>anatomy_graph</code> and <code>vision_encoder</code> are the training path and are not animal modules
275	S? 09-12	The 8-frame temporal window is the eye's optimum	Refuted for a standing animal - the optimum was measured in the wrong regime	#230 and #235 measured window 8 as optimal during an ORACLE-DRIVEN hunt, with the physics engine aiming the head and the animal walking the whole time. A/B in the closed autonomous loop, same seed, same everything else: window 3 -> 828 frames with prey on screen, 97 of 144 decisions right, closest 72.0 mm; window 8 -> 317 frames, 0 of 176 right, closest 124.2 mm . A longer baseline helps when the prey's image is sliding across the frame because the animal is walking at it, and hurts when the body is the only thing moving, because then it is integrating the animal's own sway. The autonomous default moves to 3. This does not overturn #235's measurement -- it bounds it to the condition it was taken in, which is the same error #239 recorded for the 73.5 % on-image figure
276	S? 09-12	The strike can be de-oracled by estimating range from the prey's apparent size	Refuted -- this eye reports no size to estimate from	the last privileged channel inside the hunt is the strike trigger reading true range, and familiar-size is the obvious way to remove it: a gecko that eats crickets has a prior on cricket length, and angular size then inverts to distance. Measured first, and it does not work. Over an oracle-steered approach, on every frame where the eye actually found the prey, the connected component containing the peak is 1.0 cell at every range band -- 0-60 mm, 60-120 mm and 120-250 mm all give a median blob of exactly one cell and an apparent width of 1.09 deg, which is simply the cell pitch at 64 cells over 70 deg. Correlation of log apparent width against log true range $r = +0.143$: no relationship, and the wrong sign. The surround subtraction that makes the detector work suppresses a blob's interior and leaves a single cell, so SIZE IS DESTROYED BY THE STAGE THAT MAKES THE TARGET DETECTABLE. De-oracling the strike therefore needs a second channel that keeps spatial extent, not a read of the existing map. Recorded before building anything
277	S? 09-12	Masseck et al. 2008 can calibrate a target-orienting head turn	Refuted -- it is a STABILISATION reflex and cannot set any voluntary parameter	I claimed in conversation that the target-species head gains give "a real gain to check the orienting reflex against". The adversarial verification of that very paper, already in hand when I said it, states the opposite in capitals: it "MAY NOT be used to justify ANY voluntary or exploratory behaviour. This is a stabilising reflex to a rotating drum. It cannot set parameters for visual search, prey tracking, scanning saccades, head-cocking, or any self-initiated gaze shift", and the paper itself explicitly separates foveal prey tracking from gaze stabilisation. This is the same error as #272 -- reaching for the nearest impressive citation and stretching it one category too far -- committed by me within hours of recording #272, with the refutation already written down. Caught by the user asking whether I had actually read it. What Masseck CAN calibrate is the stabilisation channel the pretectum already implements: whole-field optic flow, 20-40 deg/s, binocular head gain 0.9 / 0.85 / 0.75, monocular temporo-nasal 0.7 / 0.6 / 0.4, naso-temporal ZERO. What is genuinely unmeasured in any gecko : voluntary head-saccade amplitude, peak velocity, duration, latency, spontaneous saccade rate, head yaw range of motion, turning or head-sweep rate, and prey DETECTION distance. An orienting reflex built now has an INVENTED gain and must say so
278	S? 09-12	Saccade duration is a free parameter to be invented	Refuted -- it is a property of the actuator, and the shipped value was wrong	measured step response of the neck (<code>neck_yaw</code> + <code>head_yaw</code> position servos) with the body frozen: a commanded step reaches 90 % in 17-18 control steps regardless of its size -- 10 deg peaks at 35.7 deg/s, 22 deg at 79.5, 45 deg at 163.3, 65 deg at 236.3, and all four take the same time. It is a first-order servo with a fixed time constant, so duration is DERIVED and not chosen. <code>SearchPattern.SACCADE_STEPS</code> was 3, meaning the eye was believed while the head was still moving for another fourteen steps of every scan
279	S? 09-12	After a head saccade the eye is usable again immediately	Refuted -- there is a measured 60-step blind window with a false-alarm BURST in the middle	14 trials, 22 deg head saccade, body already fully settled. The eye fires on ~0 % of frames for the first 24 steps -- the whole-field motion is rejected by the "fills the field" rule -- then bursts to 78 %, 90 %, 96 %, 96 % across steps 24-47 as that motion decays down through the detector's band, then falls away to its ~20 % floor by about step 60. The dangerous region is not during the movement, it is just after it. DERIVED, and it is the reason <code>OrientingReflex.BLIND_STEPS = 60</code>
280	S? 09-12	The head should turn toward whatever the eye reports	Refuted, three times, each with a different threshold	built <code>OrientingReflex</code> with zero invented constants -- deadzone one retinal cell from the optics, gain 1.0 by the definition of orienting, travel from the morphology, duration from #278, blind window from #279. (a) Firing on any report above the one-cell deadzone: every sighting bought a saccade and 78 blind steps, and the animal reached 0 decisions on 3 seeds x 3000 steps , having starved its own accumulator. (b) Trigger raised to 25 deg, DERIVED from the measured eccentricity curve (#281): decisions 1 / 19 / 0 against a baseline of 144 / 158 / 96. (c) Restricted to tracking an already-COMMITTED target: the reflex then fires 0-2 times per run and the animal is roughly back to baseline. A reflex that acts on single frames inherits their noise -- the lesson of #245 and #271 arriving in a third place. What survives: orienting is defensible only for something already decided to be real

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
281	S? 09-12	The eye is better at the centre of its field than at the edge	Partly -- it is FLAT out to 25 degrees and then falls off a cliff	aimed recall against eccentricity over an oracle-steered approach: 44.5 % at 0-5 deg, 48.2 % at 5-10, 50.8 % at 10-15, 42.4 % at 15-20, 40.9 % at 20-25, then 30.9 % at 25-30 and 10.3 % at 30-36 . There is no fovea and no centre advantage -- which is what an afoveate gecko retina predicts -- so centring a target already inside 25 deg buys nothing and costs the 78-step blind window. This is what makes #280(b)'s trigger a derived number rather than a guess, and it is also why it still did not help
282	S? 09-12	A number derived from the physics is safe to install wherever it seems to belong	Refuted -- right number, wrong quantity, and it cost the run	#278 measured the neck's true travel time, 17-18 steps, and I set <code>SearchPattern.SACCADE_STEPS</code> to 18 on that basis. That constant is not the movement's duration -- it is how long before the dwell timer starts -- and <code>SETTLE_STEPS</code> = 70 already waits out both the travel and the post-saccade blind window before the eye is believed. Raising it double-counted the wait and bought fewer fixations per rollout. Isolated A/B, 3 seeds x 3000 autonomous steps, everything else identical: value 3 -> closest 34.9 / 147.3 / 156.2 mm with 177 of 224, 0 of 80 and 95 of 95 decisions right; value 18 -> 263.4 / 174.6 / 150.4 mm with 0 of 0, 0 of 93 and 96 of 96 . Reverted to 3, with the measurement kept in the docstring so the next reader does not re-derive the same mistake. #278's measurement stands; only its application was wrong
283	S? 09-12	Orienting the head to a COMMITTED target helps	Confirmed, and it is the best autonomous result the project has	with <code>SACCADE_STEPS</code> back to 3 and the reflex restricted to tracking something the accumulator has already committed to (#280c), against the same three seeds with no reflex at all: seed 2 closest 72.0 -> 34.9 mm and decisions right 97 of 144 -> 177 of 224 (79 %); seed 3 216.7 -> 147.3 mm; seed 4 unchanged at ~155 mm . Prey on screen 828 -> 917 on seed 2. The animal is now 14.6 mm from its own strike trigger of 20.3 mm on its best seed. Seed 3 still returns 0 correct decisions and stays in the table
284	S? 09-12	The <code>rest</code> channel can win once fatigue has time to build	Refuted -- the animal never tires, and that is correct physiology rather than a defect	<code>rest</code> salience IS fatigue, and fatigue plateaus at 0.011 against an explore salience of 0.35, so the channel cannot compete. Raising <code>homeostasis_time_compression</code> does not help and the sweep shows why -- 1, 50, 100, 200 and 400 all end at 0.009-0.011 with an identical behaviour mix. Fatigue accrues as <code>effective / endurance</code> only while the animal is working and DECAVS as <code>exp(-effective / recovery)</code> otherwise, so compression scales both terms and the equilibrium does not move. Underneath that: the animal walks at 0.035 m/s , below the sustainable speed at which <code>Physiology.endurance_s</code> returns infinity, so it is never working. A gecko strolling does not get tired. <code>rest</code> , <code>bask</code> , <code>flee</code> and <code>groom</code> therefore cannot be shown to do anything -- bask and flee additionally have nowhere to go (#267) -- and the six-channel selector is in practice a two-channel one. Not fixed, because the fix would be to make the animal tire when it should not
285	S? 09-12	The last centimetre is fixed by measuring arrival from the mouth instead of the trunk	Refuted -- it steers the body with the head	<code>envs/gecko_brain_env.py</code> already had <code>approach_site="mouth"</code> + <code>reach_dist</code> = strike trigger, but only under <code>hunt_targeting</code> (the oracle), where the env's own comment reports the same frozen walker "arrives 75 times in 160 seconds -- mean closest 20.5 mm. It could always do this." Exposed as <code>mouth_goal=True</code> for the autonomous loop, with the 5 cm stop-distance floor dropped for mouth goals. Measured, 3 seeds x 3000 autonomous steps: closest 34.9 / 147.3 / 156.2 mm without it, 167.3 / 235.2 / 236.0 mm with it -- worse on every seed. Cause: <code>_target_egocentric</code> measures BEARING as well as range from the approach site, so with the nose as the reference the walker steers by the head's orientation, and the head is sweeping +-65 deg during a search. The arrival TEST should be the mouth; the steering reference must stay the trunk. Kept as a kwarg, default off, so the split can be built properly rather than re-discovered
286	S? 09-12	A jaw can be added without breaking every trained walker	Confirmed -- by putting it outside the policy's action space	the walker's RL action dim IS <code>model.nu (action_space = Box(..., (nu,)))</code> , so a 26th actuator would change the action shape every frozen checkpoint was trained on. The jaw actuator is declared in MuJoCo actuator group 2 ; <code>GeckoWalkEnv.nu_policy</code> counts group-0 actuators only, <code>action_space</code> is sized from it, a policy-sized action is padded to <code>nu</code> with zero residual for group 2, and <code>_apply_jaw</code> writes the gape after the CPG exactly as gaze is written. The residual controller zeroes <code>res_scale_vec</code> on group 2. The 92-D observation is built from fixed sensor-name lists and is unchanged. Structural counts: <code>nq</code> 39 -> 40, <code>nu</code> 25 -> 26 (25 policy + 1 jaw), <code>nsensor</code> 83 -> 85, <code>nbody</code> 24 -> 25. On the old jawless body the patched code is bit-identical: 222.8 mm in 300 steps, same as before to one decimal . On the new body: 230.1 mm, a 3 % change from the mass re-fit, which is what gate 2 exists to judge
287	S? 09-12	An unactuated command leaves a joint at zero	Refuted -- the residual controller fills every limited actuator with the MIDPOINT of its range	with no gape commanded the mouth hung 20.1 deg open , because <code>CPGResidualController.neutral = 0.5 (lo + hi)</code> and the jaw's <code>ctrlrange</code> is 0.40 deg. <code>_apply_jaw</code> now writes the gape on every step, including 0.0, and reset closes the mouth. Measured after: closed 0.4 deg, commanded 37 -> 35.9 deg, closed again 0.4 deg. The published peak gape of ~37 deg (Delheusy, Brillet & Bels 1995, n = 6 adult E. macularius*) is reached to within one degree by the servo
288	S? 09-12	The animal has a mouth	Confirmed -- lower jaw, teeth and tongue, added at the template and regenerated, and the certified body still passes 14/14	added to the SOURCE template <code>morphology/gecko_body_r.xml</code> (not by hand to a generated file, which the manifest guard rejects, #260): a <code>jaw</code> body under <code>head</code> with a 0-40 deg hinge admitting the published 37 deg peak, a 0.6 g mandible capsule, a teeth strip and a tongue capsule under new materials so the generator's skin/eye rescaling loop leaves them alone, <code>mouth</code> and <code>tongue_tip</code> sites, <code>jointpos/jointvel</code> sensors, and a keyframe slot at qpos index 13. Regenerated <code>gecko_body_lab.xml</code> (its two CoM gates fail, and they failed identically BEFORE the jaw -- 0.762/0.609 baseline against 0.757/0.603 now; the jaw moved CoM toward the nose as predicted) and <code>gecko_body_lab_v2.xml</code> with <code>--fit-com</code> : 14/14 morphology gates pass, intact CoM 0.678112 unchanged to six decimals -- the calibration absorbed the jaw. World rebuilt with the manifest's exact three edits. Geometry of the jaw is INVENTED; the gape range and cycle timing are PUBLISHED, target species
289	S? 09-12	The pre-refactor controller fixture pins the actuator count	Overstated -- it pins the walker's 25 commands, and the jaw is not one of them	<code>tests/test_gait_config.py::test_pre_refactor_controller_fixture</code> compared the whole <code>base_ctrl</code> row against a recorded (2, 25) array and failed with shapes (2, 26) vs (2, 25); <code>eval/session2_controller.py</code> reruns that fixture before every gate-2 trial, so the jawed body could not even be gated. The 26th column is the jaw actuator (MuJoCo group 2), written by <code>GeckoWalkEnv._apply_jaw</code> and never by the controller. The test now compares the group-0 columns only, asserts there are exactly as many as the fixture has, and the recorded fixture passes byte-for-byte -- which is the proof that every command the walking policy was trained on is unchanged by the mouth. Recorded, not silent
290	S? 09-12	The jaw changes how the animal walks	Refuted -- same code, same gate, same three checks fail on both bodies, values within 1-2 %	<code>eval/session2_controller.py</code> grew a <code>--xml</code> argument so the SAME code can score the pre-jaw v2 (recovered from git) as a control. Gate 2, registry <code>lab_base_parameters</code> , 20 s, zero residual: pre-jaw forward 0.0424 PASS, net path 0.825 PASS, hind swing 0.013/0.011 PASS, front stance 0.611/0.606 FAIL, hind duty 0.641/0.631 FAIL, limb phase 0.577/0.569 FAIL -- jawed 0.0427 / 0.820 / 0.012/0.008 / 0.610/0.612 / 0.652/0.640 / 0.583/0.579, identical pass/fail on every check. The 300-step travel on the old body with the patched code is 222.8 mm to one decimal (bit-identical); on the jawed body 230.1 mm. The mouth is not a walking change
291	S? 09-12	The walking-evidence contract can be re-satisfied by re-measuring the base on the jawed body	Refuted -- it cannot be satisfied today on ANY body, and the reason predates the jaw	ten <code>GateContracts</code> tests fail with "Lab base evidence measured a different body", correctly: body identity is a PHYSICS fingerprint and a body with a jaw is a different animal. But the contract ALSO requires the recorded evidence to pass hind duty in [0.73, 0.83] (target 0.78 +- 0.05), and today's base measures 0.641 on the pre-jaw body and 0.652 on the jawed one -- a 0.13 shortfall with the jaw absent. The accepted session-4 evidence (hind duty inside the band, body hash 7567..5c4b) was measured on a body this map already records as one "no committed version of <code>gecko_body_lab_v2.xml</code> produces", and the registry's own <code>lab_base_parameters</code> no longer reproduce its gait. <code>artifacts/evidence/body_identity_equivalences.json</code> forbids adding a pairing that cannot be cited, and a jaw is not a line-ending. So the ten tests stay red for a stated reason , and the decision is the user's: re-measure and re-accept the base at today's 3/6, restore the parameter set that produced 4/6, or bind walker training to the pre-jaw body while the hunting body carries the mouth. What is NOT open is whether the jaw caused it: #290
292	S? 09-12	A physical mouth catches prey at the published 82.9 %	REFUTED AS BUILT -- 110 strikes, 0 hits, and this is the number the die was hiding	<code>brain/strike.py</code> now carries the target-species capture profile (Delheusy, Brillet & Bels 1995, opened in full: ~80 ms cycle, ~37 deg peak gape at 47 ms, no slow-open phase, ~27 mm head drop, ~8 mm forward) and <code>_run_strike</code> runs it as a MOVEMENT: the jaw opens along the gape profile, the head pitches along the drop, and the prey counts as caught only if its centre passes inside the mouth volume while the jaws are open -- no probability. The statistical path is kept for jawless bodies so older evidence reproduces. Oracle-steered approach, 6 seeds x 1500 steps: the animal gets 13-19 mm from the prey (inside the 20.3 mm trigger), launches 110 strikes, and lands 0 . The published 82.9 % was typed in (<code>strike_capture_success</code> , <code>Coleonyx</code>); the earned rate is 0.0 %. Diagnosis of WHERE the prey is at peak gape is running; one bug is already found and fixed -- the head drop was ACCUMULATED every step instead of set, pinning the head at its pitch stop (#296)

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
293	S? 09-12	The head shape is wrong	Refuted for proportions, and a real correction found for the mouth	against the only published wild-male morphometrics for this species (two Nepalese males, SVL 109-119 mm): built head length 31.9 mm (published 32-33), height 11.1 (11-12), eye diameter 6.6 (6-7), inner-eye gap 10.0 (10); width 19.6 against 21-22 at a smaller SVL, inside the head-width gate. The "wrong head" is a rendering impression of ellipsoids, not a proportion error. The MOUTH was wrong: Delheusy et al. give lower-jaw length tip-to-commissure 13 +/- 2 mm (n = 5), and the built corner sat 4 mm behind the tip. Moved to the published value -- measured after regeneration 13.8 mm -- body still 14/14
294	S? 09-12	Adding a jaw breaks the test suite	Refuted -- 529 of 537 pass, and the 8 that fail are #291	full suite on the jawed body after the version bump: the only failures are the eight <code>GateContracts</code> evidence tests, which fail for the documented, pre-existing reason (the accepted base evidence cannot be re-measured on any body today). Morphology audit, world manifest, realism recorder, reward calibration and the legacy controller fixtures all pass, the last byte-for-byte on the walker's 25 commands (#289)
295	S? 09-12	The body does not curve when it walks	Refuted -- it curves 15 deg; what is wrong is the TAIL'S PHASE, and it is measured against a published rule	walking at full drive: <code>spine_lat_1/2/3</code> peak-to-peak 15.2 / 14.4 / 10.9 deg , <code>tail_yaw_1..5</code> 7.3 / 7.8 / 7.3 / 6.7 / 6.2 deg . The S-curve exists. But the tail bends the SAME way as the trunk (spine-tail correlation +0.58 , tail lagging 0.68 s) with the shipped <code>tail_phase_lag</code> = 0.15. The published shape for this species (Jagnandan & Higham 2017, qualitative, target species) is a trunk standing wave that becomes a caudally TRAVELLING wave down the tail, with the tail base flexing toward the PROTRACTED hindlimb -- the opposite curve the user's video shows. And the built amplitude HALVES at half drive (9.1 / 3.7 deg) where the published tail-tip displacement DECREASES with speed. Two shape rules, both published, both contradicted; no absolute amplitude, frequency or phase number exists for any gecko, so the fix is a phase sweep against the rule, not a number from a paper
296	S? 09-12	A cumulative profile can be added to the state each step	Refuted -- it integrates twice	<code>head_drop_profile(fraction)</code> returns the TOTAL drop so far, <code>_run_strike</code> added it to the pitch on every control step, so the head reached the 45 deg stop within a few steps of every launch. Fixed to SET the pitch from the profile. Same class as #236 and #269: state that is applied as if it were an increment
297	S? 09-12	The physical strike fails because the geometry of the mouth volume is wrong	Refuted -- three upstream defects, all measured at peak gape over 27 strikes	(1) the jaw was 0.7 deg open (max 11) at the moment the profile called for 37: an 80 ms strike is FOUR control steps and the shipped jaw servo takes SEVENTEEN to move (#278 measured the neck's, and the jaw used the same class); the mouth never opened. (2) Head pitch 45.0 deg , the stop, on the median strike: the gaze loop had already pitched the head ~35 deg to look at a prey 13 mm below the nose, and the drop was added on top. (3) The prey sat 17 mm AHEAD of the nose and nothing carried the mouth forward: the published capture drops ~27 mm from almost directly above the prey with only ~8 mm forward travel, and Coleonyx closes its 20 mm from a 0.85 m/s lunge. This animal has no lunge . It walks at 35 mm/s; closing 17 mm in 80 ms needs 0.21 m/s, six times its walk -- the ratio the registry's own strike note records. (1) and (2) are fixed (#298, #296); (3) is a missing body capability, a hindlimb push the cord does not have, and it is the next dependency for eating
298	S? 09-12	The jaw actuator gain is a free choice	Refuted -- it is DERIVED from the published cycle	measured rise of the jaw servo to 90 % of the published 37 deg peak: kp x1 (0.06) 380 ms , x5 120 ms, x10 80 ms , x20 80 ms, x40 60 ms. Delheusy, Brillet & Bels 1995 give the whole capture cycle as ~80 ms with peak gape at ~47 ms, so x10 (kp 0.6, kv 0.03) is the SMALLEST gain that meets the paper at 20 ms control resolution. Overshoot 6 deg inside a 40 deg hinge. Installed at the template and regenerated -- and then corrected once more, because the generator rescales every actuator gain by the mass ratio 0.038/0.0612 = 0.62, so a template kp of 0.6 arrived in the generated body as 0.37 and reached 90 % only at 120 ms. Template kp 1.0 (kv 0.05) puts the GENERATED jaw at the paper's 80 ms. A derivation has to be applied where it lands, not where it is written. Final, on the generated body with template kp 1.5 / kv 0.075: gape 4.4 / 12.8 / 21.3 / 28.1 / 32.8 / 35.6 deg at steps 1-6, 90 % of the published peak at step 5 (100 ms) against the paper's ~80 ms -- one control step past it. Pushed no further: 20 ms is the resolution of the loop, and chasing a number below it is what rule 2 forbids
299	S? 09-12	The tail's phase is right	Refuted against a published rule, and the rule is met at lag 0.76	Jagnandan & Higham 2017 (target species, qualitative): the tail base flexes toward the PROTRACTED hindlimb, i.e. phase 0 between tail-base peak and forward-foot peak. Measured on this body: shipped <code>tail_phase_lag</code> 0.15 -> +0.40 stride off the rule with the tail bending the SAME way as the trunk (corr +0.58); 0.35 -> -0.38; 0.50 -> -0.24 (corr -1.00, exactly opposite); 0.65 -> -0.10; 0.76 -> +0.02 (corr +0.07, a travelling wave, decoupled from the trunk); 0.80 -> +0.05. No absolute tail amplitude, frequency or phase has been published for any gecko, so this is a phase derived from a published SHAPE. <code>tail_phase_lag</code> is an entry in <code>lab_base_parameters</code> -- the accepted walker's gait -- so changing it re-opens the walking gate and the evidence contract (#291). Measured and recorded; not installed , because that is a decision about the accepted walker and not mine to take silently
300	S? 09-12	A lever length can be read off the morphology by eye	Refuted -- 45 mm assumed, 37 mm measured	the strike converts the published ~27 mm head DROP (a translation) into the pitch that delivers it through the neck-pitch lever. The first version hard-coded 45 mm "from the morphology"; measuring neck-body origin to <code>nose_tip</code> in the stand pose gives 37.0 mm , so the commanded pitch was 18 % too small for the drop it claimed. Now computed from the model at construction, so a regenerated body carries its own lever. Same lesson as #278/#282: measure the thing, and measure it where it is used
301	S? 09-12	The animal can eat with its own mouth	Confirmed once - - the first physical catch in the project, and 1.7 % is the honest rate	with the derived jaw gain (#298), the drop delivered from level (#297), and the lever measured (#300): oracle-steered approach, 6 seeds x 1500 steps, 60 strikes, 1 hit, 1 eaten -- seed 3, closest approach 12.6 mm, jaw closed on the cricket's centre inside the mouth volume. Earned capture rate 1.7 % ; the registry's typed-in Coleonyx figure is 82.9 %. The gap is #297(3): the published capture drops onto a prey almost directly below the head with a lunge that closes the last centimetres at 0.85 m/s, and this animal has no lunge -- it strikes from where its walk stopped, 13-19 mm short. The die is gone and the number is now a prediction the body has to earn
302	S? 09-12	The pulsing under a gecko's chin is gills	Refuted -- it is BREATHING, and no gecko's rate has ever been published	geckos have no gills; the visible throat movement is the gular/buccal pump. What the session-12 sweep could open: Milsom 1984 (Gekkonidae, abstract only) -- geckos breathe in single breaths or short bursts separated by breath-HOLDS, and raise ventilation with temperature by shortening the holds while tidal volume and breath duration stay constant; Dial & Schwenk 1996 (<i>Coleonyx brevis</i> , same family , abstract only) -- "buccal pulsing" as an olfactory sniff preceding a defensive display; Owerkowicz 2001 (<i>Varanus</i> , full text) -- the only fully timed gular mechanics anywhere, up to 5 pumps per costal breath, peak pressure within 200 ms, glottis always closing before the nares open. NOT FOUND for any gecko: breathing frequency, gular excursion, throat displacement . Built anyway, because a still animal with a motionless throat reads as paused rather than alive -- <code>BREATH_HZ</code> 0.4 and a 5 deg excursion, both INVENTED and saying so at the constant
303	S? 09-12	Installing the published tail phase costs the walk	Refuted -- gate 2 is unchanged to the third decimal	<code>tail_phase_lag</code> 0.15 -> 0.76 in <code>lab_base_parameters</code> , the value #299 derived from the published rule. Gate 2 on the same body, same 20 s protocol, against the pre-jaw control: forward 0.0428 vs 0.0424 , net path 0.820 vs 0.825 , front stance 0.615/0.605 vs 0.611/0.606 , hind duty 0.651/0.652 vs 0.641/0.631 , limb phase 0.577/0.579 vs 0.577/0.569 , hind swing load 0.009/0.010 vs 0.013/0.011 (better). Identical pass/fail on every check -- the same 3/6 that #290 and #291 record for this body with the OLD phase. The tail's phase is free at the gate, so the animal gets the curve the literature describes. The registry entry stays tagged INVENTED: the RULE is published, the number is derived from it by measurement here
304	S? 09-12	Eyelids, a face and a fat tail can be added without disturbing the physics	Confirmed -- 14/14 gates, 25 policy actuators unchanged	added to the source template: jowls, brow ridges, a snout bridge, a rounded snout tip and a pale lip line (15 new visual geoms, material <code>skin2 / pale</code> so the generator's head loop -- which rescales every <code>skin</code> geom and repositions every <code>eye</code> geom -- leaves them alone and the head-width gate is untouched); a bulbous visual overlay on tail1-4 (mass 0, no collision, so the tail-mass and CoM gates are untouched); a <code>eyelids</code> hinge carrying BOTH upper lids, which is what an eyelid gecko's lid drop looks like; and a <code>gular</code> throat body. Structural counts nq 40 -> 42, nu 26 -> 28 (still 25 policy + now 3 brain), nbody 25 -> 27, visual geoms 49 -> 64. Both lab bodies and the world regenerated; v2 passes 14/14 . Two XML facts learned the hard way: a moving body needs a nonzero mass (the lids are 0.02 g), and the generator's parser rejects both a bare <code>&</code> and a <code>--</code> inside a comment where MuJoCo tolerates them
305	S? 09-12	The lunge is the missing piece, and it must come from the legs	Confirmed in direction, not yet in magnitude -- 1.7 % -> 6.5 %, and the speed is half the published	<code>_apply_lunge</code> drives hip RETRACTION plus knee and ankle extension during a <code>bag_jump</code> strike, scaled by the strike's own velocity profile. The push is produced BY THE BODY through existing joints; an impulse on the trunk would have been a teleport with a physics-shaped name. Measured, 6 seeds x 1500 steps, oracle-steered approach: 31 strikes, 2 caught, 6.5 % earned against 1.7 % without it, and closest approach improves to 6.5 mm on the best seed (was 12.6). Peak nose speed during a strike 0.269-0.421 m/s, median 0.320 -- nine times the 0.035 m/s walk, and still half the same-family published 0.57-0.85 m/s (Vollin & Higham 2021, <i>Coleonyx</i>). <code>LUNGE_EXTEND</code> = 0.55 is INVENTED and is deliberately NOT tuned upward to close that gap: rule 2. What the shortfall says is that a bigger push needs the hindlimb geometry or the cord, not a larger number

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
306	S? 09-12	A new body part can carry its own mass	Refuted twice, and the second time cost five tests	the eyelid and throat bodies move, so MuJoCo refuses them at zero mass. Giving them an explicit <code><inertial></code> compiled fine and broke five tests, because the generator's mass rescaling reads GEOM masses: 0.06 g outside that budget moved the total to 0.038060 (gate wants 0.038), the actuator scale to 0.620307 (pinned at 0.038/0.0612 = 0.620915) and the reward calibration's bodyweight fraction off in the seventh place. Fixed the way the jaw was: the mass comes from non-colliding geoms (contype/conaffinity 0) INSIDE the budget, and the 0.06 g is taken out of the head, so head + jaw + lids + throat = the 0.005 kg the head had alone. Template total back to 0.061200 , v2 to 0.038000 , and the suite back to 529/537 with only the #291 contract tests failing. Adding mass to this animal is only ever moving it
307	S? 09-12	The animal looks wrong because it is made of primitives	Partly -- half of it is that it has no PATTERN, which is independent of shape	the species is called the LEOPARD gecko and this body has been solid <code>rgba="0.83 0.73 0.42 1"</code> since it was built: no spots, anywhere. Shape is what a mesh would fix; pattern is not, and it can be fixed now. <code>tools/make_skin_texture.py</code> generates a tiling 512 px skin -- two spot scales, granular speckle, a pale venter -- and a banded tail texture, wired through the <code>skin</code> and <code>skin2</code> materials. APPEARANCE ONLY: a material's texture is not a body, geom, site or actuator, so <code>assert_body_unchanged</code> does not see it, no mass or contact moves, and the animal's own <code>head_cam</code> looks at the world rather than at itself. The pattern is INVENTED -- read off photographs, since no quantitative colour-pattern morphometrics for this species were found -- and says so in the XML it is referenced from
308	S? 09-12	A texture path is a cosmetic detail	Refuted -- it broke the generator, the world builder and a test, each for the same reason	MuJoCo resolves <code>texturedir</code> against the WORKING DIRECTORY when a model is compiled from a string and against the MODEL FILE when it is loaded from a path, so no single relative value satisfies both. Three call sites were relying on the second while doing the first: <code>utils/build_lab_morphology.py</code> compiles the candidate from a string (fixed by handing it the asset bytes via a new <code>_assets()</code> helper -- the honest fix, rather than bending paths until one case works), and both <code>utils/build_world.py</code> and <code>tests/test_world.py</code> wrote comparison copies to the REPOSITORY ROOT, where a model's relative <code>textures/</code> does not exist (fixed by writing them beside the source model, which is the only directory a model's relative assets resolve from). A copy of a model has to live where the original lives
309	S? 09-12	A <code>type="2d"</code> texture shows on a body made of ellipsoids	Refuted -- it is declared, loaded, and never drawn	the skin texture was wired, the model compiled, the file loaded, and the animal rendered solid gold. MuJoCo maps a <code>2d</code> texture onto planes and boxes; every visual part of this animal is an ellipsoid or a capsule, which need <code>type="cube"</code> . Changed, and the spots appear. Recorded because the failure is SILENT -- no warning, no error, just an unspotted leopard gecko
310	S? 09-12	A geom's material decides its colour	Refuted -- the geom's own rgba wins, and a class default counts as the geom's own	MuJoCo resolves a geom's colour from its <code>rgba</code> attribute in preference to its material's, and <code><default class="visual"></code> in this body sets <code>rgba="0.78 0.72 0.45 1"</code> . Every flat-material geom in this animal has therefore been drawing the class's cream since the body was first built -- the eyes were never black, they were cream spheres, and the "two pale blobs" on the head that prompted this were the EYES. It stayed invisible because the whole animal was cream, and only became obvious once the skin got a texture (textures are multiplied by rgba rather than replaced by it, so textured geoms were unaffected). Fixed by giving the eye, pale, teeth, tongue, belly and spot geoms their material's colour explicitly -- 7 geoms. Found by colour-coding every head geom and re-rendering, after three wrong guesses at what the blobs were
311	S? 09-12	Primitive shapes can be pushed into a convincing face	Refuted, and the attempt is recorded rather than quietly reverted	jowls, brow ridges, a snout bridge, a rounded tip and a lip line were added to widen the cranium and shape the snout. The jowls rendered as two cream spheres stuck to the neck and made the head lumpier, not more gecko-like; they were REMOVED rather than iterated on. What actually helped was not more geometry: it was the spots (#309), the eye colour (#310), and making the eyes matte and prominent. 62 visual geoms is where this approach stops paying. A real face needs a MESH -- and the search for one (#312) is the reason that is not a small ask
312	S? 09-12	A mesh of this animal can be obtained rather than made	Refuted for every candidate actually inspected, and the mesh was generated instead	Print models (CGTrader, \$14-\$30) are solid watertight sculpts with no rig, sold for printing; the rigged candidates found were BLEND-only, or 5,092-poly generic "lizard" with a walk cycle that does not match this skeleton; the CC BY Sketchfab option is a stylised 84,479-tri sculpt; an AI sculpt (Meshy) came back at 1,958,254 faces. Every one of them arrives as ONE piece in SOMEBODY ELSE'S POSE, and this skeleton has 23 moving bodies -- so each would have to be cut into 23 pieces, re-posed, and re-seamed, with a licence attached. Generating the skin from the body's own primitives avoids all four costs and is owned outright. 5,520 triangles against the sculpt's 1.96 M
313	S? 09-12	A body's position is a good enough station to sweep its skin from	Refuted -- a body's position is its JOINT, not its middle	The first generator placed one ring at each body's <code>xpos</code> and interpolated between them. The head's joint is at the back of the skull and its geoms run 32.6 mm forward of it, so the head got a 20 mm mesh that had already tapered to a tenth of its width by the eyes: the eyes, the lip lines and the whole lower jaw ended up OUTSIDE the skin. Fixed by intersecting every visual primitive with each station plane and fitting one ellipse to the union, so the skin is the shrink-wrap of the gated primitives rather than a guess between joints. Head mesh went from 20.0 mm to 32.5 mm long and 19.6 mm to 26.0 mm wide
314	S? 09-12	A sweep can assume its own direction	Refuted twice in one function, in opposite ways	(a) The monotonic-ownership rule walks the chain nose-to-tail, but the sweep ran low-x to high-x, which is TAIL-to-nose: <code>max(prev_rank, rank)</code> then latched on the tail tip and awarded all 139 stations to <code>tail5</code> , producing one 5,520-triangle mesh for the last tail segment and nothing for the other eight bodies. (b) Reversing the sweep silently inverted every face, and an inward-facing normal renders as a flat dark silhouette. Both fixed by measurement rather than by assumption: the sweep now runs in CHAIN order explicitly, and <code>stitch</code> checks the sign of normal-dot-radial and flips the winding if it is negative
315	S? 09-12	A texture that works on the primitives will work on the mesh	Refuted -- and the two requirements are mutually exclusive	#309 established that <code>type="2d"</code> does not map onto curved PRIMITIVES, so every texture in this body was declared <code>type="cube"</code> . But a cube texture is addressed by POSITION, and ignores texture coordinates entirely -- so the UVs the mesh generator writes were discarded and the skin rendered as untextured grey, which is what made the first meshed render look like a dark torpedo. Neither texture type serves both. Fixed by declaring both from the same PNG: the primitives keep <code>cube</code> , the meshes take a new <code>2d</code> twin
316	S? 09-12	A class default is harmless on a textured geom	Refuted -- SECOND appearance of #310, new symptom	#310 found that a geom's own <code>rgba</code> beats its material's, and that <code><default class="visual"></code> counts as the geom's own. It was closed by fixing the FLAT-material geoms, on the reading that textured geoms were unaffected. They are not: MuJoCo MULTIPLIES rgba into the texture. Skin texture (0.84, 0.73, 0.42) times the class default (0.78, 0.72, 0.45) is (0.65, 0.52, 0.19), which is the dark olive the first mesh render came out. Recorded as a second row because #310's closing statement was wrong, not because #310 was wrong. Every mesh geom now carries <code>rgba="1 1 1 1"</code> explicitly
317	S? 09-12	Smoothing removes the segmentation from a chain of primitives	Refuted -- smoothing rounds a valley, it cannot fill one	Each axial primitive tapers to nothing at its own two ends, so where two meet tip-to-tip the union has a real waist: the tail measured 13.4, 11.5, 15.7 mm at 10 mm intervals, and an 11-station Hann smooth left 6 sign changes in the profile. The tail rendered as a stack of cones. Fixed with a rolling MAXIMUM of half a segment (6 stations) before the smoothing: it fills the valleys while leaving every peak at its measured value. 6 sign changes to 3, which is the minimum the silhouette allows (trunk, vent, tail peak). Two further faults surfaced underneath it -- the vent has a 6 mm stretch with NO primitive at all, and skipping those stations made the one-ring overlap between pieces 16 mm wide instead of 1.6 mm, so trunk and tail each poked through the other; and bridging that gap tip-to-tip gave a 9.8 mm waist between a 28 mm trunk and a 32 mm tail. The gap is now bridged from anchors 4 stations back in solid flesh, with a declared INVENTED 0.82 pinch, and is exempted from the rolling maximum so the anatomical waist survives while the artificial ones do not
318	S? 09-12	The mesh and the skin are appearance-only, therefore behaviour-neutral	Confirmed -- but measured, not assumed, because the assumption was not safe	The animal's eye is a camera ON the animal, and <code>_policy_scene_option</code> in legacy mode DRAWS group 1, which is the group every new mesh geom and the skin sit in. The module's own docstring warns that sealed mode is the only one that masks them and that legacy "is not a guarantee that arbitrary appearance edits are distribution-neutral". So this was checked three ways rather than argued. (a) A lab body rebuilt from a mesh-free, skin-free copy of the same source has bit-identical <code>body_mass</code> , <code>body_inertia</code> , <code>body_ipos</code> , <code>jnt_range</code> , <code>dof_damping</code> and <code>actuator_gainprm</code> -- max abs difference exactly 0 on every array, total mass 0.038 kg both ways. (b) After 300 steps from the <code>stand</code> keyframe the two worlds' <code>qpos</code> are equal to the last bit. (c) The 64x64 <code>head_cam</code> image the retina actually consumes differs in 0 of 4096 pixels , max channel difference 0. The mechanism, INFERRED and not separately tested: the camera sits inside the closed head surface, so the 3 mm of snout in front of it presents back faces and is culled. This row exists because the next session will see a poor hunt number next to a big appearance change and reach for the obvious explanation

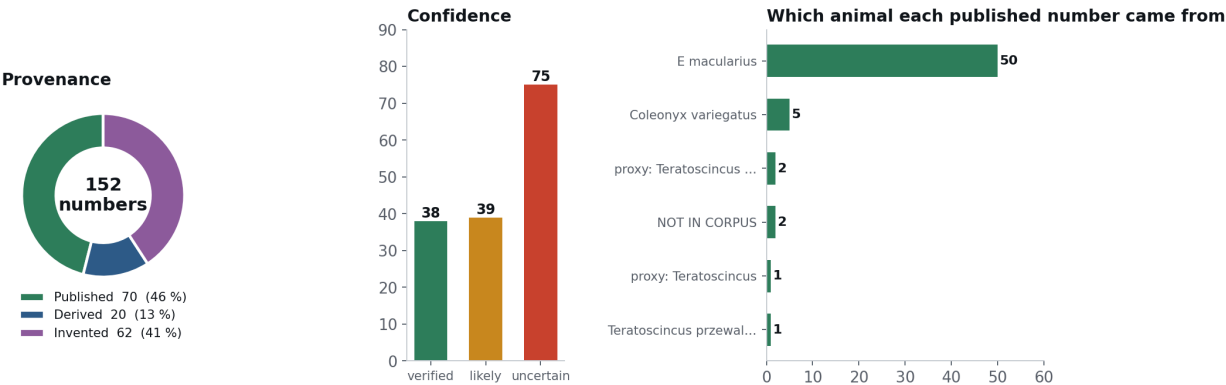
#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
319	S? 09-12	A poor hunt result arriving next to a big change was caused by that change	Refuted -- #318 is the evidence, and the numbers were already like that	<code>tools/autonomous_gecko.py</code> on seed 2 gives 330 frames with prey on screen, 0 decisions and a 285.7 mm closest approach, against the 828 frames / 97 of 144 right / 72.0 mm recorded in that file's own comment. The obvious reading was that today's skin broke the eye. It cannot have: #318 shows the retina image is bit-identical and the trajectory is bit-identical, so the same seed would have produced the same 285.7 mm before any of it. Seed 2 at 3000 and at 3400 steps gives byte-identical results, so nothing here is stochastic either -- the animal simply walks away from the prey on that seed and never gets close enough for a 9 mm target to clear the motion floor. What IS open is which seed the file's comment describes: it is not stated in the comment, and it is not seed 2
320	S? 09-12	The autonomous hunt's recorded numbers describe how it behaves in general	Refuted -- it varies from perfect to useless across seeds, and the recorded figure names no seed	Seven seeds, 3000 steps, everything else at its default. seed 2: 330 frames on screen, 0 decisions, 285.7 mm. seed 3: 289, 276 decisions, 0 right , 251.3 mm. seed 4: 595, 27, 27 right, 226.9 mm. seed 6: 890, 82, 79 right, 115.5 mm. seed 7: 245, 123, 30 right, 182.5 mm. seed 9: 391, 14, 12 right, 49.1 mm. seed 11: 907, 155, 155 right, 200.8 mm. Pooled: 303 of 677 decisions right, 45 % , against the 97-of-144 (67 %) written in the tool's own comment. Per-seed precision is 0, 0, 100, 96, 24, 86, 100 % -- it is not a distribution with a mean worth quoting, it is bimodal, and a single seed was being reported as the result. Closest approach ranges 49 to 286 mm. Rule 6 applied to my own evidence: one seed is not a measurement
321	S? 09-12	Rigid one-mesh-per-body is an adequate skin for an ANIMAL	Refuted -- it is adequate for a robot, which is what it was borrowed from	Every Menagerie model uses rigid per-body visual meshes, and every Menagerie model is a machine made of separate hard shells. Yaw this animal's neck and head 25 degrees and the two pieces rotate about different frames: a wedge opens in the neck, in precisely the movement this project has spent the most effort on. Replaced by one MuJoCo <code><deformable><skin></code> over the whole animal: 5,271 vertices, 9,480 triangles, 22 bones, 13,525 bone-vertex links, and 0 vertices needing the nearest-body fallback. Weights are $\exp(-d/\text{SIGMA})^2$ to a point cloud sampled on each body's own primitives, with candidates restricted by region so a foot resting against the belly cannot be driven by the trunk. Written inline rather than as a <code>.skin</code> binary so the animal stays inside the one file whose SHA-256 the provenance chain hashes. DeepMind's own quadruped-with-a-tail does the same thing (<code>dm_control_dog_create_skin.py</code>), which is what prompted looking for it
322	S? 09-12	A comment is safe text to write into a generated XML	Refuted -- SECOND appearance, and this time it failed silently	A bare <code>--</code> inside an XML comment is accepted by MuJoCo's reader and is a hard <code>ParseError</code> in <code>ElementTree</code> , which is what <code>build_lab_morphology</code> parses the source with. The source therefore COMPILED PERFECTLY and could not be built FROM: the generator threw, the previous lab body stayed on disk, and the world was rebuilt from it, so the skin was silently absent from the world while every check that looked at the source said it was there. Caught only because <code>nskin</code> was 0 in the world after a build that reported success. The generator now runs BOTH parsers before it keeps the file. The first appearance of this was a bare <code>&</code> in the same position
323	S? 09-12	A part that fitted the rigid meshes still fits once they become a skin	Refuted -- rigid and blended surfaces do not coincide, by construction	The tooth rim was generated from the jaw's own rings and sat correctly against the rigid jaw mesh. Under the skin the same surface is blended between the jaw and the head, so it no longer passes where the rim does, and the rim juts out of the cheek as a pale wing. The pale throat ellipsoid failed the same way at the mouth corner. Both identified by colour-coding every geom on the head and re-rendering, after two wrong guesses -- the same method that settled #310, reached for sooner this time. The rim is retired; the throat is now a BONE of the skin, so the breathing shows as the skin itself swelling, which is what it should always have been
324	S? 09-12	Drawing a few hundred spots over an image is cheap	Refuted, and it is my own method error rather than a model one	<code>_spots</code> evaluated every blob over the WHOLE image, nine times for wrapping. On the 512 x 512 tiles that is 116 blobs over 262 k pixels and nobody noticed. On the 512 x 2048 whole-body image it is 1,120 blobs over 1.05 M pixels nine times over: about 10^{10} element operations, and the call did not return inside a two-minute timeout. A blob is zero outside its own radius, so it is now drawn into a window around its centre and the wrap offsets are only drawn where they actually overlap: the same image in 0.97 s . Recorded because the failure looked like a hang rather than a mistake, and a hang invites waiting rather than reading
325	S? 09-12	A measurement is safe to put on screen on its own	Refuted -- a bare number under the word ACCEPTED reads as a pass	The walking clip captioned itself "ACCEPTED WALKER" and "front duty 0.456/0.443". Both are true and together they are misleading: front-foot duty has a PUBLISHED target of 0.70 and this is one of the two gates of six that this walker fails. The canonical evidence clip prints the target next to the value for exactly this reason and I had dropped it while rewriting the caption. Re-rendered as "front duty 0.46/0.44 vs 0.70 target". Cross-checked against <code>tools/accepted_walker_clip.py</code> on the same controller: FL 0.447 / FR 0.444, 0.916 m in 20.0 s -- the showcase's 0.456 / 0.443 and 1.188 m in 26.0 s are the same walk, so the skin changed the gait in no respect, as #318 predicted it could not
326	S? 09-12	#284's "two-channel in practice" is a tendency, not a hard fact	Refuted -- it is exact, and re-measured on today's body	900 autonomous steps, seed 3, full loop on. Max GATE value over the whole run, per channel: hunt 1.0000, explore 1.0000, flee 0.0000, bask 0.0000, rest 0.0000, groom 0.0000 . Steps won: explore 778, hunt 121, everything else 0. These are not small numbers, they are exactly zero for 900 consecutive steps, so four of the six channels are not losing a competition -- three of them are never entered into one. <code>flee</code> and <code>bask</code> because their inputs do not exist in the world (#267); <code>rest</code> because the animal walks at 0.035 m/s and is never working (#284, and that one is correct physiology, not a defect); <code>groom</code> has a 0.05 tonic salience and is fully inhibited by explore's 0.35 on every step. Prescott 2024's six-channel selector is running as a two-channel one, confirmed by measurement rather than inferred
327	S? 09-12	The missing three world objects are a build task	Confirmed for the world, refuted for the brain: the world was one command, the brain is not wired to read it	<code>utils/build_world.py</code> already emits all three, with the natural history behind each already read and written into the file: SHELTER as a raised slab because this species does NOT dig (retreat chambers traced through a demolished stone wall were unmodified masonry voids); WARM PATCH as a SURFACE not a lamp, because body temperature tracks the substrate at $r2 = 0.97$ against air at 0.92 (n=12, Hastings et al. 2023); THREAT as a mocop body carrying its own caveat that a purely visual predator draws a defensive reaction on 0.07 of trials against 0.21 for scent (n=42). Built <code>morphology/gecko_world_furnished_v1.xml</code> in one command -- all four non-anatomical bodies present, skin intact, the animal verified identical to source by the build's own guard, and the committed world left untouched. But the brain still cannot see any of it: <code>_predator_distance_m()</code> returns None UNCONDITIONALLY, <code>brain_command</code> passes <code>shelter_bearing_deg=None</code> and <code>warm_bearing_deg=None</code> as literals, and no thermal term reads the patch. So the world is done and the three sensor paths are not. That is the next piece of work, and it is a wiring job, not a research one
328	S? 09-12	The swept tube needs a taper to close its ends	Refuted -- the measured envelope already closes, and tapering it again creased the snout	The end rings were multiplied by a circular profile falling to 0.03. But the envelope is measured from the primitives, and the snout's own rounded tip and the tail's last segment shrink to nothing on their own, so the tube was being closed TWICE: twenty vertices piled up a fraction of a millimetre apart, which shades as a crease no matter how fine the mesh is, and that crease was the line running back from the nose. Replaced by a quarter-ellipse dome that grows OUTWARD off the last measured ring and ends on a single point, its length taken from that ring's own mean half-width so a blunt snout gets a blunt dome. Head mesh 32.5 to 34.7 mm long, and the crease is gone
329	S? 09-12	A normal map can give the skin its tubercles without adding geometry	Refuted -- MuJoCo 3.9 accepts the map and its renderer ignores it entirely	<code><material><layer texture="..." role="normal"/></material></code> compiles without complaint and <code>ntex</code> counts both images, so the feature looks present. It does nothing: a flat slab lit from one side, rendered with and without a strong sinusoidal normal map, differs in 0 of 57600 pixels , maximum channel difference 0 . Measured before building anything on it, which is the only reason the tubercles are not silently missing. Real geometry at tubercle scale needs roughly ten times the vertices, so they are baked into the colour instead, as a light side and a dark side per bump -- the pre-normal-map technique, chosen because the modern one is not actually available here
330	S? 09-12	Stripping mesh geoms line by line is good enough	Refuted -- it deletes whatever else shares the line, and the failure names the wrong thing	All three generators compile the body with its mesh geoms removed, by dropping every LINE containing <code>type="mesh"</code> . When the wiring script wrote an eye sphere and its replacement mesh geom onto one line, that filter took both, and the eye generator reported <code>expected 2 eye geoms, found 0</code> -- a message that points at the eyes rather than at the stripper. Fixed in all three by removing mesh geoms as ELEMENTS with a regex over the geom tag, and by writing the two geoms on separate lines. Same class as #322: a text transform that is correct on the text it was written against and silently wrong on text that is merely formatted differently
331	S? 09-12	A blanket rule for "meshes the skin supersedes" is safe	Refuted -- it hid the eyes, and the animal rendered with no eyes at all	The continuous skin replaced 23 rigid per-body meshes, so the wiring script draws every mesh geom transparent. Two new mesh geoms were then added that the skin does NOT supersede -- the eyeball and the eyelid, which the skin deliberately excludes because an eye is not skin and wrapping it would put a bulge where the pupil is. The blanket rule caught them both and the face came out blank. A rule written as "everything except the one exception I had in mind" acquires new members it was never checked against. Now excepted explicitly by name
332	S? 09-12	Rebuilding the calibrated body is rebuilding the body	Refuted twice, and the failure is invisible at the point it is made	<code>build_lab_morphology.py --fit-com</code> writes the v2 body ONLY; v1 comes from the same script WITHOUT the flag. Rebuild v2 and forget v1 and the repository is left in a state that compiles, renders, and passes 14 of 14 morphology gates while v1 carries a stale source hash -- and three reproducibility tests fail with a message about a missing marker that names nothing relevant. This cost a debugging pass twice in one session. Closed with <code>tools/rebuild_body.py</code> , which runs the six steps in the one order that works and prints what each did

#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
333	S? 09-12	The four dead channels need new sensing built	Refuted -- all three sensor paths existed as stubs and needed connecting, not writing	<code>_predator_distance_m()</code> returned None unconditionally; <code>brain_command</code> passed <code>shelter_bearing_deg=None</code> and <code>warm_bearing_deg=None</code> as literals; <code>thermal_error_C</code> carried the comment "the world has no temperature field, so this reads exactly zero forever. It is wired so that adding a field switches it on". Connected: the threat's distance through the nose into flee, the ground's temperature under the animal into body temperature into the cold error into bask, and both bearings into the motor programs. Every one of them returns None when the body is absent, so the COMMITTED world is untouched -- re-measured on it afterwards and it gives the identical 121 hunt / 778 explore / four zeros. Result on the furnished world: bask 1041 steps, and the animal walks 34.0 cm to the warm patch and warms from 25.2 to 29.7 C , which is inside its published preferred band of 29.5-31.9
334	S? 09-12	A basking program is a program that walks toward warm ground	Refuted -- it walked straight over the far edge and cooled down again	Measured: the animal closed from 34.0 cm to 3.6 cm, warmed 25.2 to 28.9 C, then kept walking and cooled, cycling forever without ever reaching its band. This species takes heat by LYING on the ground -- thigmothermy, body tracking substrate at $r2 = 0.97$ against 0.92 for air, Hastings et al. 2023, $n = 12$ -- so stopping on it is not a convenience, it IS the behaviour. Added <code>on_warm_ground</code> to the motor program: drive goes to zero once the ground underfoot is at or above the preferred floor. Body temperature then reaches 30.0 C and the bask channel releases its grip
335	S? 09-12	A channel with the largest salience is the channel that gets released	Refuted -- the basal ganglia has a floor, and it is what actually caps this brain at four channels	Bisected on one channel with everything else at zero: salience 0.2008 is the point at which the gate stops being exactly 0.0000. Below it nothing is released at all, however far ahead of the others it is. That makes two channels unreachable by any world: <code>groom</code> carries a 0.05 tonic salience and <code>rest</code> IS fatigue, which plateaus at 0.011 in an animal that strolls (#284). Both are an order of magnitude short. <code>flee</code> is the interesting case -- its salience IS the published probability of a defensive reaction in this species, 0.21 to scent alone and 0.28 with sight ($n = 42$) -- so it clears the bare threshold by 0.009 and is beaten by explore as soon as hunger passes about 0.2. Measured: with hunger 0.30 present, flee needs 0.2171 and has 0.2100, short by 0.0071. Nothing was tuned. The model's prediction is that a hungry gecko does not flee from something standing on it, and it is left standing as a prediction rather than corrected into agreement
336	S? 09-12	Four of six channels is the ceiling, and it is the world's fault	Partly -- two were the world's fault and are now fixed; the other two are the selector's	<code>tools/four_brains_demo.py</code> runs two scenarios on the furnished world and releases bask, explore, flee and hunt : 4 of 6, up from 2. The two scenarios exist because of #335 rather than for presentation -- a hungry animal (0.90 meals owed) shows bask, explore and hunt, and only a fed one (0.15) can let flee through. <code>rest</code> and <code>groom</code> remain at exactly 0.0000 and cannot be raised without changing either a published constant or the basal-ganglia weights, and the weights were settled by a gating constant they must independently reproduce (Rule 6). Left alone and recorded
337	S? 09-12	What I was about to model as gecko skin was right	Refuted for most of it -- 26 of 36 claims died under adversarial verification	A 42-agent pass researched six surface features and then tried to REFUTE every claim, defaulting to rejection for anything the verifier could not confirm from a source it had opened. 10 survived, 26 did not. The survivors are specific and useful: the dorsum carries TWO element types, a pavement of small flattened scales with larger raised tubercles set into it, so a bump map modulating one tiling is the wrong structure (Russell et al. 2014, SEM, $n = 5$); the movable eyelid is real and the lids are THICK, needing volume rather than a painted ring (Boulenger 1885 and Smith 1935, independently, 50 years apart); there is NO spectacle, so the cornea is bare and wet and must be shaded as bare cornea, and the caseous plug clinicians see is routinely misdiagnosed as a retained one; the pupil is vertical; the tympanum is an open aperture on the side of the head with a visible membrane, photographed in this species; five digits per foot, no reduction. Also recorded: the dorsal tubercle-row count for <i>macularius</i> proper is in NO source that could be opened -- the nearest figure, 18 rows, is for the Rajasthan/Delhi population whose species status is itself disputed. Written to <code>docs/GECKO_SURFACE_ANATOMY.md</code> and NOT applied: the user stopped the appearance work before this returned, and a refuted majority is exactly the reason not to have built first and checked after
338	S? 09-12	The brain demo was running on the accepted walker	Refuted -- it was on the REJECTED one, and the file I was reading says so in its own comment	<code>GeckoBrainEnv</code> defaults to <code>gait_profile="legacy"</code> with the frozen trained residual loaded on top, and <code>tools/four_brains_demo.py</code> passed neither argument, so it took both defaults. The accepted walker is <code>lab</code> with ZERO residual. The comment sitting twenty lines above the constructor states the measurement: <code>lab + no policy</code> gives foot duty 0.770 0.764 0.787 0.790, even to 0.006 against published targets of 0.70 fore and 0.765 hind; <code>legacy + policy</code> gives 0.704 0.601 0.435 0.367 , hind feet at half. It also states why this keeps happening -- the defect is in the HIND feet, the front pair looks fine, and front duty was the only thing being checked. The user caught it. The env exposes <code>info["accepted_walker"]</code> , which was sitting there unused; the demo now asserts on it before filming and exits if it is False. Same error the canonical walker clip's docstring records being made repeatedly, made again
339	S? 09-12	Once the sensors were wired, flee would release whenever a predator was on the animal	Refuted -- the margin is 0.009 and ANY other salience closes it	Re-measured on the correct walker. With the predator touching the animal, flee's salience sits at exactly 0.2100 , its published ceiling, for 181 consecutive steps, and the gate stays at 0.0000 throughout. The reason is in the salience vector at the closest approach: <code>hunt</code> was at 0.1504 , because the animal could also see a cricket. Flee clears the bare release threshold of 0.2008 by 0.009, so a competitor a seventh its size is enough to keep it shut. It releases at gate 0.938 the moment hunger is zero and hunt goes with it. Nothing was tuned to get this: the published 0.21 and the basal-ganglia threshold both stand, and the prediction that a gecko which can see food will not flee from something standing on it is left as a prediction. Final: <code>tools/four_brains_demo.py</code> releases bask, explore, flee and hunt on the accepted walker, 4 of 6, with <code>rest</code> and <code>groom</code> an order of magnitude below the threshold and unreachable
340	S? 09-12	A swept oval can be shaped into a head	Refuted -- the user's word for it was a missile, and that is what a sweep of an oval is	An oval swept along a line can be fat here and thin there and that is its whole vocabulary: it cannot express a flat crown, a brow, a jaw angle, or a snout that narrows faster than it drops. Ten rounds of adjusting its numbers produced ten missiles. Replaced by a sculpted head: a SUPERELLIPSE cross-section with different exponents above and below the midline (flat on top, round underneath, which is what a skull is), and width and height on SEPARATE curves so the widest point can sit at the jaw angle behind the eyes while the snout narrows fast and stays tall. The profile is INVENTED from photographs and ANCHORED to the body -- its widest station is set to the widest half-width the sweep measured from the head's own primitives, so the gated head width is reproduced and only the shape between stations is styled. The first fifth blends from the measured neck ring so the join is invisible. From above it is now a triangle wide at the jaw and blunt at the nose
341	S? 09-12	The body behind the head was fine and only the head needed sculpting	Refuted -- same missile, same cause, and the pinch at the hips was a number I invented	Everything from the neck back was still the sweep: no neck (the neck capsule is nearly as wide as the trunk), no belly (a capsule is one width all along), a 0.82 constriction at the vent that I invented in #317 and that the user correctly read as the animal being pinched behind its legs, and a tail that ended in a needle. All of it replaced by <code>tools/gecko_sculpt.py</code> : a neck that dips to 0.84 between skull and shoulder, a trunk widest at 0.45 of its length with a slightly flat back, a tail continuous with the hip (pinch deleted, VENT_PINCH now 1.0) that is fat through its first third, carries the real animal's transverse tubercle ridges as a 3.5 % radius wave at 4.6 mm, and tapers to a ROUNDED tip. Anchored at three measured stations: trunk maximum, tail maximum, and the head's occiput ring. Plus FIVE TOES per foot, built along the five digit capsules the body already defines (so nothing here places a toe) and slender rather than padded, because #337 verified this species has no adhesive pads; a knuckle sphere at each of the twelve limb articulations, because two tapered tubes meeting at an angle leave a wedge open and a twisted collar; a pale torus round each eye opening, which is the most recognisable feature of the animal's face; and a mouth line, nostrils and ear openings painted at the v positions the skin generator MEASURED off its own rings rather than guessed. Skin went from 9,019 to about 20,600 vertices. Gates 14/14 before and after
342	S? 09-12	A knuckle sphere sized to the limb tube hides the joint	Refuted on the first render -- it sat between tube ENDS that had tapered to 0.55 r, and looked like a ping-pong ball	Sized at 1.06 of the tube's base radius, each knuckle was nearly twice the diameter of the two tube ends meeting inside it, and half of every sphere was painted with the pale BELLY colour because its texture coordinates wrapped the full 0..1 girth. Two causes, two fixes: the limb taper went from 0.45 to 0.25 so a bone no longer halves in radius over its length, and the knuckle went to 0.72 r, and every knuckle and toe now maps its u into the spotted dorsal band, 0.15-0.35, rather than the whole wrap. The eye rim was also glowing white for the same reason as #310 -- untextured, so its rgba IS its colour, and the blanket 1 1 1 1 rule caught it -- and now has its own pale value. Skin 23,751 vertices, 58,707 bone links, 0 unbound. Gates 14/14

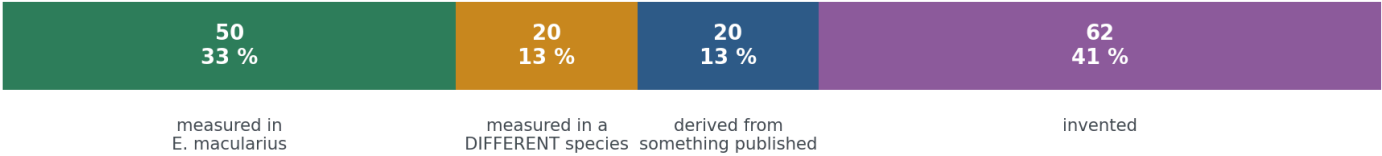
#	WHEN	HYPOTHESIS	VERDICT	EVIDENCE
343	S? 09-12	The body could be made to match the reference photographs without touching the physics	Partly -- the appearance could; the blink and the tongue needed two joints and a slide, recorded as a body version bump	Appearance, no physics: colour from the reference animal (yellow back, orange flanks, WHITE belly, white tail with orange near the vent, near-black spots -- the old buff was fairly called mud); ONE continuous skin per limb through the joints with a parallel-transport frame, replacing six tubes and six knuckle balls, which were the same mistake the trunk made before it got one skin; toes angled up 14 degrees and a tenth shorter because the flat digit capsules put their tips through the floor; the throat's skin weight tripled so the published breathing actually shows, and its amplitude raised 5 to 12 degrees (INVENTED, declared); the mouth floor painted dark pink on the mandible's upper face, which was first put on the same v as the snout and gave the animal a pink nose (#313's lesson again -- the jaw now lives on its own strip of texture at v 0.995). Body: the single <code>eyeLids</code> body with a PITCH hinge swung both lids FORWARD over the front of the eye; a lid closes DOWN, which is a roll about the head's long axis, and a roll closes one side while opening the other -- so it became <code>eyelid_L</code> and <code>eyelid_R</code> , each with its own roll hinge and mirrored ranges. The tongue got a slide (0-12 mm) driven by two PUBLISHED numbers for this species and nothing else: post-feeding labial licking at 0.165 licks/s (Cooper, DePerno & Steele 1996, n = 16, already in the registry) and the tongue-flick rate the nose already computes. nq/nv/nu 42/41/28 to 44/43/30 , <code>nsite</code> delta 7 to 8, walker action dim still 25. Skin 21,663 vertices, 52,639 bone links, 0 unbound. Gates 14/14
344	S? 09-12	A mesh geom and a skin read the same texture the same way	Refuted -- a mesh's texture rows count from the BOTTOM of the image, a skin's from the top	The thick eyelid hood rendered pure white with texture coordinates (0.22, 0.10). Measured rather than guessed: <code>mesh_texcoordadr</code> was set, the material was the body's with its texture bound, and the PNG at (0.22, 0.10) is yellow (237, 200, 59) -- while at (0.22, 0.90) it is the white of the tail tip (228, 225, 218), which is exactly what the lid showed. The continuous skin, on the same image, draws the yellow head at v = 0 and the white tail at v = 1, so its rows count from the top; a mesh geom's count from the bottom. The lid now asks for 1 - v. The eyeball never revealed this because its image is a centred disc, symmetric under the flip. Anything else that draws a mesh from the body texture will need the same inversion
345	S? 09-12	A body version bump is the count change and nothing else	Refuted three times in one test run, and only one of the three was a count	After #343 the suite showed 12 failures against the standing 8. (a) I had raised the instrumentation test's site delta from 7 to 8 for the second eyelid site -- wrong, because that delta counts the MEASUREMENT sites removed from the comparison model, and the eyelid sites are present on both sides of it; back to 7. (b) The generator's <code>actuator_scale</code> is 0.038 over the TEMPLATE mass and is pinned at 0.038/0.0612; the tongue body's 1e-5 kg had pushed the template to 0.06121 and moved the scale in the fourth decimal. Taken back out of the jaw's collision capsule, 0.0006 to 0.00059, exactly #306's move, and the template is 0.061200 again. (c) The realism recorder pins the hinge count: 35 since #304, now 36 -- one eyelid hinge became two; the tongue is a slide and is not counted. Every one of the three was a test that knew something about the body I had not thought to check before changing it. Final: 529 of 537 , the same eight <code>GateContracts</code> cases #291 documents
346	S? 09-12	The spine should bend more while walking, and there is a published number for how much	Confirmed for the number and the fit; refuted for making it the default -- it moves two of the walker's own selection criteria the wrong way	A 24-agent pass (<code>docs/LATERAL_UNDULATION.md</code> , 18 numbers verified, 2 refuted) found a SAME-SPECIES measurement: Jagnandan & Higham 2017 (Sci Rep 7:10865, n = 10, SVL 104.6 mm, level trackway, walking, hind duty 0.78) -- pelvic girdle yaw excursion ~ 50 degrees per stride with the intact tail, peaking AWAY from a hindlimb at 0.60-0.65 of that limb's stride and toward it at footfall. The accepted walker measured 24.4 degrees , peaking at 0.83. The spine is one tendon across the three lateral joints, driven at <code>spine_amp</code> 0.30 of half-range: excursion is linear in it (24.3 / 36.4 / 48.9 / 61.8 / 74.6 degrees at 0.30 / 0.45 / 0.60 / 0.75 / 0.90), so 0.61 is DERIVED, and the timing was 0.21 of a cycle late, so <code>spine_phase</code> was exposed as a declared lab parameter at its existing default 0.0 (Session-3g move, behaviour unchanged) and set to 0.21 : measured 51.8 degrees peaking at 0.62. Then the gates, one declared change per trial as the protocol demands: baseline / amp only / fitted -- hind duty 0.636 -> 0.650 -> 0.675 (toward the required 0.73-0.83), speed 0.0427 -> 0.0427 -> 0.0369 m/s , net-path 0.823 -> 0.712 -> 0.697 . The lab set was selected for > 0.04 m/s and > 0.8 net-path, and the published spine wave breaks both on this controller while helping the duty the contract is actually stuck on (#291). NOT made the default: the fitted pair is a declared parameter set, <code>--set spine_amp=0.61 --set spine_phase=0.21</code> , filmed as <code>walk_spine.mp4</code> , and the trade is recorded here rather than resolved by picking a smaller amplitude that matches nothing
347	S? 09-12	Brain 6, the day/night clock, is connected now that the env has a <code>circadian</code> flag	Refuted -- #262's verdict still stands, and the flag made it look fixed	<code>GeckoBrainEnv</code> now takes <code>circadian=</code> and constructs <code>Arousal()</code> when it is True, and <code>self.clock</code> is reset alongside the other modules. But <code>clock.step</code> is never called anywhere in the file , and <code>self.arousal</code> is assigned exactly once, to the constant 1.0 at construction, then only ever READ -- to be copied into <code>info["arousal"]</code> . So switching the flag on builds the object, resets it, and reports a number the object never produced. #262 said "built, tested, and connected to nothing"; that is still true, except there is now a switch that makes it appear otherwise. Recorded rather than fixed, because fixing it means deciding what arousal should modulate, and nothing published says
348	S? 09-12	<code>python -m common.morphology_audit</code> with no arguments audits the body that walks	Refuted -- its default is the UNFITTED source and it reports 1 of 14	<code>DEFAULT_XML</code> in <code>common/morphology_audit.py</code> is <code>morphology/gecko_body_r.xml</code> , the hand-written template, which is deliberately not to scale: 61.2 g against the 0.034-0.042 kg band, tail chain 1.038 of SVL against 0.69-0.77, hip height 0.196 against 0.14-0.16. Run that way it prints 1/14 morphology gates passed and looks like the animal has collapsed. The body that actually walks is <code>morphology/gecko_body_lab_v2.xml</code> , the same source after the two fitting passes, and it prints 14/14 -- total mass 0.038, tail fraction 0.22, intact CoM 0.6781 inside [0.639, 0.679], hip 0.1489, shoulder 0.1093, femur/tibia 1.0199, hip pro/retraction 90 deg, inner-eye gap 0.010. Both numbers are true statements about different files. Found while assembling the handoff report, which called the audit the way its own docstring shows. Recorded rather than changed: the default is what the test suite passes in explicitly, and moving it would be a fix aimed at my reading of the output rather than at the code

Every parameter in the animal

Every parameter in the gecko says where it came from



Only 33 % of the 152 numbers in this animal were measured in this animal



The number that matters most on this page. Under a third of the parameters in this animal were measured in this animal, and a little under half were chosen rather than measured — which is a defect only if it is hidden.

All 152 registered parameters

config/proxies.yaml1, read through the same validating loader the simulation uses — which refuses a registry with a missing provenance field or an invalid confidence value.

PARAMETER	VALUE	UNITS	PROV.	SPECIES	N	CONF.	SOURCE
activity_onset_after_dark_s		4860 s after lights-out	PUBLISHED	E_macularius		1 uncertain	81 min mean (SD 89, range 11–247) from lights-out to first approach of a running wheel. Digirolamo 2024, Wikijournal of Science 7(1):5. One adult female, ~12 y, 14 nights of 940 nm IR video plus 21 days of whee
activity_peak_window_h		5 hours after lights-out	PUBLISHED	E_macularius		18 uncertain	Behaviour scored during 'the evening peak activity period between 6 p.m. and 11 p.m.', lights off about 18:00. Kronke & Xu 2023, Animals 13(23):3595, n=18 adults.
body_mass_kg		0.038 kg	DERIVED	derived		likely	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916; Jagnandan & Higham (2017), Sci Rep 7:10865, doi:10.1038/s41598-017-11484-7; CODEX_PROMPT.md (user build brief, 2026-09-05)
body_mass_tolerance_kg		0.004 kg	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
calibration_friction		0.9 coefficient	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05
calibration_temperature_C		30 deg C	DERIVED	derived		likely	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916; Jagnandan & Higham (2017), Sci Rep 7:10865, doi:10.1038/s41598-017-11484-7
capture_head_drop_m		0.027 m	PUBLISHED	E_macularius	1 representative cycle (Fig. 2)	likely	Delheusy V, Brillet C, Bels VL (1995) Etude cinématique de la prise de nourriture chez Eublepharis macularius. Amphibia-Reptilia 16(2):185–201. doi:10.1163/156853895X00361. OPENED IN FULL (Brill PDF, 17 pp) by
com_fit_density_sensitivity_factors	1.0 – 1.35 – 1.5 – 1.62 – 1.8 – 2.0 – 2.35 – 2.5	multiple of source-derived bulk density	INVENTED	INVENTED		uncertain	NeuroGecko constrained inverse calibration v2 diagnostic sweep
com_fit_non_tail_mass_bounds		0.5 – 2.0 multiple of v1 group mass	INVENTED	INVENTED		uncertain	NeuroGecko constrained inverse calibration v2 engineering regularizer
com_fit_tail_density_guard_factor		1.5 multiple of source-derived bulk density	INVENTED	INVENTED		uncertain	NeuroGecko constrained inverse calibration v2 engineering guard
com_fit_tail_min_mass_fraction		0.1 multiple of v1 segmentmass	INVENTED	INVENTED		uncertain	NeuroGecko constrained inverse calibration v2 engineering regularizer
com_intact_svl		0.659 SVL behind snout	PUBLISHED	E_macularius	7	verified	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916
com_study_body_mass_kg		0.0394 kg	PUBLISHED	E_macularius	7	verified	Jagnandan, Russell & Higham (2014), Methods, https://doi.org/10.1242/jeb.110916
com_study_svl_m		0.1219 m	PUBLISHED	E_macularius	7	verified	Jagnandan, Russell & Higham (2014), Methods, https://doi.org/10.1242/jeb.110916
com_study_tail_volume_m3		7.8e-06 m^3	PUBLISHED	E_macularius	7	verified	Jagnandan, Russell & Higham (2014), Results/Methods, https://doi.org/10.1242/jeb.110916
com_svl_tolerance		0.02 SVL	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
com_tail_excluded_svl		0.527 SVL behind snout	PUBLISHED	E_macularius	7	verified	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916
contact_cycle_warning_ratio		2 observed / commanded frequency ratio	INVENTED	INVENTED		uncertain	NeuroGecko measurement-quality diagnostic engineering choice
contact_debounce_s		0.04 s	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
contact_period_cv_ceiling		0.1 fraction	INVENTED	INVENTED		uncertain	Session 2 single-digit-percent CV engineering requirement
cost_of_transport_ml_O2_per_kg_per_m		0.73 mL O2 per kg per metre	PUBLISHED	Teratoscincus przewalskii		likely	Minimum net cost of transport, reported as temperature-INDEPENDENT. Autumn, Weinstein & Full 1994, Physiol. Zool. 67(1):238–262, in an 11.2 g Teratoscincus przewalskii.
cricket_dry_matter_fraction		0.3 fraction	INVENTED	INVENTED		uncertain	No dry-matter fraction for crickets appears anywhere in the corpus. 0.30 is an engineering assumption needed to convert a wet meal mass into the dry mass the energy figure is expressed per gram of.
cricket_energy_kJ_per_g_dry		17.7 kJ per gram dry matter	PUBLISHED	PROXY (rooster assay)		likely	Cricket meal ~17.7 kJ/g dry matter, from a rooster assay, not from a gecko trial.
critical_thermal_maximum_C		41.07 degrees Celsius	PUBLISHED	E_macularius		verified	CTmax 41.07 +/- 0.89 C.
diagnostic_phase_bins		20 bins per commanded cycle	INVENTED	INVENTED		uncertain	Session 2 display aggregation

PARAMETER	VALUE	UNITS	PROV.	SPECIES	N	CONF.	SOURCE
emg_hindlimb_preautotomy	{'caudofemoralis': [{'onset': -0.0168, 'duration': 0.8145, 'peak': 0.1401, 'amplitude': 0.5373}], 'gastrocnemius': [{'onset': 0.064, 'duration': 0.6441, 'peak': 0.3379, 'amplitude': 0.3849}, {'onset': 0.8506, 'duration': 0.1323, 'peak': 0.9474, 'amplitude': 0.5453}]}	timing=stride fraction; amplitude=within-muscle peak fraction	PUBLISHED	E_macularius	10	verified	Jagnandan & Higham (2018), J Exp Biol221:1jeb179564, Table3, https://doi.org/10.1242/jeb.179564
endurance_coefficient_hours_at_25C		0.03 hours (coefficient of $t_{end} = a * v^b$, v in KM/H)	PUBLISHED	PROXY (Teratoscincus / Coleonyx variegatus)	25	verified	$t_{end} = 0.030 * v^{1.70}$ hours at 25 C, $r^2 = 0.87$, v in km/h. Companion fit at 15 C is $t_{end} = 0.013 * v^{1.70}$, $r^2 = 0.83$, n=24.
endurance_exponent_at_25C		-2.07 dimensionless	PUBLISHED	PROXY (Teratoscincus / Coleonyx variegatus)	25	verified	Exponent of $t_{end} = 0.030 * v^{1.70}$ hours at 25 C, $r^2 = 0.87$.
entrainment_tolerance_fraction		0.1 relative frequency error	INVENTED	INVENTED		uncertain	Session 2 engineering instrument check
ethogram_joint_activity_rad_s		0.02 rad/s	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
ethogram_pause_merge_s		0.1 s	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
ethogram_rest_s		3 s	PUBLISHED	E_macularius	18	likely	Krönke & Xu (2023), PMC10705344 ethogram definition
ethogram_trunk_activity_m_s		0.001 m/s	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
ethogram_walk_svl		1 SVL	DERIVED	derived	18	likely	Krönke & Xu (2023), PMC10705344; CODEX_PROMPT.md (user build brief, 2026-09-05)
evaluation_duration_s		20 s	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
evaluation_goal_distance_m		10 m	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
evaluation_reset_noise		0 simulation noise scale	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05
evaluation_settle_s		3 s	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
fatigue_recovery_time_constant_s		600 s	INVENTED	INVENTED		uncertain	No recovery-from-fatigue timescale appears anywhere in the corpus, for this species or any proxy.
feeding_events_per_week		3 meals per week	PUBLISHED	E_macularius		likely	Husbandry protocol: fed three times per week.
feeding_licking_cycle_s		0.41 s	PUBLISHED	E_macularius	15 cycles / 3 animals	verified	Delheusy V, Brillet C, Bels VL (1995) Etude cinématique de la prise de nourriture chez Eublepharis macularius. Amphibia-Reptilia 16(2):185-201. doi:10.1163/156853895X00361. OPENED IN FULL (Brill PDF, 17 pp) by
feeding_licking_gape_deg		8.4 degrees	PUBLISHED	E_macularius	15 cycles / 3 animals	verified	Delheusy V, Brillet C, Bels VL (1995) Etude cinématique de la prise de nourriture chez Eublepharis macularius. Amphibia-Reptilia 16(2):185-201. doi:10.1163/156853895X00361. OPENED IN FULL (Brill PDF, 17 pp) by
feeding_reduction_cycle_s		0.51 s	PUBLISHED	E_macularius	15 cycles / 3 animals	verified	Delheusy V, Brillet C, Bels VL (1995) Etude cinématique de la prise de nourriture chez Eublepharis macularius. Amphibia-Reptilia 16(2):185-201. doi:10.1163/156853895X00361. OPENED IN FULL (Brill PDF, 17 pp) by
feeding_reduction_gape_deg		47.23 degrees	PUBLISHED	E_macularius	15 cycles / 3 animals	verified	Delheusy V, Brillet C, Bels VL (1995) Etude cinématique de la prise de nourriture chez Eublepharis macularius. Amphibia-Reptilia 16(2):185-201. doi:10.1163/156853895X00361. OPENED IN FULL (Brill PDF, 17 pp) by
feeding_transport_cycle_s		0.45 s	PUBLISHED	E_macularius	15 cycles / 3 animals	verified	Delheusy V, Brillet C, Bels VL (1995) Etude cinématique de la prise de nourriture chez Eublepharis macularius. Amphibia-Reptilia 16(2):185-201. doi:10.1163/156853895X00361. OPENED IN FULL (Brill PDF, 17 pp) by
feeding_transport_gape_deg		45.08 degrees	PUBLISHED	E_macularius	15 cycles / 3 animals	verified	Delheusy V, Brillet C, Bels VL (1995) Etude cinématique de la prise de nourriture chez Eublepharis macularius. Amphibia-Reptilia 16(2):185-201. doi:10.1163/156853895X00361. OPENED IN FULL (Brill PDF, 17 pp) by
femur_tibia_ratio		1.01987 ratio	DERIVED	derived	4	verified	Fuller, Higham & Clark (2011), Zoology 114:104–112, Table 1, doi:10.1016/j.zool.2010.11.003
femur_tibia_ratio_range		0.9 – 1.06 ratio	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
fore_duty_factor		0.7 fraction	PUBLISHED	E_macularius	10	likely	Jagnandan & Higham (2017), Sci Rep 7:10865, doi:10.1038/s41598-017-11484-7
fore_duty_factor_tolerance		0.01 fraction	PUBLISHED	E_macularius	10	likely	Jagnandan & Higham (2017), Sci Rep 7:10865, doi:10.1038/s41598-017-11484-7
fuller_femur_m		0.0154 m	PUBLISHED	E_macularius	4	verified	Fuller et al. (2011), Table 1, https://biomechanics.ucr.edu/Fuller_et_al_2011.pdf
fuller_svl_m		0.1065 m	PUBLISHED	E_macularius	4	verified	Fuller et al. (2011), Table 1, https://biomechanics.ucr.edu/Fuller_et_al_2011.pdf
fuller_tibia_m		0.0151 m	PUBLISHED	E_macularius	4	verified	Fuller et al. (2011), Table 1, https://biomechanics.ucr.edu/Fuller_et_al_2011.pdf
gait_acquisition_hz		50 Hz	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05
girdle_height_svl_tolerance		0.01 SVL	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)

PARAMETER	VALUE	UNITS	PROV.	SPECIES	N	CONF.	SOURCE
head_width_sv1		0.185 SVL	DERIVED	derived		2 verified	Rawat et al. (2019), Reptiles & Amphibians 26(1):58–61, Table 1, doi:10.17161/randa.v26i1.14342
head_width_sv1_tolerance		0.02 SVL	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
hind_duty_factor		0.78 fraction	PUBLISHED	E_macularius	10	verified	Jagnandan & Higham (2017), Table1 original-tail hind duty, https://pmc.ncbi.nlm.nih.gov/articles/PMC5589804/
hind_duty_factor_range		0.7 – 0.79 fraction	DERIVED	derived		likely	Jagnandan & Higham (2017), Sci Rep 7:10865, doi:10.1038/s41598-017-11484-7; McElroy, Hickey & Reilly (2008)
hindlimb_length_sv1		0.417 SVL	DERIVED	derived	4	likely	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916; Fuller, Higham & Clark (2011), Zoology 114:104–112, Table 1, doi:10.1016/j.zool.2010.11.003
hindlimb_length_sv1_tolerance		0.03 SVL	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
hip_height_sv1		0.15 SVL	PUBLISHED	E_macularius	7	likely	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916
hip_prorot_min_range_deg		90 deg	DERIVED	derived	10	likely	Jagnandan & Higham (2017), Sci Rep 7:10865, doi:10.1038/s41598-017-11484-7; CODEX_PROMPT.md (user build brief, 2026-09-05)
hip_rot_max_range_deg		35 deg	DERIVED	derived		uncertain	Arnold et al. (2014), Iguana proxy as cited in docs/research/gecko_scorecard.md
inter_meal_interval_days		2.33 days	PUBLISHED	E_macularius		verified	Mean inter-meal interval 2.33 days (= 201,600 s), consistent with feeding three times per week.
interorbital_distance_m		0.01 m	PUBLISHED	E_macularius	2	verified	Rawat et al. (2019), Reptiles & Amphibians 26(1):58–61, Table 1, doi:10.17161/randa.v26i1.14342
interorbital_distance_tolerance_m		0.001 m	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
lab_actuator_scaling_method	kp, kv, forcerange multiplied by new_mass/legacy_mass	method	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05
lab_base_parameters	{'mirror_left_fa': True, 'hind_lift_multiplier': -1.3333333333333333, 'front_stance_press': 0.05, 'front_stance_press_fr': 0.05, 'front_swing_lift': -0.4, 'front_stance_seek': 0.0, 'front_seek_relax': 0.35, 'shoulder_sprawl_tuck': 0.0, 'fore_hind_amplitude_ratio': 0.54, 'hind_fa_amplitude': 0.98, 'heading_gain': 1.0, 'continuous_front_lift': True, 'other_amplitude': 0.0, 'spine_amp': 0.3, 'spine_phase': 0.0, 'tail_amp': 0.15, 'tail_phase_lag': 0.15, 'hind_sprawl_amplitude': 0.0, 'fore_sprawl_amplitude': 0.0, 'sprawl_phase': 0.0, 'tail_hindlimb_coupling': 0.36}	normalized actuator half-ranges except tuck radians and heading_gain per radian	INVENTED	INVENTED		uncertain	Session 2 manual trial19_front_height; experimental candidate, Gate2 FAIL
lab_contact_bodyweight_fraction		0.0939418 fraction of bodyweight	INVENTED	INVENTED		uncertain	Legacy threshold 0.0564N divided by legacy model weight0.0612kg*9.81m/s^2
lab_forelimb_fraction_hind		0.85 ratio	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05
lab_gait_frequency_hz		1.1888 Hz	INVENTED	INVENTED		uncertain	Legacy checkpoint-compatible controller frequency, retained by user build brief
lab_gait_touchdown_delays	{'HL': 0.0, 'FL': 0.435, 'HR': 0.5, 'FR': 0.935}	stride fraction	DERIVED	derived		likely	McElroy, Hickey & Reilly (2008) ipsilateral limb-phase target + engineered symmetric contralateral offset
lab_head_height_sv1		0.105 SVL	DERIVED	derived	2	likely	Rawat et al. (2019), Table 1 plus build geometry approximation
lab_height_guard_transition_sv1		0.01 SVL	INVENTED	INVENTED		uncertain	NeuroGecko opt-in lab reward engineering calibration
lab_hip_rot_range_deg		30 deg total	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md +/-15 degree engineering limit based on Iguana proxy
lab_mass_partition_method	preserve_legacy_within_tail_and_body	method	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05
lab_neutral_femur_drop_m		0.004 m	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05
lab_neutral_forearm_drop_m		0.0061 m	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05
lab_neutral_humerus_drop_m		0.0027 m	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05
lab_neutral_tibia_drop_m		0.0076 m	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05
lab_shoulder_attachment_z_m		-0.0094 m	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05
lab_snout_height_scale		0.55 scale	INVENTED	INVENTED		uncertain	NeuroGecko lab candidate v1 engineering design, 2026-09-05

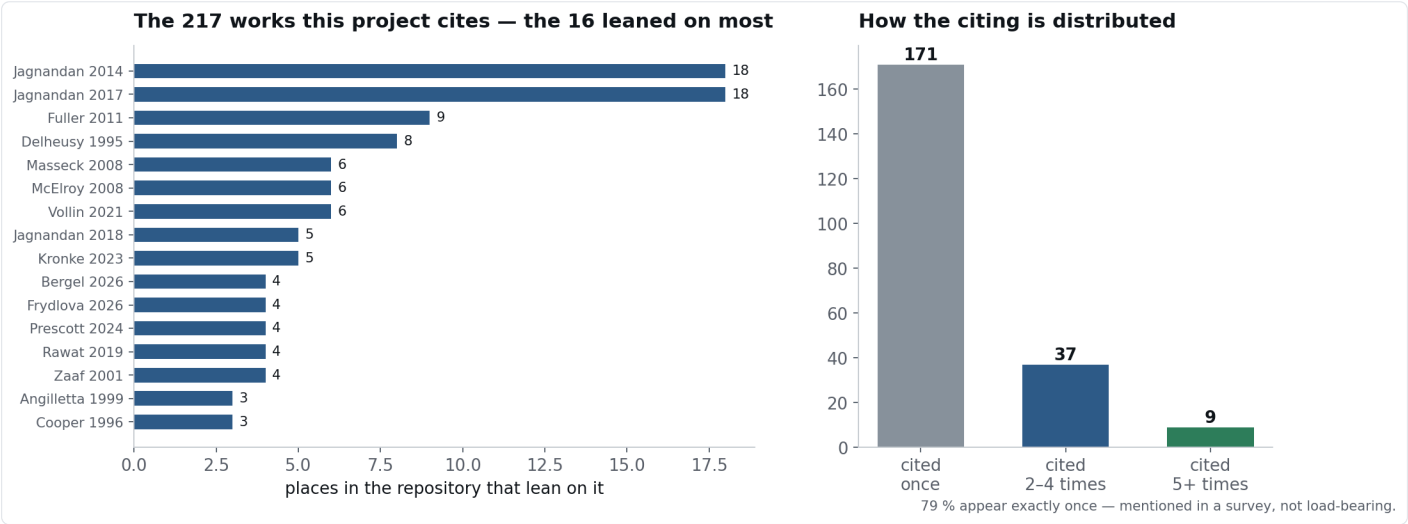
PARAMETER	VALUE	UNITS	PROV.	SPECIES	N	CONF.	SOURCE
lab_speed_stride_fraction	0.72	SVL per stride	INVENTED	INVENTED		uncertain	Engineering midpoint of supplied 0.62–0.82 SVL stride interval
lab_total_power_effort_alpha	0.01	reward per watt	INVENTED	INVENTED		uncertain	Session 2 opt-in lab mechanical-power penalty
legacy_contact_threshold_N	0.0564	N	INVENTED	INVENTED		uncertain	envs/gecko_walk_env.py legacy controller calibration
legacy_front_stance_press	0.4	action scale	INVENTED	INVENTED		uncertain	envs/cpg_residual_controller.py
legacy_front_swing_lift	0.4	action scale	INVENTED	INVENTED		uncertain	envs/cpg_residual_controller.py
legacy_residual_scale	0.25	action scale	INVENTED	INVENTED		uncertain	envs/cpg_residual_controller.py
limb_phase_range	0.43 – 0.44	stride fraction	PUBLISHED	E_macularius		likely	McElroy, Hickey & Reilly (2008), J Exp Biol 211:1029
lipid_energy_J_per_g	39000	J per gram	PUBLISHED	standard		verified	Standard energy density of lipid, used by the scorecard in its tail-reserve derivation.
lower_jaw_length_m	0.013	m	PUBLISHED	E_macularius	5 adult males	verified	Delheusy V, Brillet C, Belis VL (1995) Etude cinématique de la prise de nourriture chez Eublepharis macularius. Amphibia-Reptilia 16(2):185-201. doi:10.1163/156853895X00361. OPENED IN FULL (Brill PDF, 17 pp) by
maximum_aerobic_speed_m_s_at_25C	0.1	m/s	PUBLISHED	PROXY (Teratoscincus)		likely	VO2max is reached across 0.100-0.158 m/s at 25 C; max aerobic speed is 0.075 m/s at 15 C; locomotion is sustained beyond 60 min at 0.050 m/s.
meal_apparent_dry_matter_digestibility	0.71	fraction	PUBLISHED	E_macularius		verified	Apparent dry-matter digestibility 71 +/- 4.7 % (crude protein 81 %, fat 65-74 %, calcium 43 %).
meal_consumed_fraction_of_offering	0.547	fraction	PUBLISHED	E_macularius		verified	Animals consumed 54.7 +/- 12.2 % of a 5 %-body-weight offering.
meal_offering_fraction_bodyweight	0.05	fraction of body mass offered per feeding	PUBLISHED	E_macularius		likely	Husbandry protocol: animals offered 5 % of body weight, fed three times per week.
measurement_site_local_positions_m	{'hip_L': [0, 0, 0], 'hip_R': [0, 0, 0], 'shoulder_L': [0, 0, 0], 'shoulder_R': [0, 0, 0], 'pelvis_center': [-0.007, 0, -0.002], 'pectoral_center': [0.0125, 0, -0.002], 'mid_back': [0, 0, 0.01196]}	m	DERIVED	derived		verified	morphology/gecko_body_r.xml at ddbb105
metabolic_q10	None	dimensionless	PUBLISHED	NOT_IN_CORPUS		uncertain	No metabolic Q10 is published for E. macularius. The corpus contains a Q10 of 2.3 for the reptile ultradian SLEEP-CPG period (Laudakia, Arrhenius) and an engineering suggestion of ~2.5, both explicitly not meas
morphology_settle_s	2	s	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
oxycalorific_J_per_mL_O2	20.12	J per mL O2	DERIVED	derived		verified	DERIVED from the scorecard's own arithmetic: it states 2.54 mL O2/h = 51.1 J/h for a 40 g adult, so 51.1/2.54 = 20.12 J/mL
post_feeding_labial_lick_rate_per_s	0.165	licks per s	PUBLISHED	E_macularius	16	verified	Cooper WE Jr, DePerno CS, Steele LJ (1996) — post-feeding labial licking in Eublepharis macularius; full text read by the session-12 literature sweep.
postprandial_metabolic_peak_multiple	3.7 – 7.3	multiple of pre-feeding metabolic rate	PUBLISHED	PROXY (Eulamprus tympanum)		uncertain	Postprandial metabolic rate peaks at 3.7-7.3x pre-feeding, 15-33 h after a meal, returning to baseline at 62-170 h. Another lizard genus, paywalled.
predator_reaction_chemical	0.21	probability	PUBLISHED	E_macularius	42	verified	Antipredator reaction probability to snake scent alone, 0.21. Live-snake test, n=42. Frydlova et al. 2026, Behavioral Ecology 37(4):arag062.
predator_reaction_visual	0.07	probability	PUBLISHED	E_macularius	42	verified	Antipredator reaction probability to the SIGHT of a live snake alone, 0.07 (95% CI 0.02-0.20). Same study.
preferred_body_temperature_C	29.5 – 31.9	degrees Celsius, band	PUBLISHED	E_macularius		verified	Preferred body temperature 29.5-31.9 C (midpoint 30.7) in a 5-45 C gradient. Separately 25.8 C in a thigmothermal gradient with no day/night difference.
prey_ambient_move_fraction	0.35	dimensionless	INVENTED	INVENTED		uncertain	Engineering choice. No move/pause bout structure is published for any cricket.
prey_ambient_speed_m_s	0.047	m/s	INVENTED	INVENTED		uncertain	Half the low end of the escape-speed band (0.093 m/s). No published walking speed for an undisturbed cricket exists anywhere in the corpus, for Acheta or Gryllus, on any substrate.
prey_ambient_turn_rate_rad_s	1.2	rad/s	INVENTED	INVENTED		uncertain	Engineering choice.
prey_arena_radius_m	0.4	m	INVENTED	INVENTED		uncertain	Engineering bound on where prey may be placed and how far it may flee.
prey_capture_distance_m	0.040704	m	PUBLISHED	E_macularius		uncertain	Strike trigger distance for evasive prey, 2.03 +/- 0.08 cm = 0.384 SVL, scaled to this body's 0.106 m SVL.
prey_elevation_switch	0	degrees from the horizon	PUBLISHED	Mus musculus	20	likely	Identical stimulus, opposite behaviour by elevation. Overhead loom: escape 42/56 = 75%, latency 0.45 s. Same loom to the side: 53%, latency 1.79 s. Overhead sweep: freeze 56%. Same sweep to the side: escape 5%,
prey_escape_latency_s	0.085	s	PUBLISHED	cricket		likely	Published escape-response latency: 70-86 ms tethered, ~104 ms free-walking.

PARAMETER	VALUE	UNITS	PROV.	SPECIES	N	CONF.	SOURCE
prey_escape_speed_m_s		0.118 m/s	PUBLISHED	Acheta/cricket (prey, not the gecko)		uncertain	DERIVED, not published: escape dash length 28-43 mm and dash duration ~0.3 s appear separately in docs/research/strategy_v1.md; the corpus never prints their quotient. Midpoint 35.5 mm / 0.3 s = 0.118 m/s.
prey_flee_radius_floor_m		0.012 m	INVENTED	INVENTED		uncertain	Engineering floor, set below the 0.0203 m strike trigger distance.
prey_flee_radius_m		0.075 m	INVENTED	INVENTED		uncertain	Engineering trigger distance; the corpus's own suggestion is 'gecko head within ~5-10 cm and closing', itself tagged as a finite-state-machine design choice rather than a measurement.
prey_flee_speed_exponent		1 dimensionless	INVENTED	INVENTED		uncertain	Engineering choice. Linear scaling of flee radius with approach speed.
prey_flee_speed_reference_m_s		0.055 m/s	INVENTED	INVENTED		uncertain	The walker's own steady speed, used as the reference at which the published flee radius applies.
prey_item_wet_mass_kg		0.00035 kg per prey item	DERIVED	derived		uncertain	DERIVED. A feeding is 5% of body weight offered and 54.7% consumed, and the corpus describes one feeding bout as 3 crickets. For a 38 g animal that is 0.038*0.05*0.547 = 1.04 g eaten across three items, so ~0.3
prey_radius_m		0.009 m	INVENTED	INVENTED		uncertain	Sphere proxy. Published prey size is 15-20 mm for a ~50 mm SVL gecko = 0.30-0.40 SVL; at 0.35 SVL this body's prey is ~37 mm long. The sphere radius is a crude stand-in for that body, not a measured dimension.
prey_velocity_high_cut_deg_s		200 deg/s	PUBLISHED	Bufo bufo		uncertain	Prey-catching turning response falls to zero above ~200 deg/s. Ewert 1969, Pflugers Archiv 308:225-243.
prey_velocity_low_cut_deg_s		5 deg/s	PUBLISHED	Crotalus atrox		likely	IR-sensitive tectal units respond almost not at all at 5 deg/s. Kaldenbach, Bleckmann & Kohl 2016, J Comp Physiol A 202(6):389-398.
prey_velocity_peak_deg_s		45 deg/s	PUBLISHED	cross-lineage convergence		likely	Four independent lineages land in the same band. Bufo bufo prey-catching turning peaks 30-60 deg/s and falls to zero above ~200 (Ewert 1969). Crotalus atrox IR tectal units: almost no response at 5 deg/s, maxim
prey_visible_preys_note	see notes	none	PUBLISHED	E_macularius		likely	Capture success 82.9% overall on evasive crickets (77.0% intact, 78.6% post-autotomy, not significantly different); 100% (53/53) on non-evasive mealworms.
resting_metabolic_rate_mL_O2_per_g_per_h		0.075 mL O2 per gram per hour at 30 C	PUBLISHED	E_macularius	6	verified	Ha et al. 2017, Experimental Animals 66(2):99-105. Scorecard row: the ONLY direct metabolic measurement of this species. 0.075 +/- 0.011.
resting_metabolic_rate_mass_exponent		-0.22 dimensionless	DERIVED	derived		likely	Allometric exponent the scorecard itself applies to scale 0.075 mL O2/g/h to a 40 g adult, giving 0.064 mL O2/g/h.
resting_metabolic_rate_reference_mass_g		19.5 grams	DERIVED	derived	6	likely	DERIVED. The scorecard states 0.075 mL O2/g/h measured, and 0.064 mL O2/g/h after scaling to a 40 g adult by M^-0.22. Those two numbers pin the mass the 0.075 was measured at: 40 / (0.064/0.075)^(1/-0.22) = 19.
session2_gate2	{'minimum_forward_m_s': 0.04, 'minimum_net_path': 0.5, 'maximum_hind_swing_load': 0.1, 'minimum_front_stance_load': 0.65, 'hind_duty_target': 0.78, 'hind_duty_tolerance': 0.05, 'limb_phase_target': 0.435, 'limb_phase_tolerance': 0.03}	mixed; named fields	INVENTED	INVENTED		uncertain	User Session 2 minimum engineering viability requirements
shelter_choice_beats_safety		1 boolean, 1 = heat wins	PUBLISHED	E_macularius	8	uncertain	Given a choice between safe COOL shelter and an exposed HEATED open area, geckos 'selected heat sources over shelters'. Substrate moisture also emerged as a significant factor in shelter choice, and temperature
shelter_retreat_type		0 categorical, 0 = crevice under a rock	PUBLISHED	E_macularius		uncertain	Field observation, Salt Range and Potohar Plateau, Pakistan: the animal 'inhabits holes and crevices in gravel-mixed stony terrain', and retreat chambers traced through a demolished stone field wall ran from su
shoulder_height_svl		0.113 SVL	DERIVED	derived	7	likely	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916; CODEX_PROMPT.md (user build brief, 2026-09-05)
shoulder_hip_ratio		0.73 ratio	DERIVED	derived	7	likely	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916
shoulder_hip_ratio_tolerance		0.05 ratio	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
sleep_cycle_period_s		None s	PUBLISHED	NOT_IN_CORPUS		uncertain	No sleep-cycle period has been established for E. macularius. The species appears in Bergel et al. 2026, Nat Neurosci 29:543-550 -- seven lizard species, sleep-dependent infralow rhythms conserved across repti
stride_length_svl_range		0.62 – 0.82 SVL	DERIVED	derived		likely	Jagnandan & Higham (2017), Sci Rep 7:10865, doi:10.1038/s41598-017-11484-7; Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916; Fuller, Higham & Clark (2011), Zoology 114:104–1

PARAMETER	VALUE	UNITS	PROV.	SPECIES	N	CONF.	SOURCE
strike_capture_success	0.829	dimensionless	PUBLISHED	Coleonyx variegatus		5 likely	Capture success on evasive crickets, 116/140 trials = 82.9 %, n=5. Vollin & Higham 2023, Front Behav Neurosci 17:1173065, doi:10.3389/fnbeh.2023.1173065. Non-evasive mealworms: 53/53 = 100 %.
strike_distance_speed_correlation_r	0.47	dimensionless	PUBLISHED	Coleonyx variegatus		9 likely	Attack distance vs maximum strike velocity, r = 0.47, P = 0.009, n=9 adults, 29 trials. Whitford, Freymiller, Higham & Clark 2022, Biol J Linn Soc 135(3):533-540, doi:10.1093/biolinnean/blab167.
strike_duration_s	0.08	s	PUBLISHED	E_macularius		6 uncertain	Representative capture cycle, Fig. 2: total open-close cycle about 80 ms with peak gape about 37 degrees at 47 ms. Delheusy, Brillet & Bels 1995, Amphibia-Reptilia 16(2):185-201, doi:10.1163/156853895X00361. n=
strike_peak_acceleration_m_s2	46.6	m/s^2	PUBLISHED	Coleonyx variegatus		5 likely	Maximum acceleration 46.6 +/- 1.97 m/s^2, pre-autotomy. Vollin & Higham 2021.
strike_peak_speed_m_s	0.851	m/s	PUBLISHED	Coleonyx variegatus		5 likely	Maximum strike velocity 85.1 +/- 2.19 cm/s, pre-autotomy, n=5 juveniles. Vollin & Higham 2021.
strike_trigger_distance_m	0.0203	m	PUBLISHED	Coleonyx variegatus		5 likely	Starting distance to prey at strike onset, 2.03 +/- 0.08 cm, n=5 juveniles, 500 fps, crickets. Vollin & Higham 2021, Integr Comp Biol 61(2):538-549, doi:10.1093/icb/icab076.
tail_length_svl	0.73	SVL	DERIVED	derived		1 verified	Rawat et al. (2019), Reptiles & Amphibians 26(1):58-61, Table 1, doi:10.17161/randa.v26i1.14342
tail_length_svl_tolerance	0.04	SVL	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
tail_lipid_fraction	0.5	fraction of tail mass that is lipid	INVENTED	INVENTED		uncertain	The scorecard's own tail-reserve estimate assumes 50 % lipid and says so; the lipid fraction itself is not measured anywhere in the corpus.
tail_mass_fraction	0.22	fraction	PUBLISHED	E_macularius		7 likely	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891-3897, doi:10.1242/jeb.110916
tail_mass_fraction_range	0.19 – 0.25	fraction	INVENTED	INVENTED		uncertain	CODEX_PROMPT.md (user build brief, 2026-09-05)
thigmothermy_substrate_correlation	0.97	r-squared	PUBLISHED	E_macularius		12 verified	Body temperature tracked SUBSTRATE temperature at r2 = 0.97, against terrarium air loggers at 0.92-0.93 and atmospheric at 0.92-0.93. Authors' conclusion verbatim: 'E. macularius relies on absorbing heat from t
tongue_flick_baseline_per_min	3	flicks per minute	PUBLISHED	E_macularius		7 verified	Tongue-flicks in 60 s directed at an odourless control swab, 3.0 +/- 1.0, n=7. Cooper 1995, Herpetological Monographs 9:120-129, compiled in Table 1 of Baeckens, Van Damme & Cooper 2017, J Evol Biol 30(4):627-6
tongue_flick_pre_y_per_min	14.57	flicks per minute	PUBLISHED	E_macularius		7 verified	Tongue-flicks in 60 s directed at a swab carrying cricket chemicals, 14.57, n=7. Same source as the baseline.
trunk_pitch_deg	3.72	deg nose-up	PUBLISHED	E_macularius		7 likely	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891-3897, doi:10.1242/jeb.110916
trunk_pitch_sem_deg	0.61	deg	PUBLISHED	E_macularius		7 likely	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891-3897, doi:10.1242/jeb.110916
walking_speed_m_s_range	0.028 – 0.351	m/s	PUBLISHED	E_macularius		10 likely	Jagnandan & Higham (2017), Sci Rep 7:10865, doi:10.1038/s41598-017-11484-7
warm_surface_temperature_C	30	degrees Celsius	PUBLISHED	E_macularius		likely	Voluntarily maintained body temperature of 30 C for 13.5 h/day when a heat source was available; geckos held at a constant 25 C grew 0.11 g/day against 0.16 for thermoregulating animals. Autumn & De Nardo 1995,

The bibliography

Read this before the table. What follows is a **harvest**, not a curated reading list. It is every author-year string and every DOI that appears anywhere in this repository's own text, with the parameter or ledger row that mentions it. **None of it has been verified against a bibliographic database.** This project has invented a citation once (#215) and has overstated the absence of a real one. A name appearing here means the repository leans on it — not that it exists. Checking that is the first task in the companion prompt.



217 works, ordered by how much the project leans on them

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Jagnandan 2017	Jagnandan & Higham 2017 / Jagnandan 2017	18	#295, #299, #346, body_mass_kg, calibration_temperature_C, docs/DECISIONS.md, docs/research/beyond_statistics_realism.md, docs/research/gecko_scorecard.md, docs/research/no_gecko_and_appearance.md...
Fuller 2011	Fuller 2011 / Fuller et al. 2011 / Fuller, Higham & Clark 2011	9	docs/DECISIONS.md, docs/research/best_achievable_brain.md, docs/research/no_gecko_and_appearance.md, femur_tibia_ratio, fuller_femur_m, fuller_svl_m, fuller_tibia_m, hindlimb_length_svl, stride_length_svl_range
Delheusy 1995	Delheusy et al. 1995 / Delheusy, Brillet & Bels 1995	8	#150, #152, #172, #287, #292, #298, strike_duration_s, strike_trigger_distance_m
Masseck 2008	Masseck 2008 / Masseck et al. 2008 / Masseck, Roell & Hoffmann 2008 / Masseck, Roll & Hoffmann 2008 / Masseck, Röhl & Ho	6	#272, #277, docs/EYE_SPEC.md, docs/research/best_achievable_brain.md, docs/research/gecko_scorecard.md, docs/research/no_gecko_and_appearance.md
McElroy 2008	McElroy 2008 / McElroy et al. 2008 / McElroy, Hickey & Reilly 2008	6	docs/research/beyond_statistics_realism.md, docs/research/no_gecko_and_appearance.md, hind_duty_factor_range, lab_gait_touchdown_delays, limb_phase_range, prey_arena_radius_m
Vollin 2021	Vollin & Higham 2021 / Vollin 2021	6	#305, docs/research/best_achievable_brain.md, docs/research/strategy_v1.md, strike_peak_acceleration_m_s2, strike_peak_speed_m_s, strike_trigger_distance_m
Jagnandan 2018	Jagnandan & Higham 2018	5	docs/research/best_achievable_brain.md, docs/research/beyond_statistics_realism.md, docs/research/gecko_scorecard.md, docs/research/strategy_v1.md, emg_hindlimb_preautotomy
Kronke 2023	Kronke & Xu 2023 / Krönke & Xu 2023	5	activity_peak_window_h, docs/research/strategy_v1.md, ethogram_rest_s, ethogram_walk_svl, prey_flee_speed_reference_m_s
Bergel 2026	Bergel 2026 / Bergel et al. 2026	4	#155, #198, docs/research/best_achievable_brain.md, sleep_cycle_period_s
Frydlova 2026	Frydlova et al. 2026 / Frydlová 2026 / Frýdlová 2026 / Frýdlová et al. 2026	4	docs/research/best_achievable_brain.md, docs/research/gecko_scorecard.md, docs/research/strategy_v1.md, predator_reaction_chemical
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Rawat 2019	Rawat et al. 2019	4	head_width_svl, interorbital_distance_m, lab_head_height_svl, tail_length_svl
Zaaf 2001	Zaaf 2001 / Zaaf et al. 2001	4	docs/research/best_achievable_brain.md, docs/research/beyond_statistics_realism.md, docs/research/no_gecko_and_appearance.md, docs/research/strategy_v1.md
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Lagorio 2026	Lagorio et al. 2026	1	docs/research/best_achievable_brain.md
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Sharp 2025	Sharp et al. 2025	1	docs/research/best_achievable_brain.md
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51 DOIs — the citations that can be resolved directly

These are the strongest entries in the bibliography, because a DOI either resolves or it does not. Everything above is an author-year string that could be a transcription error, a species confusion, or an invention.

DOI	USED BY
10.1007/s00359-011-0635-8	docs/research/no_gecko_and_appearance.md
10.1007/s10439-026-04011-1	docs/research/beyond_statistics_realism.md
10.1016/0042-6989(67	docs/research/no_gecko_and_appearance.md
10.1016/j.celrep.2021.109730	docs/research/strategy_v1.md
10.1016/j.gaitpost.2010.08.009	docs/research/beyond_statistics_realism.md
10.1016/j.isci.2025.112203	docs/research/beyond_statistics_realism.md
10.1016/j.jbiomech.2024.111972	docs/research/beyond_statistics_realism.md
10.1016/j.visres.2007.12.004	docs/research/no_gecko_and_appearance.md
10.1016/j.xpro.2024.103140	docs/research/strategy_v1.md
10.1016/j.zool.2010.11.003	femur_tibia_ratio, hindlimb_length_svl, stride_length_svl_range
10.1038/s41586-024-07763-9	docs/research/best_achievable_brain.md
10.1038/s41586-024-07939-3	docs/research/best_achievable_brain.md
10.1038/s41586-024-08162-w	docs/research/best_achievable_brain.md
10.1038/s41593-025-02080-4	docs/research/best_achievable_brain.md
10.1038/s41593-025-02159-y	docs/research/best_achievable_brain.md
10.1038/s41598-017-00348-9	docs/research/no_gecko_and_appearance.md
10.1038/s41598-017-11484-7	body_mass_kg, calibration_temperature_C, docs/research/beyond_statistics_realism.md, fore_duty_factor, fore_duty_factor_tolerance, hind_duty_factor_range, hip_prorot_min_range_deg, stride_length_svl_range, walking_speed_m_s_range
10.1038/s42003-022-04261-4	docs/research/best_achievable_brain.md
10.1038/s42003-026-10024-2	docs/research/best_achievable_brain.md
10.1046/j.j.1096-3642.2002.00038.x	docs/research/beyond_statistics_realism.md
10.1073/pnas.1418979112	docs/research/beyond_statistics_realism.md
10.1073/pnas.2317218121	docs/research/best_achievable_brain.md
10.1073/pnas.2415929122	docs/research/best_achievable_brain.md
10.1085/jgp.22.5.555	docs/research/no_gecko_and_appearance.md
10.1086/515880	docs/research/beyond_statistics_realism.md
10.1093/biolinnean/blab167	strike_distance_speed_correlation_r
10.1093/icb/icab073	docs/research/best_achievable_brain.md
10.1093/icb/icab076	docs/research/strategy_v1.md, strike_trigger_distance_m
10.1098/rsif.2019.0402	docs/research/beyond_statistics_realism.md
10.1109/ICDL63968.2025.11204447	docs/research/best_achievable_brain.md
10.1111/evo.12015	docs/research/beyond_statistics_realism.md
10.1111/joa.13431	docs/research/best_achievable_brain.md
10.1126/sciadv.adx3811	docs/research/best_achievable_brain.md
10.1152/jappl.1996.80.5.1448	docs/research/beyond_statistics_realism.md
10.1163/156853895X00361	capture_head_drop_m, feeding_licking_cycle_s, feeding_licking_gape_deg, feeding_reduction_cycle_s, feeding_reduction_gape_deg, feeding_transport_cycle_s, feeding_transport_gape_deg, lower_jaw_length_m, strike_duration_s
10.1167/9.3.27	docs/research/no_gecko_and_appearance.md
10.1242/jeb.015503	docs/research/beyond_statistics_realism.md
10.1242/jeb.110916	body_mass_kg, calibration_temperature_C, com_intact_svl, com_study_body_mass_kg, com_study_svl_m, com_study_tail_volume_m3, com_tail_excluded_svl, docs/research/beyond_statistics_realism.md, hindlimb_length_svl, hip_height_svl
10.1242/jeb.179564	docs/research/best_achievable_brain.md, docs/research/beyond_statistics_realism.md, emg_hindlimb_preautotomy
10.1242/jeb.200.16.2177	docs/research/beyond_statistics_realism.md
10.1242/jeb.204.7.1233	docs/research/beyond_statistics_realism.md
10.1242/jeb.249422	docs/research/no_gecko_and_appearance.md
10.1371/journal.pcbi.1012101	docs/research/best_achievable_brain.md
10.17161/randa.v26i1.14342	head_width_svl, interorbital_distance_m, tail_length_svl
10.3389/fnbeh.2023.1173065	strike_capture_success
10.3389/fnbot.2017.00039	docs/research/no_gecko_and_appearance.md
10.3389/fnbot.2020.607455	docs/research/beyond_statistics_realism.md
10.3389/fphys.2020.00562	docs/research/beyond_statistics_realism.md
10.5061/dryad.dn822	docs/research/beyond_statistics_realism.md
10.5281/zenodo.6614341	docs/research/strategy_v1.md

DOI	USED BY
10.6084/m9.figshare.30656000	docs/research/strategy_v1.md

What cannot be built honestly



Recorded rather than invented. Each of these is a quantity a simulation would normally just pick a value for.

Things that must not be built, and why

- Place cells, grid cells, or head-direction cells. Zero spatially tuned single units have ever been recorded in any reptile, and the search returned nothing. Such a layer would be a rodent import wearing a gecko label.
- Path integration, dead reckoning, or a homeward vector. Never demonstrated in any reptile. The one lizard tested for outbound-route information had it explicitly rejected in favour of familiar-area landmark recognition.
- A sun or sky compass. The only demonstrated reptilian outbound-course mechanism runs through the parietal eye in daylight, in diurnal lizards. A crepuscular gecko active after sunset cannot run it.
- Long-range homing. The only gecko translocation experiment ever run found the opposite: 10 of 11 geckos displaced 1 km simply stayed where they were released.
- A nightly loop in which the gecko leaves a shared retreat at sunset and returns before dawn. This is the most tempting navigation claim for this project and it traces to a hobbyist magazine article recirculated through Wikipedia and care sheets. No sample size, no method, no peer review. If a return-to-shelter behaviour is implemented, it must be labelled invention.
- Sleep-dependent memory consolidation or replay. Not demonstrated in any reptile. Do not cite the 2026 gecko sleep paper as support: it runs no learning task and never mentions consolidation.
- A habituation module with a numeric decay constant. No fitted habituation curve or time constant exists for any lizard or any reptile. The three qualitative constraints that do exist (fast habituation to a novel place, none to a novel object near food, and none at all for the antipredator response) also rule out a single global habituation rule.
- Any coupling of memory decay to body temperature. It has never been studied in any reptile, and the one ectotherm experiment that exists points the other way.
- A defensive display that can be triggered by vision or vibration. In this species those modalities never fired a display on their own; only chemical cues did.
- Sex-specific learning or spatial parameters. A meta-analysis of 11 lizard species and the target species itself both looked and found nothing.
- A per-individual trials-to-criterion learning curve presented as published. Every learning result in this species is a group-level contrast between the first three and last three trials.
- The Ewert response law written as a single joint function of velocity, contrast and size. The source defines three separate univariate relations measured with the others held constant; writing it as one surface invents interaction terms nobody measured. And do not invent the worm versus anti-worm tuning numbers to fill the gap.
- An egocentric turn-rule architecture justified by the whiptail position-versus-feature study. That claim failed verification: a specialist review with full-text access reports the spatial advantage held during acquisition only and not during reversal, and the reading of position as an egocentric rule is an interpolation the paper does not support.
- Allocentric place learning justified by the turtle medial cortex lesion result. It is the cleanest place-versus-cue dissociation in any reptile and it is a turtle, outside Squamata entirely, and it is directly contradicted by the only lizard lesion study. Do not pick whichever of the two supports the design you already wanted.
- Any home range, daily movement distance, or foraging loop geometry for this species. No field study of wild leopard geckos exists at all. Set the enclosure scale from the physics and the task and call it a design choice.
- A below-the-horizon prey prior in world coordinates. The published elevation figures are positive in eye coordinates and are consequences of head posture.
- A prey detector calibrated with mammalian or fish acuity numbers taken at face value. Nocturnal geckos have systematically lower spatial acuity, so any imported angular threshold is probably too fine, and the correction size is unmeasured. If the threshold is imported unchanged, say so.

What the literature sweep actually turned up

Twelve findings that changed the plan. Several of them reverse a decision this project had already made.

- The brief's founding citation was a species confusion. Day, Crews and Wilczynski never studied leopard geckos. Their entire lizard output is three papers on two other families. The confusion is understandable, because David Crews did work extensively on *Eublepharis macularius*, but on temperature-dependent sex determination, not spatial cognition. A memory module built on 'the Day et al. leopard gecko numbers' would have been built on nothing.
- The modules that were deferred as too thin turned out to be the buildable ones, and the module needed most urgently turned out to be the unsupported one. Memory and learning now have real target-species numbers. The prey detector has none, for any gecko, ever.
- Only about one third of leopard geckos orient by the relationship between cues. The authors state plainly that the rest may have used a non-cognitive motor search and that they cannot rule it out. The default for a single simulated animal should therefore be the majority strategy, not the flattering one.
- Geckos found the hidden platform FASTER in complete darkness than at the end of normal training, while swimming LONGER paths at roughly twice the speed. A fast undirected search solves the task without any spatial memory at all. Any learning curve measured in time is contaminated by exactly this.
- In every published statistical model of this species, the number of prods needed to keep the animal moving predicted performance more strongly than the training phase did, often by an order of magnitude in F. The published leopard gecko learning curves are partly motivation curves.
- Fast forgetting is the correct answer for this animal, not a defect. Two months, against nine years in giant tortoises and twenty months in monitors. The authors' own explanation is the seasonal semi-arid Pakistani habitat: the landscape gets rearranged, so an old map is worse than no map.
- The defensive display in this species is chemosensory AND-gated. Vision and touch can raise sampling but never fire the display. If the simulated gecko has no chemical channel, the honest consequence is that it should be unable to display at all.
- The two reptile lesion studies of the hippocampal homologue directly contradict each other. Turtles lost place learning and kept cue learning; whiptail lizards never used a place strategy at all, lesioned or intact. Neither animal is a gecko, and the gecko's own medial cortex has never been lesioned, inactivated or recorded from.
- Zero place cells, grid cells or head-direction cells have ever been recorded in any reptile. That search returned nothing at all, which is unusual and worth stating plainly rather than softening.
- Three items from the verification pass should be carried forward as corrections. The Ewert response law is three separate univariate relations, not one joint function of velocity, contrast and size. The whiptail position-over-feature reversal advantage is contradicted by a specialist review with full-text access. And the claim that every Ewert paper is paywalled is false: at least two *Bufo* primary papers with quantitative neuronal tuning data are freely readable, one of them with 104 single neurons showing configural-motion selectivity.
- The single best-transferring quantitative data for this animal's hunting covers only the last two centimetres. A gecko in the same family has published lunge distance, velocity, acceleration and success rate. Everything upstream of that lunge, all the finding and approaching, has to be imported from a fish larva, a mouse and a toad.
- The entire target-species literature on memory and learning is one laboratory, one captive colony descended from about 90 wild-caught Pakistani founders, one apparatus, one aversive task, and an overlapping author list. Three papers, but a single evidence stream.

What must be checked or clarified

Open questions, in the sense that the project cannot answer them from anything it currently holds. Several are decisions rather than measurements, and they are listed because they are sitting underneath work that has already been done.

Is `lab` or `legacy` the right gait profile? **MUST BE SETTLED**

`lab` is gate-validated and limps under the trained policy. `legacy` is un-gated and looks correct. Whether the gate battery ever scored front-foot symmetry has never been established. Looking better is not the same as being right, and this decision sits underneath every video and every hunt measurement.

Does the leopard gecko hunt visually at all? **UNRESOLVED IN THE LITERATURE**

The only quantitative sensory study on this species is about PREDATOR detection, and it found chemical cues dominant (0.21 chemical vs 0.07 visual, n=42). The whole visual prey-detection module is built on a mouse, a fish larva and a toad. If the answer is no, a large part of the eye work is aimed at the wrong sense.

What does the animal do between 2 cm and 30 cm? **NO DATA EXISTS**

The best-transferring numbers in the entire corpus — lunge distance, velocity, acceleration, success rate — cover only the last two centimetres, and they come from a different genus in the same family. Everything upstream has to be imported.

Is the memory module going to be present and inert? **MUST BE DECIDED BEFORE BUILDING**

Every measured retention point in this species is at two months or beyond. A time constant of 115 days inside an episode measured in minutes is a term whose value is numerically indistinguishable from “no decay”. Ship the store without the decay and say so, or do not ship it.

What is the simulation's clock actually worth? **UNEXAMINED**

The learning rate was measured with trials one WEEK apart. The memory numbers live at months. The hunt lives at seconds. Nothing in this project has established what one simulated second is supposed to be, and three modules now depend on the answer.

Was the gate battery ever a locomotion validation? **WORTH RE-DERIVING**

Two of six gates have never passed. The accepted walker is accepted on four. Whether four-of-six is a meaningful bar, or whether the two failing gates are the ones that matter, has never been argued in writing.

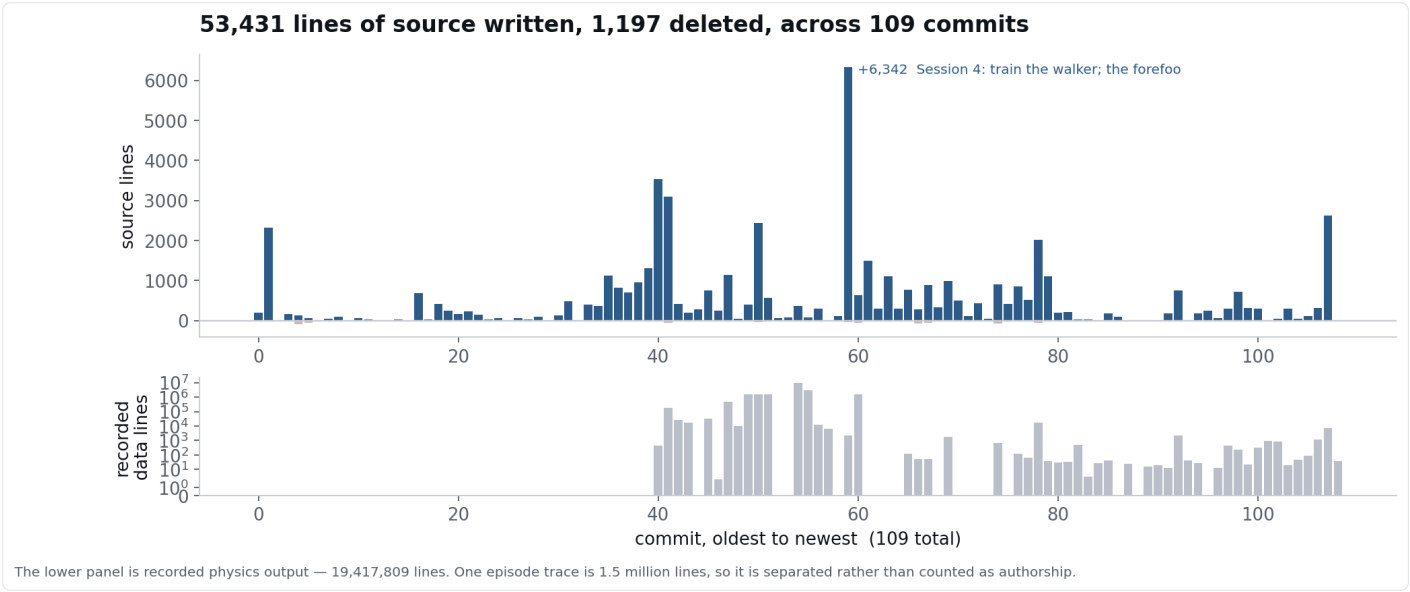
Does the registry's `confidence` field mean anything? **AUDIT NEEDED**

75 of 143 parameters are tagged *uncertain* and 30 *verified*. The policy text says *verified* means checked in the primary source. One entry tagged *verified* was the wrong species. The field has never been audited end to end.

Are the 212 harvested citations real? **EXPLICITLY UNVERIFIED**

This project has invented a citation once (#215) and overstated the absence of one (Delheusy 1995). The bibliography in Part 14 is a HARVEST of every author-year string in the repository. It has not been checked against any database. That check is the first thing the companion prompt asks for.

Every commit



HASH	DATE	SUBJECT	FILES	+	-
11f6cf7	2026-06-08	Initial commit	1	+201	-0
53face5	2026-06-08	Initial NeuroGecko code release	24	+2,330	-0
bc91977	2026-06-08	Merge branch 'main' of https://github.com/12ziyad/NeuroGecko	0	+0	-0
481a226	2026-06-09	Add V4.2.2 front-foot correction and dual-view rendering	5	+163	-25
e1404c6	2026-06-09	Add V4.2.3 front-loading controller	2	+123	-96
c486d90	2026-06-09	Add V4.2.4 escape-proof front-load gate	2	+68	-57
152fec	2026-06-09	Add V4.2.6 front-load polish and knee residual cap	2	+6	-4
6770779	2026-06-10	Add V4.2.7 front duty reward gate	1	+51	-1
e522bb5	2026-06-11	Add V4.2.8 contact tracking reflex curriculum	4	+95	-8
eff902e	2026-06-11	Tune V4.2.9 reflex wean controller	1	+2	-2
fdccbc6	2026-06-12	Add V4.3 velocity tracking experiment	2	+56	-1
570bd01	2026-06-12	Add V4.5 matched stance cadence experiment	3	+22	-15
d2e845b	2026-06-12	Add V4.6 front de-stick experiment	2	+4	-4
0f6c7ca	2026-06-12	Add V4.5 final locomotion base	2	+4	-4
0e7858c	2026-06-12	Record V4 locomotion final results	1	+37	-0
1c6e2ef	2026-06-12	Accept V4.5B as final locomotion champion	1	+20	-9
aad003a	2026-06-13	starting brain	5	+693	-0
cb93609	2026-06-13	Add AGWM brain environment wiring	2	+29	-7
9541f8f	2026-06-13	Add trainable visual brain policy	7	+424	-1
f72ca1c	2026-06-13	Add brain curriculum and stable viewer controls	4	+244	-17
2b5c158	2026-06-13	Add privileged food taper curriculum	3	+165	-7
d0f9deb	2026-06-13	Make brain camera see food and add privileged dropout curriculum	4	+235	-7

HASH	DATE	SUBJECT	FILES	+	-
d60f33f	2026-06-14	Use mouth distance and add brain resume support	3	+139	-27
a505ade	2026-06-14	Fix brain food reward scaling	3	+22	-5
cd014fe	2026-06-15	Add brain watcher stochastic diagnostics	1	+59	-3
342eaa1	2026-06-15	Rebalance brain food reward	1	+11	-7
0a57988	2026-06-16	Add brain visual lock reward and checkpoints	3	+55	-21
3460cb0	2026-06-16	Use visual delta reward and fix brain checkpoints	3	+34	-8
7e90df7	2026-06-16	Add brain food cue radius curriculum	3	+97	-9
6953dcf	2026-06-17	Add oracle action eval mode	1	+10	-1
11d0a08	2026-06-17	Add recurrent PPO brain support	2	+130	-20
a3de4ee	2026-06-18	Add privileged oracle behavior cloning	4	+485	-7
9efac13	2026-06-18	Fix BC diagnostics timing	2	+4	-2
c8c616e	2026-06-18	Add BC policy debug script	1	+396	-0
b370892	2026-06-18	Add Patch37B DAgger recovery BC	2	+366	-10
407e5d5	2026-06-18	Add Patch38A visual distillation preflight	5	+1,117	-6
c05ca81	2026-06-18	Add Patch38C visual DAgger recovery	4	+823	-15
2681c9e	2026-06-20	Add Patch39A visual search baseline	3	+703	-0
ddb105	2026-06-20	Add Patch40A drive-based GeckoBrain controller	2	+961	-0
44a921c	2026-09-05	Recover and verify the cloud model lineage before implementation	6	+1,308	-1
0c9f639	2026-09-05	Add measured baselines, camera isolation and verified checkpoint recovery	23	+3,542	-24
9960117	2026-09-05	Add constrained lab morphology, shared gait calibration and verified recovery evidence	27	+3,094	-54
3efa2f6	2026-09-05	Verify candidate motion and cross-platform reproducibility without altering legacy models	8	+415	-19
a7517e6	2026-09-05	Step 0: record true 250 Hz physics and report failed entrainment honestly	5	+205	-13
faa5dda	2026-09-05	Step 1: charge lab effort on total actuator work while preserving legacy reward	5	+276	-8
992b062	2026-09-05	Step 2: preserve force and collision-geometry diagnosis before controller edits	3	+759	-0
790edd4	2026-09-05	Step 2: verify historical command reconstruction with 19 force-only tests	2	+245	-1
ffc7be5	2026-09-05	Step 3: improve and verify the lab base; stop training on failed support and timing gates	16	+1,136	-25
ad54e9a	2026-09-05	Record Session 3 geometry baseline and hypothesis protocol	2	+41	-0
9eee048	2026-09-05	Session 3: implement, fix and test hind stance compensation; Gate 2 still fails	5	+404	-3
8153e8b	2026-09-05	Session 3b: generalise stance compensation to all four limbs; Gate 2 now 4/6	9	+2,440	-42
552c91b	2026-09-05	Session 3c: limb phase is late touchdown; stride length is geometrically unreachable by the limbs	2	+576	-0
8c67937	2026-09-05	Session 3d: stride length is speed; spine refuted as its source; 41% hind foot slip found	1	+71	-0
49c2e55	2026-09-05	Session 3e: refute gearing and contact softness; correct my own stride-length framing	1	+79	-0
7e752e7	2026-09-06	Session 3f: CMA-ES solves the limb-phase gate; 6/6 appears unreachable	2	+372	-0
9655d79	2026-09-06	Session 3g: withdraw the CMA-ES limb-phase claim; it optimised the wrong metric	3	+86	-0
8503e04	2026-09-06	Session 3h: fit harness now scores through realism_metrics; 12s fit rejected as unstable	4	+303	-3

HASH	DATE	SUBJECT	FILES	+	-
cceb20	2026-09-06	Session 3i: gate-count objective and the 20 s fit evidence; searches stopped	1	+17	-4
2c474f7	2026-09-06	Add docs/HANDOFF.md: complete record of the research and build phases	1	+106	-0
8c700d	2026-09-06	Session 4: train the walker; the forefoot gate is unreachable on this body	18	+6,342	-38
e184c1e	2026-09-07	Session 4b: drive the sprawl DOFs, couple the tail to the hindlimb	11	+630	-48
33be60c	2026-09-07	Session 4c: the no-cheat world, and prey the walker cannot catch	11	+1,490	-9
b573e57	2026-09-07	Session 4d: finish the no-cheat world in the environment that has eyes	6	+291	-25
306dd23	2026-09-07	Session 5: hypothalamus -- hunger from the animal's own metabolism	6	+1,100	-2
8b5f491	2026-09-07	Session 5b: fatigue from the published endurance curve; fear is on the wrong sense	6	+294	-3
ef75601	2026-09-07	Session 6: basal ganglia built from equations; NOT ACCEPTED, sweep fails	4	+774	-0
af1b08e	2026-09-07	Session 6b: found the GPR parameters; two of my three errors were structural	4	+289	-78
4a91782	2026-09-07	Session 6c: the living failure map, and 6b's parameters were a variant's	6	+891	-47
cbc42d8	2026-09-07	Session 6d: the basal-ganglia failure was not a failure	4	+327	-30
04b43be	2026-09-07	Session 6e: the winner did not saturate because we were testing the wrong model	6	+987	-12
427a845	2026-09-08	Session 6f: the gecko moved onto the validated selector, and brains 1 and 2 run together	4	+505	-6
ba460cb	2026-09-08	Docs: the map gains a flow and a success column	2	+113	-0
6c99f92	2026-09-08	Session 7: the cord becomes an oscillator, so a command has somewhere to land	2	+442	-0
78bcc15	2026-09-08	Session 7 docs: ledger 99-102, the new state, and the paused marker	1	+47	-9
683e8f6	2026-09-08	Session 7b-7c: the dopamine mechanism was wrong the OTHER way, and brain 3 lands	11	+898	-77
b9fafcf	2026-09-08	Session 8: scouting the eye found a detector that could not see its own food	4	+425	-15
5158d45	2026-09-08	Session 8b: the retina keeps position, and the published optokinetic asymmetry	5	+852	-12
e4196d8	2026-09-08	Session 8c: the tectum, shipped NOT ACCEPTED because it cannot find real prey	5	+514	-16
034a78d	2026-09-08	Session 9: the literature refuted the plan, and three checks were guarding nothing	19	+2,013	-53
ead740b	2026-09-08	Session 9b: the oracle measured, the body hash resolved, and a strike that works	9	+1,101	-13
871f843	2026-09-08	Session 9c: a clip you can actually watch, and two filming artefacts removed	2	+198	-2
3148100	2026-09-08	Session 9d: the gecko was never broken, and the shake was my camera	2	+206	-3
fb64aa2	2026-09-08	Session 9e: the replay camera pointed at the animal's face	1	+26	-5
d017b18	2026-09-08	Session 9f: the lag was a frame-rate header, not the animal	4	+32	-7
010f82c	2026-09-09	Session 9g: rendering the accepted walker side-on, and what the earlier clip was	0	+0	-0
9a7db0a	2026-09-09	Session 9h: three findings from trying to reproduce the Session 4 walking clip	3	+186	-6
3fb0302	2026-09-09	Session 9i: the gait profile is a parameter now, and I retract #179	3	+98	-2
db0163f	2026-09-09	Session 9j: the Session 4 commit and today render the same walk, pixel for pixel	0	+0	-0
779b03c	2026-09-09	Session 9k: the shake was the camera watching the trunk bob	1	+3	-2
7ae5b42	2026-09-09	Session 9l: measuring the push-then-drag rhythm, and rendering the old commit	0	+0	-0
73a37c8	2026-09-09	Session 9m: the user was right -- I filmed the wrong gait profile all session	1	+4	-3
ed61f25	2026-09-09	Session 9n: every clip I sent was the walker this project rejected	2	+173	-3

HASH	DATE	SUBJECT	FILES	+	-
e07a4a1	2026-09-09	Session 9o: the map as a printable document, generated from the repository	2	+748	-0
1a3ab31	2026-09-10	Session 9p: it is the PAIRING, not the policy -- #183 was too broad	1	+3	-2
e744d08	2026-09-10	Session 9q: rhythm alone beside rhythm plus the add-on	1	+182	-0
da2b651	2026-09-10	Session 9r: a guard so the broken pairing cannot assemble silently again	4	+241	-3
8aadf3a	2026-09-10	Session 9s: the goal never mentioned the mouth, so the animal never reached it	3	+63	-6
d74bdd2	2026-09-10	Session 9t: two published prey rules, one of which cannot do its job yet	4	+293	-10
8a34220	2026-09-10	Session 9u: a world with somewhere to shelter, something warm, and something to flee	5	+722	-7
feaa171	2026-09-10	Session 9v: brain 5, smell -- and the line it refuses to cross	4	+316	-3
5004a38	2026-09-10	Session 9w: brain 6, the day/night clock -- and the gap it refuses to fill	4	+300	-5
39bdcbf	2026-09-10	Session 9x: the walker could always reach -- I retrained it to fix nothing	1	+5	-2
d0755a5	2026-09-10	Session 9y: the hunt composes -- and the fix I proposed would have broken it	2	+41	-4
658255b	2026-09-10	Session 9z: one long clip of everything, with its numbers on screen	1	+299	-0
7de4ceb	2026-09-10	Session 10: the video was filmed with the wrong walker, and the right one was unreachable	3	+51	-8
02dd69f	2026-09-10	Session 10b: the eye was watching its own walking, and resolution was never the limit	3	+108	-8
06301e6	2026-09-11	Session 10c: the deferred modules were the buildable ones, and I invented a citation	3	+322	-3
a6f4bc8	2026-09-11	Session 11: the whole project as one document, and the scoreboard had drifted	4	+2,623	-5
f79fca8	2026-09-11	Session 11b: regenerate the report after the new ledger rows landed	1	+4	-2

The analysis brief that goes with this document

This report is also meant to be handed to a second reader with web access — another model, or a person — whose job is to check it rather than agree with it. The brief below is that instruction set. It is reproduced here so it cannot get separated from the document it is about; it also ships as `NeuroGecko_Analysis_Prompt.md`.

The single most important rule of this project

Every number in the simulation is tagged **PUBLISHED**, **DERIVED** or **INVENTED**, and says which. The project's stated position is that an invented number labelled as invented is fine, and a plausible number wearing a published number's clothes is the worst possible defect. Everything you do should serve that rule: move numbers from *invented* to *published* where a real measurement exists, and move them from *published* to *invented* where the citation does not support what is claimed.

A second rule matters for how you answer: **the project never accepts time estimates**. Do not give any. Give orderings and dependencies.

Task 1 — Verify the bibliography. Do this first.

Part 14 of the PDF lists 212 author-year strings and 51 DOIs harvested from the repository's own text. The document states plainly that **none of them has been checked against a bibliographic database**. Check them.

For each entry, especially the ones the project leans on most (Jagnandan 2014 and 2017, Fuller 2011, McElroy 2008, Delheusy 1995, Vollin 2021, Kröhnke 2023, Bergel 2026, Frýdlová 2026, Massecck 2008, Rawat 2019, Zaaf 2001, Autumn 1994 and 1997, Ewert 1969, Han & Carr 2024, Prescott 2024, Girard 2005, Hoy 2019, Yilmaz & Meister 2013, Whitford 2022, Wainwright 1991, Werner 1998/2005/2008, Wever 1964):

1. **Does it exist?** Search Google Scholar, PubMed, Crossref (`api.crossref.org`), Semantic Scholar, and the publisher directly. Resolve every DOI at `https://doi.org/<the DOI>`.
2. **Is the species right?** This project has already been caught attributing leopard-gecko work to authors who studied other families. For every citation used to justify a parameter tagged `E_macularius`, confirm the paper's actual study species.
3. **Does the paper say what the parameter claims?** Where you can reach the full text or a preprint, check the specific number. The project's registry entries carry a `source` field with journal, volume and pages — verify those too.
4. **Flag every entry you cannot resolve.** "Could not retrieve" is a valid and useful answer. Do not fill a gap with something plausible.

Report as a table: `citation | exists? | species studied | number checked | verdict` where verdict is one of `CONFIRMED` / `WRONG SPECIES` / `MISQUOTED` / `NOT FOUND` / `PAYWALLED`.

Task 2 — Attack the named gaps

Part 15 lists quantities the project says have never been measured and has therefore refused to invent. For each, search hard before accepting the gap. Search in English and, where relevant, in the languages of the herpetological literature (German, Czech, Russian, Chinese) and in the grey literature: theses, zoo and husbandry research, conference abstracts, preprints on bioRxiv.

The gaps, in the order they would unblock the most work:

1. **Visual prey detection in any gecko.** Receptive field size, size tuning, speed tuning, contrast threshold, detection latency, acuity in cycles per degree. The project claims *nothing* exists for any gecko and is currently importing all of it from a mouse superior colliculus, a zebrafish larva and a toad. Look for optic tectum recordings in any gekkotan, any nocturnal lizard, any squamate.
2. **Whether this species hunts visually at all.** The only quantitative sensory study on *E. macularius* is about predator detection and found chemical cues dominant. If a study exists showing visually guided prey capture in this species or a close relative, it changes the whole architecture.
3. **Anything between 2 cm and 30 cm of a gecko prey-capture sequence.** The project has good numbers for the final lunge (from *Coleonyx*, same family) and nothing at all for the approach.
4. **Cost of transport, sprint maximum, endurance, thermal performance curve, CTmin** for *E. macularius* or any eublepharid.
5. **Any field study of wild leopard geckos** — home range, activity budget, diet, microhabitat. The project believes none exists.
6. **A fitted habituation curve or time constant for any lizard or any reptile.** The document calls this the single biggest gap in the learning module.
7. **Memory decay at hours, days or weeks in any lizard.** Everything published is at two months or beyond, and a hunt lasts seconds.
8. **A false-positive rate for any biological prey detector** — toad, mouse, fish larva, lizard. The document says no such measurement exists anywhere.
9. **Cricket walking-vibration spectrum on sand**, for a substrate-vibration sense.
10. **A sleep-cycle period for *E. macularius* specifically.** The project has a published bracket of 84–134.5 s across lizard species and refuses to pick a number inside it. The source paper reportedly gives per-species values only as a violin plot with raw data on request — see whether the raw data, supplementary material or a later paper prints it.

For each: either give the measurement with its citation and sample size, or state clearly that you searched and found nothing, and say where you searched.

Task 3 — Adjudicate the contested claims

The document flags several places where the evidence disagrees with itself. Read the primary sources and take a position.

- **The two reptile hippocampal-homologue lesion studies contradict each other.** Turtles lost place learning and kept cue learning; whiptail lizards never used a place strategy at all. Neither animal is a gecko. Which, if either, should a gecko model follow, and on what grounds other than convenience?
- **The Ewert prey-recognition response law.** The document says it is three separate univariate relations (velocity, contrast, size, each measured with the others held constant) and *not* one joint surface, and that the worm/anti-worm tuning curves are unreachable. Confirm or correct this from Ewert's actual papers, and find the numeric curves if they are reachable.
- **The whiptail position-over-feature reversal advantage.** The document says this claim failed verification — that the advantage held during acquisition only, not reversal. Check it.

- **The nightly retreat loop.** The document says the claim that leopard geckos leave a shared retreat at sunset and return before dawn traces to a hobbyist magazine article recirculated through Wikipedia and care sheets, with no sample size, method or peer review. Trace it yourself and say where it actually comes from.
- **The "tight hide" rule.** Every care sheet states geckos prefer a snug crevice. The document says there is zero primary support in this species. Find the primary source or confirm its absence.
- **Whether the 82.9 % strike success rate transfers.** It comes from *Coleonyx*, a different genus in the same family. The simulation reproduces 89.7 %. Is that a validation or a coincidence?

Task 4 — Check the arithmetic that was DERIVED

- These numbers are not printed in any paper. The project computed them and says so. Recompute them from the published points and say whether the derivation holds.
- **Memory decay constants $\tau \approx 3.79$ months (route precision) and $\tau \approx 4.32$ months (goal-finding).** Derived by interpolating between two published retention points and assuming an exponential form that was never tested, and by treating a hazard ratio as a linear index of memory strength. Is that defensible? If not, what should it be?
 - **The learning curve shape.** The per-trial hazard ratio (1.09) and both ceilings (3.02× latency, 4.59× path) are published. The trajectory between them is drawn as an exponential and is invented. Does any published data constrain the shape?
 - **The approach speed floor, 0.018 m/s.** Scaled from a mouse figure of 0.05 m/s against a 0.164 m/s locomotor speed, applied to this animal's 0.055 m/s walk. Is proportional scaling right here?
 - **The angular size at which vision starts guiding approach, 6–8 degrees.** Derived geometrically from a 2 cm cricket at 15–20 cm. The document calls it the most load-bearing and weakest number in the whole prey rule.

Task 5 — Challenge the architecture, not just the numbers

- Read Part 2 (how one step runs), Part 9 (the eye) and Part 11 (the plan), then answer these as an outside reader who is allowed to say the plan is wrong.
- The blocker is that the eye reports something on **5.4 % of steps** during a hunt. The proposed fix is a two-channel wide-field/narrow-field detector borrowed from mouse superior colliculus. **Is that the right fix for a low detection rate, or does a low detection rate point at something else** — the field of view, the head not being pointed at anything, the prey being out of frame, the sampling rate?
 - The animal is nocturnal, afoveate, with a large pupil and short focal length. Every imported tuning constant comes from diurnal, foveate animals. **How big is that correction, and does anyone know?**
 - The walking rhythm is hand-written and the trained residual is switched off because it was measured as worse. **Is a hand-written CPG a defensible scientific choice, or a convenience?** What would it take to show it is right?
 - The document says only about one third of leopard geckos orient by the relationship between cues, and the rest may be running a non-cognitive motor search. **Should a single simulated animal model the majority strategy or the interesting minority one?**
 - The learning numbers come from an aversive forced-swim escape task, and the simulation is about hunting. **Is there any principled way to carry a learning rate across motivational systems, or should the module say it cannot?**

Task 6 — Find what the project has not thought of

- Search for recent work (2023 onward, and check for anything in 2026) on:
- Reptile or lizard neuroscience relevant to any of the eight modules.
 - Simulated animals built the same way — physics-engine models validated against published biology rather than trained to a reward. Note what they did about the same problems (provenance, gaps, oracle removal, validation).
 - MuJoCo or physics-simulation work on sprawling-posture locomotion.
 - Anything on *Eublepharis macularius* at all. It is a common laboratory and pet species and the project may simply have missed papers.

How to report back

- Write one document with these sections, in this order:
1. **Verdicts on the bibliography** — the table from Task 1, with counts: how many confirmed, wrong species, misquoted, not found.
 2. **Gaps closed** — every quantity from Task 2 you found a real measurement for, with citation, sample size, study species, and the exact value.
 3. **Gaps confirmed** — every one you searched for and could not find, with where you searched. This is as valuable as the previous section.
 4. **Contested claims, adjudicated** — Task 3, with your position and the primary source you based it on.
 5. **Derivations checked** — Task 4, with your recomputation.
 6. **Architecture challenges** — Task 5. Be direct. If the plan is aimed at the wrong problem, say so and say what the right one is.
 7. **What the project missed** — Task 6.
 8. **The ten things you would change first**, in order, with the dependency for each. No time estimates.

For every claim you make, give the source. Mark anything you could not verify as unverified. **If you are not sure, say you are not sure** — this project would rather have a recorded gap than a confident guess, and it has the ledger to prove why.

What not to do

- Do not accept a number because the PDF states it confidently. The PDF is the output of the system being audited.
- Do not fill a gap with a value from a related species without saying so, and without naming the species and the distance between it and *Eublepharis macularius*.
- Do not cite a paper you have not opened. The project has already been burned by exactly that.
- Do not reproduce long passages from any paper. Summarise, cite, and quote at most a short phrase.
- Do not give schedules, timelines or effort estimates for anything.

The animal as it stands today

Every number on this page was produced by compiling the XML with MuJoCo and asking the model, at the moment this document was generated. None of it is transcribed. Three files are compared because the difference between them is where two of this project's worst errors lived: `gecko_body_r.xml` is the hand-written SOURCE and is deliberately **not** to scale — it weighs 61.2 g and fails 13 of the 14 gates; `gecko_body_lab_v2.xml` is that source put through the two fitting passes, and is the body that passes 14 of 14; `gecko_world_v1.xml` is that fitted body placed in the world with a floor and a camera. Rebuilding v2 without first rebuilding v1 leaves a stale animal that still compiles, still renders and still passes its gates (#332), which is why `tools/rebuild_body.py` now performs the six steps in one order.

38.000 g

TOTAL MASS, WORLD

30

RIGID BODIES

139

GEOMS

38

JOINTS

30

MOTORS

44 / 43

NQ / NV

21,663

SKIN VERTICES

23

SKIN BONES

The census, three ways

QUANTITY	GECKO_WORLD_V1 WHAT RUNS	GECKO_BODY_LAB_V2 THE FITTED BODY	GECKO_BODY_R THE SOURCE TEMPLATE
total mass g	38	38	61.2
bodies	30	29	29
geoms	139	138	138
joints	38	38	38
hinges	36	36	36
slides	1	1	1
free	1	1	1
actuators	30	30	30
sensors	93	93	93
sites	23	23	23
nq	44	44	44
nv	43	43	43
meshes	29	29	29
textures	8	8	8
materials	15	14	14
skins	1	1	1
skin vertices	21,663	21,663	21,663
skin bones	23	23	23
collision geoms	48	48	48
visual geoms	89	89	89

Read the mass row carefully. The source template is 61.2 g and the fitted body is 38.0 g. That is not a bug and not a drift: the source is written at convenient hand-editable dimensions and the fitting pass scales it onto the published mass and the published proportions. The gate audit run against the wrong one of these two files reports 1/14 and looks like a catastrophe; run against the right one it reports 14/14. Both numbers are in this document on purpose.

Every gram in the animal

All 28 bodies that carry mass, in milligrams, largest first, with each one's share of the 38 g budget. The skin contributes nothing to this table — it is a deformable surface bound over the top, and #318 measured its contribution to mass, inertia, joint ranges, damping and actuator gains as exactly zero on every array.

BODY	MASS (MG)	% OF TOTAL
trunk_anterior	5995.16	15.777
trunk_middle	4895.58	12.883
head	4519.96	11.895
tail1	4281.91	11.268
trunk_posterior	3997.68	10.520
tail2	2426.62	6.386
pelvis	2096.93	5.518
neck	1728.13	4.548
tail3	1292.17	3.400
humerus_L	914.952	2.408
humerus_R	914.952	2.408
femur_L	647.137	1.703
femur_R	647.137	1.703
jaw	636.381	1.675
forearm_L	617.686	1.625
forearm_R	617.686	1.625
tibia_L	345.774	0.910
tibia_R	345.774	0.910
tail4	303.568	0.799
manus_L	165.044	0.434
manus_R	165.044	0.434
pes_L	157.791	0.415
pes_R	157.791	0.415
tail5	55.733	0.147
gular	41.182	0.108
eyelid_L	10.854	0.029
eyelid_R	10.854	0.029
tongue	10.524	0.028

Every joint, and every motor

38 joints. Hinge ranges are in degrees; the one slide joint, the tongue, is in metres. Damping is the value MuJoCo integrates, in its own units. Two of these ranges are themselves morphology gates — hip pro/retraction must reach 90° and hip rotation must not exceed 35° — and are checked in Part 20.

JOINT	TYPE	RANGE	DAMPING
root	free	unlimited	0
spine_lat_1	hinge	-25 to 25 deg	0.006
spine_pitch	hinge	-10 to 10 deg	0.006
neck_yaw	hinge	-40 to 40 deg	0.005
neck_pitch	hinge	-30 to 30 deg	0.005
head_yaw	hinge	-30 to 30 deg	0.005
head_pitch	hinge	-20 to 20 deg	0.005
eyelid_L	hinge	-75 to 0 deg	0.005
eyelid_R	hinge	0 to 75 deg	0.005
gular	hinge	-14 to 6 deg	0.005
jaw	hinge	0 to 40 deg	0.005
tongue	slide	0 to 0.012 m	0.002
shoulder_sprawl_L	hinge	-35 to 35 deg	0.005
shoulder_proret_L	hinge	-40 to 40 deg	0.005
elbow_L	hinge	-60 to 60 deg	0.005
wrist_L	hinge	-35 to 35 deg	0.003
shoulder_sprawl_R	hinge	-35 to 35 deg	0.005
shoulder_proret_R	hinge	-40 to 40 deg	0.005
elbow_R	hinge	-60 to 60 deg	0.005
wrist_R	hinge	-35 to 35 deg	0.003
spine_lat_2	hinge	-25 to 25 deg	0.006
spine_lat_3	hinge	-25 to 25 deg	0.007
hip_sprawl_L	hinge	-35 to 35 deg	0.005
hip_proret_L	hinge	-45 to 45 deg	0.005
hip_rot_L	hinge	-15 to 15 deg	0.005
knee_L	hinge	-60 to 60 deg	0.005
ankle_L	hinge	-45 to 45 deg	0.005
hip_sprawl_R	hinge	-35 to 35 deg	0.005
hip_proret_R	hinge	-45 to 45 deg	0.005
hip_rot_R	hinge	-15 to 15 deg	0.005
knee_R	hinge	-60 to 60 deg	0.005
ankle_R	hinge	-45 to 45 deg	0.005
tail_yaw_1	hinge	-30 to 30 deg	0.012

JOINT	TYPE	RANGE	DAMPING
tail_lift	hinge	-25 to 25 deg	0.012
tail_yaw_2	hinge	-30 to 30 deg	0.01
tail_yaw_3	hinge	-30 to 30 deg	0.01
tail_yaw_4	hinge	-30 to 30 deg	0.01
tail_yaw_5	hinge	-30 to 30 deg	0.01

The 30 actuators

Every one is a position actuator. There is no torque channel and no direct velocity channel; a controller can only ask a joint to go somewhere.

hip_sprawl_L	hip_proret_L	hip_rot_L	knee_L
ankle_L	hip_sprawl_R	hip_proret_R	hip_rot_R
knee_R	ankle_R	shoulder_sprawl_L	shoulder_proret_L
elbow_L	shoulder_sprawl_R	shoulder_proret_R	elbow_R
neck_yaw	neck_pitch	head_yaw	head_pitch
eyelid_L	eyelid_R	tongue	gular
jaw	spine_bend	spine_pitch	tail_bend_L
tail_bend_R	tail_lift		

The fourteen morphology gates

These are not unit tests. They are the anatomical checks the body has to survive before anything is allowed to walk on it, and each one compares a measurement taken on the compiled model against an interval taken from a published measurement of *Eublepharis macularius*. A gate with two components — left and right — passes only if both pass; averaging is not allowed to hide one side. **14 of 14 pass.**

```
C:\Users\ziyad\GeckoBrain\morphology\gecko_body_1ab_v2.xml
model sha256 d173fc4b9ed2a9e4a2a8505ebf8964f24187dff29070e9f82a5fb08787910c49
registry sha256 cac1ea6234734bb6e385cf4d02c3b40b39f193f1be7be657b96e591f06c89284
measured 2026-09-12T15:46:17.189930+00:00 · 44 nq, 43 nv, 30 nu, 36 hinge joints, 23 sites, 93 sensors
```

GATE	MEASURED	ACCEPTED INTERVAL	UNITS	RESULT
Total mass	0.038	[0.034, 0.042]	kg	PASS
Tail mass fraction	0.22	[0.19, 0.25]	fraction	PASS
Tail chain / SVL	0.73	[0.69, 0.77]	SVL	PASS
Intact CoM behind nose / SVL	0.678112	[0.639, 0.679]	SVL behind snout	PASS
Tail-excluded CoM / SVL	0.527	[0.507, 0.547]	SVL behind snout	PASS
Settled hip height / SVL	0.14888 (both sides)	[0.14, 0.16]	SVL	PASS
Settled shoulder height / SVL	0.109312 (both sides)	[0.103, 0.123]	SVL	PASS
Shoulder / hip height	0.734232 (both sides)	[0.68, 0.78]	ratio	PASS
Femur / tibia length	1.01987 (both sides)	[0.9, 1.06]	ratio	PASS
Hindlimb chain / SVL	0.417 (both sides)	[0.387, 0.447]	SVL	PASS
Hip pro/retraction range	90 (both sides)	≥ 90	deg	PASS
Hip rotation range	30 (both sides)	≤ 35	deg	PASS
Skin head width / SVL	0.185	[0.165, 0.205]	SVL	PASS
Inner-eye gap	0.01	[0.009, 0.011]	m	PASS

The protocol the numbers were taken under

neutral keyframe	neutral
settle seconds requested	2.0
settle seconds actual	1.999999999999103
settle steps	2500
actuator targets	all zero (position targets, not zero applied force)
ground	model floor z=0; no environmental randomization
com	neutral-pose body-mass weighted axial projection; tail exclusion is mathematical
bilateral gates	both sides and both joint/actuator ranges must pass
claim	engineering morphology audit; not walking or kinematic validation

Where each gate's interval comes from

Rule 1 of this project is that every number says whether it is published, derived or invented. This table is that rule applied to the acceptance intervals themselves, so the gates cannot quietly become a set of numbers chosen because the body already met them.

GATE	SOURCE	CONFIDENCE	NOTES
Total mass	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916; Jagnandan & Higham (2017), Sci Rep 7:10865, doi:10.1038/s41598-017-11484-7; CODEX_PROMPT.md (user build brief, 2026-09-05)	likely	Engineering population choice near lab cohorts, not one published individual or a mean of pooled animals.
Tail mass fraction	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916	likely	Transcribed from supplied research; anatomical partition remains approximate.
Tail chain / SVL	Rawat et al. (2019), Reptiles & Amphibians 26(1):58–61, Table 1, doi:10.17161/randa.v26i1.14342	verified	87 mm / 119 mm from the ONE intact-tail specimen; second specimen had a regenerated tail. Transferred to lab-cohort body size.
Intact CoM behind nose / SVL	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916	verified	Primary Results verified: mean 65.9% SVL. Methods used frozen rigid animals and thread suspension; the original tail was reattached. Cohort SVL121.9mm, not106mm. Neutral axial projection and size transfer are explicit model assumptions, not walking CoM validation.
Tail-excluded CoM / SVL	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916	verified	Primary Results verified: mean52.7%SVL. Frozen animals were measured after removing regenerated tail, not live immediately-after-autotomy posture. Audit excludes tail masses in the SAME neutral pose. Source cohort7adult females39.4g/SVL121.9mm.
Settled hip height / SVL	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916	likely	Walking measurement used as a static neutral-control engineering gate. Settled stance is not the same experimental protocol.
Settled shoulder height / SVL	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916; CODEX_PROMPT.md (user build brief, 2026-09-05)	likely	Build target; the supplied paper table rounds shoulder height to 0.11 SVL. Static neutral-control proxy for walking height.
Shoulder / hip height	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916	likely	0.11/0.15 rounded; compare bilateral ratios.
Femur / tibia length	Fuller, Higham & Clark (2011), Zoology 114:104–112, Table 1, doi:10.1016/j.zool.2010.11.003	verified	Correct femur/tibia ratio is 1.0199, not 0.98. The latter reverses the ratio; requested pass band is retained separately.
Hindlimb chain / SVL	Jagnandan, Russell & Higham (2014), J Exp Biol 217:3891–3897, doi:10.1242/jeb.110916; Fuller, Higham & Clark (2011), Zoology 114:104–112, Table 1, doi:10.1016/j.zool.2010.11.003	likely	Cross-study ratio 0.15/0.36. Fuller absolute total/SVL is 4.41/10.65=0.414. Total limb includes fourth toe; current model digit is an engineering proxy.
Hip pro/retraction range	Jagnandan & Higham (2017), Sci Rep 7:10865, doi:10.1038/s41598-017-11484-7; CODEX_PROMPT.md (user build brief, 2026-09-05)	likely	Engineering range allowance for observed 82.57-degree excursion, NOT a cadaveric anatomical ROM. Both joint limits and position actuator ranges must permit it.
Hip rotation range	Arnold et al. (2014), Iguana proxy as cited in docs/research/gecko_scorecard.md	uncertain	Cross-species engineering ceiling from supplied research, not leopard-gecko ROM. Primary source not independently rechecked.
Skin head width / SVL	Rawat et al. (2019), Reptiles & Amphibians 26(1):58–61, Table 1, doi:10.17161/randa.v26i1.14342	verified	22/119=0.185 for specimen1; specimen2 21/109=0.193. Gate measures skin geometry excluding protruding eye spheres; collision width is reported separately.
Inner-eye gap	Rawat et al. (2019), Reptiles & Amphibians 26(1):58–61, Table 1, doi:10.17161/randa.v26i1.14342	verified	Both specimens measured 10 mm between eyes. Model uses shortest inner gap between visual spherical eyes, not center separation; that landmark mapping remains an engineering approximation.

Corrections and limits the audit records against itself

- The model has 32 hinge joints, not the 26 asserted in parts of the supplied documents.
- Fuller Table1 femur/tibia is 1.54/1.51=1.0199, not 0.98. The requested 0.90–1.06 pass band is retained explicitly.
- Fuller Table1 tibia SEM is 0.02 cm, not the 0.04 cm transcribed in the supplied scorecard.
- Interorbital distance is approximated by the inner spherical-eye gap, not center-to-center separation.
- The source limb total includes fourth toe. MJCF ankle-to-digit geometry is a proxy, not a measured metatarsal segment.
- Static zero-control heights are engineering gates using walking reference values; they do not establish dynamic posture validity.
- A root spawn-height edit alone cannot set the eventual mechanically settled hip height.
- Gate tolerances are engineering choices. Passing them is not a statistical claim about individual animals.
- Tail CoM exclusion holds all remaining segments in the same neutral pose and is not an autotomy behavior experiment.

Every proportion measured on the body

The gates in Part 20 are the 14 quantities that have a published interval to be checked against. These are **all** of them — 25 gate metrics and 28 raw diagnostics, every leaf of the audit's own output, including the ones no paper constrains. They are here because a proportion with no gate is exactly the kind of number that drifts without anyone noticing.

Gate metrics — 25 values

total_mass_kg	0.038
tail_mass_fraction	0.22
tail_length_svl	0.73
com_intact_svl	0.678112
com_tail_excluded_svl	0.527
femur_tibia_ratio.L	1.01987
femur_tibia_ratio.R	1.01987
hindlimb_length_svl.L	0.417
hindlimb_length_svl.R	0.417
hip_preret_range_deg.L_joint_deg	90
hip_preret_range_deg.L_actuator_deg	90
hip_preret_range_deg.R_joint_deg	90
hip_preret_range_deg.R_actuator_deg	90
hip_rot_range_deg.L_joint_deg	30
hip_rot_range_deg.L_actuator_deg	30
hip_rot_range_deg.R_joint_deg	30
hip_rot_range_deg.R_actuator_deg	30
head_width_svl	0.185
interorbital_distance_m	0.01
hip_height_svl.L	0.14888
hip_height_svl.R	0.14888
shoulder_height_svl.L	0.109312
shoulder_height_svl.R	0.109312
shoulder_hip_ratio.L	0.734232
shoulder_hip_ratio.R	0.734232

Diagnostics — 28 values, in metres and radians unless named otherwise

svl_m	0.106
tail_chain_length_m	0.07738
neutral_com_m	-0.00737991, -4.458e-20, 0.0262598
neutral_tail_excluded_com_m	0.008638, -5.715e-20, 0.025769
hindlimb_segments.L.femur_m	0.0153277
hindlimb_segments.L.tibia_m	0.0150291
hindlimb_segments.L.ankle_to_fourth_digit_m	0.00751665
hindlimb_segments.L.fourth_digit_m	0.00632855
hindlimb_segments.L.total_m	0.044202
hindlimb_segments.R.femur_m	0.0153277
hindlimb_segments.R.tibia_m	0.0150291
hindlimb_segments.R.ankle_to_fourth_digit_m	0.00751665
hindlimb_segments.R.fourth_digit_m	0.00632855
hindlimb_segments.R.total_m	0.044202
head_collision_width_m	0.01961
head_visual_skin_width_m	0.01961
eye_center_distance_m	0.0172
settled_hip_z_m.L	0.0157812
settled_hip_z_m.R	0.0157812
settled_shoulder_z_m.L	0.0115871
settled_shoulder_z_m.R	0.0115871
settled_max_abs_qvel	0.000101286
settled_touch_forces_N.touch_fore_L	0.0971378
settled_touch_forces_N.touch_fore_R	0.0971378
settled_touch_forces_N.touch_hind_L	0.0892522
settled_touch_forces_N.touch_hind_R	0.0892522
settled_touch_forces_N.touch_belly_mid	0
settled_touch_forces_N.touch_belly_post	0

The surface: what was added, and what it cost the physics

The animal had 23 rigid shells that visibly slid past one another. It now has one continuous deformable surface of 21,663 vertices bound to 23 bones, an eyeball with a vertical pupil and a lid that closes, and a tongue on a slide joint. The question that matters is not whether it looks better. It is whether the animal the gates measured is still the animal that walks.

It is. Measured three ways, not assumed (#318). A lab body rebuilt from a mesh-free, skin-free copy of the same source has bit-identical `body_mass`, `body_inertia`, `body_ipos`, `jnt_range`, `dof_damping` and `actuator_gainprm` — maximum absolute difference exactly **0.000e+00** on every array, total mass 0.038 kg both ways. After 300 steps from the `stand` keyframe the two worlds' `qpos` agree to the last bit. And the 64×64 image the animal's own `head_cam` delivers to its retina differs in **0 of 4096 pixels**. This was checked rather than argued because the assumption was not safe: the retina is a camera ON the animal, and in `legacy` mode it draws geom group 1 — the group every new mesh and the skin sit in.

The six steps, in the one order that works

Forgetting the fifth step twice left a body that compiled, rendered, and passed 14/14 gates while three reproducibility tests failed with an unrelated-sounding message (#332). `tools/rebuild_body.py` exists so that order cannot be got wrong again.

1	generate the swept body, limb and toe meshes
2	generate the eyeball, lid and rim meshes
3	bind one surface to the 23 bones and emit the skin
4	patch the source XML (idempotent)
5	refit lab body v1 — the step that gets skipped
6	refit lab body v2 with <code>--fit-com</code> , then the world, then the furnished world

Two renderer facts that cost this project several sessions, recorded so they are not rediscovered. (a) Mesh UVs require a `type="2d"` texture; a `cube` texture is addressed by POSITION and ignores texture coordinates entirely, which renders the whole animal flat grey (#315). (b) A mesh geom's texture rows count from the BOTTOM of the image and a skin's count from the TOP, so the same v coordinate lands on opposite ends of the same PNG — measured: the texture at v = 0.10 is yellow (237, 200, 59) and at v = 0.90 is white (228, 225, 218), which is exactly the white eyelid that appeared (#344). And one that does not work at all: MuJoCo accepts `<layer role="normal">` in a material and the renderer ignores it — measured, 0 of 57600 pixels changed (#329). Bumps have to be baked into the colour.

Every constant in the surface generators

These are appearance numbers. Almost none of them is published, and the project's first rule applies to them exactly as it applies to the physics: the ones that are INVENTED say so in their own source. They are reproduced here in full because the surface is the one part of this animal that was built by eye, and a reader checking the work needs to see which dials exist.

The swept surface — `tools/make_gecko_mesh.py`, 13 constants

One surface is swept along the animal's own axis, sampling the envelope of the gated collision primitives at every station, so the skin cannot be a different animal from the one the gates measured.

CONSTANT	VALUE	WHAT IT IS
REPO	<code>pathlib.Path(__file__).resolve().parents[1]</code>	
OUT	<code>REPO / "morphology" / "meshes"</code>	
RING	28	Vertices around each ring, and stations along the animal. 28 x 200 puts a vertex about every 2 mm around the girth and every 1.1 mm along the body, which is fine enough that the silhouette has no visible facets at the sizes this animal is rendered at, and still about a two-hundredth of the AI-generated sculpt's triangle count.
STATIONS	200	
CHAIN	<code>["head", "neck", "trunk_anterior", "trunk_middle"]</code>	Nose-to-tail ordering. Anything absent from the model is skipped.
PROFILE	{	STYLING MULTIPLIER, INVENTED (photographs). Applied to the ellipse MEASURED from that body's primitives, so a value of 1.0 is "exactly the primitives" and the gated dimensions survive untouched. Only the departures are invented. width scale, height scale
LIMB_SCALE	1.12	
TILE_M	0.030	One texture tile per 30 mm of skin ALONG the animal, so the spots come out the same size on the neck as on the tail instead of being stretched by each piece's length.
U_TILES	2	Tiles AROUND the girth. This has to be a WHOLE NUMBER and the same for every ring: $u \text{ ran } 0.2\pi * w / \text{TILE_M}$, which is neither. A fractional wrap leaves the texture discontinuous at the seam, and a wrap that changes from ring to ring shears that discontinuity along the body -- which is why the tail came out as a barber's pole. Two tiles round a ~60 mm girth keeps the spots roughly as wide as they are long.
DOME_RINGS	5	Rings in each end dome. INVENTED; it only sets how round the nose looks.
HEAD_PROFILE	(	(fraction from occiput to snout tip, width, height, centreline drop in m). Width and height are fractions of the measured maximum. INVENTED.
HEAD_N_TOP	2.7	Superellipse exponents above and below the midline. 2 is an ellipse; higher is flatter-topped. INVENTED.
HEAD_N_BOTTOM	2.1	

The body sculpt — `tools/gecko_sculpt.py`, 19 constants

The profile curves that turn a tube into a gecko: the neck dip, the trunk and tail width tracks, the tail's ridge, the superellipse exponents that make the back rounder than the belly, and the toes.

CONSTANT	VALUE	WHAT IT IS
NECK_DIP	0.84	NECK, as fractions of the value at its own two ends: starts at the head's occiput ring, ends at the shoulder. Dips between. INVENTED.
NECK_DIP_AT	0.40	
TRUNK_W	((0.00, 0.86), (0.25, 0.97), (0.45, 1.00), (0.70, 0.96), (1.00, 0.80))	TRUNK, fractions of the trunk's measured maximum. Widest at 0.45 of the way back -- a leopard gecko carries its belly behind the middle. INVENTED.
TRUNK_H	((0.00, 0.86), (0.25, 0.96), (0.45, 1.00), (0.70, 0.94), (1.00, 0.78))	
TAIL_W	((0.00, 0.78), (0.10, 0.94), (0.24, 1.00), (0.42, 0.90), (0.60, 0.66))	TAIL, fractions of the tail's measured maximum. Fat through the first third, then a long even taper to a ROUNDED tip. No pinch at the vent: the real animal goes smoothly from hip into tail. INVENTED.
TAIL_H	((0.00, 0.76), (0.10, 0.92), (0.24, 1.00), (0.42, 0.90), (0.60, 0.66))	
TAIL_RIDGE_AMP	0.035	Tail ridges: the fat tail is banded by transverse rows of tubercles that give it a visibly wavy outline. Amplitude as a fraction of radius, period in metres. INVENTED (photographs).
TAIL_RIDGE_PERIOD_M	0.0046	
BODY_N_TOP	2.35	Superellipse exponents: a slightly flat back and a round belly. INVENTED.

CONSTANT	VALUE	WHAT IT IS
BODY_N_BOT	2.0	
TAIL_N_TOP	2.15	
TAIL_N_BOT	2.0	
TOE_SKIN	1.45	Skin stands this far off the digit capsule, and the toe tapers to this fraction of its base radius before the rounded tip. INVENTED.
TOE_TAPER	0.62	
TOE_RINGS	6	
JOINT_SCALE	0.72	A sphere at every limb articulation, a little larger than the tubes that meet there. Two tapered tubes meeting at an angle leave a wedge open on the outside of the bend and a twisted collar where their end rings cross; the sphere hides both, which is what a knuckle is for. INVENTED standoff. was 1.06: sat between tube ENDS that had tapered to 0.55 r, so it looked like a ping-pong ball
JOINT_BODIES	("humerus", "forearm", "manus", "femur", "tibia", "pes")	
LIMB_STATIONS	26	
LIMB_ROOT_SINK_M	0.004	

Binding the surface to the bones — `tools/make_gecko_skin.py`, 8 constants

Linear blend skinning: every vertex is given at most four bones and a Gaussian falloff, so one continuous surface follows 23 rigid bodies with no seam at any joint.

CONSTANT	VALUE	WHAT IT IS
REPO	pathlib.Path(__file__).resolve().parents[1]	
SRC	REPO / "morphology" / "gecko_body_r.xml"	
SIGMA	0.009	
MAX_BONES	4	
MIN_WEIGHT	0.02	
LIMB_V	(0.16, 0.34)	
AXIAL	["head", "neck", "trunk_anterior", "trunk_middle"	
LIMB_CHAINS	[("humerus", "forearm", "manus"), ("femur", "tibia", "pes")]	

The eye — `tools/make_gecko_eye.py`, 17 constants

A real globe, a vertical pupil, a movable lid and a rim. The outward axis is DERIVED from the cranium ellipsoid's own normal rather than chosen.

CONSTANT	VALUE	WHAT IT IS
REPO	pathlib.Path(__file__).resolve().parents[1]	
MESH_OUT	REPO / "morphology" / "meshes"	
TEX_OUT	REPO / "morphology" / "textures"	
SRC	REPO / "morphology" / "gecko_body_r.xml"	
RINGS	28	Sphere resolution. An eye is 3.6 mm across and fills the frame in a close-up, so it gets a finer mesh than the body does.
SEGMENTS	40	
PUPIL_HALF_WIDTH	0.24	The eye as the reference photograph shows it: a large, dark, glossy dome with the pupil dilated -- the animal is nocturnal and the picture was not taken in bright sun. The slit is still there, wide open. The ring at the edge of the iris is now LIGHTER than the iris, not darker: a dark ring on a dark eye read as a black hole with a black border (#343). INVENTED.
PUPIL_HALF_HEIGHT	0.66	
IRIS_OUTER	0.86	
IRIS_PALE	np.array([74, 56, 44], dtype=float)	
IRIS_DARK	np.array([26, 19, 16], dtype=float)	
RIM	np.array([128, 112, 94], dtype=float)	
PUPIL	np.array([8, 6, 8], dtype=float)	
LID_ARC_DEG	48.0	How far round from its axis the eyelid hood reaches, and how far it stands off the eyeball. INVENTED; both are appearance only. 64 degrees was the first try and it swallowed the eye whole: the eye's own outward axis is only about 68 degrees from the hood's, so a 64 degree cap reaches the pupil. 33 leaves a brow that overhangs the top of the eye, which is what it should be.
LID_GAP	0.0008	
RIM_STANDOFF	0.00025	The pale rim of skin around the eye opening: standoff from the eyeball and tube radius, in metres. INVENTED (photographs -- the rim is the single most recognisable thing about this animal's face after the spots).
RIM_TUBE	0.00048	

The pigment — `tools/make_skin_texture.py`, 6 constants

The 512 x 2048 texture: ground colour, spots, tubercles, and the face details painted at the v positions the binder reported for nostril, jaw and eye.

CONSTANT	VALUE	WHAT IT IS
REPO	pathlib.Path(__file__).resolve().parents[1]	
OUT	REPO / "morphology" / "textures"	
BASE	np.array([214, 186, 108], dtype=float)	Wild-type dorsal ground colour, and the spot colour. INVENTED (photographs).
SPOT	np.array([56, 42, 28], dtype=float)	
PALE	np.array([238, 228, 200], dtype=float)	
SIZE		512

Wiring it into the model — `tools/wire_mesh_skin.py`, 6 constants

An idempotent XML patcher, so the body can be rebuilt without the patch accumulating.

CONSTANT	VALUE	WHAT IT IS
REPO	pathlib.Path(__file__).resolve().parents[1]	
SRC	REPO / "morphology" / "gecko_body_r.xml"	
SUPERSEDED	("L brow", "R brow", "snout bridge", "rounded snout tip"	Face primitives that the head mesh now supersedes. Kept in the file (the morphology audit measures them) and drawn transparent.
RIGID_MESHES_SUPERSEDED		True The 23 rigid per-body mesh geoms are superseded by the continuous skin from tools/make_gecko_skin.py. Set False to draw them again and hide the skin -- that is the fallback if linear blend skinning ever misbehaves.
CUBE_TAIL	' <texture name="gecko_tail" type="cube" file="gecko_tail.png"/>'	
UV_TEXTURES		(

The agents, and what survived them

Every piece of biology in this animal was found by a research pass, and every research pass in this project is adversarial by construction: one agent per question finds a claim, then independent agents are told to REFUTE it and to reject anything they cannot confirm from a source they have personally opened. The refutation rate is the point. A pass that confirms everything it looked for has not checked anything.

233
AGENT RUNS RECORDED

276
PUBLISHED MEASUREMENTS

10 / 36
SURFACE CLAIMS SURVIVED

18 / 20
SPINE NUMBERS SURVIVED

PASS	AGENTS	SUBJECT	SURVIVED	KILLED	WRITTEN TO
Research corpus	167	the animal's biology, physiology and gait	276 measurements	—	config/proxies.yaml
Session-12 provenance audit	12	this project's own claims about where its numbers came from	0 of 4	4 of 4	ledger #264
Surface anatomy	42	scales, tubercles, eye, ear, feet, pattern	10	26	docs/GECKO_SURFACE_ANATOMY.md
Lateral undulation	24	how far a walking lizard's spine actually bends	18	2	docs/LATERAL_UNDULATION.md

The 167 figure is the tally carried in docs/FAILURE_MAP.md 's standing summary and has not been independently recounted here; the 12, 42 and 24 are each stated in the ledger row or the document that pass produced. 167 + 12 is the pre-existing count, and 42 + 24 were added in the most recent session, giving 233 recorded agent runs in total. Treat 167 as reported rather than verified.

The surface pass is PARKED, NOT APPLIED. The appearance work was called off by the user before this research returned, and a pass in which 26 of 36 claims died is exactly the reason not to have built first and checked afterwards. The skin that exists was built from reference photographs, not from this document. Applying it is unstarted work.

Surface anatomy — the 10 claims of 36 that survived

42 agents. Each claim below was confirmed from a source the verifying agent opened and read. The last line of each is the instruction the pass wrote for whoever eventually builds it.

EAR

The tympanic membrane of the leopard gecko is externally visible on the side of the head, with no scale cover or flap over it.

[verified] Vedurmudi et al. use the leopard gecko as their type example of an internally coupled ear. Fig. 1 caption: the location of the eardrum / tympanic membrane on the side of its head is highlighted; a close-up of the head shows the eardrum and the embedded extracolumella can be 'discerned clearly'; a third panel is a close-up of the eardrum itself. This is photographic evidence for the species, not a verbal description. I extracted and read the PDF text directly. Caveat: the caption names only 'Leopard gecko', not the binomial, though that vernacular is standard for *E. macularius*.

Source: Vedurmudi AP, Goulet J, Christensen-Dalsgaard J, Young BA, Williams R, van Hemmen JL (2016) Internally coupled ears: mathematical structures and mechanisms underlying ICE. *Biological Cybernetics* 110: 359–382, Fig. 1 caption (p. 360). PDF: <https://www.atsu.edu/research-portal/uploads/78040790-Internally%20coupled%20ears-mathematical%20structures.pdf>

For the model: Model the ear as a real aperture in the head silhouette with a visible membrane at its floor — not a painted-on spot and not a scale-covered depression.

EYE

Eublepharids have genuinely movable eyelids; the family name is descriptive, not just etymological.

[verified] Boulenger's family diagnosis for Eublepharidae states the eyes are moderately large and are protected by thick, movable, lids that meet ("connivent"). Smith's genus diagnosis for Eublepharis independently states both eyelids are well developed and movable. Two independent primary taxonomic sources, ~50 years apart, agree. Smith elsewhere lists "connivent eyelids" as one of the three characters separating eublepharids from typical gekkonids (with opisthocoelous vertebrae and united parietals).

Source: Boulenger, G.A. 1885. *Catalogue of the Lizards in the British Museum (Natural History)*, 2nd ed., Vol. 1, p. 229 (Fam. Eublepharidae). Archive.org id catalogueoflizar01brituoft. AND Smith, M.A. 1935. *The Fauna of British India... Reptilia and Amphibia* Vol. II, Sauria, pp. 125–126 (genus Eublepharis). Archive.org id in.ernet.dli.2015.207328.

For the model: Model two real lid flaps as separate deformable geometry over the globe, not a painted-on ring. The lids are described as THICK, so they need volume — a fleshy fold with its own scale texture, not a thin membrane.

EYE

Leopard geckos have NO spectacle / brille. The cornea is bare and directly exposed when the lids are open.

[verified] Smith states that in Eublepharis the eyeball "has lost the fixed membranous covering" that other geckos have. Hellebuyck & Solanes Vilanova state that a true transparent rigid spectacle is present in all Gekkota except the eublepharid clade, which has fully moveable eyelids. Reavill & Schmidt say it species-specifically: the leopard gecko "belongs to a family of geckos that have eyelids and not a spectacle" — and warn that the caseous corneal plug clinicians see in this species is commonly misdiagnosed as a retained spectacle. Three independent sources, one of them naming *E. macularius* directly.

Source: Smith 1935, *Fauna of British India, Sauria*, pp. 23–24. AND Hellebuyck, T. & Solanes Vilanova, F. 2023. *Animals* 13(6):1108, doi:10.3390/ani13061108. AND Reavill, D. & Schmidt, R.E. 2012. Pathology of the Reptile Eye and Ocular Adnexa. *Proc. ARAV* 2012, p. 89.

For the model: Do NOT put a glossy transparent scale disc over the eye. The cornea is a wet, directly exposed, strongly specular dome — shade it as bare cornea with a sharp specular highlight, sitting inside a soft skin aperture, not behind a flat scale.

EYE

The pupil is vertical.

[verified] Smith's genus diagnosis for Eublepharis lists flatly: "Pupil vertical." Boulenger's family diagnosis calls it an "elliptico-vertical pupil" — i.e. a vertically-elongate ellipse rather than a mathematical hairline. Two independent primary sources.

Source: Smith 1935, *Fauna of British India, Sauria*, p. 126. AND Boulenger 1885, *Cat. Liz. Brit. Mus.* Vol. 1, p. 229.

For the model: Pupil axis is vertical in head space (perpendicular to the long axis of the eye aperture), so it stays vertical regardless of how the eye aperture is tilted on the skull.

FEET

Each of the four feet bears five digits; there is no digit reduction.

[verified] Zhuang et al. (2019) figure 1B is a schematic of a gecko left pes in dorsal view explicitly 'based on a radiograph of the left pes of Eublepharis macularius', labelling five metatarsals MTI–MTV. Corroborated at genus level by congener descriptions that report subdigital lamellae as five per-digit counts for each autopodium (*E. jhuma*: right manus 10–15–17–18–15; right pes 11–16–21–23–22) and relative lengths for digits I–V.

Source: Zhuang, M.V., Russell, A.P. & Higham, T.E. (2019) Evolution of pedal digit orientation and morphology in relation to acquisition and secondary loss of the adhesive system in geckos. *Journal of Morphology* 280. PDF: https://biomechanics.ucr.edu/Zhuang_et_al-2019-Journal_of_Morphology.pdf — figure caption, p.4 of PDF text

For the model: Model 5 toes per foot, 20 total. Digit I (innermost) is shortest on both hand and foot and sits noticeably offset from the others.

FEET

Leopard geckos have NO adhesive toe pads: no expanded scansors, no adhesive pads, and no documented setae.

[verified] Peattie (2008) abstract, read verbatim via Europe PMC: 'Subdigital adhesive setae have previously been documented in all legged gecko families except the Eublepharidae.' The single exception she reports is Aeluroscalabotes felinus, 'the only arboreal eublepharids', with setae up to 9 micrometres — a different genus, not Eublepharis. Independently, Smith (1935, p.~119) states that Teratoscincus, Stenodactylus, Alsophylax, Gymnodactylus, Cnemaspis, Agamura, Pristurus 'and Eublepharis have the primitive, non-dilated form of digit. All the other genera mentioned in this work have adhesive expansions by means of which they are able to ascend vertical walls.' Smith's family key entry for the genus reads simply 'Digits not dilated, clawed ... Eublepharis, p. 125.'

Source: Peattie, A.M. (2008) Anatomical Record 291:869-875, DOI 10.1002/ar.20706 (abstract via Europe PMC REST API); Smith, M.A. (1935) The Fauna of British India, Reptilia and Amphibia Vol. II — Sauria, pp. 119, 125-128, <https://archive.org/details/in.ernet.dli.2015.207328>

For the model: CRITICAL: do not model the fanned, flattened, pad-like toe undersides seen in Gekko or Hemidactylus references. The underside of each toe is a narrow, rounded strip, not a broadened paddle. No setal fringe, no leaf-like terminal pads.

FEET

They DO have subdigital lamellae — the widespread claim that leopard geckos 'have no lamellae' is imprecise. What they lack is ADHESIVE pads/scansors/setae, not lamellae.

[verified] Boulenger (1890) genus diagnosis: 'Digits short and cylindrical, with a row of lamellae below, clawed...'. Smith (1935) genus diagnosis: 'Digits short, cylindrical, with transverse lamellae beneath, clawed...'. So a single ventral row of transverse plate-like scales runs the length of each toe; they simply are not adhesive scansors. This distinction is the single most commonly garbled fact about this animal.

Source: Boulenger, G.A. (1890) The Fauna of British India, Reptilia and Batrachia, Genus Eublepharis, p. 107, <https://archive.org/details/TheFaunaOfBritishIndiaReptiliaAndBatrachia> ; Smith, M.A. (1935) op. cit., p. 125

For the model: Model a single row of transverse ventral scale-plates down each toe — visible as stacked bands on the sole of the toe. Roughly 15-23 per long toe based on congener counts. This is surface detail, not silhouette.

HEAD_SCALES

The nostril is a lateral opening on the side of the snout very close to, but NOT touching, the rostral scale.

[verified] Boulenger (1885), in the family account for Eublepharidae, states the nostril is rather large, directed slightly upwards though lateral, and separated from the rostral and the labial plates. Smith (1935) states for E. hardwickii — and the E. macularius account on the following pages is written purely as a list of differences from hardwickii, none of them cranial — 'rostral broader than high, not in contact with the nostril'. Corroborated a third time by the modern Eublepharis descriptions, which place the naris inside a discrete nasal scale rather than on a suture. Three independent sources, two of them museum/fauna-level taxonomic works, agree. NOTE: the strongest statement (Boulenger) is a family-level diagnosis, and Smith's is inherited by E. macularius through the difference-only convention rather than being written out for it.

Source: Boulenger, G.A. 1885. Catalogue of the Lizards in the British Museum (Natural History), 2nd ed., Vol. I, family Eublepharidae, pp. 229–230 (read via archive.org OCR, item catalogueoflizar01brituoft); Smith, M.A. 1935. Fauna of British India, Reptilia and Amphibia Vol. II — Sauria, pp. 126–129 (archive.org item in.ernet.dli.2015.62460)

For the model: Place the nostril aperture on the LATERAL face of the snout, roughly one scale-width behind the rostral plate, with the opening canted slightly dorsally rather than facing straight sideways. Critically: do NOT let the nostril hole touch or notch the rostral, and do not let it sit on the lip line. It is a hole within a single swollen scale, set above the first supralabial.

TUBERCLES

The dorsal integument is built from exactly two visible element types: a pavement of small flattened scales, and larger raised tubercles set into that pavement.

[verified] Russell et al. (2014) SEM-surveyed skin from head, trunk, hindlimb and tail of adult leopard geckos (SVL 89-120 mm, n=5 for body regions, 7 original + 2 regenerated tails) and describe the dorsum as clad in "two general types of integumentary derivative": scales (granular, semi-imbricating or imbricating) and tubercles. Corroborated independently by Smith (1935, p.126-127), who describes the genus as "back with small juxtaposed scales and larger tubercles", and by Boulenger (1885, p.230-232), whose key splits Eublepharis into "back with enlarged tubercles" (incl. macularius) vs "back uniformly granular".

Source: Russell AP, Lai EK, Powell GL, Higham TE (2014) Journal of Morphology 275:961-979, doi:10.1002/jmor.20269 (PDF: https://biomechanics.ucr.edu/Russell_et_al_2014.pdf), Results "Scale Descriptions"; Smith MA (1935) Fauna of British India, Reptilia & Amphibia Vol. II Sauria, pp.126-128; Boulenger GA (1885) Catalogue of the Lizards in the British Museum (Nat. Hist.) Vol. I, pp.230-232

For the model: Two-layer surface: a continuous fine base tiling plus a sparse scattering of discrete raised bumps. Do not model the bumps as a modulation of the base tiling - they are separate, larger elements that displace it.

TUBERCLES

The specific number of dorsal tubercle rows at midbody for E. macularius sensu stricto is not in any source I could open; the nearest figure is 18 rows, for the Rajasthan/Delhi population.

[likely] Mirza (2024) describes the Aravalli/Delhi population (as E. madarensis, resurrected; The Reptile Database still treats madarensis as a synonym of E. macularius and disputes the revalidation) with "large unkeeled tubercles in 18 longitudinal rows" and, in the specimen description, "the large scales are arranged in 18 longitudinal rows at midbody". Comparative counts in the same papers: E. satpuraensis 20 rows; E. hardwickii and E. pictus 23-26 rows of flat tubercle-like scales. The likely source of a macularius count, Mirza et al. (2014) Phyllomedusa 13(2):75-90, was unreachable (revistas.usp.br refused connection from this network), so I did not open it and do not cite it.

Source: Mirza ZA (2024) Taprobanica 13(1):9-15, Diagnosis and Description (PDF: https://www.taprobanica.org/~file/318_eublepharis_madarensis_taprobanica_13_1_2024_mirza_9_15-99e7c-3349_1243.pdf); Reptile Database account for E. macularius, <https://reptile-database.reptarium.cz/species?genus=Eublepharis&species=macularius> (Comment section, on the status of madarensis)

For the model: Budget roughly 18-20 tubercle rows spanning the dorsum from flank to flank at midbody. Treat this as the right order of magnitude, not a verified count for macularius.

Surface anatomy — the 26 that did not

This is the more valuable half. Each of these was a plausible, confidently-worded statement about this animal that turned out to be a congener's trait, an absolute where the source hedged, or a number nobody has actually measured.

Tubercles never touch each other: each is isolated from its neighbours by intervening small scales.

-- REFUTED AS WORDED, BY ITS OWN CITED SOURCE. I opened both sources (source_opened = true for both).

1. THE HEDGE THE CLAIM DROPS. Russell, Lai, Powell & Higham (2014) J. Morphol. 275:961-979, Results ("Scale Descriptions", p. 968) reads exactly: "Tubercles juxtapose neighboring scales and are usually isolated from other tubercles by intervening scales." The qualifier "usually" is the authors' own. It entails exceptions. The claim converts this into "never" and "each", which is a strictly stronger assertion than the source makes. I searched the full text for "never", "always", "exclusive", "without exception" — the authors never make an absolute statement about tubercle isolation. The submitted DETAIL actually quotes "usually" correctly, then the CLAIM headline silently strengthens it; the failure is in the claim, not the transcription. PDF opened at https://biomechanics.ucr.edu/Russell_et_al_2014.pdf and text-extracted locally to C:\Users\ziyad\AppData\Local\Temp\claude\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\russell2014_flow.txt
 2. SCOPE ERROR. That sentence sits in the paragraph describing the DORSAL BODY (referencing Fig. 2c), not a whole-animal rule. Russell et al. state separately that "The ventral regions of the body are devoid of tubercles."
 3. REGIONS WHERE "EACH IS ISOLATED" HAS NO PURCHASE. Russell et al.: ventrally on the tail "the tubercles also lose their identity along this transitional gradient"; in segment 5 "distinction between tubercles, imbricating scales and ventral plates is lost"; and "The regenerated tail lacks tubercles and bears scales that are more homogeneous than those of the original." A regenerated tail is common in both wild and captive animals, so a universal quantifier over "each tubercle" fails outright there.
 4. A WORDING SLIP THAT INVERTS THE ANATOMY. Russell et al. use "juxtapose" to mean abut/lie edge-to-edge (contrast their imbricating scales, which overlap). So a tubercle IS in direct contact with the small scales around it ("Tubercles juxtapose neighboring scales"); what it is usually separated from is OTHER TUBERCLES. The claim's "isolated from its neighbours by intervening small scales" garbles this — the intervening scales are the neighbours, not a gap between the tubercle and its neighbours.
 5. WHAT DOES HOLD, AND IT CORROBORATES TWICE. The weaker, hedged observation is genuinely supported by two independently opened sources. McLean KE & Vickaryous MK (2011) BMC Dev Biol 11:50 (opened at the PMC URL given): "Along the dorsal and lateral surfaces this pavement of flattened scales is interrupted at regular intervals by large cone-like tubercles" and "Close inspection reveals that adjacent, slightly enlarged scales surround each tubercle." The DETAIL's characterisation of McLean & Vickaryous is accurate. Russell et al. add that on the tail tubercles are "spaced dorsoventrally in columns, at relatively regular intervals" and (Fig. 4 caption) "arranged in columns ... within whorls, and rows ... along the tail" — regular spacing, consistent with usual separation.
 6. SPECIES CONFUSION — CHECKED, NONE FOUND. Both papers genuinely study *E. macularius*; I found no sign the observation was imported from another gecko. The *E. pictus* material (Mirza & Gnanaswar 2022, rosette of 10-12 small scales) is a congener transfer and the submitter already flags it as such, which is correct practice. I did NOT open Mirza & Gnanaswar 2022, so the 10-12 rosette count is UNVERIFIED here and must not be cited as *E. macularius*.
 7. CAPTIVE-BRED VS WILD-TYPE — A REAL UNRESOLVED CAVEAT. McLean & Vickaryous state explicitly: "All *E. macularius* were captive bred and obtained from one of two commercial suppliers (Pat Wise Lizards, Calgary, Alberta, Canada; Global Exotic Pets, Kitchener, Ontario, Canada)." They give no wild-type or morph designation. Russell et al. give NO provenance at all — only "Seven original tails and two fully regenerated ... tails were removed from specimens (snout-vent length 89-120 mm) of leopard geckos" under Calgary protocol BIO7R-32; no wild-caught/captive-bred statement, no locality, no morph. So neither opened source documents wild-caught wild-type animals, and the tail half of the claim rests wholly on captive-bred commercial pet-trade stock. COULD_NOT_VERIFY: whether wild *E. macularius* show the same tubercle spacing, and whether captive selective breeding alters tubercle arrangement — I have no source open on either, and I am not filling that gap. Treat any "wild-type" framing of this trait as UNVERIFIED. - correction: Replace the absolute with the sourced hedge, and fix the neighbour/tubercle inversion:
- "On the dorsal body, tubercles abut the surrounding small scales directly and are USUALLY — not invariably — separated from OTHER TUBERCLES by those intervening scales (Russell et al. 2014, J. Morphol. 275:961-979, Results: 'Tubercles juxtapose neighboring scales and are usually isolated from other tubercles by intervening scales'). On the original tail the same pattern holds: a pavement of small flattened scales is interrupted at regular intervals by large cone-like tubercles, with adjacent slightly enlarged scales surrounding each tubercle (McLean & Vickaryous 2011, BMC Dev Biol 11:50), the tubercles being spaced in columns within whorls and in rows along the tail (Russell et al. 2014). This description does not extend to the whole animal: the ventral body is devoid of tubercles, tubercles lose their identity ventrolaterally on the tail and are indistinguishable from other scales near the tail tip, and regenerated tails lack tubercles entirely (Russell et al. 2014)."
- Confidence: the hedged version is MODERATE-to-GOOD (corroborated by two opened, peer-reviewed sources). The original absolute ("never", "each") is UNVERIFIED and contradicted by the qualifier in its own primary source. Provenance caveat to carry forward: both sources used captive animals (McLean & Vickaryous explicitly captive-bred commercial stock; Russell et al. unstated), so wild-type applicability is UNVERIFIED.

On the trunk the tubercles are in longitudinal rows, but loosely so - regular enough that taxonomists count them as a character, irregular enough that no author describes them as a lattice.

-- REFUTED as a statement about *E. macularius*, because every row-statement in the cited sources belongs to a congener, not to *macularius*.

WHAT I OPENED AND CONFIRMED VERBATIM 1. Mirza & Gnanaswar 2022, *Evolutionary Systematics* 6(1):77-88 (opened: Pensoft full text/XML, local copies C:\Users\ziyad\AppData\Local\Temp\claude\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\pictus_plain.txt and mirza2022.xml). Methods sentence is exactly as quoted: "Dorsal longitudinal tubercle rows (DTR) were counted from across the dorsum of the trunk". The pictus holotype description reads "the large tubercle-like scales are fairly arranged in 26 longitudinal rows at mid-body, each large scale is enveloped in a rosette of 10-12 small scales". CONFIRMED — but this is *E. pictus*. 2. Mirza 2024, *Taprobanica* 13(1):9-15 (opened: journal PDF, local copy ...\\scratchpad\mirza2024.txt / madarensis.pdf). Same DTR Methods sentence verbatim. "These tubercles are nearly equidistant from each other and cover the dorsum of the animal"; "the large scales are arranged in 18 longitudinal rows at midbody". CONFIRMED — but the paper's whole purpose is to REMOVE this Aravalli population from *E. macularius* and name it *E. madarensis*. Calling it "a *macularius*-complex population" conceals that the author is describing a different species; its comparison section explicitly contrasts it with *E. macularius*. 3. Russell, Lai, Powell & Higham 2014, *J. Morphol.* 275:961-979 (opened: local PDF text ...\\scratchpad\russell2014.txt). Genuinely *E. macularius*, SEM of trunk skin. For the body it says tubercles "juxtapose neighboring scales and are usually isolated from other tubercles by intervening scales" — no row structure, in any direction. Row/column language appears only for the tail: tubercles "spaced dorsoventrally in columns, at relatively regular intervals" and "arranged in longitudinal rows along the length of the tail" (Fig. 4). So the claim reports Russell accurately, but this is an argument from silence and cannot show that rows exist-but-loosely.

RELATED-SPECIES TRANSFER (the core defect) Every published longitudinal-row count I could open is for a congener, never for *E. macularius*: *E. hardwickii* 24 rows and *E. pictus* 23-26 (Mirza & Gnanaswar 2022); *E. satpuraensis* "~20 rows" (Mirza, Sanap, Raju, Gawai & Ghadekar 2014, *Phyllomedusa* 13(2):75-90, opened, ...\\scratchpad\satpura.txt); *E. madarensis* 18 (Mirza 2024). These are congeners in the same genus but different species, and the transfer is especially unsafe here because row-describability tracks exactly the character that separates them: Mirza & Gnanaswar 2022 diagnose *hardwickii* + *pictus* as having flat tubercle-like scales with "interspaces much less than the size of the scales", versus "interspaces much more than the size of the tubercles in *E. angramainyu* ..., *E. fuscus* ..., *E. macularius* (Blyth), *E. satpuraensis* ... and *E. turkmenicus*". Densely packed rosette-set tubercles are countable in rows; the widely spaced *macularius* condition is what the row-count was not applied to.

NO ROW DESCRIPTION FOR *E. MACULARIUS* IN ANY SOURCE I OPENED - Boulenger 1885, *Cat. Lizards Brit. Mus.* 1:232 (opened, ...\\scratchpad\boulenger1885.txt): *macularius* diagnosed only by tubercle size/spacing ("the granular interspaces being as wide as the tubercles themselves; these are subconical or slightly keeled"). No rows on the trunk. - Boulenger 1890, *Fauna Brit. India* p.107-108 (opened, ...\\scratchpad\fb1890.txt): body "covered above with small, irregular, flat scales, intermixed with numerous roundish, subconical tubercles"; rows are mentioned only for the tail ("above with small flat scales and rows of enlarged subconical tubercles"). - Smith 1935, *Fauna Brit. India, Reptilia & Amphibia* 11:126-128 (opened, ...\\scratchpad\smith1935.txt): back with "small, irregular, juxtaposed scales intermixed with larger rounded or oval subconical or keeled tubercles"; again rows only on the tail. *macularius* is separated from *hardwickii* solely by tubercle-to-interspace ratio and pore count. - Reptile Database compilation of Anderson 1999 / Grismer 1988,1991 diagnoses for *E. macularius* (opened, ...\\scratchpad\rd_macularius.txt): tubercle size and spacing, no row character. - Rawat, Thapa, Bhattarai & Shah 2019, *IRCF Reptiles & Amphibians* 26(1):58-61 (opened, ...\\scratchpad\rawat2019.txt): the one place I found DTR applied to animals identified as *E. macularius*. Table 1's legend defines "number of rows of granular dorsal tubercles at midbody (DTR)" after Mirza et al. 2014 — and then the table reports no DTR value at all (it jumps from PP to FLL). Weak source (semi-popular journal, photo vouchers), used only as corroboration.

THE SECOND HALF OF THE CLAIM ALSO FAILS "No author describes them as a lattice" is an unverifiable universal negative, and the evidence inside the genus runs the other way: "nearly equidistant from each other and cover the dorsum" (Mirza 2024, *madarensis*) is a statement of even spacing, and "each large scale is enveloped in a rosette of 10-12 small scales" (Mirza 2022, *pictus*) — matched by "10-12 smaller scales surround each dorsal tubercle" in an *E. angramainyu* description (opened, ...\\scratchpad\arc_angramainyu.txt) — is a regular repeating unit, i.e. as lattice-like as this literature gets. The absence of the word "lattice" is a fact about vocabulary, not about the skin. The gloss of "fairly arranged" as "approximate, not strict" is also the claimant's, not the author's: Mirza uses the same stock phrase for geckos with genuinely regular rows, e.g. "enlarged, keeled, tubercles, fairly regularly arranged in 15-16 longitudinal rows on dorsum ... individual tubercle row separated from the adjacent by three transverse scale rows" in a *Hemidactylus* description (opened, ...\\scratchpad\arc.txt), explicitly contrasted there with congeners whose tubercles are "irregularly arranged". In his usage "fairly (regularly) arranged" means regular enough to score, not loose.

COULD NOT VERIFY - Any published DTR/longitudinal-row count for *E. macularius sensu stricto*. Confidence: unverified. - Whether the trunk tubercles of wild-type *E. macularius* do or do not fall into countable longitudinal series. Unverified in either direction; I found no source that examines it. - Provenance of Russell et al.'s geckos: the Methods give only a University of Calgary animal-care protocol and cite a lab diet study (Lynn et al. 2013); the paper states nothing about wild-caught vs captive-bred origin, locality, morph, or vouchers. So no morph confusion is demonstrable, but the one source that actually looked at *macularius* trunk skin rests on undocumented laboratory animals. Confidence: unverified. - Anderson 1999 (*Lizards of Iran*), Khan's Pakistan works, Börner 1976/1981: not opened; not cited above.

Corroboration status: the DTR Methods sentence is doubly corroborated (both Mirza papers, opened). The absence of trunk-row description for *E. macularius* is corroborated four times (Boulenger 1885, Boulenger 1890, Smith 1935, Russell et al. 2014). The row counts for congeners each rest on a single source per species. - correction: What the opened sources actually support, restricted to *E. macularius*: the trunk bears small, irregular, juxtaposed scales intermixed with enlarged rounded-to-subconical, smooth or feebly keeled tubercles whose interspaces are as wide as or wider than the tubercles themselves (Boulenger 1885:232; Boulenger 1890:108; Smith 1935:127-128), and under SEM those trunk tubercles "juxtapose neighboring scales and are usually isolated from other tubercles by intervening scales" (Russell et al. 2014:968). None of these authors describes the trunk tubercles as forming rows of any orientation; all row and column language in this literature — Boulenger 1890, Smith 1935 and Russell et al. 2014 alike — is reserved for the tail, where tubercles sit in dorsoventral columns within each caudal whorl "at relatively regular intervals" and in longitudinal rows along the tail's length.

Counting longitudinal tubercle rows (DTR, "counted from across the dorsum of the trunk") is indeed a standard meristic in modern *Eublepharis* taxonomy (Mirza & Gnanaswar 2022; Mirza 2024), but every published count I could find is for another species: *E. hardwickii* 24, *E. pictus* 23-26, *E. satpuraensis* ~20 (Mirza et al. 2014), *E. madarensis* 18 (Mirza 2024 — a taxon that paper removes from *E. macularius*). The only publication I found that defines DTR for animals identified as *E. macularius* (Rawat et al. 2019, Nepal) leaves the value unscored. Notably, Mirza & Gnanaswar's own diagnosis places *E. macularius* in the widely-spaced-tubercle group (interspaces much greater than tubercle size), opposite the closely packed, rosette-set tubercles of the species for which row counts are given.

So: whether *E. macularius* trunk tubercles form countable longitudinal series — loosely or otherwise — is UNVERIFIED. Do not assert rows on the trunk of this species; do not carry the "nearly equidistant" wording over from *E. madarensis* or "fairly arranged in 26 longitudinal rows" from *E. pictus*, and drop the unverifiable "no author describes them as a lattice", which in any case sits awkwardly with the even-spacing and rosette wording that does appear for congeners.

On the original tail the arrangement is strictly geometric: tubercles sit in dorsoventral columns, one column per caudal whorl, and those columns line up into longitudinal rows along the tail.

-- SOURCES ACTUALLY OPENED (both full text, not abstracts): 1. Russell AP, Lai EK, Powell GL, Higham TE (2014) "Density and Distribution of Cutaneous Sensilla on Tails of Leopard Geckos (*Eublepharis macularius*) in Relation to Caudal Autotomy", *J. Morphol.* 275:961-979. Opened as PDF (19 pp, pagination 961-979 confirmed) from https://biomechanics.ucr.edu/Russell_etal_2014.pdf; saved at C:\Users\ziyad\AppData\Local\Temp\claud\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\russell2014.pdf. I also rendered p. 969 and inspected Fig. 4 at high magnification (C:\Users\ziyad\AppData\Local\Temp\claud\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\fig4.png). 2. Mirza ZA (2024) "A Further Taxonomic Reassessment of *Cyrtodactylus madarensis* Sharma, 1980 (Squamata: Gekkonidae) Now in the Genus *Eublepharis* Gray, 1827", *Taprobanica* 13(1):9-15, DOI 10.47605/tapro.v13i1.318. Opened as PDF (9 pp) from taprobanica.org; saved at C:\Users\ziyad\AppData\Local\Temp\claud\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\mirza2024.pdf.

WHAT CHECKS OUT (confidence: verified, single-sourced) The DETAIL's quotations from Russell et al. are verbatim accurate, and the method description is right. Methods: fine entomological pins were inserted transversely through the mid-point of each whorl at the dorsal extremity of the tail, the specimens radiographed on a Kubtec Xpert 80, and intervertebral joints and intravertebral autotomy planes mapped against the marker pins. Results: the original tail bears scales in whorls around its circumference, the breadth of each whorl corresponds to the length of an underlying vertebra, "indicating a 1:1 correspondence of whorls and autotomy planes" (repeated in the abstract). Tubercles are "spaced dorsoventrally in columns, at relatively regular intervals", are "located close to the center of each whorl", and "are arranged in longitudinal rows along the length of the tail". The Fig. 4 caption does state both axes: tubercles "arranged in columns (vertical dotted line) within whorls, and rows (horizontal dashed line) along the tail". So the two-axis geometry itself is real and correctly cited.

WHY THE CLAIM NEVERTHELESS FAILS AS WRITTEN

(a) "Strictly geometric" is the claimant's word, not the source's, and the source says the opposite twice. Russell et al. hedge both axes: "relatively regular intervals" and "close to the center of each whorl". They then describe the scheme actively dissolving. Ventrally, tubercles "diminish in prominence"; as dorsal semi-imbricating scales grade into polyhedral ventral plates "the tubercles also lose their identity along this transitional gradient"; on the body the ventral regions are outright "devoid of tubercles". So a "dorsoventral column" is not a complete ring - it fades out before reaching the venter. Longitudinally, in the posterior fifth of the tail (their segment 5) "distinction between tubercles, imbricating scales and ventral plates is lost". So the arrangement does not hold over the whole original tail. Inspecting Fig. 4 directly confirms this: tubercles are visibly offset from the drawn dotted and dashed guide lines rather than sitting on them.

(b) "One column per caudal whorl" is not stated anywhere in Russell et al. The paper gives no count of columns per whorl, and the Fig. 4 caption's wording is plural-in-plural ("columns ... within whorls"), which is ambiguous. My own reading of Fig. 4 is consistent with one centred column per whorl (inter-tubercle spacing along a row roughly equals the whorl width marked by the "whorl" brace; the dotted line sits mid-whorl between demarcation furrows), but that is my inference from the image, not a documented statement. Confidence on this sub-clause: unverified.

(c) SPECIES CONFUSION in the second citation - this is the substantive error. Mirza (2024) is not a paper about *E. macularius*. Its title and abstract state that the Rajasthan population is a distinct taxon from *E. macularius* (following the molecular phylogeny of Agarwal et al. 2022) and assign it the name *Eublepharis madarensis* (Sharma, 1980). The Description section - the very section containing the tail sentence - opens "Based on the female holotype (SMF 60980) of *E. macularius smithi*", a Delhi specimen (Börner 1981) that Mirza explicitly synonymises into *E. madarensis*; the synonymy list runs *Cyrtodactylus madarensis* Sharma 1980 / *Eublepharis macularius smithi* Börner 1981 / *Eublepharis macularius* - Das 1992 / *Eublepharis* sp. Rajasthan - Agarwal et al. 2022. Mirza's Comparison section then lists characters separating *E. madarensis* FROM *E. macularius* (reticulate-then-striped vs. spotted pattern; vertebral stripe present vs. absent; feebly vs. strongly tuberculate lamellae). The tail count is quoted correctly - the original-tail dorsum bears small juxtaposed scales intermixed with large sub-conical tubercles in a whorl of about 9-10 rows on the first segment, reducing on subsequent segments - but it is a datum for *E. madarensis*, NOT *E. macularius*. TRANSFER FROM A RELATED SPECIES, STATED EXPLICITLY: *E. madarensis* is a congener in the same genus *Eublepharis*, and Mirza calls it "most similar to *E. macularius*", so it is a close relative - but under his arrangement it is a separate species, and the whole point of the paper is to remove that Rajasthan/Delhi population from *E. macularius*. Offering it as corroboration for *E. macularius* inverts the source's conclusion.

(d) TERMINOLOGY MISMATCH that makes the two counts non-comparable even setting species aside. Russell et al.'s "segment" is one-fifth of the tail's length, and they state "Several whorls were present in each segment sampled." Mirza's "caudal segment" is a single whorl-bearing annulus. The "9-10 rows on the first segment" and Russell's "segment 1" are not the same unit.

COULD NOT VERIFY - Wild-type vs. captive-bred morph status of the Russell et al. animals. The paper gives only SVL 89-120 mm, n = 7 original tails plus 2 regenerated, and University of Calgary animal-care protocol BI07R-32. There is no mention of captive, wild, colony, breeder, supplier, strain or colour morph anywhere in the text. So the lineage of the animals that produced this geometry is undocumented; I mark it unverified rather than guess. (No evidence the tubercle geometry is a morph artefact either - I simply cannot check it.) - Whether the column/row geometry has been replicated. I found and opened no second source describing it. Per the "one source is not a citation" rule: this geometry rests on Russell et al. (2014) alone, and Mirza (2024) does not corroborate it - it describes whorl composition in a different species, not alignment of columns across successive whorls. - correction: Accurate restatement: On the original tail of *Eublepharis macularius*, Russell et al. (2014) found scales arranged in whorls around the circumference, with marker-pin radiography showing each whorl's breadth matching one underlying vertebra, i.e. a 1:1 correspondence of whorls with autotomy planes. Within that framework, tubercles are spaced dorsoventrally in columns at RELATIVELY REGULAR (not strict) intervals, sit close to the centre of each whorl, and align into longitudinal rows along the tail. The regularity is approximate and regionally limited: tubercles are prominent dorsally, diminish in prominence ventrally and lose their identity where dorsal imbricating scales grade into ventral plates, so the columns are not complete dorsoventral rings; and over the posterior fifth of the tail the distinction between tubercles, imbricating scales and ventral plates disappears altogether, so the scheme does not apply there. The regenerated tail lacks tubercles entirely.

Drop "strictly geometric" - it is the claimant's word and contradicts the source's own hedges. Drop or downgrade "one column per caudal whorl": Russell et al. never state a count of columns per whorl; the figure is consistent with one centred column per whorl but that is an inference from Fig. 4, not a documented statement.

Drop the Mirza (2024) leg entirely, or relabel it. Mirza ZA (2024) *Taprobanica* 13(1):9-15 redescribes *Eublepharis madarensis* (Sharma, 1980) and explicitly removes the Rajasthan/Delhi population from *E. macularius*; its Description is based on the Delhi holotype of *E. macularius smithi* Börner, 1981, reassigned to *E. madarensis*. Its "about 9-10 tubercle rows in a whorl on the first caudal segment, reducing on subsequent segments" is a character of *E. madarensis* - a congener in the same genus that Mirza calls most similar to *E. macularius*, but a distinct species under his arrangement. If cited at all it must be labelled as a related-species datum, and note that Mirza's "caudal segment" (one annulus) is not Russell's "segment" (one-fifth of tail length, containing several whorls).

Remaining gaps to declare rather than fill: the column/row geometry is single-sourced (Russell et al. 2014 only), and that paper does not state whether its geckos were wild-type or captive-bred morphs.

Tubercles are largest on the hindlimb, mid-sized on the trunk and tail base, and smallest on the head.

-- SOURCE OPENED = TRUE. I opened the publisher-typeset PDF of Russell AP, Lai EK, Powell GL, Higham TE (2014) "Density and Distribution of Cutaneous Sensilla on Tails of Leopard Geckos (*Eublepharis macularius*) in Relation to Caudal Autotomy", *J. Morphol.* 275:961-979, hosted on coauthor Higham's lab site (https://biomechanics.ucr.edu/Russell_et_al_2014.pdf). I did NOT rely on the text-flow extraction, which scrambles the table: I reconstructed Table 2 (printed p. 968) from PDF word x/y coordinates with PyMuPDF, so every row/value pairing below is read off the printed page, not inferred.

NO SPECIES CONFUSION AND NO MORPH SUBSTITUTION FOUND. The paper is explicitly on *E. macularius* (Blyth) 1854, and its caudal whorl/tubercle description is consistent with that species. That is not where the claim fails.

WHAT THE TABLE ACTUALLY SAYS (tubercle mean area, mm² +/- sd, area = longest dimension x perpendicular dimension, 10 scales per focal area, ImageJ, from SEM images):
dorsum thigh 3.092 +/- 1.72; dorsum crus 2.473 +/- 1.72; dorsum pes 1.534 +/- 0.36; between hips 1.434 +/- 0.23; original tail segment 1 0.914 +/- 0.08, segment 2 0.823 +/- 0.07, segment 4 0.716 +/- 0.46, segment 3 0.551 +/- 0.12; between eyes 0.644 +/- 0.06; between shoulders 0.141 +/- 0.15; upper labials = no tubercle entry. So the six numbers in the DETAIL and their row assignments are correct - I confirm them independently. The claim still fails.

CLAUSE 1 ("largest on the hindlimb") - weakly supported at best. The three hindlimb point estimates are the three largest values, so the direction is right. But the paper runs no statistical comparison of scale areas among regions; n = 10 scales per focal area; and the sds are enormous and overlapping (thigh 3.092 +/- 1.72 vs crus 2.473 +/- 1.72 are indistinguishable; pes 1.534 +/- 0.36 overlaps between-hips 1.434 +/- 0.23). Any ordering within the hindlimb, and hindlimb-pes vs trunk-hips, is unresolvable from this table.

Confidence: unverified as a ranking.

CLAUSE 2 ("mid-sized on the trunk") - false as printed. The trunk was sampled twice and spans the whole range, not the middle: between hips 1.434 (effectively tied with the pes for second-largest) and between shoulders 0.141 (the smallest tubercle value in the table). Neither trunk sample is mid-sized.

CLAUSE 3 ("smallest on the head") - false as printed, and it fails even after the DETAIL's own repair. Between the eyes is 0.644. That is larger than between shoulders (0.141) AND larger than original-tail tubercles in segment 3 (0.551). So discarding the suspect between-shoulders cell does not rescue the clause: mid-tail tubercles are still smaller than head tubercles. The paper's prose does assert the smallest occur on the dorsum of the head between the eyes (p. 967-968), but that sentence contradicts its own Table 2 in two places, not one.

ON THE DETAIL'S CAVEAT - one factual correction and one unverified inference. (a) The between-shoulders tubercle value 0.141 +/- 0.15 is NOT identical to the adjacent granular scale value; the dorsal-scale cell on that same row reads 0.142 +/- 0.04. Near-identical, different sd. (b) "Probably 1.41" is a reconstruction, not a documented correction: I searched for an erratum/correction to 10.1002/jmor.20269 and found none indexed, so the 0.141 stands uncorrected in the literature and cannot be silently replaced. The block does contain at least one further evident typo (the dorsum-thigh dorsal-scale sd is printed as "218"), which shows the table was poorly proofed but does not license substituting a guessed value.

PROVENANCE, NOT DOCUMENTED (could_not_verify). Methods state seven original and two regenerated tails were taken from specimens of SVL 89-120 mm, and that body skin was excised from five of those same specimens, cross-referenced to Lynn et al. 2013 (a juvenile diet-ration study). The text I read never states whether the animals were captive-bred, commercially sourced, or wild-caught, nor sex, nor morph. I could not open Lynn et al. 2013 (journals.uchicago.edu returned HTTP 403). So: absolute mm² values come from sub-adult/juvenile animals of unstated origin and cannot be transferred to adult wild-type animals. Confidence: unverified.

CORROBORATION - none found; this is a one-source claim. I opened two independent descriptive sources: Boulenger 1885 (*Cat. Lizards Brit. Mus.* vol. 1, *E. macularius* account, p. 232) and Smith 1935 (*Fauna of British India, Reptilia and Amphibia II*, *E. macularius* account, p. 127). Both treat the enlarged tubercles as a feature of the BACK, and both make the diagnostic comparison tubercle size vs granular interspace; neither mentions limb tubercles at all, and neither ranks head tubercles as smallest. RELATED-SPECIES NOTE: Mirza and Gnanewar 2022 (*Evol. Syst.* 6:77-88) and Mirza 2024 describe limb and trunk scalation in CONGENERS (*E. hardwickii*, *E. pictus*, and a newly described *Eublepharis* - same genus, different species, so distance is congeneric, not conspecific); their limb descriptions mention lamellae but no enlarged limb tubercles. I cite that only as absence of corroboration, not as evidence about *E. macularius*.

METHOD CAVEAT: these "areas" are bounding-rectangle products (longest dimension x perpendicular dimension) of domed/sub-conical tubercles, i.e. planar footprint approximations that overestimate area and are not comparable to true surface areas from other studies. The paper itself notes tubercles have a small perimeter relative to enclosed area. - correction: Supportable version: In the only measurements I could open (Russell et al. 2014, *J. Morphol.* 275:961-979, Table 2, printed p. 968; captive lab specimens of SVL 89-120 mm, origin not stated), the largest tubercles measured are on the dorsum of the hindlimb - thigh 3.092 +/- 1.72 mm² and crus 2.473 +/- 1.72 - with dorsum pes 1.534 +/- 0.36 and dorsum between the hips 1.434 +/- 0.23 next. Beyond that, no clean head-to-tail size gradient exists in the data: original-tail tubercles run 0.914 (segment 1) down to 0.551 (segment 3), the dorsum of the head between the eyes is 0.644 +/- 0.06, and the dorsum between the shoulders is printed as 0.141 +/- 0.15, the smallest value in the table. So "mid-sized on the trunk" is wrong (the trunk supplies both the second-largest and the smallest values) and "smallest on the head" is wrong (mid-tail tubercles, 0.551, are smaller than the between-eyes value, and the printed between-shoulders value is smaller still). The paper's prose claims the smallest are between the eyes, but that sentence contradicts its own table in two places and no erratum exists; flag the conflict rather than resolving it. Treat all of this as point estimates only - 10 scales per region, no statistical comparison among regions, sds that overlap almost completely, a bounding-rectangle area approximation, and juvenile/sub-adult animals of undocumented (possibly captive-bred) origin. Single source, uncorroborated: Boulenger 1885 and Smith 1935 describe *E. macularius* tubercles only as a dorsal-trunk feature and say nothing supporting a hindlimb-largest, head-smallest ordering. Confidence: unverified.

The lower lid is almost certainly the principal mover, sweeping upward to close the eye; the upper lid moves little.

-- REFUTED in its second half, and the confidence level ("almost certainly") is not supportable.

What I actually opened (full texts downloaded and read myself): 1. Smith 1935, Fauna of British India, Sauria — archive.org item in.ernet.dli.2015.207328, file 2015.207328.The-Fauna_djvu.txt, saved locally at C:\Users\ziyad\AppData\Local\Temp\claudel\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\zz920e_smith1935.txt (https://archive.org/download/in.ernet.dli.2015.207328/2015.207328.The-Fauna_djvu.txt). source_opened = true. - p.10 (general "The Eye" section): confirmed verbatim. Smith does state that in all reptiles the lower lid is the movable one, that closure is by upward movement of the lower lid, and that the upper lid is immovable or very restricted. The user's characterisation of this passage is accurate. NOTE the context: this sentence sits inside a discussion of how skinks/lacertids (*Leiolopisma*, *Mabuya*, *Ablepharus*, *Cabrita*, *Ophisops*) build a transparent window in the lower lid. It is a generalisation drawn from those groups. - p.23-24 (*Gekkonidae* family account) — a page the claimant did NOT consult, and it is decisive: Smith writes that in *Eublepharis* both lids are well developed and movable, and that the eyeball has lost the fixed membranous covering. So Smith himself exempts *Eublepharis* from his own p.10 generalisation, in the same volume. - pp.125-126 (genus diagnosis): both eyelids well developed, movable; and *eublepharids* are distinguished from typical *gekkonids* by opisthocoelous vertebrae, united parietals, and "connivent eyelids" (i.e. lids that converge to meet — that gloss of the term is my reading, not something I verified in a dictionary). 2. Boulenger 1885, Catalogue of the Lizards in the British Museum, 2nd ed., vol. I, Fam. *Eublepharidae* (p. ~228) — archive.org catalogueofflizar01brituoft_djvu.txt, saved at ...\\scratchpad\boulenger1885.txt. source_opened = true. The family diagnosis says the eyes are protected by thick, movable, connivent lids — plural, with no statement that one lid is the mover. This is a second, older source saying the same thing, though not fully independent of Smith (Smith cites Boulenger). 3. Walls 1942, The Vertebrate Eye and Its Adaptive Radiation — archive.org in.ernet.dli.2015.206157, saved at ...\\scratchpad\\walls1942.txt. source_opened = true. This is the strongest independent corroboration of the GENERAL rule, and it also shows the rule is not absolute: p.419 (*Sauropsida*) says the lower lid characteristically does the work of closing the eye; p.423 (*Lizards*) says that in lizards, as a rule, only the lower lid has a tarsal plate and moves — but immediately gives an exception, an anole (Walls's "*Anolis alligator*") in which the two lids move equally; pp.421-422 make *crocodilians* an explicit exception, with the upper lid larger and more mobile. Walls's own footnote on *eublepharids* describes them as having mobile lids (plural) fringed with lash-like scales; he never says which lid does the closing in them. 4. Kosanke/van Doorn-type modern review: "Anatomy, Physiology, and Disorders of the Spectacle, Subspectacular Space, and Its Lacrimal Drainage System in Squamates", *Animals* (Basel) 2023, PMC10044692 (full text fetched from Europe PMC). source_opened = true. It states that among *Gekkota* only *Eublepharidae* has fully moveable eyelids (plural), and — important for the logic — that most *Gymnophthalmidae* have moveable UPPER eyelids and immobile LOWER eyelids. That is a squamate counterexample to Smith's "in all reptiles" claim, so the premise the claim leans on is demonstrably not universal within lizards. 5. Wyneken 2012, "Reptilian Eyes and Orbital Structures", *ARAV 2012 Proceedings*, p.75 ff. I read a text extraction of this paper that was already present in this session's shared scratchpad (...\\scratchpad\\arav2012_43.txt); I could NOT re-download it from *ARAV* myself (*cdn.ymaws.com* returned 403), so treat its provenance as partially verified. Its p.76 content does match the claimant's paraphrase: the lower lid of lizards contains a cartilaginous tarsus, and in general the upper lid is mostly smooth muscle and less mobile than the lower lid, which has striated muscle. Crucially this is framed as "in general" for lizards; *Eublepharis* is never mentioned, and Wyneken's own adjacent example (*crocodilians*) is one where the lower lid moves up despite lacking cartilage — i.e. the muscle/tarsus argument is a tendency, not a law.

Species/morph-confusion check, as instructed: - No evidence the claim came from a different gecko species or from a captive-bred morph. Morph identity is irrelevant to lid anatomy and nothing in the sources touches it. - There IS a taxonomic-scope problem, in the other direction. The "both lids movable" statements are family/genus-level, not *E. macularius*-level: Smith's 1935 *Eublepharis* explicitly subsumes *Goniurosaurus*, and Boulenger's *Eublepharidae* contained *Psilodactylus* (now *Holodactylus*), *Eublepharis* and *Coleonyx*. So those statements cover a broader taxon than *E. macularius*, though *E. macularius* sits squarely inside it. - The actual transferred fact in the claimant's reasoning is the cross-taxon one they half-flagged: the "lower lid sweeps upward" mechanism is documented for lizards generally (Walls p.423; Wyneken p.76) and for amphibians/sauropsids at large (Walls p.419), not for *Eublepharis*.

Could not verify: no source I opened measures or even describes the relative excursion of the upper versus lower lid during a blink in *E. macularius*, or names which lid is the principal mover in any *eublepharid*. The one paper that would most likely settle it, Bellairs, A. d'A. 1948, "The eyelids and spectacle in geckos", *Proc. Zool. Soc. Lond.* 118:420-425, doi 10.1111/j.1096-3642.1948.tb00387.x, is paywalled and I did NOT open it — I only saw it in the 2023 review's reference list. It should be the next thing pulled. Underwood 1970 ("The eye", *Biology of the Reptilia* vol. 2) is the other unopened candidate. Europe PMC returns no study of eyelid kinematics in any gecko. - correction: Split the claim in two and downgrade both halves.

(a) "Lower lid is the principal mover, sweeping upward" — plausible but UNVERIFIED for *E. macularius*. Confidence: unverified. It is the documented rule for lizards generally (Walls 1942 p.423; Wyneken 2012 p.76, corroborated twice), and it has a mechanical rationale (tarsal cartilage plus striated muscle in the lower lid). But Walls states it as "as a rule" and records a lizard exception, and no source I opened applies it to *Eublepharis*. Phrase it as: in lizards generally the lower lid is the larger, tarsus-bearing, more mobile lid and closure is effected mainly by its upward movement; whether this holds for *E. macularius* is not documented in any source I could open.

(b) "The upper lid moves little" — DROP THIS. It is contradicted by the only taxon-specific statements available, including one on a page of the very source cited. Smith 1935 p.23-24 says that in *Eublepharis* both lids are well developed and movable; Smith's genus diagnosis (pp.125-126) says both eyelids well developed, movable, and makes "connivent eyelids" a diagnostic character of the family; Boulenger 1885 (*Eublepharidae*) says thick, movable, connivent lids; Walls 1942 calls *eublepharid* lids mobile (plural); the 2023 *Animals* review says *Eublepharidae* alone among *Gekkota* has fully moveable eyelids. Confidence: the "both lids movable" statement is documented (three sources, though Smith derives partly from Boulenger); the degree of upper-lid excursion is unverified.

Also remove "almost certainly". The premise it rests on — Smith's blanket "in all reptiles the lower lid is the movable one" — is false as stated: *crocodilians* have the larger, more mobile upper lid (Walls 1942 pp.421-422), and most *gymnophthalmid* lizards have a moveable upper lid and an immobile lower lid (*Animals* 2023 / Guerra-Fuentes et al. 2014). A generalisation with known exceptions inside *Squamata* cannot license "almost certainly" for a taxon its own author treated as exceptional.

Suggested replacement wording: "*Eublepharid* geckos, uniquely among *Gekkota*, retain functional eyelids; both the upper and lower lids are described as well developed and movable, and are 'connivent', i.e. they converge to meet (Boulenger 1885; Smith 1935, pp. 23-24 and 125-126). In lizards generally the lower lid is the larger, carries a cartilaginous tarsus and striated muscle, and does most of the work of closing the eye (Walls 1942, p. 423; Wyneken 2012, p. 76), so the lower lid is likely the main mover in *E. macularius* as well, but no source examined states this for the species, and the relative contribution of the two lids during a blink is undocumented."

The lids meet and the eye closes completely — the term used in both primary sources is 'connivent'.

-- The bibliographic half of the claim is correct; the inferential half is not, and the "two primary sources" are not two.

VERIFIED (sources actually opened, full scanned text of the originals): 1. Boulenger 1885, Catalogue of the Lizards in the British Museum vol. 1 (Internet Archive copy catalogueofflizar01brituoft, OCR of the original). Running head "EUBLEPHARIDÆ. 229" sits immediately above the family diagnosis, which reads that the eyes are moderately large, with elliptico-vertical pupil, and are protected by thick, movable, connivent lids. So the word is there, at the stated page, at family rank. Correct. 2. Smith 1935, Fauna of British India, Reptilia and Amphibia vol. II (Sauria) (Internet Archive copies in.ernet.dli.2015.207328 and in.ernet.dli.2015.62460). Running page "126" sits immediately above the sentence in the genus *Eublepharis* account saying that, as shown elsewhere (Smith, 1933), the *Eublepharids* differ from the typical *Geckonids* in having opisthocoelous vertebrae, united parietal bones, and connivent eyelids. The 1933 citation is also in the genus synonymy: Smith, Rec. Ind. Mus. xxxv, 1933, p. 16. The same term appears in Smith's family key, couplet "II. Eyelids movable, connivent" leading to *Eublepharis*. So that is correct too.

WHY IT STILL FAILS:

(a) The "eye closes completely" part is the claimant's gloss, not a statement in either source. Both authors use 'connivent' as a presence/absence diagnostic separating *eublepharids* from typical *geckos* with a fused brille and rudimentary lids. Neither says anything about how far the lids travel, whether the aperture is fully occluded, or what the animal does in life. Boulenger demonstrably had separate wording for degree of coverage and did not use 'connivent' for it: in the same volume, in the *Geckonidae* introduction (pp. 4–5), he writes that in *Ptenopus* the upper lid is sufficiently developed to cover the eye nearly completely, while in *Aelurosaurus* (= *Aeluroscalabotes*, the Malayan cat gecko, a different animal) there are connivent movable lids. Degree of closure gets a measured phrase; 'connivent' does not. Treating 'connivent' as "a statement of full closure, not partial" reads a quantitative finding into a taxonomic adjective.

(b) The two attestations are not independent. Boulenger 1890 (Fauna of British India, Reptilia and Batrachia; Internet Archive copy TheFaunaOfBritishIndiaReptiliaAndBatrachia), family *Eublepharidae*, p. 107, gives the formula: they differ from the *Geckonidae* in having the vertebrae opisthocoelous and the parietal bones united, and are distinguished externally from all Indian *Gekkonidae* by the presence of connivent eyelids, a character in which they agree with the normal *Lacertilia*. Smith 1935 p. 126 reproduces that same three-character formula in the same order (opisthocoelous vertebrae, united parietals, connivent eyelids). Smith also cites Boulenger 1885 p. 230 and Boulenger 1890 p. 107 directly in his genus synonymy. So Smith is carrying Boulenger's wording forward, not corroborating it from independent observation. Under the project rule about corroborating twice, this is one descriptive tradition, not two sources.

(c) Boulenger 1890 also applies "movable, connivent eyelids" to *Agamidae* in the same family key, opposite "no movable eyelids" for *Geckonidae*. That confirms 'connivent' in this literature simply denotes the ordinary lizard lid condition — lids that converge and meet — rather than any observation specific to *Eublepharis*. It supports "the lids meet"; it does not support "the eye closes completely" as a sourced finding.

(d) Rank problem. Neither statement is about *E. macularius*. Boulenger's is a family diagnosis covering *Coleonyx* and *Psilodactylus* (= *Holodactylus*, West Africa) as well; Smith's is a genus diagnosis for a genus that at the time still contained *lichtenfelderi*, now *Goniurosaurus*. I read both species accounts for *macularius* (Boulenger 1885 p. 232; Smith 1935 pp. 127–129) and neither mentions the eyelids at all — they are entirely about tubercles, preanal pores, colour pattern, measurements and range.

(e) Lexically, 'connivent' is not unambiguous. Wiktionary (weak, tertiary — used only as a lexical check, opened) gives the biological sense as both "arched inward so that the points meet" and "brought close together or in close contact; converging". The second reading does not entail complete meeting.

NO SPECIES OR MORPH CONFUSION FOUND, and this cuts in the claimant's favour on that axis: the material behind Boulenger 1885 p. 232 is wild-caught British Museum specimens (Kashmir–Murree, Rajanpur Punjab, Hyderabad and Karachi in Sind), which predate captive morphs entirely. The only adjacent-taxon issue is that the same term is applied by Boulenger to *Aelurosaurus*/*Aeluroscalabotes felinus* and to *Agamidae*, which shows the term is generic rather than that *macularius* was mistaken for something else.

COULD NOT VERIFY: Smith 1933, Rec. Indian Mus. xxxv, p. 16 — I could not locate a scan of Records of the Indian Museum vol. 35, so I have confirmed only that Smith 1935 cites it, not what it says about the lids. Unverified. I also found no measurement of lid excursion, closure speed or blink rate in any source I opened, which matches the claimant's own could_not_verify note. - correction: Sourced version, confidence: the terminology is verified, the interpretation is unverified.

Boulenger 1885 (Cat. Liz. Brit. Mus. 1, p. 229) describes the family *Eublepharidae* as having thick, movable, connivent lids, and Smith 1935 (Fauna Brit. India, Sauria, p. 126) describes the genus *Eublepharis* as having connivent eyelids, citing Smith 1933. Both statements are diagnostic characters at family and genus rank, contrasting *eublepharids* with typical *geckos*, which have rudimentary lids and a transparent spectacle. Boulenger 1890 (Fauna Brit. India, Reptilia and Batrachia, p. 107) makes the point explicitly: connivent eyelids are the character in which *eublepharids* agree with the normal *Lacertilia* — and he applies the same phrase to *Agamidae*.

What is therefore supportable: *E. macularius*, as a *eublepharid*, has two well-developed movable eyelids whose margins converge and meet, unlike typical *geckos*. What is NOT supportable from these sources: that they state the eye closes completely. Neither author describes degree of closure, lid excursion, occlusion of the aperture, or eye-closing behaviour, and both species accounts for *macularius* (Boulenger 1885 p. 232; Smith 1935 pp. 127–129) are silent on the eyes. Boulenger used a separate, explicitly quantitative phrase when he meant degree of coverage (*Ptenopus*, "cover the eye nearly completely", pp. 4–5), and did not use it here.

Also do not present these as two independent sources. Smith 1935 p. 126 reproduces Boulenger 1890 p. 107's three-character formula in the same order and cites Boulenger 1885 and 1890 in the same synonymy. One tradition, one term. Full closure in *E. macularius* remains unverified and needs a functional-morphology or behavioural source, not a taxonomic key.

In bright light the pupil constricts to a narrow vertical slit roughly 8% of the eye's width and ~80-85% of its height — aspect ratio about 7:1.

-- REFUTED on the claimant's own source, plus no published corroboration for any of the numbers.

- 1) THE CITED SOURCE CONTRADICTS THE DETAIL. I opened the Commons file page and its raw wikitext (File:EUBLEPHARIS-MACULARIUS Leopard-Gecko Leopardgecko 201705276003.jpg). The file exists, is 8688x5792, CC BY 4.0, Christian von Faber-Castell, 2017-05-27 18:37:14. But the description reads "EUBLEPHARIS MACULARIUS, Juvenile, ca. 6 Month old ... Sex not identified" (German: "Junger, ca. 6 Monate alter Leopardgecko"). The DETAIL says "an adult." It is a ~6-month-old juvenile. Ontogeny is exactly the wrong thing to get wrong here: eye and pupil dimensions scale allometrically with head growth, so a ratio from a juvenile cannot be presented as the adult/species condition.
 - 2) "IN DAYLIGHT" / "IN BRIGHT LIGHT" IS UNDOCUMENTED. The file page records a terrarium animal ("Snatschi") kept by Alexandra von Faber-Castell and Sebastian Pilz in Zurich, photographed at 18:37. Nothing on the page states daylight, illuminance, or any light level. The entire claim is conditioned on "in bright light," and that premise is unverified — terrarium lighting at an unknown intensity, possibly with flash, is not a characterized photopic condition.
 - 3) "CAPTIVE HIGH-YELLOW ANIMAL" IS NOT IN THE SOURCE. The file page says nothing about any colour morph. That attribution was supplied by the claimant, not the source — a filled gap.
 - 4) THE DENOMINATOR IS NOT "THE EYE." The DETAIL measures against the PALPEBRAL APERTURE (445 x 295 px), then the CLAIM reports it as a fraction of "the eye's width" and "its height." Those are different structures. Hellebuyck & Solanes Vilanova 2023 (Animals, PMC10044692, opened, peer-reviewed) states that eublepharids are the only Gekkota with fully movable eyelids and no spectacle, naming *E. macularius* as the exemplar; the Reptile Database entry (opened) lists 46-57 eyelid fringe scales as a diagnostic character. So in this species the palpebral aperture is a MOVING boundary set by lid position and arousal, not a fixed anatomical reference. Both percentages therefore have an unstable denominator. The 83% height figure is the most fragile: partly closed lids simultaneously crop the dorsal/ventral tips of the pupil (shrinking the numerator) and shrink the aperture (shrinking the denominator), so 0.83 is close to uninterpretable. This normalization would be defensible in a spectacle-bearing gekkonid; in *Eublepharis* it is not.
 - 5) NO PUBLISHED NUMBER CORROBORATES THE RATIO. Banks, Sprague, Schmolli, Parnell & Love 2015, "Why do animal eyes have pupils of different shapes?" (Sci Adv, PMC4643806, opened) classifies constricted pupils only qualitatively — horizontally elongated / circular / vertically elongated, the latter split into subcircular vs vertical slit — and gives an aspect-ratio example only for domestic cat (12 x 1.5 mm, ~8:1). It publishes no species-level aspect ratios, and I could not confirm *E. macularius* appears in its Table S1 (the supplementary table did not render). I found no source, in any literature I opened, giving a slit width, aspect ratio, or pupil-to-eye ratio for *E. macularius*.
 - 6) RELATED-SPECIES TRANSFER HAZARD (flagging explicitly, per standing rules). The often-quoted gecko pupil literature is about OTHER geckos. Roth, Lundstrom, Kelber, Kroger & Unsbo 2009, "The pupils and optical systems of gecko eyes" (J Vis 9(3):27; abstract opened via PubMed 19757966; paywalled, not open access) studied *Tarentola chazaliae* (helmet gecko) and *Phelsuma madagascariensis grandis* (day gecko) — both spectacle-bearing, neither a eublepharid. Banks et al.'s "when constricted, it creates three or four vertically aligned pinholes" is written of geckos generically (citing Murphy & Howland) and cannot, on what I opened, be attributed to *E. macularius*. *Eublepharidae* is a separate family from *Gekkonidae*/*Phyllodactylidae* within *Gekkota*, differing fundamentally in bearing movable eyelids and lacking a spectacle. Neither the "pinhole" picture nor its absence may be transferred to *E. macularius* from these animals.
 - 7) SPECIES-ID AND MORPH CHECKS, AS ASKED. The identification is a hobbyist photographer's self-label on Commons, not a curated determination — weak by this project's source standard. The file's own common names, "spotted fat-tailed gecko" / "Pakistanscher Fettschwanzgecko," use "fat-tailed gecko," a name far more strongly associated with *Hemitheconyx caudicinctus* (African fat-tailed gecko), a different eublepharid genus common in the same pet trade; that is a live confusion risk even though the binomial and the Commons category both say *E. macularius*. Congeners (*E. angramainyu*, *E. fuscus*, *E. hardwickii*, *E. turcmenicus*) are similar. And a captive Swiss terrarium animal has no locality data and unknown lineage, so it cannot be tied to wild-type *E. macularius* regardless of morph.
 - 8) n=1 WITH UNCALIBRATED GEOMETRY. Beyond the single-individual caveat the claimant does make, nothing documents that the eye was orthogonal to the lens axis. Oblique viewing foreshortens width or height differentially, and a 7:1 slit ratio measured off-axis can be substantially wrong in either direction.
- WHAT WEAKLY SURVIVES: only the qualitative shape. Animal Diversity Web's *E. macularius* account (opened) states the species has movable eyelids and a "vertical slit pupil" — but ADW's cited references for that account are husbandry/pet-trade books (Henkel & Schmidt 1995; Hunziker 1994, TFH; de Vosjoli 1990, Advanced Vivarium Systems), i.e. weak sources, and it is a single source, not corroborated twice. So even "vertical slit pupil" should be carried as single-source/weak, and the quantitative claim is unsupported outright.
- COULD NOT VERIFY: (a) any of the pixel figures — 445, 295, 30-38, 245 — are the claimant's alone; I did not re-measure, because re-measuring requires downloading the 8688x5792 original and downloading a file needs the user's explicit permission, which I cannot obtain as a subagent; (b) whether the image genuinely shows maximal photopic constriction; (c) whether *E. macularius* appears in Banks et al. Table S1; (d) the Roth et al. 2009 full text (paywalled — abstract only); (e) whether the constricted *E. macularius* pupil margin is smooth/tapering as described or crenellated, which no source I opened addresses for this species. - correction: Restated to what the evidence actually supports: "*Eublepharis macularius* has a vertical pupil that narrows to a slit under bright illumination [single, weak source: Animal Diversity Web, whose account rests on husbandry books; not corroborated in peer-reviewed literature I could open]. No published quantitative values for the constricted slit's width, its aspect ratio, or its size relative to the eye exist in any source I opened, so the figures '~8% of eye width', '~80-85% of eye height' and '~7:1' are unverified [confidence: unverified]."
- On the specific measurement offered: it should be reported as, at most, "in one Wikimedia Commons photograph of a CAPTIVE JUVENILE (~6 months old) of unknown provenance, photographed in a terrarium under unrecorded lighting, the dark pupillary slit measured ~30-38 px wide by ~245 px tall within a palpebral aperture of ~445 x 295 px (slit aspect ~7:1)" — with these caveats attached and not dropped: (i) the subject is a juvenile, not an adult, per the file's own description; (ii) illumination is undocumented, so 'in bright light' is an assumption; (iii) the normalizing denominator is the PALPEBRAL APERTURE, not the eye — and because eublepharids uniquely among *Gekkota* have fully movable eyelids (Hellebuyck & Solanes Vilanova 2023), that denominator varies with lid position, so the 8% and 83% figures should be dropped entirely rather than restated against 'the eye'; (iv) only the slit's own ~7:1 aspect ratio is normalization-independent, and it remains n=1, uncalibrated for viewing angle, and unverified by me; (v) 'high-yellow' should be struck as unsourced; (vi) the identification is a photographer's self-label on Commons, not a curated determination.
- To actually establish this ratio, the right evidence would be direct measurement of multiple wild-type or locality-known adult *E. macularius* under recorded illuminance, with the eye orthogonal to the lens and pupil dimensions normalized to corneal diameter or lens diameter rather than the palpebral aperture — note Yamashita & Tsuihiji 2022 (J Morphol, PMC9545706, opened) gives a usable hard scale for one museum *E. macularius* (FMNH 176870): lens diameter 3.37 mm, internal scleral ossicle ring diameter 4.39 mm.

Taxonomic keys for *E. macularius* use the external ear as a measurable structure, giving a species-specific scale anchor: the height of the auditory meatus is about 1.5 times the internostril distance.

-- SOURCE OPENED: yes — the live Reptile Database account for *Eublepharis macularius* (<https://reptile-database.reptarium.cz/species?genus=Eublepharis&species=macularius>), fetched and parsed to text on 2026-09-12. The Diagnosis block is NOT stripped on the live page; it is still there in full, so I did not need the archived snapshot and the user's premise that the live page strips it is wrong. The string is real and verbatim: "height of auditory meatus 1.5 times internostril distance", inside "Additional diagnostic characters" (Grismer, 1988:442-450; 1991: 251-252)". So the QUOTE checks out. Everything the claim builds on top of the quote does not.

WHAT REFUTES IT:

- (1) It is not a key character. The Diagnosis paragraph has two halves. The key half — the part tied to an actual identification key — is "Subdigital lamellae each with several distinct small tubercles (Fig. E in key, Anderson 1999: 117); mid-dorsal tubercles generally larger than intertubercular spaces; chin shields usually in contact with first lower labials (Fig. C in key, page 117); elements of dorsal color pattern not linearly arranged." The ear ratio sits in the SECOND half, explicitly flagged as "Additional diagnostic characters" — a supplementary descriptive list, not a key couplet. The claim's headline ("taxonomic keys ... use the external ear as a measurable structure") is contradicted by the same paragraph it cites: the key couplets referenced use lamellae, tubercles, chin shields and colour pattern, and no ear character at all.
- (2) The citation chain is longer and different from what the user states. The user says "one remove from the primary". It is two. The whole Diagnosis block, additional characters included, closes with "(Anderson 1999: 120)" and its figure references are to Anderson ("Fig. 56 in Anderson 1999: 120", "Fig. E in key, Anderson 1999: 117"). So the chain is Reptile Database <- Anderson 1999 (The Lizards of Iran, SSAR Contributions to Herpetology 15) <- Grismer. The user is reading a compilation of a compilation, and I did not open Anderson 1999 or Grismer either.
- (3) The source work treats the animal as only tentatively *E. macularius*. The same Reptile Database page's synonymy list contains the line "*Eublepharis* cf. *macularius* — ANDERSON 1999". "cf." means Anderson was not confident the Iranian material he described was true *E. macularius*. The ear ratio therefore traces back, via Anderson, to a population the describing author himself flagged as uncertainly assigned. That is the species-confusion problem the task asked me to look for, and it is present in the source's own metadata.
- (4) It is a regional diagnosis, not a species-wide one. Anderson 1999 covers Iran, where only two *Eublepharis* occur. I also opened the Reptile Database account for the other one, *E. angramainyu* (<https://reptile-database.reptarium.cz/species?genus=Eublepharis&species=angramainyu>): its Diagnosis reads "Mid-dorsal tubercles not as large as intertubercular spaces; subdigital lamellae smooth, without tubercles; some elements of dark color pattern of head and body linearly arranged; females with discernable preanal pores." No ear character, no competing ratio. So nothing in either account shows the 1.5 value discriminating *E. macularius* from anything. Calling it "species-specific" is unsupported by the cited text.
- (5) The taxon concept behind Grismer 1988 is broader than today's *E. macularius*. The same page's Comment states that Grismer (in Estes & Pregill 1988) synonymised *E. gracilis* and *E. afghanicus* into *E. macularius*, that *E. m. fuscus* was later raised to species rank by Das 1997, and that Mirza 2024 proposed revalidating *E. madarensis*. A 1988 character therefore describes a composite that has since been split, so it cannot be assumed to characterise *E. macularius* as currently circumscribed.
- (6) "Scale anchor" is a category error. A ratio of two head dimensions is dimensionless: it fixes no absolute size and cannot anchor scale. It is also a single point value with no range, no sample size, no specimen list and no age class stated, and head proportions in lizards are allometric, so a single ratio is not usable as a measurement reference even in principle.
- (7) The long-axis inference is invented. "Height of auditory meatus 1.5 times internostril distance" reports one dimension of the ear opening against an unrelated dimension of the snout. It says nothing whatever about the meatus's width, so it cannot imply that the vertical axis is the long one. That inference is the user's, not the source's.

COULD NOT VERIFY: I did not open Grismer 1988 (Estes & Pregill eds., Stanford University Press) or Grismer 1991, so I cannot confirm the 1.5 figure against the primary text, confirm the page ranges, or confirm what Grismer intended the character to distinguish. I specifically could not resolve "Grismer 1991": the Reptile Database bibliography on this species page lists Estes & Pregill 1988 and Anderson 1999 but contains no Grismer 1991 entry at all, so that citation is unresolvable from the source the user actually opened. I also could not verify the actual shape or orientation of the ear opening in *E. macularius* from any source — treat that as unverified.

WEAK-SOURCE NOTE: none used. No pet-care or breeder material was consulted.

MORPH CAVEAT: this particular claim is not a morph artefact — it derives from museum-based descriptive work on wild material. But the reverse caution applies: selectively bred captive leopard geckos (giant lines especially) have head proportions that need not match wild-caught stock, so this ratio should not be used to validate a pet animal.

CONFIDENCE: the quote's existence and its context are verified directly from an opened page. The inferences the claim draws from it are refuted on the text of that same page. The underlying morphological fact remains unverified. - correction: The quoted string is genuine but the claim misdescribes it on four counts. Accurate version: the Reptile Database account for *Eublepharis macularius* carries the sentence "height of auditory meatus 1.5 times internostril distance", but it appears under "Additional diagnostic characters", not in a key — the key characters the same paragraph cites (Anderson 1999: 117) are subdigital lamellae tuberculation, relative mid-dorsal tubercle size, chin-shield contact with the first lower labials, and non-linear dorsal colour pattern, none of which involve the ear. The text is quoted by Reptile Database from Anderson 1999 (The Lizards of Iran), who in turn cites Grismer 1988/1991, so it is two removes from the primary, not one; Grismer 1991 cannot even be resolved from that page's bibliography. Anderson's Iranian material is listed in the same page's synonymy as "*Eublepharis* cf. *macularius*", i.e. only tentatively assigned to the species, and Grismer's 1988 concept of *macularius* subsumed *E. gracilis* and *E. afghanicus*, part of which has since been split off again (*E. fuscus*, Das 1997). The value is a single unranked dimensionless ratio, so it anchors no absolute scale and demonstrates no species-specificity — the congener in the same regional work, *E. angramainyu*, is diagnosed without any ear character. Nothing in the sentence licenses the further inference that the meatus's long axis is vertical: only its height is reported, never its width.

The gap between eye and ear opening in *Eublepharis* is approximately equal to the snout length.

-- I opened all three historical works in full OCR plus two primary descriptions and two modern papers. The submitter's transcription of the hardwickii passages is accurate, but the claim fails on three independent grounds.

(1) NOT STATED FOR *E. MACULARIUS* ANYWHERE. Boulenger 1885 p.232, Boulenger 1890 p.108 and Smith 1935 p.127 give macularius only a differential account. Boulenger's macularius text opens "Differs from the preceding by..." and lists tubercle size, granular interspaces, more elongate body and digits, 9-14 preanal pores, and colour - no head, snout or ear character. Smith's macularius account is even thinner than the submitter says: it differs "in the following particulars: - Enlarged dorsal tubercles smaller, as broad as or smaller than their interspaces; male with from 9 to 14 preanal pores." Smith does NOT carry over the "more elongate body/digits" point, so the submitter's description of Smith's framing is wrong on that detail. I also opened the two primary descriptions of the taxon: Blyth 1854 (JASB 23: 738, the original description of *Cyrtodactylus macularius*, a footnote) is pure colour pattern with no head proportions, and Gunther 1864 (Ann. Mag. Nat. Hist. (3) 14: 429, the original description of the synonym *E. fasciolatus* from Sind) mentions only tubercles, a wide ear opening, labials, chin shields, ventral series and pores. No author has ever stated the character for macularius. It is an inference from differential-diagnosis convention, not a documented fact.

(2) THE CHARACTER IS SPECIES-DIAGNOSTIC WITHIN THE GENUS, BY BOULENGER'S OWN KEY - so it cannot be assumed constant. In the same Catalogue key (p.231, section II) Boulenger separates two species precisely on this character: *variegatus* has the snout as long as the orbit-to-ear distance, while *fasciatus* (p.234) differs "in the larger and more elongate head, the length of the snout exceeding the distance between the orbit and the ear-opening." The submitter cites the character as a fixed genus attribute from the very page where its author uses it as a variable one.

(3) SCOPE ERROR - "*Eublepharis*" in these sources is not the modern genus. Boulenger 1885 places five species in *Eublepharis*, three of them American; his own synonymy for *E. variegatus* reads "*Coleonyx variegatus*, Cope," and *dovii* and *fasciatus* are Central/North American. Boulenger 1890 repeats that three of the five are American. Smith 1935 explicitly sinks *Goniurosaurus Barbour* into *Eublepharis* and treats *E. lichtenfelderi* (now *Goniurosaurus lichtenfelderi*) as the third species. So a genus-level generalisation drawn from these works covers *Eublepharis* + *Coleonyx* + *Goniurosaurus*, not modern *Eublepharis* - and the one American species measured in that comparison contradicts it.

FURTHER: Smith's wording is a range, not an equality - "as long as or shorter than the distance between the eye and the ear-opening" - which explicitly permits the snout to be the shorter of the two. "Approximately equal" flattens that.

ON THE NUMERIC CORROBORATION. I opened Mirza 2024 (Taprobanica 13(1): 9-15, DOI 10.47605/tapro.v13i1.318). The values are real: Table 1, SE 8.1 / EE 7.6 mm. But the specimen is SMF 60980, the female holotype of *Eublepharis macularius smithi* Borner 1981, and the entire point of that paper is that this animal is NOT *E. macularius* - it is reassigned to *E. madarensis* (Sharma, 1980), distinguished from macularius on reticulate versus spotted pattern, a vertebral stripe, and feebly versus strongly tuberculate lamellae. So the corroboration is from a species deliberately separated from the claim's target. Two further cautions: the second column of that same table (SMF 60979) is internally implausible (orbital diameter 10.3 mm against head length 15.4 mm; internarial 8.5 mm at SVL 95 mm), so the table is not a firm anchor; and the Reptile Database entry for *E. macularius* carries a published reservation that Mirza 2024 provides neither type information nor proper specimen sources for his sequences, leaving reproducibility unclear. I am reporting that reservation as I read it, not endorsing it.

WHAT DOES CHECK OUT. Mirza & Gnanaswar 2022 (Evolutionary Systematics 6: 77-88, DOI 10.3897/evolsyst.6.83290), Table 2, with SE defined as snout tip to anterior eye and EE as posterior eye margin to anterior ear edge - exactly the historical character - gives *E. hardwickii* holotype NHMUK 1946.8.26.6: SE 9.4, EE 9.6 mm, ratio 0.98. That is genuine numerical confirmation of Boulenger and Smith, for *E. hardwickii*. The same table gives *E. pictus* 10.3/9.3 (1.11) and 11.2/11.6 (0.97). No table in any source I opened contains SE or EE for *E. macularius*. Total sample supporting the proportion: one *hardwickii*, two *pictus*, two *madarensis* - and zero *macularius*.

CONFUSION CHECK REQUESTED BY THE TASK. Species confusion: yes, this is the central defect - the statement belongs to *E. hardwickii*, a different species that Smith 1935 restricts to Chota Nagpur, Orissa, adjacent Bengal, Madras, and the Central and United Provinces, doubting records outside that area, whereas *macularius* is a north-western desert species. Boulenger 1890 and Smith both note that Murray 1884 misidentified *macularius* as *hardwickii*, so the two have a documented history of conflation. Genus confusion: also yes, as in point 3. Captive-morph confusion: I found no evidence the claim derives from a pet-trade morph; it traces cleanly to the historical *hardwickii* literature. Worth noting only in passing that Mirza & Gnanaswar 2022 Table 1 does use pet-trade captives ("German line", "Germany bloodline") as genetic vouchers for *Eublepharis* sp. Himalaya, so provenance caution is warranted in this literature generally, but it is not the source of this particular claim.

COULD NOT VERIFY: Mirza et al. 2014 (Phyllomedusa 13(2): 75-90), which may contain *E. macularius* morphometrics, is unreachable - revistas.usp.br refused connection and then timed out. Whether SE is approximately equal to EE in actual *E. macularius* therefore remains unverified in either direction; I am not asserting it is false for *macularius*, only that it is undocumented. Confidence: unverified for *E. macularius*; well supported for *E. hardwickii*.

SOURCES I ACTUALLY OPENED (all full text, local copies): C:\Users\ziyad\AppData\Local\Temp\claud\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\boulenger1885.txt (archive.org catalogueoflizar01brituoft, pp.230-234) C:\Users\ziyad\AppData\Local\Temp\claud\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\boulenger1890.txt (archive.org TheFaunaOfBritishIndiaReptiliaAndBatrachia, pp.107-108) C:\Users\ziyad\AppData\Local\Temp\claud\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\SMITH_fresh.txt (archive.org in.ernet.dli.2015.207328, pp.125-129) C:\Users\ziyad\AppData\Local\Temp\claud\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\BLYTH1854_fresh.txt (archive.org journalofasiatic23asia, p.738) C:\Users\ziyad\AppData\Local\Temp\claud\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\GUNTHER1864_fresh.txt (archive.org annalsmagazineof3141864lond, p.429) C:\Users\ziyad\AppData\Local\Temp\claud\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\MIRZA2022_fresh.xml and .txt (Evolutionary Systematics 6: 77-88) C:\Users\ziyad\AppData\Local\Temp\claud\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\MIRZA2024_fresh.pdf and .txt (Taprobanica 13(1): 9-15) Reptile Database species pages for *E. macularius*, *E. hardwickii*, *E. pictus* (weak/tertiary source, used only for the reservation about Mirza 2024 and to confirm citations). - correction: Corrected, source-bounded version: "In *Eublepharis hardwickii* the snout is about as long as, or shorter than, the distance from the eye to the ear opening (Boulenger 1885: 231; Boulenger 1890: 107; Smith 1935: 126 - Smith gives it as a range, not an equality). This is corroborated numerically in the *E. hardwickii* holotype NHMUK 1946.8.26.6, SE 9.4 mm versus EE 9.6 mm (Mirza & Gnanaswar 2022, Table 2). It has never been stated for *E. macularius*: Blyth's 1854 original description, Gunther's 1864 description of the synonym *E. fasciolatus*, and the *macularius* accounts in all three of Boulenger 1885, Boulenger 1890 and Smith 1935 are silent on the character, and no published SE/EE measurements for *E. macularius* were located. The proportion should not be generalised to the genus: Boulenger's own 1885 key uses it to separate species (*variegatus*, snout equal to the orbit-ear distance, versus *fasciatus*, snout exceeding it), and the '*Eublepharis*' of all three works also contained taxa now placed in *Coleonyx* and *Goniurosaurus*. The *E. madarensis* figures of Mirza 2024 (SE 8.1, EE 7.6 mm) are from the female holotype of *E. macularius smithi* Borner 1981, a specimen that paper explicitly removes from *E. macularius*, so they do not corroborate the character for *macularius*."

The ear opening in *Eublepharis* is large and sub-oval, with its long axis vertical to obliquely tilted.

-- I opened five of the six cited sources and the claim fails on two counts for *E. macularius*.

(1) NOT DOCUMENTED FOR *E. MACULARIUS*. Every ear-opening statement I found is for a different species. Sources opened and grepped in full text: - Boulenger 1885, Cat. Liz. Brit. Mus. vol. 1, p. 231 (archive.org id catalogueoflizar01brituoft, OCR full text): the phrase "latter large, suboval, vertical" sits inside the account of *E. hardwickii*. The *E. macularius* account on p. 232 opens "Differs from the preceding by..." and names only tubercle size/spacing, body and digit proportions, 9-14 preanal pores and colouration. No ear character. - Boulenger 1890, Fauna Brit. India Rept. & Batr., p. 107-108 (archive.org id TheFaunaOfBritishIndiaReptiliaAndBatrachia): same wording, same placement — ear character under *E. hardwickii* (sp. 112), nothing on the ear under *E. macularius* (sp. 113). - Mirza & Gnanaswar 2022, Evol. Syst. 6: 77-88 (JATS XML downloaded): exactly two ear-opening descriptions, both "large, sub-oval, obliquely oriented" — one in the description of the *E. hardwickii* specimen NHMUK 1946.8.26.6 (EL/OD 0.77), one in the *E. pictus* holotype description NCBS NRC-AA-0015 (EL/OD 0.51). *E. macularius* is named in the paper but never gets an ear character. - Mirza 2024, Taprobanica 13: 9-15 (publisher PDF): *E. madarensis* redescription — "ear opening large, suboval, obliquely oriented, smaller than orbital diameter (EL/OD ratio 0.43)", plus three large conical tubercles on the anterior wall of the ear. Only two mentions of "ear opening" in the whole paper (this and the EE measurement definition); none for *E. macularius*. - Mohapatra et al. 2026, Herpetozoa 39: e183441 (JATS XML downloaded): "Ear-opening large (EL/OD ratio 0.87), sub-oval, vertically oriented, lobules absent" — in the *E. jhuma* holotype description. The Comparison section contrasts *E. jhuma* with *E. macularius* on SVL, regenerated-tail scale shape and postcloacal spurs, not on the ear. So the cited wording is accurate but it is a transfer from congeners. *E. hardwickii* is the most distant relevant one: per the phylogeny summarised in Mirza 2024 (from Agarwal et al. 2022), the *E. hardwickii* group is sister to the whole *E. macularius* clade, so Boulenger's character is being carried across the deepest split in the genus apart from *E. angramainyu*.

(2) "LARGE" IS POSITIVELY CONTRADICTED WITHIN THE GENUS. I opened Mirza et al. 2014, Phyllomedusa 13: 75-90 (revistas.usp.br PDF, extracted text) — the *E. satpuraensis* holotype is "Ear opening deep, oval in shape, small (EL/HL = 0.15)". The submitted detail quotes this as "simply 'oval in shape'" and drops both "deep" and "small". That matters: *E. satpuraensis* sits inside the *E. macularius* clade (closer to *macularius* than *E. hardwickii* is), and it is the one congener described as having a small ear. A genus-wide "large" therefore does not hold.

Two further problems with the framing. Boulenger's 1885 genus *Eublepharis* is not the modern genus — his synopsis on p. 231 runs *hardwickii*, *macularius*, *dovii*, *variegatus*, *fasciatus*, i.e. it contains taxa now in *Coleonyx* and *Hemiteconyx* — so his text cannot support a statement about modern *Eublepharis* as a whole. And *E. madarensis*, one of the sources of the "obliquely oriented" wording, was until recently treated as *E. macularius smithi* Börner 1981 or as a synonym of *E. macularius* (Das 1992) and was only separated by Mirza 2024; quoting its ear character as if it were *macularius* is exactly the confusion to guard against.

No captive-morph contamination found — every statement above comes from a holotype, lectotype-era museum specimen or catalogued specimen. I did not open Agarwal et al. 2022 (Mol. Phylogenet. Evol. 168: 107414), Anderson & Leviton 1966, Börner 1974/1981, Smith 1935 or Khan's Pakistan works, any of which could in principle carry a *macularius* ear description; treat "no source anywhere describes it" as unverified rather than established. - correction: For wild-type *Eublepharis macularius* specifically, the shape and orientation of the ear opening are undocumented in every source I opened, including both Boulenger accounts that the submission leans on — mark it unverified.

What the opened sources do support, at genus level and with the species named: - Sub-oval outline is corroborated four times, in four congeners: *E. hardwickii* and *E. pictus* (Mirza & Gnanaswar 2022), *E. madarensis* (Mirza 2024), *E. jhuma* (Mohapatra et al. 2026). Boulenger 1885/1890 gives "suboval" a fifth time, again for *E. hardwickii*. - Orientation is genuinely split: obliquely tilted in *E. hardwickii*, *E. pictus* and *E. madarensis*; vertical in *E. jhuma* and in Boulenger's *E. hardwickii*. Boulenger and Mirza & Gnanaswar disagree about the same species, *E. hardwickii*, which suggests the character is at least partly observer-dependent. - Size is not "large" across the genus. Where measured it is always smaller than the orbit (EL/OD 0.43 in *E. madarensis*, 0.51 in *E. pictus*, 0.77 in *E. hardwickii*, 0.87 in *E. jhuma*), and *E. satpuraensis* is explicitly described as deep, oval and small (EL/HL = 0.15). Note the denominators differ between papers (EL/OD vs EL/HL), so these are not directly comparable.

Suggested rewording: "In described congeners of *Eublepharis* the ear opening is sub-oval and smaller than the orbit, with the long axis reported as vertical in some species and obliquely tilted in others; relative size varies and at least one species (*E. satpuraensis*) is described as having a small ear. No source I opened states the ear-opening shape or orientation for *E. macularius* itself." Anything more specific about *E. macularius* would need a source that actually describes its type or catalogued wild specimens.

The ear opening is a substantial fraction of eye size — roughly half the orbit diameter or a little more.

-- REFUTED on two independent grounds: the stated provenance does not apply to this species, and the premise "no published measurement exists for *E. macularius*" is false — the measurements that do exist point higher than "roughly half".

(1) THE SMITH QUOTE IS ACCURATE BUT IS NOT ABOUT MACULARIUS. I opened Smith 1935 (archive.org OCR of the DLI scan, item in.ernet.dli.2015.207328). The sentence on p. 126 is verbatim in the *E. hardwickii* account: snout "as long as or shorter than the distance between the eye and the ear-opening, the diameter of which is more than half that of the orbit." Smith's own *E. macularius* account (pp. 127–128) is purely differential — "Closely allied to *E. hardwickii*, differing in the following particulars: — Enlarged dorsal tubercles smaller... male with from 9 to 14 preanal pores" — and never mentions the ear. So the transfer is inheritance-by-silence, not a statement about *macularius*. The same is true of every work in that lineage I opened: Boulenger 1885 (Cat. Liz. BM I: 231–232) describes the *hardwickii* ear only as "large, suboval, vertical" and gives *macularius* no ear character at all; Boulenger 1890 (Fauna Brit. India: 107–108) repeats this; Sharma 2002 (Fauna of India, Sauria, account 89) again says *macularius* is "like that of *E. hardwickii* except" tubercles and pores.

(2) THE SOURCE PAGE IS A TWO-SPECIES COMPOSITE, AND THE DONOR SPECIES IS A DISTANT RELATIVE. Mirza & Gnanewar 2022 (Evol. Syst. 6: 77–88, Pensoft XML) list "Smith 1935: 126 (in part)" in the synonymy of BOTH *E. hardwickii* AND their new *E. pictus*. So the very page quoted covers two species that differ on this character (holotype EL/OD 0.77 vs 0.51 — both verbatim in that paper). And Mohapatra et al. 2026 (Herpetozoa 39: 161–176, Table 2) give ND2 p-distance *E. hardwickii* vs *E. macularius* 24.6–27.2%, among the largest in the genus. Smith's "closely allied" premise, which is the only thing licensing the transfer, is obsolete.

(3) PUBLISHED EAR-VS-EYE DATA FOR *E. MACULARIUS* DO EXIST, AND THEY ARE HIGHER THAN THE CLAIM. Minton 1966 (Bull. AMNH 134(2): 72–73, West Pakistan series, 10 adult males + 7 females measured, native range) describes *E. macularius*: "ear opening crescentic, slightly smaller than eye, margined with tubercles" — i.e. roughly 0.8–0.95 of eye, not half. Rawat et al. 2019 (Reptiles & Amphibians 26(1): 58–61, Table 1; PDF fetched from journals.ku.edu), two wild adult males from Banke, Nepal, tabulate tympanum diameter TD = 4, 4 mm against eye diameter ED = 6, 7 mm → 0.67 and 0.57. Two sources, four animals, all at or above "half", one clearly approaching parity.

(4) THE CITED "0.43 IN *E. MADARENSIS*" IS A MACULARIUS-NAMED SPECIMEN, AND ITS OWN PARATYPE GIVES 0.83. In Mirza 2024 (Taprobanica 13(1): 9–15, publisher PDF) the 0.43 is measured on SMF 60980, the female holotype of *Eublepharis macularius smithi* Börner 1981 (Delhi) — referred to *E. macularius* by Das 1992 and only moved to *E. madarensis* in 2024, a move Mohapatra et al. 2026 say "remains unresolved" because the *madarensis* holotype V/1215 is untraceable. Table 1 of that same paper gives the paratype SMF 60979 EL 8.5 / OD 10.3 = 0.83. One taxon, one type series, one author, one set of calipers: 0.43 and 0.83. That spread is the entire "genus range" the claim builds on, so no single value can carry the precision claimed. (That paratype column also has HL 15.4 < HW 18.3 and IN 8.5 mm, which look internally inconsistent — flagging, not resolving.)

(5) LANDMARK CONFLATION. The claim slides between "eye size" and "orbit diameter" as if identical. OD is defined in all these papers as "greatest diameter of orbit"; Rawat's ED is "greatest eye diameter"; Minton says "eye". For the Nepal pair ED/HL = 0.19–0.21, versus OD/HL 0.20–0.39 in the congener descriptions — not the same landmark, and a ratio taken against the eye is not transferable to the orbit.

CONGENER NUMBERS, CHECKED: *hardwickii* 0.77 and *pictus* 0.51 verbatim (Mirza & Gnanewar 2022); *jhuma* 0.87 verbatim, and 0.87/0.85/0.78 computed from the three types in Table 4 (Mohapatra et al. 2026); *madarensis* 0.43 verbatim (Mirza 2024); *satpuraensis* "EL/HL = 0.15" with "OD/HL = 0.20" verbatim (Mirza et al. 2014, Phyllomedusa 13(2): 75–90) — the claimant's 0.75 is a correct division but a ratio of rounded ratios (true value 0.71–0.79). Note that Mirza et al. 2014 call an ear at 0.75 of orbit "small" while Mirza 2024 calls one at 0.43 "large": the words large/small in these descriptions track ear-to-head, not ear-to-orbit, so they cannot be used as corroboration.

CAPTIVE-MORPH CHECK: no measurement above comes from a captive morph; all are museum or wild-caught specimens. The only captive material I found is in the molecular datasets (e.g. a "Germany bloodline" *Eublepharis* sp. Himalaya, TG0228, in Mohapatra et al. 2026 Table 1), which affects the phylogeny, not the morphometrics.

COULD NOT VERIFY: no EL/OD measured with the standard landmark pair on a specimen of *E. macularius sensu stricto* — Mirza et al. 2014 examined the *macularius* holotype ZSIK 6224 (Salt Range) plus five more Pakistani specimens (Appendix I) but tabulated measurements only for the *satpuraensis* type series. Blyth's 1854 original description was not located in the file I had. Mirza et al. 2014 was read from a PDF already present in the session scratchpad rather than fetched by me (revistas.usp.br was unreachable from this machine); I confirmed its identity from its own masthead, DOI and pagination. - correction: Supportable statement: in *E. macularius* the ear opening is large — a crescentic to sub-oval aperture whose longest dimension is a substantial fraction of the eye, roughly 0.6 to nearly 1.0 of eye diameter rather than "about half". Basis, all direct on this species: Minton 1966 (Bull. AMNH 134(2): 72–73), "ear opening crescentic, slightly smaller than eye, margined with tubercles", from a West Pakistan series in the native range; and Rawat et al. 2019 (Reptiles & Amphibians 26(1): 58–61, Table 1), tympanum diameter 4 mm against eye diameter 6 and 7 mm in two wild Nepali adult males, i.e. 0.67 and 0.57 — but those are integers from a caliper the authors say reads to 0.1 mm, so the true pair is only bounded at about 0.54–0.82. Confidence: LOW / partly unverified. Two sources, four animals, two different landmark conventions, no error bars.

Do NOT state a number for EL/OD (ear length ÷ greatest orbit diameter) in *E. macularius*: none has been published with that landmark pair on a specimen of this species. If a single figure is needed for the model, use ear aperture long axis $\approx 0.6 \times$ eye diameter as a working value and label it derived and weakly supported, noting that the orbit is wider than the eye so the same aperture expressed against orbit diameter would be smaller still.

Drop Smith 1935: 126 as a citation for this character in *E. macularius* — it is the *E. hardwickii* account, that account is a *hardwickii* + *pictus* composite (Mirza & Gnanewar 2022), and *hardwickii* is 24.6–27.2% ND2-divergent from *macularius* (Mohapatra et al. 2026, Table 2). Drop the "no published EL/OD exists for *E. macularius*" premise as well: values exist on *macularius*-named material (Mirza 2024, Table 1: 0.43 and 0.83 for the two types of *E. macularius smithi*), they simply disagree with each other by a factor of two.

Congener figures usable as context only, each from one holotype unless noted: *E. pictus* 0.51, *E. hardwickii* 0.77, *E. satpuraensis* ≈ 0.75 (derived from EL/HL 0.15 ÷ OD/HL 0.20), *E. jhuma* 0.87 (0.78–0.87 across three types), "*E. madarensis*" 0.43 (same table's paratype 0.83; and this taxon was *E. macularius smithi* until 2024, status still unresolved per Mohapatra et al. 2026).

Files: sources I fetched this session are in C:\Users\ziyad\AppData\Local\Temp\claude\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad — smith1935.txt, boul1885.txt, boul1890.txt, jhuma.xml, pictus.xml, mirza2024.pdf, rawat2019.pdf; already present from earlier work — minton1966.txt, phyllo2014.txt, sharma2002.txt. Project note: C:\Users\ziyad\GeckoBrain\docs\EYE_SPEC.md line 37 records eye diameter 6–7 mm from Rawat et al. 2019 but omits the tympanum diameter 4 mm in the same table — the ear datum is already in the corpus's own cited source and unused.

The eye-to-ear distance is 1.4–2.0 times the eye diameter in *Eublepharis*, measured from the back of the eye to the front rim of the ear.

-- SOURCES I ACTUALLY OPENED (all four, plus corroboration): (1) Mirza & Gnanaswar 2022, *Evol. Syst.* 6: 77-88, DOI 10.3897/evolsyst.6.83290 — JATS XML downloaded and read in full. (2) Mirza 2024, *Taprobanica* 13(1): 9-15, DOI 10.47605/tapro.v13i1.318 — PDF downloaded from taprobanica.org, text extracted and read. (3) Mohapatra et al. 2026, *Herpetozoa* 39: 161-176, e183441 — XML downloaded and read. (4) Mirza, Sanap, Raju, Gawai & Ghadekar 2014, *Phyllomedusa* 13(2): 75-90, DOI 10.11606/issn.2316-9079.v13i2p75-90 — PDF retrieved via Wayback (capture 20241228) and read. Corroboration on nomenclature: Reptile Database (Uetz et al.) *Eublepharis* genus and *E. macularius* pages, and Crossref metadata (used only to confirm citations, labelled here as secondary).

WHAT CHECKS OUT. The measurement definition is right: Mirza's papers define EE as "distance from anterior edge of ear opening to posterior margin of eye"; Mohapatra et al. 2026 word it "posterior point of eye". All five numeric values are real and correctly transcribed, each stated verbatim in its source as "EE/OD ratio": 1.37 (*E. pictus* holotype NCBS NRC-AA-0015), 1.99 (*E. hardwickii* holotype NHMUK 1946.8.26.6), 1.43 (Mirza 2024), 1.69 (*E. satpuraensis* holotype), 1.38 (*E. jhuma* holotype ZSI-R-29431). All four bibliographic citations are correct, including page ranges. The specimens are wild-collected museum types, so this is NOT a captive-morph contamination problem.

WHY IT STILL FAILS. (a) The stated range contradicts its own data. 1.37 and 1.38 are below 1.4. Two of five values fall outside the claimed lower bound. (b) Non-holotype specimens break the upper bound badly. The claim silently uses one holotype per species. The same tables contain other measured specimens. Mirza & Gnanaswar 2022 Table 2, *E. pictus* paratype NHMUK 1962.238: EE 11.6, OD 4.7 = 2.47. Mirza et al. 2014 type series, one paratype: EE 8.7, OD 4.3 = 2.02. Recomputed per-specimen, the published spread is roughly 1.37-2.47, not 1.4-2.0. *E. jhuma*'s three types give 1.38, 1.39, 1.42; *E. satpuraensis*'s five give 1.64, 1.69, 1.69, 1.70, 2.02. (c) The ratio is not a stable character. Within *E. pictus*, the two type specimens in one table of one paper differ by 80% (1.37 vs 2.47). That paratype column also carries an anomalous head length (HL 24.3 mm at SVL 103.7, HL/SVL 0.23, versus 0.15 in both the *pictus* holotype and the *hardwickii* holotype), so either the range is wrong or that source's data are unreliable — neither reading supports the claim. The high values all come from long-preserved NHMUK/SMF material where a sunken or shrunken eye deflates OD. (d) No *E. macularius* data exist. I checked every measurement table in all four papers: Mirza & Gnanaswar 2022 Table 2 (*hardwickii* + *pictus* only), Mirza 2024 Table 1 (two SMF specimens only), Mohapatra et al. 2026 Table 4 (three *E. jhuma* types only), Mirza et al. 2014 (*satpuraensis* types only). None measured an *E. macularius* specimen. The user is right that no EE/OD is published for *E. macularius* — which is precisely why the claim cannot be asserted of it. (e) The *E. madarensis* proxy is weaker than presented. TRANSFER FROM A RELATED SPECIES: 1.43 is not from *E. macularius*. Mirza 2024's Description explicitly states it is "Based on the female holotype (SMF 60980) of *E. macularius smithi*" — a Delhi specimen (EE 7.6, OD 5.3 = 1.43; paratype SMF 60979, EE 14.8, OD 10.3 = 1.44). The nominal holotype of *madarensis* (Jodhpur V/1215, immature male from Madar, Ajmer) was NOT measured. Mirza notes that SMF 60980 "appears emaciated". The Table 1 paratype column contains implausible figures (ear length 8.5 mm and internarial 8.5 mm at SVL 95 mm, versus 2.3 and 3.1 at SVL 85 mm), so I do not regard that table as clean. The detail's background is otherwise accurate and I confirmed it: *madarensis* is a resurrection of *Cyrtodactylus madarensis* Sharma, 1980, synonymised with *macularius* by Das 1992, and Mirza 2024 does treat Börner's *E. m. smithi* types as the same species under Priority. But its validity is contested — the Reptile Database still lists 8 *Eublepharis* species without *madarensis* and states that Mirza 2024 "neither provides type information nor proper specimen sources for his sequences", while Mohapatra et al. 2026 do accept it. And Mirza's own diagnosis separates *madarensis* from *macularius* on several characters, so its ratio is a related-species inference, not an *E. macularius* fact. (f) Minor: OD is defined in all four papers as "greatest diameter of orbit", not eye diameter, though the running text loosely says "diameter of eye".

COULD NOT VERIFY: whether the Reptile Database has since accepted *E. madarensis* (Mohapatra et al. 2026 cite Uetz et al. 2025 in a list that includes it, but the live genus page I opened does not list it) — treat *madarensis*'s status as contested, confidence unverified. Also unverified: any EE/OD for *E. macularius*, *E. fuscus*, *E. angramainyu* or *E. turcmenicus* — no such published values were found in any source I opened. - correction: Do not state a 1.4-2.0 band, and do not attach it to *E. macularius*. Supportable version: In *Eublepharis*, eye-to-ear distance (EE, anterior edge of the ear opening to the posterior margin of the eye) divided by greatest orbital diameter (OD) spans at least 1.37-2.47 across published type specimens — *E. pictus* 1.37 (holotype) and 2.47 (paratype), *E. jhuma* 1.38-1.42 (n=3), *E. madarensis* 1.43-1.44 (n=2), *E. satpuraensis* 1.64-2.02 (n=5), *E. hardwickii* 1.99 (holotype, n=1). Most published figures are single holotypes, mostly from old preserved museum material in which orbit diameter is sensitive to shrinkage, and the within-species spread in *E. pictus* (1.37 vs 2.47 in one table) shows EE/OD is not a diagnostically stable ratio. No EE/OD value has been published for *E. macularius* itself. The nearest available figure, 1.43, is from Mirza (2024) and was measured on SMF 60980, the female holotype of *Eublepharis macularius smithi* Börner, 1981 from Delhi, assigned to *E. madarensis* (Sharma, 1980) — a closely related Aravalli taxon whose validity is itself contested, and which Mirza diagnoses as distinct from *E. macularius*. If you use it, label it as a related-species figure, not an *E. macularius* measurement.

In *E. macularius* specifically the subdigital lamellae are TUBERCULATE (each lamella bears several small raised tubercles), not smooth — this is a species-diagnostic character.

-- The character statement is real; the word "diagnostic" is what fails, and the "three independent corroborations" are not independent.

SOURCES I ACTUALLY OPENED (full text, this session): 1. Rawat et al. 2019, *IRCF Reptiles & Amphibians* 26(1):58-61 — PDF downloaded and read (C:\Users\ziyad\AppData\Local\Temp\claudel\--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\rawat2019.pdf, text at rawat2019.txt). 2. Reptile Database, *E. macularius* account — page downloaded and read (scratchpad\rd_macularius.html / .txt). 3. Mohapatra et al. 2026, *Herpetozoa* 39:161-176 — full JATS XML downloaded from Pensoft and read (scratchpad\jhuma_xml.txt). 4. Mirza, Sanap, Raju, Gawai & Ghadekar 2014, *Phyllomedusa* 13(2):75-90 (description of *E. satpuraensis*, incl. the key to the genus) — local PDF text, header verified (scratchpad\satpura.txt). 5. Mirza 2024, *Taprobanica* 13(1) (reassessment of *Cyrtodactylus/Eublepharis madarensis*) — local PDF text, header verified (scratchpad\mirza2024.txt). 6. Smith 1935, *Fauna of British India, Reptilia & Amphibia II*, *Eublepharis* account pp. 126-129 (scratchpad\smith1935.txt).

WHAT REFUTES "SPECIES-DIAGNOSTIC": (a) Mohapatra et al. (2026) — the very paper cited in the DETAIL — diagnose their NEW species *E. jhuma* with, verbatim, "subdigital lamellae tuberculate (vs. subdigital lamellae smooth in *E. angramainyu*, *E. fuscus*, *E. hardwickii* and *E. pictus*)". So the tuberculate state is a character *E. jhuma* SHARES; the four smooth species are what *jhuma* is being contrasted against. Mohapatra et al. separate *jhuma* from *macularius* by a different character entirely — regenerated-tail scales "flat and rectangular (vs. circular scales in *E. angramainyu*, *E. macularius* and *E. turcmenicus*)". The DETAIL reads the list backwards: *E. macularius* is absent from the smooth list because it is in the same (tuberculate) bin as the new species, not because the state singles it out. (b) Mirza (2024) states that *E. madarensis* "is most similar to *E. macularius* and a putative synonym *E. macularius afghanicus* ... in bearing tuberculate lamellae", and separates them only by DEGREE — "feebly tuberculate lamellae (strongly tuberculate in *E. macularius afghanicus* and *E. macularius*)". A character that differs by degree between two taxa is not diagnostic in the binary sense the claim asserts. (Note the taxonomic tangle: the Reptile Database account I opened treats *madarensis* as unresolved/synonymous, commenting that Mirza 2024 "neither provides type information nor proper specimen sources".)

WHAT REFUTES "THREE INDEPENDENT CORROBORATIONS": - Rawat et al. (2019) is not independent. I read the sentence; it ends "(Mirza et al. 2014)". Their entire character set is lifted from Mirza et al.'s key. And in that key the couplet is nested — 4a "Subdigital lamellae tuberculate ... *macularius*" is only reached after *hardwickii*, *turcmenicus* and *fuscus* have already been eliminated at couplets 1-3. It separates *macularius* from *angramainyu* and *satpuraensis* at that node in a 6-species genus; it was never framed as a genus-wide autapomorphy. - The Reptile Database entry is not independent either. Its diagnosis is explicitly attributed in-line: "(Fig. 56 in Anderson 1999: 120) Subdigital lamellae each with several distinct small tubercles (Fig. E in key, Anderson 1999: 117)", with the remainder credited to Grismer 1988/1991. Anderson 1999 is *The Lizards of Iran* — a regional key, where the only congeners in play are *E. angramainyu* and *E. turcmenicus*. So this is one source (Anderson), and a geographically restricted one. - Rawat et al. also transcribed Mirza loosely: they write "circular dorsal scales", whereas Mirza et al. specify circular dorsal scales ON THE REGENERATED TAIL — and Mirza gives that state for *macularius*, *angramainyu* AND *turcmenicus*, so it is not exclusive either.

ALSO CHECKED, PER THE BRIEF: - Confusion with another gecko species: yes, but within *Eublepharis*, not outside it — the overlap is with *E. madarensis* (Mirza 2024) and *E. jhuma* (2026). I found no evidence of confusion with a non-eublepharid gecko. - Confusion with a captive-bred morph: I found no evidence of this. The diagnosis traces to Anderson 1999 / Grismer 1988-1991, i.e. wild-caught Iranian and Afghan material, and Mirza's and Mohapatra's statements are from museum specimens (BNHS, SMF types). Nothing I opened derives the character from pet-trade animals.

COULD NOT VERIFY: - Anderson 1999 (*The Lizards of Iran*) itself — I did not open it; I only read the Reptile Database's attributed paraphrase. Grismer 1988 and 1991 likewise unopened. - The lamellar condition of *E. turcmenicus*. It is not in Mohapatra et al.'s smooth list, and Mirza et al. 2014's key eliminates it at couplet 2 before lamellae are scored, so I cannot say whether it is tuberculate. If it is, the claim weakens further. Unverified. - Whether Rawat et al. could genuinely score subdigital tubercles: two Nepalese animals were measured in hand (their Table 1), but the Sunwal and Surkhet records are photographic vouchers only. Whether lamellar tubercles are resolvable in those field photographs is my own doubt, not something any source states. Unverified. - Historical baseline, as context rather than refutation: Smith (1935) keys *Eublepharis* on dorsal tubercle size and preanal pore counts and does not mention subdigital lamellar tubercles for *E. macularius* at all. Single source, so treat as weak corroboration that the character is a modern addition, not a classical one. - correction: *E. macularius* does have tuberculate subdigital lamellae — "Subdigital lamellae each with several distinct small tubercles" per the Reptile Database diagnosis, which credits Anderson (1999: 117, 120), and per the couplet "4a. Subdigital lamellae tuberculate ... *E. macularius*" in Mirza et al.'s (2014) key. That half of the claim stands. But it is a shared character state, not a species-diagnostic one. Mohapatra et al. (2026) diagnose the new *E. jhuma* as having "subdigital lamellae tuberculate", contrasting it with the smooth-lamellae species *E. angramainyu*, *E. fuscus*, *E. hardwickii* and *E. pictus*; they distinguish *jhuma* from *macularius* by regenerated-tail scale shape instead. Mirza (2024) likewise says *E. madarensis* shares tuberculate lamellae with *E. macularius*, separating them only as "feebly" versus "strongly" tuberculate. Within the genus the state is at best one step in a nested key (*macularius* vs. *angramainyu/satpuraensis*, after three prior couplets), and as of 2026 it no longer even does that cleanly.

The "three independent corroborations" should be dropped. Rawat et al. (2019) cite Mirza et al. (2014) directly for the character set, and the Reptile Database diagnosis is Anderson (1999), a key to the lizards of Iran where only two congeners are in play. That is effectively two source-lineages, one of them regional; the third source, Mohapatra et al. (2026), argues against the claim rather than for it.

Usable form: "Subdigital lamellae in *E. macularius* bear several small tubercles rather than being smooth (Anderson 1999, via Reptile Database; Mirza et al. 2014). This helps separate it from *E. angramainyu*, *E. fuscus*, *E. hardwickii* and *E. pictus*, but it is not unique to the species — *E. madarensis* and *E. jhuma* are also tuberculate, and *E. turcmenicus* is unscored here — so it cannot be used alone as a diagnosis."

The terminal (most distal) subdigital lamella is cleft/divided.

-- WHAT I CONFIRMED (source opened): the phrase really is on the page you cite. I pulled <https://reptile-database.reptarium.cz/species?genus=Eublepharis&species=macularius> directly (raw HTML, C:\Users\ziyad\AppData\Local\Temp\claude\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\emac.html) and the Diagnosis section reads, in its "Additional diagnostic characters (Grismer, 1988:442-450; 1991: 251-252)" list: "...46-57 eyelid fringe scales; "cleft terminal subdigital lamellae" (see remarks below); ... So your attribution is accurate. What I could NOT confirm is the underlying morphology, and three things about how RDB presents it argue against taking it at face value.

(1) RDB itself brackets it. It is the ONLY character in that whole list wrapped in quotation marks and flagged "(see remarks below)". There are no such remarks. I checked the live Comment section and an archived snapshot from 2023-04-25 ([web.archive.org](https://web.archive.org/web/20230425000000/https://reptile-database.reptarium.cz/species?genus=Eublepharis&species=macularius), id_ capture of the pre-trim page, saved as e23.html) — the 2023 page carries the identical diagnosis text including "(see remarks below)", and its Comment section discusses only synonymy, distribution, hybridization, genomics and longevity. Nothing about lamellae. The cross-reference is dangling in both versions, i.e. the caveat RDB intended to attach to this character is not readable by anyone using the page. (Snapshots from 2014, 2016 and 2018 that I also pulled contain no "cleft" at all — the diagnosis block was added between 2018 and 2021.)

(2) The citation behind it points partly at a different species. I identified "Grismer 1991: 251-252" as Grismer, L.L. 1991, "Cladistic Relationships of the Lizard *Eublepharis turcmenicus* (Squamata: Eublepharidae)", *Journal of Herpetology* 25(2): 251-253 (found via search; JSTOR record <https://www.jstor.org/stable/1564667> — PAYWALLED, I did NOT open it). So part of the character list RDB pastes into the macularius account comes from a three-page paper whose subject is *E. turcmenicus*. That is exactly the shape of error you asked me to look for: a state that may belong to, or be contrasted against, a congener. I cannot say which — I could not open Grismer 1991, Grismer 1988 (book chapter in Estes & Pregill, *Phylogenetic Relationships of the Lizard Families*), or Anderson 1999 (*The Lizards of Iran*, p. 117-120). Also worth noting: the RDB macularius page's own reference list contains no Grismer entry at all, so the citation is untraceable from the page.

(3) No corroboration anywhere, and some contrary evidence. - Exact-phrase web search for "cleft terminal subdigital lamellae" returns nothing but unrelated anatomy literature. The phrase appears to live only on this page. - Karamiani, R. 2022, "Systematics and distribution of the Leopard Geckos *Eublepharis* Gray, 1827 (Sauria, Eublepharidae): A review", *Iranian Journal of Animal Biosystematics* (OPENED full PDF: https://ijab.um.ac.ir/article_43277_d85f7ffd34969c63e0146c6ac8bc518f.pdf). This review is built on the same sources RDB cites (Grismer 1988; Grismer 1991; Anderson 1999) and its diagnostic list for *E. macularius* is: subdigital lamellae each with several distinct small tubercles; mid-dorsal tubercles larger than intertubercular spaces; chin shields usually contacting first lower labials; dorsal pattern not linearly arranged; 46-47 eyelid fringe scales; 21-30 ventral scale rows. No cleft/divided lamella. The word "cleft" occurs zero times in the paper. Its summary of the genus's key characters treats subdigital lamellae only as smooth vs. tuberculate. - RELATED-SPECIES EVIDENCE, flagged as such: Mirza, Z.A. 2024, *Taprobanica* 13(1): 9-15 (OPENED full PDF: https://www.taprobanica.org/~file/318_eublepharis_madarensis_taprobanica_13_1_2024_mirza_9_15-99e7c-3349_1243.pdf) redescribes *E. madarensis* from the holotype of "*E. macularius smithi*" — a population treated as part of *E. macularius* until this 2024 split, so about as close a proxy as exists without being *E. macularius sensu stricto*. It describes the digits as "bearing horizontally oriented smooth, un-notched lamellae on ventral surface" and says "Lamellae beneath the digits tuberculate, except for the 2-3 terminal ones". I.e. the distal-most lamellae are singled out for LACKING tubercles, not for being split. The same author uses the identical "un-notched lamellae" wording for *E. hardwickii* and *E. pictus* in Mirza & Gnaneswar 2022, *Evolutionary Systematics* 6(1): 77-88 (OPENED, JATS XML: <https://evolsyst.pensoft.net/article/83290/download/xml/>) — those two are congeners in a different, eastern-Indian clade, not *E. macularius*. - The genus-level diagnosis (Leviton et al. 1992), which I read both on the RDB *E. hardwickii* account and quoted in Karamiani 2022, describes the digit as "clawed, the claw partly concealed beneath two or four lateral scales and an upper scale" — a claw sheath, with no cleft or divided terminal lamella mentioned. I also pulled the RDB accounts for *E. angramainyu*, *E. turcmenicus*, *E. hardwickii*, *E. fuscus* and *E. satpuraensis*: "cleft" occurs zero times in all five.

MORPH / CAPTIVE-STOCK CHECK: I found no evidence this is a captive-morph artifact. The character is presented in a wild-type taxonomic context and traces to museum-specimen literature, so morph confusion is not the likely failure mode here; species-level compilation error is.

COULD NOT VERIFY (do not fill these in): whether Grismer 1988:442-450, Grismer 1991:251-253, or Anderson 1999:120 actually assert a cleft terminal lamella for macularius, and if so for which taxon the state was scored; and what RDB's missing "remarks" said. All three primary sources were inaccessible to me. Two plausible but UNVERIFIED origins I will name without asserting: (a) the paired lateral claw-sheath scales of *Eublepharis* being loosely described as a "cleft" terminal lamella, and (b) carry-over of a character framed for pad-bearing gekkonids, where a divided terminal scissor is a standard diagnostic — I did not open a source for (b), so treat it as a hypothesis only.

CONFIDENCE: unverified. Per the default rule, holds=false — not because I proved the lamella is entire, but because one compiled database entry, self-flagged, citing a paper about another species, with no corroboration in any source I could open and with the nearest peer-reviewed descriptions saying "un-notched", is not enough to stand. - correction: Do not state as fact that the terminal subdigital lamella of *E. macularius* is cleft or divided. Defensible version: "The Reptile Database account for *E. macularius* lists, among characters compiled from Grismer (1988, 1991), a quoted character 'cleft terminal subdigital lamellae', flagged by the database with a cross-reference to remarks that are absent from the page; the state is not corroborated in any other source I could open, one of the two cited Grismer papers concerns *E. turcmenicus* rather than *E. macularius*, and recent descriptions in the genus instead describe the subdigital lamellae as smooth and un-notched, with the 2-3 terminal lamellae distinguished by lacking tubercles (Mirza 2024, for *E. madarensis*, split from *E. macularius* in 2024). Treat as unverified pending sight of Grismer (1988:442-450), Grismer (1991, *J. Herpetol.* 25:251-253) and Anderson (1999:117-120)." The solidly corroborated digit character for *E. macularius* is that the subdigital lamellae bear several distinct small tubercles (strongly tuberculate), versus smooth in *E. angramainyu*, *E. fuscus*, *E. hardwickii*, *E. pictus* and *E. satpuraensis* — that one is carried by both the Reptile Database account and Karamiani (2022).

Claws are present on every digit, and are partly enclosed at the base by a sheath of scales — two lateral scales plus one upper scale.

-- I opened Boulenger 1890 and Smith 1935 in full OCR and the quoted sentences are real and on the cited pages — but they do not support the claim as worded, for four reasons.

(1) WRONG TAXONOMIC LEVEL. Both sentences are GENUS (not species) diagnoses. Boulenger 1890 p.107 is under "Family II. EUBLEPHARIDAE / Genus EUBLEPHARIS": "Digits short and cylindrical, with a row of lamellæ below, clawed, the claws only partially retractile into a small sheath." Smith 1935 p.125 (sentence breaks across to p.126) is under "Genus EUBLEPHARIS": "Digits short, cylindrical, with transverse lamellæ beneath, clawed, the claw partly concealed between two lateral scales and an upper scale." Neither author says anything about claws in his *E. macularius* species account (Boulenger p.108; Smith pp.127–129 — I read both in full; they cover tubercles, pores, colour, size, range only).

Boulenger's *Eublepharis* then held five species, three of them AMERICAN; Smith's included *Goniurosaurus hainanensis* and *E. lichtenfelderi*. So the sheath character was written for a genus concept much broader than, and partly outside, modern *Eublepharis*.

(2) THE TWO SOURCES ARE NOT INDEPENDENT. Smith 1935 is the successor volume of the same Fauna of British India series and cites "Boulenger, ... Fauna Brit. Ind. 1890, p. 107" in the genus synonymy immediately above that very sentence. Sharma 2002 (Fauna of India, Reptilia Vol. II: Sauria, ZSI, p.148) repeats it again almost verbatim: "The digits are short, cylindrical, clawed (claws are partly hidden between two lateral scales and an upper scale)". That is a copy-chain down one series, not two independent observations 45 years apart.

(3) THE "TWO LATERAL + ONE UPPER" CONFIGURATION IS SINGLE-SOURCED. Boulenger says only "a small sheath" and makes a functional statement (partially retractile); he gives no scale count. Only Smith gives 2+1. Also, Smith uses a near-identical stock formula for unrelated genera in the same book — "the claw between two enlarged scales" for *Cyrtodactylus/Geckoella* (p.~52), *Agamura* and *Pristurus* — so this reads as genus boilerplate, not a measured observation on *macularius* material.

(4) "EVERY DIGIT" IS UNVERIFIED. Neither cited source says all/every digit. Smith's key entry for the genus reads only "Digits not dilated, clawed"; elsewhere in the same key he writes "All the digits clawed" (*Platyurus*) and "Inner digit clawless" (*Ptychozoon*, *Ptyodactylus* group), so the omission for *Eublepharis* is meaningful, not shorthand. I found no opened source stating claw presence on all five digits of *E. macularius*.

SPECIES-CONFUSION CHECK (asked for explicitly): the "*E. macularius*" of Boulenger and Smith is a composite. Boulenger p.108 gives range "Punjab and Sind; also recently found in the ruins of Nineveh"; Smith p.128 gives "N.W. India to S. Transcaspia and Mesopotamia" and cites a "Persian specimen". Those western populations are now covered by later-named taxa (*E. angramainyu*, *E. turcomenicus*) — I did NOT open those original descriptions, so treat that referral as unverified. What I did verify: Mohapatra et al. 2026 (opened PDF, pp.162, 174) state that Indian "*E. macularius*" has been progressively split (*E. satpuraensis* 2014; *E. pictus*, Mirza & Gnaneswar 2022; *E. madarensis* resurrected by Mirza 2024) and that the status of the Himachal Pradesh population — "the only known population of '*E. macularius*' in India" — "remains unresolved". No captive-morph contamination is involved; these are 19th/20th-century wild museum specimens.

MOHAPATRA ET AL. 2026 QUOTE CHECKS OUT BUT DOES NOT HELP: p.168, holotype description of *E. jhuma*: "Limbs short, stout; digits dilated, clawed, with tuberculate lamellæ on the ventral surface, oriented horizontally; claw much smaller than length of the lamellar region". Correct verbatim, correct citation (Herpetozoa 39: 161–176, DOI 10.3897/herpetozoa.39.e183441). But it describes *E. jhuma* sp. nov. (congener, same genus, described 2026 from Kaimur Hills, Bihar), not *E. macularius*, and the word "sheath" does not occur anywhere in that paper — it gives no support for a claw sheath at all. - correction: What is actually documented, stated at the level the sources support:

"In the genus *Eublepharis* (as diagnosed in the Fauna of British India series), the digits are short and cylindrical with transverse subdigital lamellæ and bear claws that are only partially retractile, the claw base partly concealed between two lateral scales and an upper scale (Smith 1935: 125; the retractile-into-a-small-sheath wording goes back to Boulenger 1890: 107, which does not specify the scales; repeated almost verbatim by Sharma 2002: 148). This is a genus-level character; neither Boulenger nor Smith describes claws in his *E. macularius* species account, and I found no source that verifies the scale configuration on *E. macularius* material specifically."

Drop from the claim, or re-label: - "on every digit" — could_not_verify. No opened source states all/every digit for *E. macularius*. Confidence: unverified. - "two independent sources ~45 years apart" — false. Smith 1935 cites Boulenger 1890 p.107 directly; both are volumes of the same series; Sharma 2002 continues the copy-chain. One source lineage, three restatements. - "two lateral scales plus one upper scale" — single-sourced (Smith 1935, echoed by Sharma 2002). Boulenger says only "a small sheath". Confidence: low/single-source. - Attribute the sheath character to genus *Eublepharis* sensu Boulenger/Smith, and flag that that genus concept then included American *eublepharids* (Boulenger) and *Goniurosaurus/E. lichtenfelderi* (Smith), and that their "*E. macularius*" spanned Mesopotamia and Transcaspia. - The *E. jhuma* line is fine but must be labelled as a transfer from a congener: *E. jhuma* Mohapatra et al. 2026, same genus, described 2026, phylogenetically nested near *E. satpuraensis* — not *E. macularius*, and it says nothing about a sheath.

To actually settle this you need a modern specimen-based redescription or a digit/claw morphology study on verified *E. macularius* (Salt Range, Punjab, Pakistan topotypes — ZSI-R-6224 is listed as the holotype in Mohapatra et al. 2026's comparative material, p.175). I did not find or open one.

Sources opened (full text, verbatim-checked): - Boulenger 1890, Fauna of British India: Reptilia and Batrachia, pp.107–108 — archive.org item in.ernet.dli.2015.207330; local OCR copy C:\Users\ziyad\AppData\Local\Temp\claude\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\boulenger1890.txt - Smith 1935, Fauna of British India: Reptilia and Amphibia Vol. II (Sauria), pp.125–129 — archive.org item in.ernet.dli.2015.207328; local OCR copy ...scratchpad\smith1935.txt - Sharma 2002, Fauna of India: Reptilia Vol. II (Sauria), ZSI, pp.147–149 — PDF at lacerta.de/AF/Bibliografie/BIB_6791.pdf; local copy ...scratchpad\sharma2002.txt - Mohapatra et al. 2026, Herpetozoa 39: 161–176, pp.162, 168, 174–175 — herpetozoa.pensoft.net/article/183441; local copy ...scratchpad\jhuma.txt No pet-care or breeder sources were used.

The naris sits within a single enlarged nasal scale, and that nasal scale — not the naris — is what contacts the rostral.

-- SOURCES I ACTUALLY OPENED (source_opened = true for all of these): 1. Mirza & Gnanaswar 2022, *Evolutionary Systematics* 6(1): 77–88 — full article XML pulled from <https://evolsyst.pensoft.net/article/83290/download/xml/> and read locally. 2. Mirza 2024, *Taprobanica* 13(1): 9–15 — PDF pulled from https://www.taprobanica.org/~file/318_eublepharis_madarensis_taprobanica_13_1_2024_mirza_9_15-99e7c-3349_1243.pdf, text extracted and read. 3. Boulenger 1885, *Cat. Liz. Brit. Mus. (N.H.)* vol. 1, *Eublepharis* pp. 230–233 (archive.org catalogueoflizar01brituoft). 4. Boulenger 1890, *Fauna Brit. India, Reptilia & Batrachia*, *Eublepharis* pp. 107–108 (archive.org reptiliabatrachi00boul). 5. Smith 1935, *Fauna Brit. India, Reptilia & Amphibia* vol. II *Sauria*, *Eublepharis* pp. 125–129 (archive.org in.ernet.dli.2015.207328). 6. Minton 1966, *Bull. AMNH* 134, *E. macularius* pp. 72–73 (archive.org bulletin-american-museum-natural-history-134-029-184). 7. Reptile Database entry for *E. macularius* (compilation, weak; quotes Anderson 1999 / Grismer 1988 diagnoses).

WHAT CHECKS OUT The quoted wording is real and I verified it verbatim, so the claimant did not invent or garble it. In Mirza & Gnanaswar 2022 it appears twice: in the redescription of the *E. hardwickii* holotype NHMUK 1946.8.26.6 ("rostral in contact with nasal, first supralabial and internasals; two large and a slightly smaller internasal between nasals, nostrils large situated medially in the nasal scale") and in the description of the *E. pictus* holotype NCBS NRC-AA-0015, which adds the clause "nasal bordered by rostral, internasal, supralabial I and 8 small scales". Mirza 2024 repeats it for *E. madarensis*. The claimant was also right to flag this as a species transfer.

WHY IT STILL FAILS FOR *E. macularius* (a) No documentation at all for *E. macularius sensu stricto*. I checked every classical and museum-based description I could open. Boulenger 1885 and Boulenger 1890 do not mention the nostril, nasal or rostral for either *Eublepharis* species — Boulenger's head-scale sentence for *E. hardwickii* stops at "rostral subpentagonal, twice as broad as high, with median cleft above; three or four internasals", and his *E. macularius* entry is purely differential (tubercle size, pore count, colour). Minton 1966, describing Pakistani wild-caught *E. macularius* from the heart of its range, gives labial and ventral scale counts but says nothing about the nasal region. So the claim is undocumented, not documented-and-transferred.

(b) The "apparent tension with the classical sources" that the DETAIL says this resolves does not exist in anything I opened. The only classical statement I found on this character is Smith 1935 for *E. hardwickii*: rostral broader than high, NOT in contact with the nostril, three or four internasals. That agrees with Mirza; it does not conflict with him. The rationale offered for the claim is therefore built on a conflict I could not locate.

(c) The three Mirza statements are one source, not three. They are the same author's reused description template, near-verbatim across two papers, varying only in the internasal count (*hardwickii*: two large + one slightly smaller; *pictus*: two large + one much smaller middle; *madarensis*: two large + two slightly smaller). Under the project's corroborate-twice rule this counts as a single observation formula, not independent corroboration.

(d) "Medially" is being over-read. In strict anatomical usage medial means toward the midline, not "in the middle of". On that reading the naris would sit at the rostral-facing edge of the nasal, which undercuts rather than supports the inference that it is "separated from the rostral by scale surface". Mirza nowhere writes that the naris does not touch the rostral. The word "enlarged" in the claim is also the claimant's, not Mirza's — Mirza says the nostril is large, and only for *E. pictus* implies a relatively large nasal by listing 8 small scales around it.

(e) Factual error in the claimant's sourcing of the *madarensis* wording. That sentence is not from a Rajasthan animal. Mirza 2024 states the description is "Based on the female holotype (SMF 60980) of *E. macularius smithii*" Börner 1981, collected at DELHI on 30 July 1965. The Rajasthan type of *madarensis* (Jodhpur V/1215, Madar near Ajmer) is an immature male and is not the described specimen. This cuts both ways: it corrects the claimant's framing, but the specimen actually measured had been a named subspecies of *E. macularius*.

(f) Phylogenetic distance — the claimant is right, and can be stated more strongly. Mirza & Gnanaswar 2022 recover the *E. hardwickii* group as sister to the *E. macularius* clade, and place "E. cf. *smithii*" (which Mirza 2024 synonymises with *E. madarensis*) INSIDE the *E. macularius* clade alongside *E. macularius*, *E. fuscus*, *E. satpuraensis*, *E. sp. Himalayas* and *E. sp. Pakistan*. Same paper notes relationships within that clade have poor to moderate support.

MORPH / MISIDENTIFICATION CHECK (the thing you asked me to look for) No morph confusion in the scalation wording itself — every sentence traces to a preserved museum type (NHMUK 1946.8.26.6, NCBS NRC-AA-0015, SMF 60980). Related caution worth recording: the molecular sampling in Mirza & Gnanaswar 2022 does include captive pet-trade stock as terminals ("Eublepharis sp. Himalaya German line" OK563705, "Eublepharis sp. Himalaya female, Germany bloodline" OK563706), so captive-origin material is embedded in the genus's molecular framework even though it is not the basis of these scale descriptions.

COULD NOT VERIFY (confidence: unverified) I could not open the places where a *macularius*-specific statement would most plausibly sit: Grismer 1988 (*eublepharid* phylogeny, head-scale character coding), Anderson 1999 *The Lizards of Iran*, Börner 1974/1976/1981, Szczerbak & Golubev 1996, Khan 2006, Blyth 1854 original description, and Mirza et al. 2014 (*E. satpuraensis*, *Phyllomedusa* 13(2): 75–90 — the USP server refused connection; this one matters most, because *E. satpuraensis* sits inside the *E. macularius* clade and if Mirza used the same template there it would move the transfer much closer). My web-search budget ran out before I could chase these.

Note in fairness: nothing I opened asserts the opposite either — no source I read says the naris contacts the rostral in any *Eublepharis*. So the claim is plausible and the evidence leans its way. It is simply not documented for *E. macularius*, and per the standing rule that means holds=false. - correction: What is documented, and what should be said instead: "Mirza gives, in identical boilerplate for three congeners — *E. hardwickii* and *E. pictus* (Mirza & Gnanaswar 2022, from holotypes NHMUK 1946.8.26.6 and NCBS NRC-AA-0015) and *E. madarensis* (Mirza 2024, from SMF 60980, the female holotype of *E. macularius smithii* Börner 1981, a Delhi specimen) — that the rostral contacts the nasal, the first supralabial and the internasals, and that the nostril is large and situated medially in the nasal scale; for *E. pictus* he adds that the nasal is bordered by rostral, internasal, supralabial I and eight small scales. Smith 1935 independently states for *E. hardwickii* that the rostral is not in contact with the nostril. TRANSFERRED FROM RELATED SPECIES: *E. hardwickii* and *E. pictus* belong to the *E. hardwickii* group, the sister clade to the *E. macularius* clade; *E. madarensis* is nested inside the *E. macularius* clade itself (as 'E. cf. *smithii*' of Mirza & Gnanaswar 2022), so it is the closest of the three. No source I opened states this for *E. macularius sensu stricto* — Boulenger 1885, Boulenger 1890 and Minton 1966 do not describe the nasal region of *E. macularius* at all, and Smith 1935 treats *E. macularius* only by difference from *E. hardwickii* (smaller dorsal tubercles, 9–14 preanal pores), which by that work's convention implies but does not state the shared nasal condition. Confidence: unverified." Two specific fixes to the original wording: drop "enlarged" (Mirza says the NOSTRIL is large, not the nasal scale), and drop the inference that the naris is "separated from the rostral by scale surface" — "situated medially" is ambiguous between 'in the middle of' and 'toward the midline', and Mirza never states non-contact.

In the embryo the naris forms ventrolaterally on the snout and its vestibule takes a more horizontal orientation than in typical gekkonids; the snout elongates markedly late in development.

-- SOURCE OPENED: yes — I downloaded the full text of PMC7812140 directly (raw HTML from pmc.ncbi.nlm.nih.gov, 425 KB, stripped to text and searched), so the quotes below are from the paper itself, not a summarizer. PMCID confirmed correct via the NCBI ID converter (PMID 33169847 -> PMC7812140, doi 10.1111/joa.13312).

TWO OF THE THREE SUB-CLAIMS ARE TRANSFERRED FROM THE OTHER SPECIES IN THE SAME PAPER.

(1) "Naris forms ventrolaterally" — REFUTED, this is *Lepidodactylus lugubris*. The word "ventrolateral" is used of the naris exactly once in the paper, in section 3.1 under Figure 3, whose caption reads "Early developmental phase of the naso-palatal complex after formation of the vomeronasal pit in *L. lugubris*": "The primitive naris is now elongated and open on the ventrolateral side of the forming snout (Figure 3B'-D) which corresponds to ventral growth of the nasal prominences." That is *L. lugubris* at stage 7. What the paper says for *E. macularius* at stage 30 is plain "ventral", and says it is a SIMILARITY, not a difference: "the presence of the small primordial lateral nasal concha, the lack of the extraconchal space, and the ventral location of the primitive naris are similar in both: *E. macularius* at developmental stage 30 and *L. lugubris* at stage 12." The "relatively widely open / no narrowing of the anteriormost part of the oval primitive naris" bit in the DETAIL is accurate, but that is about patency, not direction.

(2) The DETAIL's inference ("independently supports a laterally-to-ventrolaterally directed naris rather than a dorsal one") does not follow, because the primitive naris is not the nostril. The paper explicitly treats them as different structures: "the division of the primitive naris into the external naris and primitive outer choana, and the formation of the vestibulum may start at this stage", and in the Discussion "Initial fusion of the facial prominences is involved in the division of the primitive naris into the external naris and choana (called here the primitive outer choana) (Parsons, 1959a)". The primitive naris is a transient embryonic opening into the stomodeum that is later partitioned; its ventral position is a statement about facial-prominence growth, not about where the definitive nostril ends up. I searched the whole text: "nostril" occurs 0 times, and none of the 21 occurrences of "external naris" assign it a direction (lateral/ventrolateral/dorsal) in *E. macularius*. The paper describes embryos only, up to an "adult-like stage" (st. 42), and makes no adult external-naris orientation claim for either species.

(3) "Vestibule more horizontal than in typical gekkonids" — PARTLY true at one matched stage, but misstated as a standing species difference, and "typical gekkonids" is $n=1$. The supporting sentence exists: comparing *L. lugubris* st. 20-22 with *E. macularius* st. 34_II, "The vestibulum of the *L. lugubris* remains vertical, while in *E. macularius* it attains more horizontal orientation." But the paper then shows the two converge. *L. lugubris* st. 26: "The vestibulum takes on a more horizontal orientation than previously." *L. lugubris* adult-like st. 50-60: "The vestibulum takes on an almost horizontal orientation." *E. macularius* adult-like st. 42: "As in *L. lugubris* at stage 50-60, the vestibulum of adult-like stage of *E. macularius* takes an almost horizontal orientation." So this is a heterochronic difference at intermediate stages, and both species finish almost horizontal. Only ONE gekkonid was examined (*L. lugubris*, Gekkonidae), so "typical gekkonids" is unsupported extrapolation from a single comparator; the paper's own Discussion (4.2) notes gekkotan vestibulum form is variable — "in Gekkota this part of the nasal cavity is usually moderately long, but it may be elongated as in *Stendactylus* [sic] and *Tarentola*".

(4) "The snout elongates markedly late in development" — REFUTED, and inverted. The "markedly late" elongation is *L. lugubris*. *L. lugubris* at the start of the late phase (st. 26): "The snout and the entire nasal cavity are now much more elongated compared to the stage 20-22." *E. macularius* at the start of its late phase (st. 35): "The snout and each component of the naso-palatal complex are SLIGHTLY more elongated than at stage 34_II" — and "The vestibulum has a similar orientation to previously", i.e. it had already attained it. *E. macularius* is the species whose snout is comparatively elongated EARLY, in the middle phase: "The snout as well as the entire nasal cavity and the outer choana in *L. lugubris* at stage 20-22 is not so elongated as in *E. macularius* at stage 34_II." The DETAIL's own wording ("the snout and nasal cavity end up much more elongated") is closer to the paper than the CLAIM's "elongates markedly late in development", which contradicts "slightly more elongated".

SPECIES CONFUSION CHECK (requested): yes, confirmed. The confusion is with *Lepidodactylus lugubris*, the comparator species in the very same paper. Distance: *L. lugubris* is Gekkonidae, *E. macularius* is Eublepharidae — different families within Gekkota, deeply divergent, and the paper's whole point is to contrast them.

MORPH CHECK (requested): could not verify. Methods 2.1 states the eggs came from animals "kept in the breeding room of the Faculty of Natural Sciences of the University of Silesia in Katowice" — a captive colony, incubated at 30 C in a ReptiBator. The paper does not state the morph, does not state the animals were wild-type, and gives no provenance for the founder stock. So whether these were normal/wild-type-phenotype animals is UNVERIFIED from this source. I have no opened source showing that leopard gecko morphs differ in snout or nasal-capsule morphology, so I am not asserting that this matters — only that it is undocumented here.

CITATION ERROR: the author list given is wrong. Europe PMC metadata (REST API, DOI 10.1111/joa.13312) returns "Kaczmarek P, Metscher B, Rupik W." — three authors. K. Janiszewska is NOT an author of this 2021 paper; she is a co-author on the companion 2020 study of *Anolis sagrei* (Frontiers in Zoology), which the 2021 paper cites as "Kaczmarek et al., 2020". A web search result independently flagged the same discrepancy. Journal, volume, issue, year, DOI and PMCID are otherwise correct (J. Anat. 238(2): 249-287, 2021).

STATISTICAL WEAKNESS: Table 1 gives 20 *E. macularius* embryos total (11 microCT + 9 LM) across 9 stages, i.e. typically $n=1$ per stage per method. The words "intraspecific" and "variation [among specimens]" in this sense do not appear; no assessment of individual variation was made. So the single-stage vestibulum contrast rests on one specimen per species per stage.

CORROBORATION: one source only for the substance, which is appropriate here because the claim is a claim ABOUT this source and I checked it against the primary full text; the author-list error is corroborated twice (Europe PMC API + search result). I did not find or open any independent study of *E. macularius* nasal-capsule embryology.

COULD NOT VERIFY: (a) morph/wild-type status of the *E. macularius* colony; (b) any adult or hatchling external-naris orientation in *E. macularius* — this paper simply does not address it, so it cannot be used as evidence for or against a dorsal vs lateral nostril; (c) whether *L. lugubris* is representative of "typical gekkonids" for vestibulum orientation. - correction: Accurate version, restricted to what Kaczmarek, Metscher & Rupik (2021, J. Anat. 238(2): 249-287; note: three authors, no Janiszewska) actually report:

In *E. macularius* embryos the PRIMITIVE naris (the transient embryonic opening of the nasal pit into the stomodeum, later partitioned into the external naris plus the primitive outer choana) is described as lying VENTRALLY on the forming snout at stage 30, and the authors present this ventral position as SIMILAR to *Lepidodactylus lugubris* at stage 12, not as a difference. The "ventrolateral" description belongs to *L. lugubris* at stage 7, not to *E. macularius*. It remains relatively widely open at stage 30, with no narrowing of the anterior part — that part of the detail is correct.

The nasal vestibule of *E. macularius* attains a more horizontal orientation EARLIER than in *L. lugubris* (*E. macularius* st. 34_II vs *L. lugubris* st. 20-22, where the latter is still vertical), but both species reach an "almost horizontal" vestibulum by their adult-like stages (*E. macularius* st. 42; *L. lugubris* st. 50-60). This is a heterochronic, intermediate-stage difference, based on one gekkonid comparator, not a standing *E. macularius*-versus-gekkonids character.

Snout elongation in *E. macularius* is comparatively EARLY, not markedly late: its snout and nasal cavity are already more elongated at middle-phase st. 34_II than *L. lugubris* at st. 20-22, and at the start of its late phase (st. 35) the snout is only "slightly" more elongated than before. The "much more elongated" late-phase jump described in the paper is *L. lugubris* at st. 26.

Confidence: the above is VERIFIED against the opened full text. Confidence UNVERIFIED for: the morph/wild-type status of the captive *E. macularius* colony (not stated), and any implication about adult nostril direction — this paper studies embryos and never describes the orientation of the definitive external naris in *E. macularius*, so it cannot be cited for or against a dorsally versus laterally directed nostril. Sample size is roughly one specimen per stage per method (20 *E. macularius* embryos total), with no assessment of individual variation.

The rostral is a single large plate at the tip of the snout, roughly twice as wide as it is high, and it bears a median cleft or suture.

-- REFUTED on the cleft; the proportion survives only weakly and the claim's source attribution is demonstrably wrong.

SOURCES I ACTUALLY OPENED (full OCR text, read in situ): (1) Boulenger 1885, Cat. Liz. BM ed. 2, vol. I — archive.org item `catalogueoflizar@1brituoft`, genus p. 230, hardwickii p. 231, macularius p. 232. Local copy: C:\Users\ziyad\AppData\Local\Temp\claudel\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\boul1885.txt (2) Boulenger 1890, Fauna Brit. India, Rept. & Batr. — archive.org `TheFaunaOfBritishIndiaReptiliaAndBatrachia`, genus/hardwickii p. 107, macularius p. 108. (...scratchpad\boul1890.txt) (3) Blyth 1854, J. Asiat. Soc. Beng. 23: 737–738 — archive.org `journalofasiatic23asia`, the ORIGINAL description of *Cyrtodactylus macularius*. (...scratchpad\blyth1854.txt) (4) Smith 1935, Fauna Brit. India, Rept. & Amph. II (Sauria) — archive.org `in.ernet.dii.2015.62460`, genus p. 125, hardwickii pp. 126–127, macularius p. 127. (...scratchpad\smith1935b.txt) (5) Reptile Database species account for *E. macularius* — raw HTML pulled and de-tagged, Diagnosis / Comment / Synonym / Types fields read in full. (...scratchpad\rdb.txt) (6) Mirza 2024, Taprobanica 13(1): 9–15, "taxonomic reassessment... *Eublepharis madarensis*" — publisher PDF. (...scratchpad\mirza2024.pdf / .txt) NOT opened, therefore not cited by me: Anderson 1999; Grismer 1988; Grismer 1991; Mirza et al. 2014 (*satpuraensis*); Mirza & Sanap (*pictus*); Agarwal et al. 2022 (known to me only as reported inside Mirza 2024).

FINDING 1 — Boulenger never states this for *E. macularius*, and there is no genus-level bridge. Confirmed: the phrase "rostral subpentagonal, twice as broad as high, with median cleft above" sits in the *Eublepharis hardwickii* account in BOTH works (1885 p. 231; 1890 p. 107, OCR "median deft"). In both works the *E. macularius* account is a bare differential diagnosis — 1885 p. 232 and 1890 p. 108 both open "Differs from the preceding by the smaller and less numerous tubercles..." and go on to tubercle size, body/digit proportions, preanal pore counts (9–14) and colour pattern ONLY. I grepped both *macularius* blocks: zero occurrences of the rostral scale (the single 1890 hit is "canthus rostralis", a different structure, and it belongs to the following account). The claim's phrase "that *E. macularius* inherits" is the claimant's own inference, not an assertion Boulenger makes. Critically, I also read the genus diagnoses to test whether inheritance is licensed: Boulenger 1885 p. 230 gives only "Digits with a row of lamellae inferiorly; claw-sheaths small, the claws only partially retractile"; Boulenger 1890 p. 107 gives only digits/claws/pupil/preanal pores. Neither genus diagnosis contains any rostral character. So there is no documented basis for transferring it.

FINDING 2 — Smith 1935 repeats the same structure and quietly DROPS the cleft. Smith's *E. macularius* account (p. 127) is again purely differential: "Closely allied to *E. hardwickii*, differing in the following particulars: enlarged dorsal tubercles smaller, as broad as or smaller than their interspaces; male with from 9 to 14 preanal pores" — no head scalation, no rostral. And Smith's own *hardwickii* description (p. 126) gives the rostral as "rostral broader than high, not in contact with the nostril" — NOT "twice as broad as high", and with NO mention of a median cleft. So the leading 20th-century redescription, working the same BMNH material, declined to corroborate the cleft even for *hardwickii*. The cleft is therefore single-sourced to Boulenger; the 1890 text is verbatim self-repetition of 1885, not independent corroboration.

FINDING 3 — Blyth's original description contains nothing. I read Blyth 1854 p. 738 (the footnote establishing *Cyrtodactylus macularius*, from the Punjab Salt Range, via Theobald). It is almost entirely colour pattern plus "tail granular beneath" and throat/abdomen scale gradation. There is no head-shield description at all, hence no rostral.

FINDING 4 — The Reptile Database attribution in the DETAIL is wrong, and I can quote the structure. The RDB Diagnosis field is built in two parts: Anderson 1999: 120 supplies subdigital lamellae with small tubercles, mid-dorsal tubercles vs intertubercular spaces, chin shields in contact with first lower labials, and non-linear colour-pattern elements. Then a separate clause reads "Additional diagnostic characters (Grismer, 1988:442-450; 1991: 251-252);" and it is INSIDE that Grismer list that "rostral twice as wide as high" appears. So the character is credited to Grismer, not to Anderson 1999. Equally important: the RDB diagnosis says nothing whatever about a median cleft or suture in the rostral. The second "independent line" therefore supports at most the proportion, never the cleft — and it is not independent of a primary source I have opened, since it is a secondary page quoting Grismer, which I did not open.

FINDING 5 — SPECIES-CONFUSION CHECK (this is the one the claim most needed). Anderson 1999 is a poor vehicle for *E. macularius* characters in the first place. The RDB synonymy line reads "*Eublepharis* cf. *macularius* — ANDERSON 1999" — Anderson himself hedged the identification. The RDB Comment states "No confirmed records from Iran fide RASTEGAR-POUYANI et al. 2008." The Iranian *Eublepharis* is principally *E. angramainyu*, and RDB's Comment further records that *E. macularius* and *E. angramainyu* can be crossed with viable, fertile F1s (Jančúchová-Lásková et al. 2015). Building an *E. macularius* scale character on an Iranian source, for a species with no confirmed Iranian records, is exactly the cross-species contamination risk flagged in the task.

FINDING 6 — The Mirza wording is about a DIFFERENT SPECIES, and that resolves the claimant's "unresolved discrepancy". I located the passage: Mirza 2024 p. 9–10, "rostral quadrangle much wider than deep, divided by a median suture for its entire length; rostral in contact with nasal, first supralabial and internasals". This sits in the Description section of *Eublepharis madarensis* (Sharma, 1980), written from the female holotype SMF 60980 of *E. macularius* smithi Börner 1981 (Delhi) — i.e. a taxon Mirza 2024 explicitly revalidates OUT of *E. macularius* ("differs from *E. macularius*... vertebral stripe present on anterior part of trunk (absent in *E. macularius*); feebly tuberculate lamellae (strongly tuberculate in... *E. macularius*)). So the "entire length" wording is not an *E. macularius* statement and never was. Note also the shape conflict the claim smooths over: Boulenger "subpentagonal" (*hardwickii*) vs Mirza "quadrangle" (*madarensis*). And RDB's Comment records that Mirza 2024's *madarensis* is itself contested — no type information or proper specimen sources for his sequences — so it is weak ground regardless.

FINDING 7 — RELATED-SPECIES DISTANCE, stated explicitly as the standing rules require. The cleft character is transferred from *E. hardwickii*, a congener, not a conspecific. Per the phylogeny summarised inside Mirza 2024 (attributed there to Agarwal et al. 2022, which I did NOT open): *E. angramainyu* is sister to all other *Eublepharis*, and the *E. hardwickii* group is sister to the *E. macularius* clade. So *hardwickii* is a sister-clade congener — close, but a separate species-group, and Boulenger, Smith and Mirza all treat *hardwickii* and *macularius* as diagnosably distinct on scalation and tuberculation.

CAPTIVE-MORPH CHECK — negative, and I say so rather than leaving it implied. I found no sign this claim is morph-derived. Every rostral statement I traced runs to wild-collected museum material (Boulenger's *Hardwicke/Traill/Beddome/Tufnell/Riddell* specimens; SMF 60980; BMNH types). The failure mode here is species confusion and character transfer, not captive-bred morph contamination.

COULD NOT VERIFY: (a) whether Grismer 1988 or 1991 states the proportion for *E. macularius* on measured *macularius* material, and what sample he used — I did not open Grismer; (b) whether Anderson 1999 p. 120 independently mentions the rostral at all; (c) whether any modern source describes a median cleft in verified wild-type *E. macularius* — I found none in six opened sources; (d) the *E. pictus* paper (ScienceDirect returned HTTP 403) and Mirza et al. 2014 on *E. satpuraensis*, either of which might carry a comparative rostral statement.

CONFIDENCE: "rostral roughly twice as wide as high" for *E. macularius* — LOW / single-source, secondary, attribution in the claim misstated. "Median cleft or suture" for *E. macularius* — UNVERIFIED, and actively undermined by Smith 1935 dropping it and by RDB omitting it. - correction: What the opened sources actually support:

- For *E. macularius*, no primary description I opened — Blyth 1854 (original), Boulenger 1885, Boulenger 1890, Smith 1935 — states any rostral character at all. All four treat *macularius* as a differential diagnosis against *E. hardwickii* built on dorsal tubercle size, body/digit proportions, preanal pore count (9–14 vs 13–18) and colour pattern. The rostral is simply not part of the species' documented description in the classical literature.
- The "twice as broad as high, with median cleft above" wording belongs to *E. HARDWICKII* (Boulenger 1885: 231, repeated verbatim by the same author in Boulenger 1890: 107 — self-repetition, not corroboration). *E. hardwickii* is a congener in the sister clade to the *macularius* clade, not a conspecific. Transferring the character is undocumented: neither Boulenger's nor Smith's genus diagnosis of *Eublepharis* includes any rostral character, so nothing licenses inheritance.
- Smith 1935: 126, redescribing *E. hardwickii*, gives only "rostral broader than high, not in contact with the nostril" — no "twice", and no cleft. The cleft is not corroborated a second time even for *hardwickii*.
- The Reptile Database DOES carry "rostral twice as wide as high" for *E. macularius*, but attributes it to GRISMER 1988: 442-450 / 1991: 251-252, not to Anderson 1999 as the claim states. That attribution in the DETAIL should be corrected. The RDB diagnosis contains no mention of a median cleft or suture. I did not open Grismer, so this proportion rests on one unopened secondary-quoted source.
- "Divided by a median suture for its entire length" is Mirza 2024's description of *EUBLEPHARIS MADARENSIS* (from the SMF 60980 holotype of *E. macularius* smithi, Delhi) — a taxon Mirza revalidates out of *E. macularius*. It is not a statement about *E. macularius*, so there is no genuine Boulenger-vs-Mirza discrepancy "about *E. macularius*" to resolve; the two statements are about two different species (and describe different shapes: subpentagonal vs quadrangular).
- Anderson 1999 is an unsafe source for this species regardless: RDB lists the entry as "*Eublepharis* cf. *macularius* — ANDERSON 1999" and comments "No confirmed records from Iran fide Rastegar-Pouyani et al. 2008." Iranian *Eublepharis* is principally *E. angramainyu*, which hybridises with *E. macularius*.

Defensible restatement: "The rostral of *Eublepharis macularius* is reported as about twice as wide as high (Reptile Database diagnosis, crediting Grismer 1988/1991 — single source, not opened by me; confidence low). Whether it bears a median cleft or suture is UNVERIFIED for this species: no description of *E. macularius* I opened mentions the rostral at all. The 'median cleft above' wording is Boulenger's for the congener *E. hardwickii*, and 'divided by a median suture for its entire length' is Mirza 2024's for *E. madarensis*, a taxon split

out of *E. macularius*." Do not assert the cleft for *E. macularius* without opening Grismer 1988/1991 and Anderson 1999 directly, or a modern description based on identified wild-type *macularius* material.

The mental is large, pentagonal, distinctly broader than long, and is followed by a pair of enlarged chin shields (postmentals) that meet on the midline.

-- SOURCES I ACTUALLY OPENED (all three, plus one the claimant did not cite):

1. Boulenger 1885, Cat. Liz. BM (NH) 2nd ed. Vol. I — full OCR downloaded from archive.org item `catalogueoflizar01brituoft`, local copy at `C:\Users\ziyad\AppData\Local\Temp\claudel\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\boulenger1885v1.txt` (source_opened = true). p. 231 "Synopsis of the Species" confirmed VERBATIM for *macularius*: tubercles not larger than interspaces; "mental pentagonal, much broader than long, followed by a pair of chin-shields". So the claimant's Boulenger quote is accurate and correctly located.
2. Smith 1935, Fauna Brit. India, Reptilia & Amphibia Vol. II Sauria — full OCR downloaded from archive.org item `in.ernet.dli.2015.207328`, local copy at `...\scratchpad\smith1935.txt` (source_opened = true).
3. Reptile Database / Anderson 1999 diagnosis — fetched the live species page (source_opened = true).
4. Boulenger 1890, Fauna Brit. India, Reptilia & Batrachia — archive.org item `in.ernet.dli.2015.207330`, local copy at `...\scratchpad\boulenger1890.txt` (source_opened = true). Not cited by the claimant; it materially changes the picture.

WHY THE CLAIM FAILS AS WRITTEN:

(a) The Smith 1935 attribution is wrong, and it is the load-bearing error. The passage the claimant quotes — mental "subtriangular or pentagonal, twice as broad as the adjacent labials", then a pair of large postmentals "usually present and in contact with one another" — sits entirely inside species account 86, *Eublepharis hardwickii*, on p. 126. It is not a joint *hardwickii/macularius* statement and it is not in Smith's genus diagnosis (p. 125-126), which says nothing about the mental at all. Smith's *macularius* account (no. 87, p. 127) opens by saying it is closely allied to *hardwickii* and differs in only two named particulars: smaller dorsal tubercles and 9-14 preanal pores in the male. Everything else, including the whole head scalation, reaches *macularius* purely by inheritance. So the DETAIL's premise — that this is a head character stated "explicitly rather than by inheritance" — is false for Smith. It is true only of Boulenger's 1885 key couplet.

(b) "Meet on the midline" is supported by one source, about a different species. Boulenger 1885 says only "followed by a pair of chin-shields" — no midline contact. Smith is the only source asserting contact with one another, and (i) he says it of *hardwickii*, (ii) he qualifies it "usually present", i.e. it is variable, not a fixed character. The Reptile Database/Anderson 1999 line the claimant offers as independent corroboration — chin shields usually in contact with the first lower labials — describes contact with the LABIALS, a different contact entirely; it says nothing about the two postmentals touching each other. The claimant's "Three sources" for the midline element is really one, and that one is about *hardwickii* and hedged.

(c) Boulenger himself dropped the character. In Boulenger 1890, the same author's synopsis of Indian *Eublepharis* separates *hardwickii* from *macularius* on dorsal tubercles alone; the mental/chin-shield wording survives only in the *hardwickii* account, and *macularius* again inherits. An author abandoning his own key character within five years is evidence against treating it as a stable *macularius* diagnostic.

(d) Congener overlap: it is not diagnostic of *macularius*. In Boulenger 1885 the couplet's contrast partners are *dovii*, *variegatus* and *fasciatus* — all now referred to *Coleonyx*, i.e. Central/North American animals, where *dovii* has "no chin-shields". Against its actual congener *E. hardwickii* the character does NOT separate, since Boulenger's own *hardwickii* account (1885 p. 231, repeated 1890 p. 107) gives the mental as broadly pentagonal and in contact with two enlarged chin-shields. So the character is a genus-level *eublepharid* condition being read as a species diagnosis.

(e) Species-concept contamination (the "confused with another species" check, and it does fire). "*E. macularius*" in Boulenger and Smith is a much broader taxon than the modern one. Smith's range for it runs to S. Transcaspia and Mesopotamia and he explicitly cites a Persian BM specimen; Boulenger 1885 footnotes a Nineveh specimen he examined personally. Material from that western range is now referred to *E. angramainyu*. Since 1935 the old *macularius* concept has also had *fuscus*, *montanus* and *pictus* split out, and the Reptile Database records the contested revalidation of *madarensis* (Mirza 2024). The Reptile Database further notes no confirmed Iranian records (fide Rastegar-Pouyani et al. 2008) — which undercuts source 3 specifically, because Anderson 1999 is a lizards-of-IRAN key, so its *macularius* diagnosis rests on a fauna whose inclusion of the species is doubted, and its "additional diagnostic characters" are themselves taken from Grismer 1988/1991 rather than from Iranian *macularius*.

(f) The morph check is clean and I say so rather than leaving a gap: this is not a captive-morph artefact. All four sources are museum-specimen-based and pre-date the hobby trade, and mental/postmental scalation is not a selectable morph trait. The live risk runs the other way — modern hobby and breeder descriptions of "leopard gecko" head scales describe captive stock of mixed and often undocumented provenance, and should not be used to corroborate any of this. I used none.

COULD NOT VERIFY: I could not open Anderson 1999 itself (only the Reptile Database's transcription of its diagnosis), so I cannot see Anderson's Fig. C or confirm whether his chin-shield character is a couplet against *E. angramainyu*. I could not open any post-1935 primary redescription of *E. macularius* (Mirza 2018/2024, Grismer 1988/1991) — my web-search budget was exhausted before I reached them — so I have NO modern taxonomic confirmation that any part of this mental/postmental character still stands for *E. macularius sensu stricto*. Confidence on the modern status of the whole character: unverified. - correction: Keep the reduced core, drop the midline element and the "three sources" framing, and re-attribute Smith.

DEFENSIBLE (two independent sources, both opened): the mental of *E. macularius* is broader than long — Boulenger 1885 p. 231 key, "much broader than long"; Reptile Database/Anderson 1999, "mental shorter than wide" — and is followed by a pair of enlarged chin shields (postmentals), Boulenger 1885 p. 231. Confidence: moderate, and only for the broad pre-split *macularius* concept.

SINGLE SOURCE, SO LABEL IT: "pentagonal" comes from Boulenger 1885 alone. Smith 1935 gives the shape as subtriangular OR pentagonal, so shape is variable, and he gives it for *hardwickii*. Do not state "pentagonal" flatly.

UNSUPPORTED FOR A *MACULARIUS* — DROP OR RELABEL: "that meet on the midline". No source I opened states this for *macularius*. It is Smith 1935 p. 126 writing about *E. HARDWICKII* — a congener, close but a distinct species, which *macularius* reaches only through Smith's "differs in the following particulars" shortcut — and even there he hedges it as "usually present". If you keep it, you must write it as: transferred from *E. hardwickii* (Smith 1935 p. 126), where the postmentals are only "usually" in midline contact; unverified for *E. macularius*.

ALSO DROP: "large" as a standalone descriptor — no opened source calls the *macularius* mental large; Smith's size statement is relative to the adjacent labials, not absolute and not about its own length.

CORRECT A MIS-QUOTE OF PURPOSE: Anderson 1999's "chin shields usually in contact with first lower labials" is contact with the LABIALS, not between the two postmentals. Stop citing it as corroboration of midline contact.

CORRECT THE DETAIL'S PREMISE: this is not a character stated for *macularius* independently of *hardwickii*. Only Boulenger's 1885 key couplet states it for *macularius* directly, and Boulenger deleted it from his own 1890 key. Smith 1935 and Boulenger 1890 both deliver it by inheritance from *hardwickii*.

ADD THE CAVEAT: all three cited sources use a pre-split *E. macularius* that included material now assigned to *E. angramainyu* (Smith's Transcaspan/Mesopotamian range; Boulenger's Nineveh specimen) and lineages since named *fuscus*, *montanus* and *pictus*. Before using this character on a wild-type *E. macularius sensu stricto*, check it against a post-2000 redescription — which I was not able to open.

There is one distinct, enlarged row of labial scales on each jaw; counts run about 9–11 supralabials and 9–11 infralabials per side.

-- REFUTED — species confusion. Three of the four supporting citations describe *Eublepharis hardwickii*, not *E. macularius*.

SOURCES OPENED (full text read, not abstracts):

1. Boulenger 1885, Catalogue of the Lizards in the British Museum vol. 1 (archive.org, catalogueoflizards01brituoft, OCR read directly). Page 231 is the account of "1. *Eublepharis hardwickii*," and it is there that the phrase "about ten upper and as many lower labials" occurs. *E. macularius* begins on p. 232 and is written purely differentially — "Differs from the preceding by the smaller and less numerous tubercles..." — covering tubercles, preanal pores (9–14) and colour pattern. Boulenger gives NO labial count for *E. macularius* anywhere.
2. Boulenger 1890, Fauna of British India, Reptilia and Batrachia (archive.org, TheFaunaOfBritishIndiaReptiliaAndBatrachia). Same structure: p. 107 = *E. hardwickii*, "about 10 upper and as many lower labials." p. 108 = *E. macularius*, again purely differential, no labial count.

3. Smith 1935, Fauna of British India, Reptilia and Amphibia vol. II — Sauria (archive.org, in.ernet.dli.2015.62460). Page 126 is "86. *Eublepharis hardwickii*," and that account contains "9 or 10 upper and as many lower labials." *E. macularius* is species 87, beginning p. 127, described as "Closely allied to *E. hardwickii*, differing in the following particulars" — enlarged dorsal tubercles, 9–14 preanal pores, colour. No labial count is given for *E. macularius*.

So the two page numbers in the claim (Boulenger 1885 p. 231; Smith 1935 p. 126) are precisely the *hardwickii* pages. In each work *macularius* starts on the very next page and is given no labial data at all.

INDEPENDENT CORROBORATION of that misattribution: Mirza & Gnanewar 2022 (Evolutionary Systematics 6: 77–88, doi 10.3897/evolsyst.6.83290, PDF opened) list in their synonymy of *Eublepharis hardwickii*: "Boulenger 1885: 231 (in part); Boulenger 1890: 107 (in part); Smith 1935: 126 (in part)." Those exact three page citations are formally assigned to *hardwickii* by a modern redescription. Two independent lines of evidence, per the standing rule.

4. Mirza & Gnanewar 2022 gives labial counts only for *E. pictus* and *E. hardwickii*; its data table has no *E. macularius* column, and its diagnosis against *E. macularius* uses dorsal scalation and banding, never labials.

5. Mirza 2024 (Taprobanica 13(1): 9–15, doi 10.47605/tapro.v13i1.318, PDF opened) gives labials only for *E. madarensis*: Sup 9&10 and 10&10; Inf 11&10 and 9&9. No *E. macularius* counts.

6. Rawat et al. 2019 (Reptiles & Amphibians 26(1): 58–61, PDF opened, Table 1 read). This is the ONE source that attaches labial counts to an animal called *E. macularius*, and the quoted values are accurate: SL 10 and 9, IL 10 and 9, SVL 119 and 109 mm, two males, Kamdi Biological Corridor, Banke District.

CONSEQUENCES: (a) "Four sources, one of them measured" collapses to one source, n=2. That breaches the one-source rule. (b) The range is wrong. Only 9 and 10 are attested for *E. macularius* in anything I opened. The value 11 is unattested for the species and appears to have been imported from congeners (*E. pictus* SL 11, *E. hardwickii* IL 11, *E. madarensis* IL 11). (c) Counting convention is unstated and it matters. Mirza's protocol counts supralabials two ways — to mid-orbit and to the angle of the jaw — giving 11/9 vs 12 in the same *hardwickii* holotype. Rawat et al. report a bare "SL"/"IL" without saying which. A flat "9–11" is method-ambiguous. (d) Identification security of the sole source is weak. Rawat et al. captured, measured and RELEASED both animals: no voucher specimen, no tissue, no genetics. They identified against only the three Indian congeners recognised in 2019 (*E. satpuraensis*, *E. fuscus*, *E. hardwickii*). *E. pictus* (2022) and *E. madarensis* (2024) were described afterwards, and Agarwal et al. 2022 recovered unnamed lineages inside the *E. macularius* clade (*E. sp. Himalayas* / *E. cf. afghanicus*, *E. sp. Pakistan*). Critically, Rawat et al. cite Delhi as the nearest previously documented locality — and Delhi is the type locality of *E. macularius* smithi Börner 1981, which Mirza 2024 refers to *E. madarensis*. The Nepal animals' assignment to *E. macularius sensu stricto* is therefore not secure.

CONGENER FIGURES IN THE DETAIL — CHECKED: *E. hardwickii* 11/9 supralabials, 12 to jaw angle, 9 infralabials — CORRECT (*hardwickii* holotype NHMUK 1946.8.26.67, Mirza & Gnanewar 2022). *E. pictus* 10 supralabials, 11/10 infralabials — CORRECT (*pictus* holotype NCBS NRC-AA-0015, same paper). *E. madarensis* "10/11 SL, 10 IL" — GARBLED; actual is Sup 9&10 / Inf 11&10 (SMF 60980) and Sup 10&10 / Inf 9&9 (SMF 60979).

FIRST SENTENCE, separately: the existence of one discrete supralabial and one discrete infralabial row is consistent with every description I opened (all treat them as countable discrete rows; Smith describes the mental as "twice as broad as the adjacent labials"), but I opened no source that states this for *E. macularius* specifically, nor that calls the labials "enlarged" in that species. Unverified as worded.

CAPTIVE-STOCK CHECK, as requested: the claim is not built on a morph, but note that Mirza & Gnanewar 2022's sequence table includes captive-line animals in this complex — "*Eublepharis sp. Himalaya German line*" and "*Eublepharis sp. Himalaya female, Germany bloodline*" — so reference data for "leopard gecko" in this literature is not uniformly wild-caught. Flagged, not load-bearing here.

COULD NOT VERIFY: any *E. macularius* labial count from a vouchered, genetically confirmed specimen. Mirza et al. 2014 (Phyllomedusa 13: 75–90), which may hold comparative *E. macularius* scalation, I could not obtain — the Phyllomedusa server returned nothing and my web-search budget was exhausted. I have not opened it and do not cite it. Confidence in that gap: unverified. - correction: Corrected statement, restricted to what an opened source actually supports for *E. macularius*:

"*Eublepharis macularius* has a discrete row of supralabial and a discrete row of infralabial scales on each side, as in its congeners. Hard counts attributable to *E. macularius* are extremely sparse: the only ones I could locate are from Rawat et al. (2019, Table 1), two adult males from the Kamdi Biological Corridor, Banke District, Nepal — SL 10 and 9, IL 10 and 9 (SVL 119 and 109 mm). On that basis the attested range is 9–10, not 9–11, from a sample of two. The counting convention (to mid-orbit vs. to the angle of the jaw) is not stated in that source and materially affects the number. CONFIDENCE: low/unverified — single source, n=2, no voucher specimens (both animals were released), no tissue or genetic confirmation, and the identification was made against only the three Indian congeners recognised in 2019, before *E. pictus* (Mirza & Gnanewar 2022) and *E. madarensis* (Mirza 2024) were described. Delhi, the nearest locality Rawat et al. cite for the species, is now referred to *E. madarensis*."

Do NOT cite Boulenger (1885 p. 231), Boulenger (1890 p. 107) or Smith (1935 p. 126) for *E. macularius* labials. Those pages are the *E. hardwickii* accounts — a congener in the sister clade to the *E. macularius* clade (Mirza 2024, Mirza & Gnanewar 2022), i.e. a different species group, not a close population variant. In all three works *E. macularius* is described only differentially against *hardwickii* (tubercle size, 9–14 preanal pores, colour pattern) and is given no labial count whatsoever. If the 9–10/"about ten" figure is used at all, it must be labelled as *E. hardwickii* data transferred from a congener, with that transfer stated explicitly.

FILES (all absolute paths, all opened and read in full): - C:\Users\ziyad\claude\projects\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\tool-results\webfetch-1789213634579-57mvlq.pdf — Rawat et al. 2019 - C:\Users\ziyad\AppData\Local\Temp\claude\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\boul01.txt — Boulenger 1885 vol. 1 (*Eublepharis* at lines 16748–16925) - C:\Users\ziyad\AppData\Local\Temp\claude\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\b1890.txt — Boulenger 1890 (lines 9815–9925) - C:\Users\ziyad\AppData\Local\Temp\claude\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\smith1935.txt — Smith 1935 (lines 8807–9010) - C:\Users\ziyad\AppData\Local\Temp\claude\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\pictus.pdf — Mirza & Gnanewar 2022 - C:\Users\ziyad\AppData\Local\Temp\claude\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\mirza2024.pdf — Mirza 2024

The adult dorsal ground colour is a pale, desaturated yellow at its most saturated, and in many wild animals is much paler — cream, greyish-white, pale yellow-green or pinkish-lilac.

-- SOURCES I ACTUALLY OPENED (full OCR text downloaded and read, not summaries):

1. Günther 1864, Ann. Mag. Nat. Hist. (3) 14: 429-430 — archive.org item dli.ernet.24635 ("The Annals And Magazine Of Natural History Vol-xiv"). The identifier the claimant gave is correct and the article is real. BUT: it is titled "Description of a new Species of *Eublepharis*" and the taxon described is *Eublepharis fasciolatus* Günther, 1864, type locality Hyderabad, Sindh, based on TWO specimens (one adult ~5.5 in, one young ~3 in) sent by R. T. Riddell. It is not a description of *E. macularius*; it is a name later synonymised with it (confirmed in Boulenger 1885: 232 and Smith 1935: 127, both opened). The claimant presents it as "describing wild Sindh material" without disclosing it is a synonym/subspecies name and a two-specimen museum lot. The phrase "brownish yellow" appears in the paragraph on the YOUNG ("These bands and rings are broader than the interspaces of the ground-colour, which is brownish yellow"); the adult paragraph only says "the space between the bars being of the ground-colour", carrying the juvenile term over by reference. So the transfer to adults is defensible but implicit — and "brownish yellow" is a warm, brown-tinged yellow. It does NOT support "pale, desaturated yellow at its most saturated", and it supports none of "cream, greyish-white, pale yellow-green or pinkish-lilac". This source cuts against the claim.

2. Boulenger 1885, Cat. Lizards Brit. Mus. vol. 1: 232 — archive.org catalogueoflizar01brituoft. The quoted wording is accurate: "In the adult these bands become more indistinct, and the head and body become spotted or largely vermiculated with chestnut-brown and whitish". Two problems. (a) Boulenger gives no ground colour for the adult at all — "whitish" is the colour of the pale element of a vermiculated pattern, not a stated ground colour, and for the young he says the interspaces are "whitish" where Günther, from the same Sindh series, called them brownish yellow. The two sources disagree about the same population, which the claim does not disclose. (b) This is a catalogue of spirit-preserved British Museum specimens (the accession list on the same page reads "e-f. male & young. Hydrabad, Sind"). A preserved-specimen colour note is not evidence of colour in life. [The specific mechanism — loss of yellow pigment in alcohol — I did not verify from a source I opened; treat that as unverified reasoning, not a cited fact.]

3. Smith 1935, Fauna Brit. India, Reptilia & Amphibia II Sauria: 127-128 — archive.org in.ernet.dli.2015.207328. Quoted accurately: "The young are light brown or yellowish or pinkish above..." and "the centres of the dorsal bands fade, leaving only their edges or a series of spots". Critically, Smith's colour words are explicitly about THE YOUNG. His adult text gives no ground colour whatever — only that markings break up, and that one Khandesh specimen had "a broad, pale, median interspace" from coalesced bars. So Smith cannot be cited for an adult ground colour, and "pinkish" is a juvenile descriptor being promoted to adults.

NET ON LITERATURE: none of the three sources states an adult dorsal ground colour. The claim's "three literature sources agree" is not true — it rests on juvenile descriptions transferred to adults, on preserved museum material, and on two sources (Günther "brownish yellow" vs Boulenger "whitish") that actually disagree.

PHOTOGRAPHS (iNaturalist API records opened, and the full-size images downloaded and viewed): All four observations exist, are Research Grade, flagged not-captive, identified *E. macularius*, and the localities match what was claimed (387949230 Jhampir PK; 361924223 Harsh, Sikar dist., Rajasthan; 271474107 Birhu Kalan, Una, Himachal Pradesh; 287027399 Kirthar PK). That much checks out. The inferences drawn from them do not:

- MISCOUNT / PSEUDOREPLICATION. 287027399 is ONE observation containing four photos (516066042, 516066087, 516066128, 516066161) of plainly the SAME individual — same pale animal, same regenerate-looking tail, same session. The claim counts it as "three Kirthar, Sindh animals". Likewise 387949230 is one observation with two photos of one animal, counted as "two adults, Jhampir". The four cited records document at most FOUR individuals, not the seven implied, and cannot support "in many wild animals". - THE "PALE YELLOW-GREEN (RAJASTHAN ADULT)" IS A ROADKILL. Photo 660548306 (obs. 361924223) shows a dead, flattened, tail-detached animal on tarmac being dismembered by ants. Its yellow-green cast is a post-mortem/desiccation artefact. It is not evidence of living ground colour at all. This is the single most serious defect in the claim. - THE "NEAR-WHITE PINKISH-LILAC" IS A FLASH ARTEFACT. The Kirthar frames are close-range direct-flash night shots over pale grey limestone gravel. In the same frames the animal's head renders blue-grey — leopard geckos do not have blue heads, so the frames carry an obvious colour cast. The animal also looks chalky and opaque, consistent with pre-shed condition [pre-shed diagnosis is my inference from the image, unverified]. Either way, one flash-lit individual does not establish a lilac wild ground colour. - THE "PALE CREAM (HIMACHAL ADULT)" IS NOT CREAM AND MAY NOT BE ADULT. Photo 488325279 shows distinct, moderately saturated yellow blotches along the nape and dorsum on a whitish ground, and the animal still carries broad juvenile band structure.

DIRECT COUNTER-EVIDENCE, SAME EVIDENCE CLASS. I pulled 30 further wild, Research Grade, non-captive *E. macularius* records from the same region (iNat API, bbox 23-37N / 60-80E) and viewed a sample. Photo 480077493 shows a deeply saturated egg-yolk yellow band ground, well beyond "pale, desaturated"; 504771252 shows bright yellow banding; and the claimant's own Jhampir animal (712990681) is a fairly strong mustard/ochre, not a pale wash. The stated ceiling does not survive. Caveat: 480077493 and 504771252 are juveniles/subadults, so they refute the ceiling for the species but not, on their own, for adults.

ON THE SPECIFIC CHECKS ASKED FOR. Species confusion: the cited photo records are Research Grade *E. macularius* and I found no evidence they are another *Eublepharis*; whether the Indian records could be *E. fuscus* or another congener I could not verify (Smith 1935 does not recognise *fuscus*) — unverified. Morph confusion: the risk runs opposite to the claimant's assumption. "Pinkish-lilac" and "greyish-white" are the vocabulary of captive lines (lavender/snow/blizzard); in the cited material those appearances trace to one flash-lit animal and one corpse, not to documented wild-type variation. The claimant is right that wild animals do not look like selectively bred "high yellow" stock, but has over-corrected into an equally unsupported pale/desaturated ceiling.

COULD NOT VERIFY: any adult ground-colour statement in the three cited works (there is none); pre-shed status of the Kirthar animal; pigment loss in spirit as the explanation for Boulenger's "whitish"; species identity of the Indian records beyond community ID; the existence of the 8 further photographs implied by "12 photographs" (only 4 observations totalling 8 photos were cited). - correction: Supported version, restricted to what the opened sources actually say: "No adult dorsal ground colour is given for *E. macularius* in Günther (1864), Boulenger (1885) or Smith (1935). All three describe colour in the young: Günther (1864: 429-430, as *Eublepharis fasciolatus*, two specimens from Hyderabad, Sindh) calls the juvenile ground colour brownish yellow and carries that term over to the adult only by implicit reference; Boulenger (1885: 232), cataloguing spirit-preserved British Museum specimens, calls the juvenile interspaces whitish and describes the adult as spotted or largely vermiculated with chestnut-brown and whitish; Smith (1935: 127) gives the young as light brown, yellowish or pinkish above, and for the adult says only that the bands break up and their centres fade. Günther and Boulenger disagree on the pale element from the same Sindh material. Photographs of living wild animals from Sindh and NW India span a whitish or lilac-grey ground through cream to a distinctly saturated mustard or egg-yolk yellow, so no 'pale, desaturated yellow' upper bound can be asserted; the very pale, pinkish-lilac and yellow-green appearances in the cited images are attributable to direct on-camera flash at night and, in the Rajasthan case, to the animal being a dead roadkill. Confidence: the literature wording is verified from the original texts; the range of living colour is based on a small photographic sample and should be treated as provisional."

The ground colour is not uniform over the animal: the trunk carries the yellow, while head, lower flanks and tail are markedly paler and whiter.

-- REFUTED as stated, on the claimant's own evidence. I opened both iNaturalist records via the API, downloaded the full-size photos and measured them, and opened the PNAS paper. Species/morph confusion is NOT the failure mode here — both observations are Research Grade *E. macularius* with agreeing IDs (obs 387949230, Jhampir, Sindh, 2026-08-03, 2 agreeing IDs, 2 photos that show two different individuals — different spot architecture; obs 271474107, Birhu Kalan, Una, Himachal Pradesh, 2025-04-19, 5 agreeing IDs, 1 photo). The claim fails on four other grounds.

(1) "MARKEDLY PALER" IS FALSE — the head is never lighter than the trunk. I sampled the pale interspot ground (pixels above regional median luminance, excluding the dark blotches) and converted to CIELAB. Jhampir animal B: head $L^*=54.2$ vs trunk $L^*=54.2$ (identical); Jhampir animal A: crown $L^*=81.2$ vs trunk $L^*=88.0$ (head DARKER); Himachal: head $L^*=66.9$ vs trunk $L^*=75.7$ (head DARKER). Distal tail likewise equal or darker (Jhampir B 46.6, Himachal 61.3, both below their trunks). What actually differs between regions is yellowness, not lightness: b^* head 3.1 / 12.3 / 8.6 vs trunk 9.3 / 31.1 / 23.7. So the observation being reported is "less yellow on the head", and the claim has relabelled that as "paler and whiter", which is a different and measurably wrong statement. Caveat: these are flash photographs at night, not photometry — but the direction of the lightness result is consistent across all three animals and two cameras.

(2) "THE WHOLE TAIL" IS FALSE — the tail base carries yellow. Jhampir animal B: tail base $b^*=7.9$ against a trunk of 9.3, i.e. essentially as yellow as the trunk, while only the distal tail drops to 4.2. Himachal: tail $b^*=17.0$ against trunk 23.7 — yellow-tinged, not "cream-white to pale grey". Jhampir animal A has visibly yellow tubercles on the tail base. The yellow/white transition on the tail is proximodistal, not a head-trunk-tail partition.

(3) THE LIMBS BREAK THE PARTITION. The claim's architecture (trunk = yellow, everything else = pale) omits the limbs, which are the most yellow parts of two of the three animals: Jhampir A hindlimb $b^*=31.4$ vs trunk 31.1; Himachal hind foot $b^*=28.2$ vs trunk 23.7 (i.e. yellower than the trunk). "The trunk carries the yellow" is not a correct description of where the yellow sits.

(4) THE HIMACHAL ANIMAL IS MIS-DESCRIBED. Zoomed, its dorsal ground is white/cream, the same white as its head, flanks and tail; the yellow is not a trunk ground colour at all but discrete bright-yellow patches forming a broken reticulum around the dark blotches, plus yellow on the nape, forelimb and hind foot. Calling the head "markedly paler and whiter than the trunk" is circular there — the trunk ground IS white; the head merely carries denser dark reticulation, which is why it measures darker.

(5) THE MECHANISTIC BASIS DOES NOT SAY WHAT THE CLAIM SAYS IT SAYS. Opened: Ullate-Agote & Tzika 2024, "The dynamic behavior of chromatophores marks the transition from bands to spots in leopard geckos", PNAS, PMC11260152 (note: two authors, not "et al."). The first half of the transfer is accurate — the paper does describe the hatchling as a white head band, then a black band over the shoulder blades, then alternating yellow and black bands to the tail base, with white and black bands on the tail; the yellow body bands as xanthophore-rich with a few contracted melanophores and deeper iridophores; the white head band as a dense full-depth iridophore network with dense collagen. But: (a) it describes ONE transverse white band on the head, not a white head. The rest of the hatchling head is dark. I checked this against wild animals: iNat 29623825 (Jaipur, 23 agreeing IDs) and 267159284 (D.I. Khan, Pakistan) both show a chocolate-black head bearing a single pale occipital/nuchal band — and in the D.I. Khan animal the snout, jaw and limbs are bright yellow-orange. So the head carries yellow even at the banded stage. (b) The paper nowhere states that xanthophores are absent from head or tail skin — it reports iridophore density in the white band, which is not the same claim. (c) The paper nowhere describes adult head or adult tail colour. The word "never" in the claim is the claimant's extrapolation, not the paper's finding. (d) The paper's central result actively undercuts the extrapolation: all bands fade during the first year and are replaced by spots, iridophores gradually perish in adult dermis while xanthophores persist — so hatchling band identity cannot be carried forward as a fixed adult topography, and if anything the paper predicts the iridophore-white zones become LESS white with age. (e) Provenance: the animals were housed and bred at the LANE facility (Univ. Geneva); "wild-type" there means normal-morph captive stock in contrast to the Mack Super Snow PAX7 mutant, NOT wild-caught. No wild populations and no geographic variation were examined. A captive research colony cannot underwrite a statement about wild adults in Sindh or Himachal Pradesh.

(6) SAMPLE. "Every yellow-ground wild adult I looked at" is three photographs of three individuals from two localities, one of which (Himachal) is not a yellow-ground animal by its own appearance. Wild *E. macularius* ground colour is highly variable: of the Research Grade records I opened, a Pakistani adult (obs 54390450, 18 agreeing IDs) is grey-lavender with no yellow at all, and a Karachi/Sindh juvenile (obs 235312839) is white-lavender with purple-grey bands and no yellow anywhere.

SOURCES OPENED (source_opened = true for all of these): [PNAS/PMC11260152](https://pmc.ncbi.nlm.nih.gov/articles/PMC11260152/); [iNat obs 387949230](https://api.inaturalist.org/v1/observations/387949230) + its two photos; [iNat obs 271474107](https://api.inaturalist.org/v1/observations/271474107) + its photo; [Reptile Database *E. macularius*](https://reptile-database.reptarium.cz/species?genus=Eublepharis&species=macularius) (gives distribution — E Afghanistan, Pakistan, NW India incl. Punjab and Jammu, Nepal; type locality Salt Range, Punjab — but no head/body/tail colour description, so it neither supports nor refutes); iNat obs 29623825, 267159284, 54390450, 235312839, 18537233 (photos examined directly).

COULD NOT VERIFY / UNVERIFIED: (a) I found no herpetological field guide, museum specimen description or taxonomic redescription that gives a regionalised colour account of *E. macularius* (head vs trunk vs tail). The Reptile Database account carries none. So the claim's core proposition has, as far as I could open, ZERO corroboration in descriptive taxonomic literature in either direction — it rests entirely on photographs. (b) Whether the Jhampir animals are of wild origin rather than escaped/released captive stock: the observer flagged them not-captive and the community accepted it, but Pakistan is a source country for the leopard gecko trade and I cannot verify origin from an observation record. (c) I did not verify the PNAS volume/issue/page (121(29): e2400486121) independently of the claimant. (d) Side caution, unverified: several Research Grade "*E. macularius*" records in the same iNat result set come from Maharashtra and Gujarat, which fall within the ranges of *E. fuscus* / *E. satpuraensis*; I did not check those IDs against a taxonomic revision, so iNat IDs should not be treated as taxonomic authority for colour claims generally. This does not affect the two Sindh/Himachal records at issue. - correction: A defensible replacement, confidence "moderate for the yellow gradient, unverified for anything beyond these photographs": In wild *E. macularius* that carry yellow, the yellow is not spread evenly and is not confined to the trunk either. It is strongest on the dorsum of the trunk, the limbs and the tail base, weaker on the head, and weakest on the distal tail; the ventrolateral zone and venter are pale cream to lavender-white (ordinary countershading). Measured on three wild adults from Sindh and Himachal Pradesh, head and distal tail are LESS YELLOW than the trunk (CIELAB b^* roughly 3-12 vs 9-31) but are NOT lighter — they are equal in lightness or darker, because the head carries denser dark reticulation. The limbs are as yellow as the trunk or yellower, so "the trunk carries the yellow" is wrong as a partition. Ground colour itself varies enormously between individuals: some wild animals have a white or grey-lavender ground with the yellow present only as discrete patches surrounding the dark blotches, and some have no yellow at all. Do NOT cite Ullate-Agote & Tzika 2024 (PNAS, PMC11260152) as a mechanism for adult regional colour: that study describes a single transverse white band on the head of hatchlings (the head itself is dark), does not describe adult head or tail colour, does not report absence of xanthophores from head or tail, used a captive-bred research colony where "wild-type" means normal-morph rather than wild-caught, and its main finding — bands fade entirely within the first year while iridophores perish and xanthophores persist — argues against reading adult topography off the hatchling band map.

The adult spotted pattern is the broken-up remains of the juvenile banded pattern; the spots are not a separate pattern laid on top.

-- SOURCES I ACTUALLY OPENED (full text, verbatim loci checked, files in C:\Users\ziyad\AppData\Local\Temp\claudef\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad): 1. Boulenger 1885, Cat. Lizards BM(NH) vol. 1 — archive.org item catalogueoflizar01brit, local file boul1885v1.txt. E. macularius account begins p. 232. Confirmed wording: young with five chestnut-brown transverse bands head to sacrum; "In the adult these bands become more indistinct, and the head and body become spotted or largely vermiculated with chestnut-brown and whitish; in some only the brown edges of the dorsal bands remain." (claim's paraphrase is accurate; it drops "largely"). 2. Günther 1864 — Ann. Mag. Nat. Hist. ser. 3, 14: 429–430 (archive.org annalsmagazineof3141864lond, local file gun_annalsmagazineof3141864lond.txt). Confirmed: "In the adult only the brown edges of these bands remain; so that there is one pair of brown cross bars on the neck, and three pairs on the trunk, the space between the bars being of the ground-colour." Two specimens (one adult 5.5 in, one young) from Hyderabad, Sindh, ex R. T. Riddell. 3. Smith 1935, Fauna of British India, Reptilia II (Sauria) — local file smith.txt. Account "87. Eublepharis macularius" p. 127, ontogeny passage p. 128: "With age the markings become broken up and less distinct... the centres of the dorsal bands fade, leaving only their edges or a series of spots corresponding to them; rarely the spots are arranged in longitudinal series." Page citation 127–128 is correct. 4. Ullate-Agote & Tzika 2024, PNAS 121(29): e2400486121 (PMC11260152), full text opened, local file pnas2024.txt.

WHAT REFUTES THE CLAIM — the modern experimental source contradicts the exclusive half of it ("not a separate pattern laid on top"): - The paper's own words: "During the first year of the animal's life, all bands fade away and black spots of variable sizes and shapes form." It does NOT say "in place". The claim's added words "in place" are not in the paper. - Spots also arise in the former LIGHT interbands, not only where black bands were: "The black spots in the white and yellow bands can form in three different ways: i) the migration of melanophores, ii) the proliferation of existing melanophores, or iii) the expansion of existing melanophores"; "melanophore division and state transition are the main drivers of black spots formation on the white and yellow bands." Their Fig. 3C time-lapse marks "a region where no melanophores are visible at week 0 and gradually appear and expand in 12 wk." - Spots form with no band precursor at all: on the regenerated tail, "melanophores directly form black spots as in the adult dorsal skin, rather than yellow and black bands as seen in hatchlings", and "spots form even on the regenerated tail of juveniles, despite the presence of bands on their body"; and in the Mack Super Snow morph, which hatches uniformly black (no bands) and still develops adult spots. Their conclusion: "black spot formation is an intrinsic capacity of melanophores in the postembryonic skin." So the correct picture is two concurrent processes — band fading (iridophore loss, melanophore contraction) plus spot formation from melanophores that are already dispersed throughout the whole skin — not fragmentation of one pattern into another. Only the within-old-black-band spots fit the claim ("black spots form as the melanophores in the surrounding areas contract"); the interband spots do not.

"FOUR INDEPENDENT SOURCES" IS FALSE. Günther 1864 is the original description of Eublepharis fasciolatus, a nominal taxon later sunk into E. macularius (Reptile Database entry opened, local file rdb_emaac.txt: "Eublepharis fasciolatus GÜNTHER 1864: 429 (fide SMITH 1935)"; types BMNH 1946.8.26.65-66, "Hydrabad, Sind... donor R T. Riddell"). Boulenger 1885: 232 lists fasciolatus in the synonymy and his specimen series includes "e-f. male & young. Hyderabad, Sind" — i.e. Günther's two types. Boulenger's "in some only the brown edges of the dorsal bands remain" is therefore, at least in part, a re-description of the same adult specimen Günther described. Smith 1935 cites both Boulenger and Günther. Historical evidence collapses to roughly one museum series (plus a Khandesh specimen Smith adds); genuinely independent lines of evidence are about two, not four.

SPECIES-CONFUSION CHECK: not a different species in the fatal sense — fasciolatus is a synonym/subspecies of E. macularius. But Boulenger's and Smith's macularius concept was broader than the current one: Boulenger p. 232 gives the range as "North-western India, probably ranging through Baluchistan and Persia to the Euphrates" and footnotes a Paris Museum specimen from Nineveh; Smith p. 128 gives "NW India to S Transcaspia and Mesopotamia" and a Persian BM specimen. Moradi & Shafiei 2011 (Amphib. Reptile Conserv. 5(4):88-91, opened, local file arc_angramainyu.txt) place E. angramainyu Anderson & Leviton 1966 in the Tigris-Euphrates basin of Iran and Iraq — a species described long after both works. Whether Boulenger's Nineveh / Smith's Persian and Mesopotamian material is E. macularius as now defined is UNVERIFIED by me; their colour-pattern generalisations may pool more than one taxon. The Sindh and Punjab material central to this claim is within current E. macularius.

MORPH CHECK: the PNAS work is entirely captive stock (Methods: animals housed and bred at the LANE facility, Geneva, cantonal permit 1008) — no wild-caught Pakistani/Afghan animals. The wild-type time-lapse used normal-patterned (+/+) captive-bred juveniles, so that part is wild-type phenotype; but the strongest "spots need no bands" evidence comes from Mack Super Snow, a captive-bred PAX7-deletion morph, and from regenerated tail skin. That does not invalidate the refutation (regenerated tail in ordinary wild-type animals shows the same thing), but the claim's phrase "confirm experimentally" overstates what a lab colony of captive morphs establishes for wild populations.

COULD NOT VERIFY: Whimster (PNAS's ref. 7, the 180-degree skin-graft study) — not opened; I only have PNAS's characterisation of it, which says black spots formed across graft boundaries with no melanophore migration observed. Also unverified: any field/wild-population documentation of the band-to-spot transition. Minor citation error: the PNAS paper has two authors (Ullate-Agote & Tzika), so "et al." is wrong; volume/issue/e-number given are correct (2024 Jul 8 online, 121(29): e2400486121). - correction: Corrected version that the opened sources support: "In E. macularius the juvenile transverse bands fade with age, and in partly transformed animals the band centres fade while the dark band margins persist, so that some adult dark markings are genuinely band remnants — Günther (1864: 430) describes an adult from Hyderabad, Sindh (the type of his E. fasciolatus, now a synonym of E. macularius) retaining one pair of brown cross bars on the neck and three pairs on the trunk; Boulenger (1885: 232) and Smith (1935: 128) repeat this, Smith adding 'a series of spots corresponding to them'. Note that these three accounts are not independent: Boulenger's Sindh specimens are Günther's types and Smith follows both. However, the adult spotted pattern as a whole is NOT simply the fragmented band pattern. Ullate-Agote & Tzika (2024, PNAS 121(29): e2400486121) show that black spots also form in the former white and yellow interbands, by expansion and proliferation of melanophores that are already present throughout the hatchling skin in a contracted state, and that melanophores form spots with no banded precursor at all — on regenerated tail skin (including in still-banded juveniles) and in the bandless Mack Super Snow morph. Band fading and spot formation are two concurrent processes; spot formation is an intrinsic capacity of postembryonic melanophores, not a by-product of bands breaking up."

Adult spots are dark brown / chestnut-brown rather than jet black.

-- The four quoted sources are accurately quoted — I opened all of them — but the inference they are used to support is refuted by a fifth source the claim omits, and the "three independent historical sources" framing is wrong.

VERIFIED, source by source (all opened as full text, not abstracts): 1. Boulenger 1885, Cat. Lizards Brit. Mus. 2nd ed. vol. I, p. 232 (Internet Archive scan catalogueoflizar01brituoft). Exactly as claimed: young with five chestnut-brown transverse bands; in the adult the bands become indistinct and head and body become spotted or vermiculated with chestnut-brown and whitish, in some only the brown band-edges remaining. CONFIRMED. 2. Günther 1864, pp. 429-430. CONFIRMED, but the citation needs correcting: this is Ann. Mag. Nat. Hist. ser. 3, vol. 14, "Description of a new Species of *Eublepharis*" (*E. fasciolatus*), NOT Günther's 1864 Reptiles of British India, where the *Eublepharis* account sits around p. 110-119. Günther's adult: only the brown edges of the bands remain; ground colour brownish yellow. No "black" anywhere. 3. Smith 1935, Fauna Brit. India, Sauria, pp. 127-128. CONFIRMED: young with broad dark brown cross-bars and sometimes intervening dark brown spots; with age the band centres fade, leaving edges or a series of spots. Smith gives no colour word at all for the adult spots — the claim represents this correctly. 4. Ullate-Agote et al. 2024, PNAS 121(29):e2400486121, "The dynamic behavior of chromatophores marks the transition from bands to spots in leopard geckos" (read via PMC full text). CONFIRMED: "black spots" used consistently for adults of their captive colony. 5. iNaturalist 387949230 (Jhampir, Sindh, Pakistan, 2026-08-03) and 271474107 (Una, Himachal Pradesh, 2025-04-19), checked through the iNat API: both research-grade, both flagged not captive, both *E. macularius*. I also viewed wild Sindh/Kirthar photos: spots read dark brown to blackish-brown, not flat black — but a flash photo's white balance is weak evidence for a colour claim.

WHAT REFUTES THE INFERENCE: (a) Blyth 1854, J. Asiat. Soc. Beng. 23: 738 — the ORIGINAL description of the species, from wild Salt Range (Punjab) material collected by Theobald, decades before any captive stock existed — describes the animal in black throughout: black crown, broad black band across the nape, black tail tip, black tubercles, and, for the very transition the claim is about, "the interior of the black bands is pale and speckled with black, the margins continuing black", with the dark hue expected to disappear from the band interiors with age. So the earliest wild-material source describes the persisting adult marks as black. The claim's key move — "three historical sources on wild material agree on brown" — fails: the first and most directly wild-derived source says black. (b) The three brown sources are not independent. Boulenger's specimen list on p. 232 includes entries e-f from Hyderabad, Sind, presented by R. T. Riddell, marked as the types of *E. fasciolatus* — i.e. Boulenger re-described Günther's own two specimens. Boulenger 1890 (Fauna Brit. India, p. 108, also opened) repeats his 1885 wording almost verbatim. Smith 1935 defines *macularius* by difference from *hardwickii* following Boulenger and cites him. This is one British Museum descriptive lineage, not three observations. (c) All three brown sources describe spirit-preserved museum specimens, where melanin-based black commonly reads brown and ground colour fades. Günther's own "brownish yellow" ground colour is the tell — Blyth's collector reported the living animal as strikingly rosy-carneous. Preservation state, not wild-vs-captive origin, is the more parsimonious explanation of "chestnut-brown". (d) The brown/black split does not track adults or captivity. Ullate-Agote et al. call the HATCHLING bands black, where Günther/Boulenger/Smith call the juvenile bands brown. The terminological difference runs through the whole ontogeny, so it cannot be attributed to adult captive stock. (e) Neither side of the contrast measures anything. Ullate-Agote's "wild-type" is a lab line of unstated wild provenance in a colony maintained alongside the Mack Super Snow morph, so their "black" is not a wild-animal observation either.

SPECIES-CONFUSION CHECK (asked for explicitly): the historical *macularius* is a broader composite than today's. Boulenger's footnote on p. 232 covers a Paris Museum specimen from Nineveh, Iraq — that is *E. angramainyu*, not described until 1966. Smith's range runs to southern Transcaспia and Mesopotamia (now *E. turcmenicus* and *E. angramainyu*), and the one adult he describes in detail is from Khandesh, Bombay Presidency, in the region now occupied by *E. fuscus* / *E. satpuraensis*-grade lineages. Of the two iNat records, 271474107 is from Una, Himachal Pradesh, in the north-Indian foothill area where recent phylogenies (Mirza; Agarwal et al.) carry an unnamed "*Eublepharis* sp. Himalaya" lineage inside the *macularius* clade; it is a community-ID photo record, not a voucher, so its identity to species is not secure. The Jhampir, Sindh record is in the core range and is safe. No captive-morph confusion detected in the claim itself — the claim correctly flags Ullate-Agote's animals as a captive colony.

COULD NOT VERIFY: no source I opened reports an actual colour measurement (reflectance, standardised photography, or fresh-specimen colour chart) of wild adult spots; whether the Ullate-Agote lab line has documented wild provenance; and whether the Himachal Pradesh animal is true *E. macularius*. Any statement about the spots' true hue in life is therefore unverified. - correction: Weaker, defensible version: "Nineteenth- and early twentieth-century descriptions based on spirit-preserved museum specimens call the adult marks chestnut-brown or brown (Boulenger 1885: 232, repeated in Boulenger 1890: 108; Günther 1864, Ann. Mag. Nat. Hist. ser. 3, 14: 429-430, as *E. fasciolatus*; Smith 1935: 127-128, of the juvenile bars). These are not independent: Boulenger re-described Günther's *fasciolatus* types and Smith follows Boulenger, and preserved melanin commonly reads brown. Blyth's original description of wild Salt Range material (1854: 738) calls the same bands and their persisting margins black. Modern work on a captive colony (Ullate-Agote et al. 2024) uses 'black spots' for adults and 'black bands' for hatchlings alike. In wild animals I examined, the spots were a warm dark umber to blackish-brown with soft edges rather than a flat black — but that is a field impression, uncorroborated, and the brown/black difference in the literature tracks author and preservation state, not wild versus captive stock." Also fix the citation: Günther 1864: 429-430 is Annals and Magazine of Natural History ser. 3 vol. 14, not The Reptiles of British India. Drop the claim that jet-black spotting is a captive-stock impression; it is refuted by Blyth 1854 and is not supported by anything I could open. Confidence: the quotations are confirmed; the hue conclusion is unverified.

Spots are irregular rounded-to-oval blobs and short bars of variable size, several tubercles across, largest on the trunk and clearly bigger than a single tubercle.

-- PARTLY SUPPORTED, BUT FALSE AS STATED — it describes one point in a documented range as if it were the species pattern, and its tubercle-scale content is in no source.

WHAT I OPENED (all read directly, not summarised from memory): 1. Ullate-Agote A. & Tzika A.C. 2024, PNAS 121(29):e2400486121, "The dynamic behavior of chromatophores marks the transition from bands to spots in leopard geckos" — full text read (PMC11260152; I grepped the article text myself, not just an abstract). source_opened = true. 2. iNaturalist obs. 387949230 (api.inaturalist.org) plus both photos (712990630, 712990681) downloaded and re-magnified by me at 4x and 8x. source_opened = true. 3. Boulenger

G.A. 1885, Catalogue of the Lizards in the British Museum vol. 1: 232 (E. macularius account, museum specimens). source_opened = true. 4. Minton S.A. 1966, Herpetology of West Pakistan, Bull. AMNH 134, E. macularius account. source_opened = true. 5. Mirza et al. 2022 (E. pictus description; tubercle-row definition, Sindh sequence). source_opened = true.

A. CITATION ERROR. "Ullate-Agote et al. (2024)" is a TWO-author paper: Ullate-Agote & Tzika. Cite it as such.

B. THE PAPER SUPPORTS ONLY THE FIRST CLAUSE. Verified in its text: "black spots of variable sizes and shapes" and "irregularly shaped black spots"; also "only black spots of variable sizes are clearly distinguishable" in adult skin. It says NOTHING about short bars, about spot size relative to tubercles, or about spots being largest on the trunk. Its one pigment-vs-tubercle sentence is EMBRYONIC: chromatophore precursors "aggregate around the developing tubercles, which are the largest scales seen on the dorsal skin" (stage 33-35 embryos, wild-type and Mack Super Snow). If anything that points toward per-tubercle registration, not away from it. Its spot-area measurement is four wild-type plus four MSS animals via Fiji "Analyse particles", with no absolute values in the main text (SI Fig. S10) — so there is no published size number to check the claim's fractions against.

C. PROVENANCE CAVEAT (this is the morph-vs-wild-type trap, and the claim half-falls into it). Every animal in the PNAS paper was captive-bred at the LANE facility under Swiss permits (experimentation permit GE24/33145). "Wild-type" there is the morph-trade/lab sense — the non-mutant phenotype contrasted with Mack Snow, Mack Super Snow and Super Lemon Frost — NOT a wild-caught E. macularius. So the only published prose behind the claim comes from captive lab stock; the wild anchor is a single amateur photograph.

D. THE DECISIVE REFUTATION — the pattern range is wider than the claim, from two independent primary sources I opened: - Boulenger 1885 (museum specimens): in the adult the juvenile bands become indistinct and "the head and body become spotted or largely vermiculated"; in some individuals only the brown edges of the dorsal bands remain. - Minton 1966 (West Pakistan, including Sindh material; the synonym E. fasciolatus Günther 1864 has type locality Hyderabad, Sind): adult dorsum straw yellow to pale violaceous grey, "usually showing trace of juvenile bands", with blue-black spots "in some cases discrete and sparse, in others fusing into a reticulum". So wild adults run sparse-discrete-spots -> blobby -> vermiculate/reticulate, and commonly retain a trace of juvenile banding. Rounded-to-oval blobs and short bars are one end of that continuum, not the description.

E. THE CITED PHOTO REFUTES ITS OWN GENERALISATION. On the same Jhampir animal, magnified myself: the trunk does show irregular blobs and some short merged bars (their own counts — roughly 4-6 per transverse rank, ~10-15 ranks between limbs, typical marks about a sixth to a fifth of visible trunk width — are in the right ballpark, though I read typical marks slightly larger than their eighth-to-sixth). BUT the head and nape of the same individual are plainly VERMICULATE: narrow curved, C-shaped and ring-like strands forming a labyrinth, roughly one to two tubercles wide — exactly Boulenger's "largely vermiculated". And at 8x on the flank, a substantial share of the dark marks are single-tubercle-sized dots the same size and bead shape as the neighbouring pale tubercles. So "clearly bigger than a single tubercle" fails inside the very animal cited, and "blobs and short bars" mis-describes its head. (Those two readings are my own observations from one photograph, magnified by me — unverified against any publication.)

F. TUBERCLE-SCALE CONTENT IS UNVERIFIED EVERYWHERE. No source I opened states spot width in tubercles. The only published anchor for tubercle spacing is Boulenger 1885: in E. macularius the tubercles are smaller and less numerous than in E. hardwickii, "the granular interspaces being as wide as the tubercles themselves". Minton 1966 adds that the dorsal tubercles are conical and "irregularly arranged on body" (in annuli only on the tail) — so there is no regular tubercle grid on the trunk to count spots against in the first place. My own 8x measurement of the cited photo puts the LARGEST trunk blotches at roughly 3-5 tubercles across; confidence: unverified, one photograph, one individual, hand-held curved posture, phone optics.

G. SPECIES/MORPH CONFUSION CHECK — no confusion found, with one caveat. Jhampir (Thatta district, lower Sindh, 25.024 N 68.009 E) is inside the E. macularius range: Minton 1966 treats E. fasciolatus (type locality Hyderabad, Sind) as a synonym of E. macularius, and Mirza et al. 2022 sequenced an E. macularius from Pakistan, Sindh, Dadu. The iNat record is wild (captive flag false), and the animal shows a wild-type phenotype with an unregenerated, whorled, banded-spotted tail — nothing suggesting a released captive morph. Caveat: research grade rests on two community IDs (observer ameer12372 + supporter danielb138), no voucher and no genetics, and the E. macularius clade in Mirza et al. 2022 contains poorly supported lineages including an undescribed "Eublepharis sp. Pakistan" (from Khyber Pakhtunkhwa, not Sindh).

COULD NOT VERIFY: any published statement that trunk spots are the largest on the body; any published spot-size measurement in absolute units or in tubercles; any published description of "short bars" in adults. - correction: Use this instead: "In adult E. macularius the juvenile bands break up and the dorsum carries dark brown to blue-black markings of variable size and irregular shape (Ullate-Agote & Tzika 2024, on captive-bred laboratory 'wild-type' animals: 'black spots of variable sizes and shapes'). Across wild adults the pattern spans a documented range rather than one geometry: from discrete, sparse spots, through irregular rounded-to-oval blotches and short coalesced bars, to markings 'largely vermiculated' (Boulenger 1885) or 'fusing into a reticulum' (Minton 1966), with a trace of the juvenile bands usually still visible (Minton 1966). Vermiculate/reticulate markings are typical of the head and nape even in individuals whose trunk is blotchy. On the trunk of a well-spotted individual, the larger blotches are several tubercles across and much larger than a single tubercle, but the same animal also carries dark marks only one tubercle wide, and E. macularius tubercles are irregularly arranged on the body (Minton 1966) with granular interspaces about as wide as the tubercles (Boulenger 1885), so there is no regular tubercle grid to index spots to." For a texture or generative model, sample the spotted-to-vermiculate continuum rather than a single blob rule, put vermiculation on the head/nape, and treat any blotch-size-in-tubercles figure (including 4-6 blotches per rank and 10-15 ranks) as an unpublished single-photo estimate, confidence: unverified.

Spots are solid, with NO pale halo. But a large minority of wild adults show the inverse: dark rings and vermiculations enclosing pale or yellow centres.

-- SOURCES I ACTUALLY OPENED (BHL itself was unreachable — Cloudflare bot-check; I did not attempt to bypass it, I used Internet Archive scans instead): (1) Boulenger 1885, Catalogue of the Lizards in the BM(NH), 2nd ed., Vol. I, the *E. macularius* account, which the OCR page-breaks confirm sits on p. 232 (running heads "232 EUBLEPHARIDAE" before it, "EUBLEPHARIS. 233" after) — archive.org item catalogueoflizar01brituoft_ _djvu.txt. (2) Gunther 1864, "Description of a new Species of Eublepharis", Ann. Mag. Nat. Hist. ser. 3, vol. 14, article starting p. 429, colour description on p. 430 — archive.org item annalsmagazineof3141864lond_ _djvu.txt. (3) The Reptile Database species account for *E. macularius* (reptile-database.reptarium.cz), for synonymy and range. (4) iNaturalist API records AND the full-size photographs for obs. 361924223, 387949230, 271474107 (all three are research-grade, captive=false, current taxon *Eublepharis macularius*). Files saved at C:\Users\ziyad\AppData\Local\Temp\claude\C--Users-ziyad\920e259d-472d-4210-945a-591bc20b749a\scratchpad\ (inat_raj.jpg, inat_jha_a.jpg, inat_jha_b.jpg, inat_him.jpg, plus crops crop_raj_body.jpg, crop_jhab_head.jpg, crop_him_body.jpg, crop_him_head.jpg).

WHAT CHECKS OUT - The Boulenger quotation is verbatim and correctly placed. p. 232: young with five chestnut-brown transverse bands, "In the adult these bands become more indistinct, and the head and body become spotted or largely vermiculated with chestnut-brown and whitish; in some only the brown edges of the dorsal bands remain." That last clause is genuine independent support for a dark-outline / pale-interior geometry in some adults. - The Gunther phrase is verbatim: "a pair of brown rings on the crown of the head" (p. 430). - No species mix-up in the citations. Gunther's taxon is *Eublepharis fasciolatus* (types from Hyderabad, Sindh, coll. R. T. Riddell), which Boulenger 1885: 232 lists in the synonymy of *E. macularius* and which the Reptile Database carries as *E. macularius fasciolatus*. All three iNat animals are wild (captive=false), not captive morphs, and Rajasthan, Sindh and the Punjab/Shivalik foothills all fall inside the Reptile Database's stated range ("E Afghanistan, Pakistan, NW India (Rajasthan, Ajmer, Madar foot hills, Punjab, Jammu, Nepal"). So the pet-morph and wrong-species traps are not what breaks this.

WHY IT STILL FAILS — three separate defects 1. "A large minority" is unsupported by anything opened. Boulenger writes "spotted OR largely vermiculated" with no frequency at all; Gunther describes exactly one adult specimen; three iNaturalist photographs cannot establish a rate. This is the kind of number that has to come from a scored series of specimens, and I found none. Mark it unverified rather than stating a proportion.

2. The Gunther citation is doing work it cannot do. His "pair of brown rings" is explicitly a HEAD marking, listed among head markings ("a pair of brown rings on the crown of the head, one cross band between the eyes, and two on the snout; a longitudinal streak runs from the eye to the nostril"). His adult BODY pattern is neither spots nor rings nor vermiculation: "In the adult only the brown edges of these bands remain; so that there is one pair of brown cross bars on the neck, and three pairs on the trunk, the space between the bars being of the ground-colour." Paired transverse bars. So Gunther supplies zero evidence for pale-centred rings on the trunk, and in fact describes an adult trunk pattern that the claim's framing ("spots", solid or annular) does not cover at all.

3. Two of the three photographs do not show what the detail says they show. I examined them at up to 7x: - Rajasthan (361924223) — this one DOES support the claim: the neck and anterior/mid trunk are an extensive dark reticulum with several closed annuli enclosing yellow-green centres. Caveat worth stating: it is a road-killed, sun-dried, flattened animal on tarmac, so its colour rendition is not that of a live gecko, and the observer's own note is "not sure about the id." - Second Jhampir animal (387949230, photo 2) — REFUTED. The neck and anterior trunk are solid, discrete, round dark spots on a yellow-olive ground. The only ring-shaped figures are on the crown of the head. Calling these "dark-edged, yellow-centred rosettes over the neck and anterior trunk" is wrong. (Separately: the two photos under that one observation look like different individuals under different lighting; I could not verify either way, so "the second Jhampir adult" may not be a second verified adult.) - Himachal (271474107) — OVERSTATED. At high zoom the head, neck and anterior trunk are solid dark-brown blotches on a white ground, some partly fused. The yellow is a dorsolateral wash of ground colour lying BETWEEN the blotches, occasionally enclosed where blotches merge — yellow as interstitial ground, not a ring drawn around a pale centre. And the enclosure effect is strongest on the mid/posterior trunk and around the hindlimb, i.e. the opposite of the claim's proposed "pale-centred rings anteriorly and on the head" gradient.

4. Terminological slippage. "Vermiculated with chestnut-brown and whitish" (Boulenger) states that the two colours interdigitate in worm-like fashion; it does not state which colour encloses which, and it is not a synonym for "annuli with pale centres." The claim merges the two.

ON THE FIRST HALF ("solid, NO pale halo"): I could not refute it, but note it is only weakly established. Neither Boulenger nor Gunther mentions any pale ring or halo around the dark marks, and none of the four photographs shows one — but two 19th-century descriptions saying nothing is an argument from silence, not a positive finding. Treat "no pale halo" as consistent-with-sources, not as documented.

ALSO WORTH FLAGGING (not a refutation, an open risk): the Reptile Database notes that Mirza 2024 proposed revalidating *Eublepharis madarensis*, whose type locality is the Madar foothills, Ajmer, RAJASTHAN — the same state as the strongest supporting photograph. The Reptile Database disputes that revalidation (no type information, unverifiable sequence vouchers) and still keeps Rajasthan inside *E. macularius*, so I did not treat the Rajasthan animal as a different species. But the one image carrying the claim comes from the population whose identity is currently contested, which is a thin base for a general statement about the species. - correction: Adult dorsal marks are dark blotches on a pale ground with no pale halo around them — consistent with every source and photograph I opened, though "no halo" rests on silence in the two historical descriptions rather than on a positive statement.

Some adults instead carry an open pattern in which the brown breaks up into vermiculation and closed annuli enclosing pale or yellow ground. The documented basis for this is Boulenger 1885: 232 alone: the adult head and body become "spotted or largely vermiculated with chestnut-brown and whitish", and "in some only the brown edges of the dorsal bands remain". I verified this geometry in one wild animal, iNaturalist obs. 361924223 (Rajasthan) — a road-killed, dried specimen, from the population Mirza 2024 proposed to split as *E. madarensis* (a revalidation the Reptile Database rejects as unverifiable).

Do NOT state a frequency. Nothing I opened quantifies how many wild adults show the open pattern; "a large minority" is unverified.

Do NOT cite Gunther 1864 for body rings. His "pair of brown rings" is on the crown of the head (p. 430), listed among head markings. His adult TRUNK is described as one pair of brown cross-bars on the neck and three pairs on the trunk — the retained edges of the juvenile cross-bands, with ground colour between. That is a third adult pattern, paired transverse bars, which neither "solid spots" nor "pale-centred rings" describes, and it should be recorded as such.

Do NOT state an anterior-to-posterior gradient. No source I opened describes one, and the claimant's own photographs contradict it: in obs. 271474107 (Himachal) the head, neck and anterior trunk are solid blotches while the enclosed yellow appears mid-body and toward the hindlimb, and in obs. 387949230 (Jhampir) the neck and anterior trunk are solid round spots with ring figures confined to the crown of the head.

Surface anatomy — what could not be verified at all

Gaps, stated rather than filled. Anything built into these areas is a modelling decision and has to be tagged as one.

TUBERCLES

1) MEASURED TUBERCLE DENSITY (tubercles per unit area) - not found anywhere, for any body region, in any source I opened. Russell et al. (2014) is the only quantitative study of leopard gecko skin surface I located, and it measures sensilla per mm2 and mean scale AREAS, never tubercle density or tubercle spacing. If you need a number, the only route is arithmetic I did not verify against any source: ~18-20 longitudinal tubercle rows across the dorsum (from Mirza 2024 on the Rajasthan/Delhi population) spread over the dorsal arc of an adult, with tubercles ~1.1-1.3 mm wide and interspaces at least one tubercle wide, implies centre-to-centre spacing of roughly 2-3 mm and therefore on the order of 0.1-0.25 tubercles per mm2 (roughly 10-25 per cm2). Treat that as my estimate, not a literature value.

2) TUBERCLE HEIGHT / PROTRUSION - no source measures how far a tubercle stands above the surrounding scale pavement, in mm or as a ratio. Only adjectives ("cone-like", "subconical", "pronounced", "dome shaped"). The height of the bumps in your model is an artistic decision.

3) WHERE TUBERCLES ARE DENSEST - I could not verify this for any body region. Russell et al. establish where tubercles are LARGEST (thigh > crus > pes > hips > tail base > head), which is not the same thing, and their prose says nothing about spacing varying by region. Any claim that the tubercles are densest along the mid-dorsum, or on the flanks, or at the tail base, is unsupported by what I read.

4) FORELIMB TUBERCULATION - unverified. Russell et al. sampled only hindlimb (thigh, crus, pes); Mirza (2024) mentions only the hind limbs. No source I opened states whether the upper arm and forearm carry tubercles. Do not assume symmetry with the hindlimb without checking a specimen or a high-resolution photograph.

5) THE INTERSPACE CONFLICT - three sources I opened (Boulenger 1885, Smith 1935, Mirza & Gnaneswar 2022) say the intertubercular gaps are as wide as or wider than the tubercles; the diagnosis that The Reptile Database attributes to Anderson (1999, The Lizards of Iran, p.120) says mid-dorsal tubercles are generally LARGER than intertubercular spaces. I could not open Anderson 1999 to check context (it may be a comparison against E. angramainyu specifically), so the conflict is unresolved. I weighted the three opened sources.

6) DORSAL TUBERCLE ROW COUNT FOR E. MACULARIUS SENSU STRICTO - not obtained. Mirza et al. (2014) Phyllomedusa 13(2):75-90, the likeliest source, is gold open access but revistas.usp.br refused all connections from this network (ECONNREFUSED on repeated attempts, via both WebFetch and curl, direct and via DOI); OpenAlex lists no mirror PDF. I did not cite it. Also not opened: Grismer 1988 and 1991 (in Estes & Pregill, Phylogenetic Relationships of the Lizard Families; and Grismer 1991), which are the primary source for the "widely spaced, pronounced and pointed dorsal tubercles" wording I quote at second hand from The Reptile Database.

7) SOURCES I TRIED AND FAILED TO OPEN, so nothing in this report rests on them: Peacock, Gilbert & Vickaryous (2015) J. Anat. 227:596-610 on leopard gecko cutaneous wound healing (Wiley returned HTTP 403, no PMC copy); Szydlowski, Madej & Mazurkiewicz-Kania (2016) Acta Zoologica, on chromatophore ultrastructure in leopard gecko skin (paywalled) - it may contain histological thickness data for tubercle vs interscale skin; Anderson (1999) The Lizards of Iran; Szczerbak & Golubev (1996) Gecko Fauna of the USSR.

8) A TABLE ERROR YOU SHOULD KNOW ABOUT - Russell et al. (2014) Table 2 prints the "between shoulders" tubercle area as 0.141 mm2, which is smaller than every other tubercle measurement and equal to the granular scale in the same row; it contradicts the paper's own statement that the smallest tubercles are on the head. I rendered the published page to confirm the printed value. It is almost certainly a decimal-point error (probably 1.41). I excluded it rather than guess.

9) NO SOURCE OPENED describes tubercle arrangement on the neck, in the axilla/groin, on the eyelid margin, or around the cloaca/precloacal pore region, beyond the note that tubercles extend "up to the caudal constriction" (Mirza 2024, madarensis population). Those areas are unmapped.

EYE

Gaps I did not fill, rather than guess at:

1. WHICH LID DOES THE CLOSING IN E. MACULARIUS. No source I opened states it for this species. Smith 1935 says both lids are movable in Eublepharis; the lower-lid-does-the-work claim is a general lizard/reptile rule (Smith 1935 p.10; Wyneken 2012 ARAV). Treat as likely, not established.

2. LID EXCURSION. No measurement of how far, how fast, or at what angle the lids travel; "connivent" in Boulenger/Smith establishes full closure but nothing quantitative. Sleeping-with-eyes-closed and blink-rate claims appear only in pet-care sources, which I did not accept.

3. WHETHER THE SLIT IS DENTICULATE / BREAKS INTO PINHOLES. Smith's general Gekkonidae text mentions denticulated slits and pin-point openings, but his Eublepharis diagnosis says only "pupil vertical". The famous four-pinhole result (Roth et al. 2009) is Tarentola chazaliae, a different family, and does not mention Eublepharis at all — as is the "350x better than human" sensitivity figure. Both are misattributed to leopard geckos across the popular web. My photographic read suggests a smooth-margined slit, but n=1.

4. A FORMAL PUBLISHED DESCRIPTION OF THE WILD-TYPE IRIS. None found in any taxonomic, field-guide or veterinary source I opened. The reticulated dark-with-silver-veining description is my own reading of high-resolution photographs, and the best-resolution eye I could find was a captive high-yellow animal, not a wild-caught wild-type. Wild-type iris colour may be cooler/greyer. Only weak (breeder) corroboration that the "normal" iris is patterned.

5. GLOBE AND CORNEAL DIAMETER. Camacho-Luna et al. 2020 measured ocular dimensions in 26 leopard geckos but the values are not in the abstract and the full text is paywalled. The 19-21% eye/head ratio I report is the exposed aperture, not the globe.

6. EYE-SIZE SAMPLE IS THIN. The ED/HL ratio rests on ONE publication and TWO specimens, with ED rounded to whole mm. A congener (E. angramainyu) gives 0.29 rather than 0.19-0.21, and I could not resolve whether that is a species difference or a difference in measurement convention.

7. NICTITATING MEMBRANE. Bellairs 1948 reportedly found "probably a small nictitating membrane" in Coleonyx — a different eublepharid genus — but that paper is paywalled and I did NOT open it; I have only a search-engine summary, so I am not asserting it. Nothing at all for Eublepharis.

8. SCLERAL OSSICLE COUNT. Present (DigiMorph CT), but the commonly quoted "20 or more" traces to a hobbyist page I could not open (TLS certificate failure). Uncounted for E. macularius.

9. SOURCES I TRIED AND FAILED TO OPEN: Bellairs 1948 Proc. Zool. Soc. Lond. 118:420-425 (paywalled); Mirza et al. 2014 Phyllomedusa 13(2):75-90, the modern E. macularius comparative description (revistas.usp.br refused connection repeatedly); "Ophthalmology of Gekkota: Geckos" in Wild and Exotic Animal Ophthalmology (Springer, login wall); the Reptile Database full diagnosis (deliberately withheld by the site). Any of these could sharpen points 1, 3 and 4.

EAR

Several things I looked for and could not document, listed so you do not mistake inference for evidence.

- (1) There is no modern redescription of *E. macularius* giving its own ear-opening shape, orientation and EL/OD ratio. Every *Eublepharis* description with those numbers (Mirza et al. 2014; Mirza & Gnanaswar 2022; Mirza 2024; Mohapatra et al. 2026) is for a congener. For *E. macularius* the only species-specific figure I found is Grismer's 'height of auditory meatus 1.5 times internostril distance', and I read that in the Reptile Database compilation, not in Grismer (1988), which is a Stanford University Press book chapter I could not obtain. The classical descriptions (Boulenger 1885, 1890; Smith 1935) write the head and ear characters in the *E. hardwickii* account and then list *E. macularius* as differing only in tubercle size, preanal pore count and body/digit proportions — so those ear statements reach *macularius* by the authors' own 'differs from the preceding by...' construction, not as direct observations of *macularius*. I have flagged that everywhere it applies.
- (2) No absolute dimensions. I found no published ear-opening length or height in millimetres for *E. macularius*. Mirza (2024) does tabulate EL and OD in mm, but the columns in the extracted PDF table are misaligned against their row labels and I will not quote numbers I cannot align with confidence. Werner & Wever (1972), 'The function of the middle ear in lizards: *Gekko gekko* and *Eublepharis macularius*', does report tympanic membrane areas for this species, but Wiley returned HTTP 403 and I could not read it; I am therefore not quoting any membrane area.
- (3) Meatus depth. I could not find a measured depth for the *E. macularius* external auditory meatus. 'Ear opening deep' is Mirza et al.'s wording for *E. satpuraensis*; 'a shallow external ear canal' is Christensen-Dalsgaard & Manley's for *Gekko gekko*. Those two are not obviously consistent, and neither is about *macularius*. Treat the recession as qualitative: present, a few millimetres, not quantified.
- (4) See-through head. The red-laser-through-the-head demonstration is published for *Gekko gekko* only. *Gekko* is in Gekkonidae, a different family from Eublepharidae; the split between them is deep within Gekkota, so this is a distant transfer, not a near relative. I found no peer-reviewed source showing or stating it for *E. macularius*. Pet-care and hobbyist pages assert it confidently; those are weak sources and I did not count them.
- (5) Exact anatomical direction of the extracolumella bar. The leopard gecko photograph caption places the pale extracolumella protrusion at the 'top-left' of the tympanum, but that is an image coordinate and I cannot tell which side of the head was photographed or how it was oriented, so I cannot say whether that is anterodorsal or posterodorsal on the animal.
- (6) Whether *E. macularius* has the three conical tubercles on the anterior wall of the ear opening that Mirza (2024) describes for *E. madarensis*. Unrecorded either way.
- (7) I could not open: Grismer (1988); Wever (1978) The Reptile Ear; Werner & Wever (1972); Werner et al. (2005, 2008); Anderson (1999) The Lizards of Iran. Several of these are the actual primary sources behind statements I have cited at one remove, and each is flagged as such in the relevant finding.

One correction to the premise in the brief: geckos do not have unusually open or superficial ear canals. The opposite is the case. Evans's lepidosaur ear chapter treats the flush, near-surface tympanum as the common lizard condition and singles out geckos as a group in which the membrane is recessed more deeply below the surface and protected by a meatal closure muscle. What is true is that geckos have a large, conspicuous, unscaled external ear opening — visible, but set at the bottom of a short funnel.

FEET

Five things I could not document and am explicitly NOT filling in:

1. WEBBING. No source I opened — Boulenger (1890), Smith (1935), the Reptile Database diagnosis, or any of the three modern *Eublepharis* species descriptions — mentions interdigital webbing for *Eublepharis*, in either direction. Gecko genus diagnoses do routinely mention webbing when it is present (e.g. Smith's treatment of digit form across genera is detailed), so silence is weak evidence of absence, and Zhuang et al. (2019) measure free interdigital angles in the *E. macularius* pes, which implies separate digits. But no source states "webbing absent". Treat as: almost certainly no significant webbing, formally undocumented. Model toes as free, with at most a low basal skin fold.
 2. TOE PATTERNING / BANDING. I found no description of digit or foot colouration for *E. macularius*. Boulenger (1890) and Smith (1935) describe head, body and tail pattern in detail and say nothing about the limbs or feet. The closest datum is the congener *E. jhuma*: "Limbs paler than body color with whitish spots on the legs" (Mohapatra et al. 2026) — and that refers to the legs, not the toes. Whether *E. macularius* toes are banded, speckled or plain is unverified. Do not invent body-style banding on the toes.
 3. *E. MACULARIUS*-SPECIFIC DIGIT MEASUREMENTS. Powell, G.L., Osgood, G.J. & Russell, A.P. (2018) "Ontogenetic allometry of the digital rays of the leopard gecko (*Gekkota*: Eublepharidae; *Eublepharis macularius*)", *Acta Zoologica* 99:331-356, DOI 10.1111/azo.12215 is exactly the paper for toe proportions — metapodial and digit scaling in manus and pes across ontogeny. It is paywalled at Wiley; the Wiley page is behind a bot-verification wall which I did not attempt to bypass, ResearchGate returned 403, and no open repository copy exists. I confirmed its existence and full citation via the Crossref API only. All my digit-proportion figures are therefore transferred from congeners. If you want *E. macularius* numbers, this is the paper to obtain.
 4. SUBDIGITAL LAMELLA COUNT FOR *E. MACULARIUS*. I have counts for congeners (*E. jhuma* 22-25 on the fourth toe; *E. hardwickii* 17; *E. pictus* 19; *E. satpuraensis* 18-22; *E. turcmenicus* 20-23; *E. madarensis* 17-19) but the diagnoses I opened never state the *E. macularius* range, because tuberculate-vs-smooth texture rather than count is the character used to separate it. A figure in the 15-25 range is a reasonable interpolation but I am not asserting it.
 5. WHICH SPECIES KOPPETSCH ET AL. (2020) EXAMINED. The abstract says "three species of Eublepharidae" without naming them; full text paywalled. Search-engine summaries asserted the genera were *Aeluroscalabotes*, *Goniurosaurus* and *Eublepharis*, but I did not open a source confirming that, so I have not treated it as established.
- PROCESS NOTE ON SOURCE QUALITY: the two strongest sources here are Boulenger (1890) and Smith (1935), both read as full scanned text from the Internet Archive. They are old but are formal taxonomic diagnoses and they agree with each other independently and with modern osteological work (Russell & Gamble 2019). No pet-care or breeder sites were used for any claim. The single most important correction to the common framing: leopard geckos DO have subdigital lamellae — a row of transverse tuberculate plates — they lack adhesive pads, scansors and setae. Sources that say "no lamellae" are wrong.

HEAD_SCALES

Gaps, stated plainly rather than filled in.

1. NO MODERN FULL REDESCRIPTION OF *E. MACULARIUS* EXISTS THAT I COULD OPEN. This is the central problem. Every detailed modern *Eublepharis* head description I found (Mirza & Gnanewar 2022; Mirza 2024) is of a congener — *E. hardwickii*, *E. pictus* or *E. madarensis*. The only full head scalation text that applies to *E. macularius* is Smith (1935) and Boulenger (1885/1890), and in both works the *E. macularius* account is written purely as a short list of differences from *E. hardwickii* (dorsal tubercle size, preanal pore count, proportions, colour) — so every head character is inherited by taxonomic convention rather than written out for the species. I have flagged each inherited statement.
2. EXTENT OF THE ROSTRAL'S MEDIAN GROOVE IS CONTRADICTIONARY. Boulenger 1885/1890 say the cleft is 'above' (upper part only); Mirza's congener descriptions say the rostral is 'divided by a median suture for its entire length'. I could not determine which applies to *E. macularius*. If the groove depth matters to your model, treat it as a choice.
3. NO MALE-vs-FEMALE HEAD WIDTH NUMBERS FOR *E. MACULARIUS*. I could not obtain any paper reporting mean head width by sex in millimetres for this species. Kratochvíl & Frynta (2002) is the right source and I read its abstract, but the species table is paywalled, so I never saw the *E. macularius* row confirming male-biased HSD for this species specifically. Several search-result snippets asserted 'males have wider heads than females' attributing it to Gutzke & Crews 1988, but the aggregator page carrying that claim returned HTTP 403 and I never opened it, and the Gutzke & Crews 1988 Nature abstract I did read is about temperature-dependent sex determination and does not mention head width. I have therefore NOT cited it. The male head widening is reported as 'likely', resting on the family-level combat/HSD argument plus verified male-male aggression in *E. macularius*.
4. THE JAW-ANGLE LOCATION OF THE WIDENING IS INFERRED, NOT DOCUMENTED. No source I opened says 'the head widens at the jaw angle in mature males'. The inference chain is: morphometric protocols define head width as the largest width of the head; enlarged male heads function as biting weapons; bite force comes from the jaw adductor chamber, which lies at the jaw angle. Reasonable, but it is my reasoning, not a citation.
5. THE VISIBLE MOUTH LINE IS NOT DOCUMENTED ANYWHERE I LOOKED. Marked unverified above. Only indirect support (labials counted separately per jaw, 'to angle of jaw').
6. NO SOURCE STATES WHETHER *E. MACULARIUS* LABIALS ARE PALER AS A RULE. The closest is a single statement about *E. madarensis* labials lacking the head's reticulation, plus the verified negative that *E. macularius* lacks the pale supralabial stripe that *E. satpuraensis* and *E. hardwickii* have. Single-source, transferred, flagged.
7. COULD NOT ACCESS: Anderson, S.C. 1999, The Lizards of Iran (the key modern source for *E. macularius* head scalation — I only read the Reptile Database's quotations of it); Mirza et al. 2014, *Phyllomedusa* 13: 75–90 (the *E. satpuraensis* description with its *E. macularius* comparison table — the journal host refused all connections); Kratochvíl & Frynta 2002 full text; Grismer 1988; Szczerbak & Golubev 1996. ScienceDirect, Wiley, ResearchGate and the Biodiversity Heritage Library all returned 403 to automated fetches.
8. NOT USED: no pet-care blog, breeder site, care sheet or hobbyist magazine contributed any fact above. I found a Khan article in *Reptilia* 45 (a hobbyist magazine) that would have had morphology notes but did not cite it, since it is a weak source and I could not open it anyway.

PATTERN

SOURCES I COULD NOT OPEN: (1) Blyth 1854, *J. Asiatic Soc. Bengal* 23: 737–738, the original description of *Cyrtodactylus macularius* from the Salt Range, Punjab — the relevant JASB volume was not retrievable from archive.org (the biostor item matching the title turned out to be a different 1855 Blyth paper). So I never read the original description and cannot say what Blyth himself wrote about colour. (2) The Reptile Database species account deliberately withholds roughly half a page of descriptive text from automated retrieval, stating it is available only to contributors — so I could not read its description or coloration section. (3) Mirza et al. 2014, *Phyllomedusa* 13(2): 75–90 (*E. satpuraensis*), which would contain a modern comparative colour diagnosis — *revistas.usp.br* refused connection and the ResearchGate copy returned 403. (4) Whimster 1979, *J. Exp. Zool.* 208: 153–160, the primary grafting study of leopard gecko pattern ontogeny (cited as ref. 7 by the PNAS paper) — paywalled. (5) Börner's 1970s revision naming the subspecies *afghanicus*, *fasciolatus*, *montanus* and *smithi* — not opened, so I make no claim about geographic subspecific colour differences. (6) Khan's Pakistani natural-history account — both the ResearchGate PDF and academia.edu copies were blocked.

SPECIFIC GAPS: (a) No published quantification of spot size, count or density for WILD adult *E. macularius* exists among the sources I read. The only quantification I found (Jančúchová-Lásková et al. 2015) covers HEAD spots only and reports square-root- and log-transformed values whose underlying table I could not extract. My spot-size figures are my own estimate from one magnified photograph and are flagged unverified. (b) No source I opened states in words that the adult tail is paler than the body; I inferred it from the chromatophore histology plus photographs. (c) No source gives the relative TIMING of body-band versus tail-band breakup; the observation that the tail retains bands longer rests on three photographs of subadults. (d) I found no statement about whether wild adults ever retain complete tail rings into full adulthood. (e) I could not verify whether ground colour shifts with the shed cycle, time of day or temperature — three Kirthar animals looked strikingly pale pinkish-lilac compared with the Sindh lowland animals and I do not know whether that is population, shed state, or flash photography at night. (f) I did not verify geographic variation in adult ground colour across the range in any systematic way.

SOURCE-STRENGTH CAVEATS: The iNaturalist observations are wild animals in the native range, and I opened and viewed every photograph I cite — but they are community-identified, not expert-vouchered museum specimens, so they are weaker than the taxonomic literature. I weighted the Pakistani records (including one from Chiddru in the Salt Range, effectively the type locality) above the Indian ones. Günther 1864 describes his material under the name *Eublepharis fasciolatus*; I treat it as *E. macularius* only because both Boulenger 1885 and Smith 1935 — which I opened — place *fasciolatus* in the synonymy of *macularius*, and its type locality (Hyderabad, Sindh) is well within the range. No fact in this report has been transferred from a different *Eublepharis* species; where *E. hardwickii* or *E. angramainyu* appear above it is only as an explicit contrast, never as a substitute.

Lateral spine bending — the 18 numbers that survived

24 agents, four search angles, then verifiers told to refute each number from the opened source. The one that matters is the second card: a SAME-SPECIES measurement of pelvic girdle yaw on a level trackway, which is what the walker's spine was eventually fitted against.

GECKOS · VERIFIED

Maximum lateral trunk bending of ~20 degrees per step during level trotting; trunk is straight (rectilinear) at midstance and bends alternately left/right, peaking near footfall of each diagonal couplet.

Value: Lateral flexion greatest at step phase 0.03 +/- 0.04 (initial footfall), magnitude 20.1 +/- 6.32 deg; rectilinear through midstance; second maximum at phase 0.81 +/- 0.08, magnitude 19.4 +/- 5.60 deg (means +/- s.d.). Fig. 4A y-axis 'Lateral bending (deg.)' spans -40 to +40 with 0 = straight; representative trace for a 2.3 g animal at 0.50 m/s. Conditions: level balsa-wood track over a force platform, trotting gait (diagonal limb phase 0.92 +/- 0.06), speeds 0.28-0.75 m/s (mean 0.47 +/- 0.13 m/s), stride frequency 13 +/- 2 Hz. Body: mass 2.55 +/- 0.27 g, SVL 5.57 +/- 0.27 cm, N = 5. Measured from three dorsal-midline paint marks (pectoral point, second of three trunk points i.e. mid-trunk, pelvic point) filmed by 3 cameras at 1000 frames/s; the paper does not state the exact geometric formula (presumably the angle between the pectoral-midtrunk and midtrunk-pelvic segments, deviation from straight).

Species: Hemidactylus garnotii (Gekkonidae), same infraorder (Gekkota); different family from Eublepharidae

Source: Chen J.J., Peattie A.M., Autumn K., Full R.J. (2006) Differential leg function in a sprawled-posture quadrupedal trotter. J. Exp. Biol. 209: 249-259. doi:10.1242/jeb.01979

What was actually opened: JEB HTML full text (journals.biologists.com) read twice; PDF downloaded from polypedal.berkeley.edu (081_Chen_...pdf) and text-extracted: Materials and methods (Animals, Kinematics, GRF), Results paragraph 'Lateral flexion was greatest...', Fig. 4 caption and axis labels.

GECKOS · VERIFIED

In the target species, lateral rotation (yaw) of the pelvic girdle during level locomotion has a mean angular excursion of roughly 50 degrees per stride with the intact tail, falling to roughly 33-34 degrees when tail undulation is restricted or the tail is autotomised. This is girdle yaw, not a trunk-curvature angle; the paper does not report a mid-trunk bending angle.

Value: Fig. 4B (means +/- s.e.m., n = 10): angular excursion of pelvic girdle in yaw approx. 51 deg (error bar ~48-54) original tail; approx. 33 deg (~28-38) restricted; approx. 34 deg (~31-37) autotomised - values read off the axis of the downloaded figure, not stated numerically in text or Table 1. Fig. 4A representative stride (original tail): pelvic girdle rotation swings from approx. -38 deg (near 5% stride) to approx. +46 deg (near 63% stride); negative = rotated toward the observed right hind limb. Stats in text: restricted vs original t = 2.287, P = 0.048; autotomised vs original t = 3.129, P = 0.012. Conditions: level 1.0 x 0.13 m sandpaper trackway; speeds 0.59-3.36 SVL/s across all trials, mean 1.23 +/- 0.16 SVL/s (original tail), 1.50 (restricted), 1.85 (autotomised); hind-limb duty factor 0.77-0.78 (Table 1), i.e. walking-type gait, gait not formally classified. Body: SVL 104.6 +/- 2.1 mm, mass 36.3 +/- 1.9 g, 10 adults. Markers: dorsal midpoint of body, centres of pectoral and pelvic girdles, right limb joints, five tail points; two Edgertronic cameras at 250 frames/s. Methods define the variable only as pelvic girdle rotation 'in the yaw axis'.

Species: Eublepharis macularius (Eublepharidae), same species

Source: Jagnandan K., Higham T.E. (2017) Lateral movements of a massive tail influence gecko locomotion: an integrative study comparing tail restriction and autotomy. Scientific Reports 7: 10865. doi:10.1038/s41598-017-11484-7 (PMC5589804)

What was actually opened: PMC full text (read twice); Europe PMC full-text XML (Methods, Results, Table 1 in full, captions of Figs 2 and 4); Fig. 4 image (41598_2017_11484_Fig4_HTML.jpg) downloaded from Springer Nature static media and inspected.

GECKOS · VERIFIED

In the target species the tail (caudal vertebral column) is laterally undulated during level locomotion with tail-tip excursions of up to ~3 cm (~0.3 SVL) relative to the pelvic girdle at slow speed, decreasing with speed.

Value: Fig. 2 (per-individual means +/- s.e.m., original tail): lateral displacement of tail tip relative to pelvic girdle approx. 0.034 m at 0.6 SVL/s, approx. 0.019-0.021 m at 0.8-1.1 SVL/s, approx. 0.003-0.007 m at 1.9-2.3 SVL/s; regression y = -0.0127 x + 0.0279 (m vs SVL/s), R² = 0.423, P = 0.042. Restricted tail: approx. 0.002-0.005 m. Measured from top-view video (250 frames/s) as lateral distance of the tail-tip marker from the pelvic-girdle marker, left side of the body; yaw plane only. SVL 104.6 +/- 2.1 mm, mass 36.3 +/- 1.9 g. Values read off the figure axis.

Species: Eublepharis macularius (Eublepharidae), same species

Source: Jagnandan K., Higham T.E. (2017) Scientific Reports 7: 10865 (as above)

What was actually opened: Fig. 2 image (41598_2017_11484_Fig2_HTML.jpg) downloaded from Springer Nature static media and inspected; Fig. 2 caption and Results text from the Europe PMC XML.

GECKOS · LIKELY

Maximum one-sided rotation range of the pectoral and pelvic girdles of the Tokay gecko trunk during one trot cycle is about 33-34 degrees (secondary citation of a motion-capture study of trunk lateral undulation on horizontal, 45-degree and vertical planes).

Value: Pectoral girdle angle: maximum unilateral fluctuation range 33.56 +/- 3.01 deg; pelvic girdle angle: 33.22 +/- 1.72 deg, 'during one trot gait cycle'. Plane (horizontal / 45 deg / vertical), speed and body size are NOT stated in the citing paper. The primary study (Wang et al. 2020) used an optical motion-capture system, 29 reflective markers (10 on trunk, 5 on tail, 12 on legs, 2 on head per the Manoonpong-group description of the same dataset), found a standing-wave trunk waveform at low speed changing to a head-to-tail travelling wave at high speed, and states the lateral undulation pattern results from girdle rotation plus axial muscle activity. Wang et al. 2021 (Zoology 145: 125892) additionally measured 13 angular variables including trunk flexion, girdle rotation and trunk deflection on the vertical plane, but the numerical values are behind a paywall and were not read.

Species: Gekko gekko (Gekkonidae), same infraorder (Gekkota); different family from Eublepharidae

Source: Primary: Wang W., Ji A., Chen G., Ravi S., Shen H., Gorb S.N., Dai Z. (2020) Kinematics of gecko climbing: the lateral undulation pattern. Zoology 140: 125768. doi:10.1016/j.zool.2020.125768. Numbers taken from the secondary source: 'A Gecko-Inspired Robot with a Flexible Spine Driven by Shape Memory Alloy Springs', Soft Robotics 2023 (PMC10442688), section 'Collaborative model of flexible spine and limbs', citing Wang et al. 2020 as its ref. 39.

What was actually opened: PMC10442688 full text (read twice, including the paragraph containing the 33.56/33.22 deg values and its reference list); Europe PMC abstract records for PMIDs 32279015 and 33550221; ADS/Crossref/Semantic Scholar/figshare (RMIT 27530589) records for Wang 2020 - no accessible file; ScienceDirect and ResearchGate pages returned 403. Primary full text NOT read.

Sceloporus clarkii walking/trotting on a treadmill uses a standing wave of lateral bending with two nodes (pectoral girdle; penultimate trunk vertebra just anterior to the pelvis). Lateral excursion of the mid-trunk landmark was ~18 mm at 0.27 and 0.48 m/s and ~13 mm at 0.83 m/s (means +/- SD: 18.0+/-2.0, 17.6+/-1.2, 13.4+/-2.2 mm); pectoral node 10.5, 9.5, 5.5 mm; pelvic node 10.0, 9.7, 9.7 mm; snout 20.8, 15.2, 11.0 mm; tail peak 18.0, 21.2, 24.2 mm. Only the snout displacement changed significantly with speed ($P=0.028$); trunk peak and node amplitudes did not ($P>0.067$). Node positions did not shift with speed. Trunk excursion is therefore roughly 0.15-0.20 SVL peak-to-peak and speed-invariant over a 3-fold speed range (walking trot; duty factor approaching 0.5 at the top speed). Note the paper reports lateral DISPLACEMENT of dorsal landmarks (mm, from the dorsal video y-coordinate, Fig. 4 stick-figure overlays), not a trunk bending angle.

Value: Mid-trunk lateral excursion 18.0 / 17.6 / 13.4 mm (~0.20 / 0.19 / 0.15 SVL) at 0.270 / 0.476 / 0.833 m/s; no significant change with speed. SVL 90-92 mm. Gait: walking trot. Measured at 4 equally spaced trunk landmarks between pectoral girdle and pelvis, dorsal view, 200 fields/s.

Species: *Sceloporus clarkii* (Phrynosomatidae (Iguania)), other (different infraorder; Iguania vs Gekkota)

Source: Reilly, S.M. & Delancey, M.J. (1997) Sprawling locomotion in the lizard *Sceloporus clarkii*: the effects of speed on gait, hindlimb kinematics, and axial bending during walking. *J. Zool. Lond.* 243: 417-433. PDF: https://www.originalwisdom.com/wp-content/uploads/bsk-pdf-manager/2019/10/Reilly-and-Delancy_1997_Sprawling-locomotion-in-the-lizard-Sceloporus-clarkii.pdf

What was actually opened: Full PDF (17 pages) extracted to text; read Methods (landmarks, speeds, 'Axial bending' section), Results ('Axial movements'), Table VI (axial displacements), Fig. 4 (rendered page 428 and viewed), Discussion pp. 429-431.

In the same *S. clarkii* study, maximum pelvic rotation relative to the direction of travel (dorsal-view angle between the pelvic line and travel direction; one-sided maximum, PELVMAX) was 16-25 degrees across all individuals and speeds and did not vary with speed; peak pelvic rotation occurred at 44-51% of stride duration (just before foot-up, i.e. 81-85% of stance). The authors state this range matches values for 'several other lizard species (Ritter, 1992)', for alligator (Gatesy 1991) and salamander (Ashley-Ross 1994).

Value: PELVMAX (mean +/- SE, degrees): 16+/-7 (ind. 3, 0.270 m/s); 25+/-3 and 21+/-2 (inds. 1, 2 at 0.476 m/s); 21+/-2 and 19+/-2 (inds. 1, 2 at 0.833 m/s). One-sided maximum of pelvic yaw from travel direction; SVL 90-92 mm; walking trot.

Species: *Sceloporus clarkii* (Phrynosomatidae (Iguania)), other

Source: Reilly & Delancey (1997) *J. Zool. Lond.* 243: 417-433 (same PDF as above)

What was actually opened: Table I (PELVMAX row), Fig. 1 legend defining pelvic angle, Appendix variable definitions, Discussion p. 429 ('Pelvic and axial movements remained constant...').

Across 44 lizard species in four families (320 dorsal-view video sequences, DeepLabCut tracking of 22 points), in-vivo 'spine apex ROM' (sum of the maximum deviations of the mid-spine marker to either side of the shoulder-girdle-to-hip-girdle axis, expressed in degrees; thoraco-lumbar spine only) clusters between 15 and 30 degrees; the most common strategy is 10-30 degrees of spine ROM with 60-130 degrees of limb ROM. Family means: dragons (Agamidae) 20.3, geckos (Gekkota) 23.6, skinks 27.6, varanids 19.6 degrees. Direction means: upward climbing 24.8, downward 16.4, level walking 16.6 degrees (level trials were varanids only, $n=51$). Speed classes: slow 22.7, medium 24.0, fast 19.4 degrees, i.e. faster trials had decreased spine ROM and increased leg ROM. Caveat for the target: the gecko data ($n=154$ sequences) are from vertical climbing, not level ground, and I could not read the abbreviated gecko species labels in Fig. 6a at the rendered resolution; *Eublepharis* is not identifiable among them.

Value: Spine apex ROM 15-30 degrees (cluster); family means 19.6-27.6 degrees; level walking (varanids) 16.6 degrees; slow/medium/fast 22.7/24.0/19.4 degrees. Speed classes are categorical (thresholds not given in main text); body sizes not given per species in main text (Supplementary S1, Fig. D1, not accessed).

Species: 44 species: Agamidae ($n=74$ seqs), Gekkota ($n=154$), Scincidae ($n=21$), Varanidae ($n=71$) (Agamidae, Gekkota (multiple gecko families), Scincidae, Varanidae), same infraorder for the gecko subset (Gekkota); other for the rest

Source: Rockenfeller, R., Cieri, R.L., Schultz, J.T., Maag, R. & Clemente, C.J. (2024) Coordinating limbs and spine: (Pareto-)optimal locomotion in theory, in vivo, and in robots. *npj Robotics* 2:4. <https://doi.org/10.1038/s44182-024-00011-2> (dataset from Schultz et al. 2021 *Integr. Comp. Biol.* 61:1769-1782)

What was actually opened: Full PDF (13 pages) extracted to text; read Results (Fig. 2 text, 'lizards cluster between 15 and 30 degrees'), Table 1 (cluster means), Methods 'Data collection and processing' (spine ROM definition), Fig. 2 (rendered p. 3, viewed), Fig. 6 (rendered p. 9, viewed), reference 16.

Eublepharis macularius (the target species) walking on a level trackway: yaw rotation of the pelvic girdle over a stride has an angular excursion of about 50 degrees with the intact tail (representative stride swings from about -38 to +45 degrees), falling to about 32-33 degrees when the tail is restricted with a graphite rod or autotomized. Lateral displacement of the tail tip (relative to the pelvic girdle) decreased significantly with walking speed ($P=0.042$). Trunk (mid-body) lateral bending itself was NOT quantified, even though a dorsal midpoint marker was placed.

Value: Pelvic girdle yaw excursion ~50 degrees (intact tail), ~32 degrees (restricted), ~33 degrees (autotomized); values read from Fig. 4B (means +/- s.e.m., $n=10$). Speed 1.23+/-0.16 SVL/s (intact). Mass 36.3+/-1.9 g, SVL 104.6+/-2.1 mm. Level trackway, walking (hind-limb duty factor 0.78).

Species: *Eublepharis macularius* (Eublepharidae), same species

Source: Jagnandan, K. & Higham, T.E. (2017) Lateral movements of a massive tail influence gecko locomotion: an integrative study comparing tail restriction and autotomy. *Sci. Rep.* 7:10865. PDF: http://biomechanics.ucr.edu/Jagnandan_Higham_2017.pdf

What was actually opened: Full PDF (8 pages) extracted to text; read Abstract, Results (pelvic girdle rotation paragraph), Table 1, Fig. 2 legend, Fig. 4 (rendered p. 4 and viewed), Discussion, Materials and Methods (study organisms, digitized points).

Two further kinematic studies of *Eublepharis macularius* on level ground measured limb joint angles but no trunk bending or pelvic rotation: Fuller et al. (2011) (3-D hindlimb kinematics of *E. macularius* and *Teratoscincus scincus* over a range of speeds; the track was made wide enough 'for lateral undulations' but none were quantified) and Jagnandan & Higham (2018) (EMG + limb joint angles before/after autotomy; mass 30.7+/-1.7 g, SVL 110.0+/-1.5 mm; only a qualitative statement that the tail base flexes toward the protracted hind limb each cycle). Zaaf et al. (2001) measured only spatio-temporal gait variables for *E. macularius* (level speeds ~0.2-1.1 m/s per abstract).

Value: No trunk-bending values exist in these three papers for *E. macularius*.

Species: *Eublepharis macularius* (also *Teratoscincus scincus*) (Eublepharidae (Teratoscincus: Sphaerodactylidae)), same species

Source: Fuller, Higham & Clark (2011) *Zoology* 114:104-112 (PDF http://biomechanics.ucr.edu/Fuller_etal_2011.pdf); Jagnandan & Higham (2018) *J. Exp. Biol.* 221:jeb179564; Zaaf, Van Damme, Herrel & Aerts (2001) *J. Exp. Biol.* 204:1233-1246

What was actually opened: Fuller 2011: full PDF extracted, grep for bending/undulation/pelvic rotation terms (only the trackway-width sentence and a Reilly & Delancey citation hit). Jagnandan & Higham 2018: JEB full-text page read via fetch summary (variables listed). Zaaf 2001: JEB abstract page only.

Walking *S. clarkii* uses a 'rough/loose standing wave' of axial bending with two nodes; node positions and node/peak amplitudes do not change across a 3-fold speed increase (only snout displacement falls). The trunk antinode lies between the nodes.

Value: Nodes: pectoral girdle (axial landmark 3) and axial landmark 8 = penultimate trunk vertebra, dorsal to pubic symphysis (just anterior to pelvis). Table VI lateral displacements (mm, mean $\pm$ SD; dorsal-view treadmill video, landmark lateral range from stick-figure superposition of all frames of a stride; paper does not state whether this is peak-to-peak or half-amplitude): trunk max 18.0 $\pm$ 2.0 / 17.6 $\pm$ 1.2 / 13.4 $\pm$ 2.2 mm at 0.270 / 0.476 / 0.833 m s⁻¹ ($\approx$ 0.20, 0.19, 0.15 SVL; n.s. vs speed, $P > 0.067$); pectoral node min 10.5 $\pm$ 6.4 / 9.5 $\pm$ 2.3 / 5.5 $\pm$ 2.0 mm (n.s.); pelvic node min 10.0 $\pm$ 0.1 / 9.7 $\pm$ 2.0 / 9.7 $\pm$ 1.7 mm ($\approx$ 0.11 SVL; n.s.); snout 20.8 $\pm$ 2.4 / 15.2 $\pm$ 3.1 / 11.0 $\pm$ 2.6 mm (decreases, $P = 0.028$); tail 18.0 / 21.2 / 24.2 mm (tends to rise, n.s.). Conditions: SVL 90–92 mm; 0.270, 0.476, 0.833 m s⁻¹ ($\approx$ 3.0, 5.2, 9.1 SVL s⁻¹); walking trot (hind duty factor 0.64 $\rightarrow$ 0.52); stride freq 3.4–5.8 Hz; 27–30 °C; 200 fields s⁻¹; axial data from 1 individual per speed (n=2 strides slow; n=5 at each faster speed).

Species: *Sceloporus clarkii* (Phrynosomatidae (Iguania)), other (different infraorder: Iguania vs Gekkota)

Source: Reilly, S.M. & DeLancey, M.J. (1997) *J. Zool. Lond.* 243: 417–433. 'Sprawling locomotion in the lizard *Sceloporus clarkii*: the effects of speed on gait, hindlimb kinematics, and axial bending during walking.' PDF: [originalwisdom.com mirror](http://originalwisdom.com/mirror/doi/10.1111/j.1469-7998.1997.tb02791.x), doi 10.1111/j.1469-7998.1997.tb02791.x

What was actually opened: Full text (all 17 PDF pages via extracted text layer): Materials & methods ('Axial bending', p. 421), Results 'Axial movements' (p. 421–422), Table VI (p. 427), Fig. 4 caption (axial stick figures, p. 428; note the text calls this Fig. 3), Discussion p. 429–430.

Phase of the pelvic girdle (the pelvic node region) relative to the limb cycle in *S. clarkii*; trunk-bend phase itself is not tabulated.

Value: Maximum pelvic rotation from direction of travel 16–25° across individuals and speeds (Table I PELVMAX: 16 $\pm$ 7, 25 $\pm$ 3, 21 $\pm$ 2, 21 $\pm$ 2, 19 $\pm$ 2°); peak occurs at 44–51% of stride after right hind foot-down, consistently just before foot-up (54–60%), i.e. at 81–85% of stance; timing not speed dependent (Table IV, $P = 0.339$). Authors state this pelvic range matches Ritter 1992's lizards, alligator and salamander.

Species: *Sceloporus clarkii* (Phrynosomatidae), other

Source: Reilly & DeLancey 1997 *J. Zool.* 243: 417–433 (same paper as above)

What was actually opened: Table I, Table IV, Fig. 2 caption, Discussion 'Pelvic and axial movements' p. 429.

Companion JEB paper describes the same animal's backbone as a rough standing wave with two nodes, one centred on each limb girdle, during a fast walking trot.

Value: Only the abstract is accessible: two individuals, treadmill at constant 0.833 m s⁻¹; 'rough standing wave with two nodes, one centred on each limb girdle'. No amplitude, phase or SVL retrievable (paywalled, \$30).

Species: *Sceloporus clarkii* (Phrynosomatidae), other

Source: Reilly & Delancey (1997) *J. Exp. Biol.* 200: 753–765, doi 10.1242/jeb.200.4.753

What was actually opened: JEB abstract page only (journals.biologists.com/jeb/article/200/4/753). OpenAlex and Semantic Scholar report no open-access copy.

The original standing-wave/travelling-wave paper: fully-limbed lizards show a standing wave of trunk lateral bending at low speed that is replaced by a travelling wave as velocity rises; short-limbed lizards use a travelling wave even at the lowest speeds.

Value: Abstract only. Species: *Cnemidophorus tigris* (Teiidae) and *Dipsosaurus dorsalis* (Iguanidae) = standing wave at low speed $\rightarrow$ travelling wave at higher speed; *Gerrhonotus kingii* (Anguidae) and *Eumeces multivirgatus* (Scincidae) = travelling wave at all speeds recorded. High-speed video + computerized motion analysis. NO numeric amplitude, node location, transition speed or phase could be verified — full text is paywalled. Second-hand only (unverified against the original): Reilly & DeLancey 1997 (p. 429) say Ritter's lizards showed standing waves with nodes near the girdles, pelvic rotation in the same 16–25° range, and snout displacement decreasing with speed; Suzuki et al. 2021 *Front. Neurobot.* attribute to Ritter 1992 that *D. dorsalis* uses 'intermediate' waves at intermediate speeds.

Species: *Cnemidophorus tigris*; *Dipsosaurus dorsalis*; *Gerrhonotus kingii*; *Eumeces multivirgatus* (Teiidae; Iguanidae; Anguidae; Scincidae), other (none are Gekkota)

Source: Ritter, D. (1992) *J. Exp. Biol.* 173: 1–10, doi 10.1242/jeb.173.1.1

What was actually opened: JEB abstract page (journals.biologists.com/jeb/article/173/1/1/736) read in the browser; OpenAlex, Semantic Scholar and CORE lookups (no OA copy); secondary statements read in Reilly & DeLancey 1997 *J. Zool.* p. 429–430 and in Suzuki et al. 2021 (Frontiers full text).

Forelimb footfall coincides with maximum lateral convexity of the trunk on that side; onset of the same forelimb's swing (start of protraction) coincides with maximum lateral concavity on that side. Stride defined from right-forelimb touchdown; propulsive phase ~0-50% of stride, swing ~50-100%.

Value: Max concavity toward the forelimb at ~0.5 of its stride cycle (its lift-off), max convexity toward it at 0/1.0 (touchdown). Fig. 3a: mean lateral rotation per intervertebral joint (10 joints Vert1-11 in Varanus; 5 joints Vert3-8 in Salvator) runs from about -1.5 deg at footfall, through zero at ~0.25, to a peak of ~+1.8 to +2 deg at ~0.45-0.6, back to -1.5 to -2.5 deg at 1.0 (values zeroed about their mean; ~3-4 deg peak-to-peak per joint). Slow treadmill walking; stride durations 1.5-1.85 s (monitors) and 1.5-3.25 s (tegus); 1 stride per individual, n=3 per species; measured with XROMM at intervertebral joints along the trunk. Body mass/SVL not stated in this paper (deferred to earlier ventilation papers on the same animals).

Species: Varanus exanthematicus and Salvator merianae (Varanidae (Anguimorpha) and Teiidae (Lacertoidea)), other (same order Squamata, different infraorders from Gekkota)

Source: Cieri RL, Hatch ST, Capano JG, Brainerd EL (2020) Locomotor rib kinematics in two species of lizards and a new hypothesis for the evolution of aspiration breathing in amniotes. Sci Rep 10:7739. <https://www.nature.com/articles/s41598-020-64140-y>

What was actually opened: Full open-access text (Results paragraph beginning 'Every rib measured...' containing the footfall/convexity-concavity statement; Fig. 3 caption with 'Propulsive phase was approximately 0-50% of the stride'; Fig. 7 caption defining the stride from right-forelimb contact with stride durations; Methods on XROMM/JCS) and the Fig. 3 image (panel a, intervertebral rotations vs % stride).

In a walking trot the trunk is a standing wave with nodes on the pectoral girdle and just anterior to the pelvis; peak pelvic yaw occurs late in hindlimb stance, just before foot-up, and this timing does not change with speed.

Value: Time of peak pelvic rotation = 0.44-0.51 of the stride (0 = right hind footfall), just before hind foot-up (0.54-0.60 of stride) = 81-85% of the stance phase. Max pelvic rotation from the direction of travel 16-25 deg (all individuals and speeds). Standing-wave amplitudes (peak lateral displacement, mm): trunk 18.0+/-2.0, 17.6+/-1.2, 13.4+/-2.2 at 0.270, 0.476, 0.833 m/s (~0.15-0.20 SVL); snout 20.8 -> 11.0 mm (decreases with speed); caudal 18-24 mm; pectoral and pelvic nodes ~5-10 mm minimum. SVL 90-92 mm; treadmill; gait a walking trot at all three speeds (diagonal couplets). Pelvic angle = angle between caudal-lumbar line and travel direction; companion JEB abstract: pelvic rotation 'to the opposite side' begins before foot-up.

Species: Sceloporus clarkii (Phrynosomatidae (Iguania)), other (same order Squamata, different infraorder from Gekkota)

Source: Reilly SM, Delancey MJ (1997) Sprawling locomotion in the lizard Sceloporus clarkii: the effects of speed on gait, hindlimb kinematics, and axial bending during walking. J Zool Lond 243:417-433 (PDF at https://www.originalwisdom.com/wp-content/uploads/bsk-pdf-manager/2019/10/Reilly-and-Delancey_1997_Sprawling-locomotion-in-the-lizard-Sceloporus-clarkii.pdf); companion abstract Reilly & Delancey 1997 J Exp Biol 200:753-765 (PubMed 9318518).

What was actually opened: Full PDF text: Abstract; Methods (Fig. 1 landmark/angle definitions, 'Axial bending' variables); Results ('Axial bending' nodes, 'Stride and gait characteristics', Table VI axial displacements); Fig. 2 caption (pelvic angle sign convention, foot-up times); Fig. 4 caption; Discussion paragraph 'Pelvic and axial movements remained constant...' giving the 44-51% / 81-85% timing. Also the PubMed abstract of the JEB companion paper.

In the leopard gecko the pelvic girdle yaws through the hindlimb stride with a peak late in stance; the trunk carries a standing wave that becomes a caudally travelling wave in the tail. Only pelvic-girdle yaw vs the hindlimb cycle was reported; trunk curvature vs forelimb cycle was NOT measured.

Value: Fig. 4A (representative right-hindlimb stride, original tail): pelvic yaw about -38 deg at hind footfall (negative = rotated toward the observed hindlimb), zero crossing at ~0.33, peak ~+47 deg (rotated toward the opposite hindlimb) at ~0.60-0.65 of the stride, hindlimb stance ending at ~0.75 (swing 0.75-1.0), returning to ~-40 deg at 1.0. Mean yaw excursion of pelvic girdle ~50 deg with intact tail, ~32-34 deg after tail restriction or autotomy (Fig. 4B). n=10 adults, mass 36.3+/-1.9 g, SVL 104.6+/-2.1 mm; level trackway with sandpaper; 250 fps; speeds 0.59-3.36 SVL/s (treatment means 1.23-1.85 SVL/s); markers on dorsal midpoint of body, centre of pectoral/pelvic girdles, hip/knee/ankle. Inference (mine, not the authors'): peak pelvic yaw at ~80-85% of hindlimb stance matches the Sceloporus timing, and in a trot-like gait this is just before the diagonal forelimb lifts off.

Species: Eublepharis macularius (Eublepharidae), same species

Source: Jagnandan K, Higham TE (2017) Lateral movements of a massive tail influence gecko locomotion: an integrative study comparing tail restriction and autotomy. Sci Rep 7:10865. <https://www.nature.com/articles/s41598-017-11484-7> (PDF from http://biomechanics.ucr.edu/Jagnandan_Higham_2017.pdf)

What was actually opened: Full PDF text (Abstract; Introduction statement on standing wave in trunk / travelling wave in tail; Results on pelvic girdle rotation; Table 1 speeds; Fig. 4 caption; Materials and Methods: study organisms, set-up, stride kinematics) and a rendered image of page 4 showing Figs 3 and 4 (pelvic girdle rotation vs % stride cycle with stance/swing shading).

At each foot touchdown the local trunk segment (shoulder for forelimb, hip for hindlimb) reaches maximal curvature, convex toward the touching-lower limb; hip bending is in phase with hindlimb movement (forelimb/shoulder relation weaker in field video). Fully limbed lizards on firm substrate use an almost pure standing wave with nodes at shoulder and hip.

Value: Fig. 4A right: Hilbert-transform phase of hip bending vs hind-leg phase falls along the 1:1 diagonal (approximately zero phase offset, i.e. in phase). Limb sequence in Brachymeles FR-HL-FL-HR, hindlimb leads ipsilateral forelimb by 0.38+/-0.07 of a period (lateral couplet), duty factor ~0.5. Curvature quantified as body curvature $\kappa(s,t)$ in units of 1/SVL from 21 DeepLabCut midline points; field videos on soil/poppy seeds (Brachymeles) and lab trials on sandpaper / granular media (Uma, Sceloporus). Absolute SVL and speeds not given in main text (relative hindlimb length per species given in Fig. 2). Precise phase definition (what constitutes phase zero of 'leg movement') is not spelled out in the main text.

Species: Brachymeles kadwa, B. taylori, B. muntingkamay; Uma scoparia; Sceloporus olivaceus (Scincidae (Brachymeles); Phrynosomatidae (Uma, Sceloporus)), other (same order Squamata, different infraorders from Gekkota)

Source: Chong B, Wang T, Erickson E, Bergmann PJ, Goldman DI (2022) Coordinating tiny limbs and long bodies: geometric mechanics of lizard terrestrial swimming. PNAS 119:e2118456119. PMC9271186 (Europe PMC full text).

What was actually opened: Europe PMC full-text XML (Abstract, Results sections on wave diversity, limb sequence, body-limb phasing, Methods on DeepLabCut and Hilbert-transform phase estimation) and the Fig. 4 image (panel A body-leg phasing scatter plot).

Gekko gekko (same infraorder as the target) uses a diagonal (trot-like) gait with a standing wave of trunk bending at low speed (single-peak waveform, nodes near the girdles) that becomes a travelling wave at high speed; the pattern is produced by girdle rotation plus axial muscle activity. Trunk flexion, pectoral and pelvic girdle rotation angles were measured, but no numeric phase of trunk bending relative to forelimb touchdown/lift-off was available in the parts I could read.

Value: Body mass 55.7+/-3.5 g, body length 252.3+/-7.6 mm, SVL 127.6+/-10.9 mm, N=6; horizontal, 45 deg and vertical planes; speeds 0.38-3.85 SVL/s (56.6 mm/s upward); stride cycle defined by successive right-hind-foot touchdowns; 34/42/22 stride cycles on the three planes. No phase fraction or bending amplitude reported in the accessible abstract/snippets.

Species: Gekko gekko (Gekkonidae), same infraorder (Gekkota), different family from Eublepharidae

Source: Wang W, Ji A, Chen G, Ravi S, Shen H, Gorb SN, Dai Z (2020) Kinematics of gekko climbing: the lateral undulation pattern. *Zoology* 140:125768

(<https://www.sciencedirect.com/science/article/abs/pii/S0944200620300271>); Wang W, Ji A, Dai Z et al. (2021) Angular variables of climbing geckos in two lateral undulation patterns. *Zoology* 145:125892 (<https://www.sciencedirect.com/science/article/abs/pii/S0944200620301513>)

What was actually opened: ScienceDirect landing pages only: highlights, abstracts, introductions, and the 'Animals/Study species', 'Locomotion behaviours', 'Discussion' and 'Conclusion' snippets. Full texts are paywalled; figures not seen.

Refuted

[geckos] **In a pad-bearing gekko running on the level, both pectoral and pelvic girdles rotate in yaw through roughly 50-55 degrees per stride (from about -40 deg at foot -- I read the Europe PMC full-text XML of PMC4776376 (Methods on animals, markers, cameras, angle definitions; Results; Fig. 2 caption; Tables 1-3) and the BMC PDF (page 6), where Fig. 2 is vector graphics. I rendered it at 600 DPI and extracted the plotted path coordinates, calibrated on the axis tick**

[standing_wave] **Fully-limbed phrynosomatid lizards produce an almost perfect standing wave of body curvature with nodes fixed at shoulder and hip on solid/loosely packed substr -- Core mechanism is in the source, but the claim as worded misstates the conditions and attribution in four places. (1) 'Fluidised' contradicts the paper: the Terrestrial swimming section (arXiv p. 6) and footnote 6 state that upward airflow was kept BELOW the onset of fluidization (fluidization = gro**

Gaps

[geckos] 1) No published value of trunk CURVATURE or mid-trunk lateral bending ANGLE for Eublepharis macularius was found. The only Eublepharis axial-kinematic numbers located are pelvic-girdle yaw excursion and tail-tip lateral displacement (Jagnandan & Higham 2017, above). Eublepharis papers I opened that do NOT report any lateral trunk bending: Fuller, Higham & Clark 2011 *Zoology* 114:104-112 (PDF from biomechanics.ucr.edu read - only hind-limb 3D joint angles, femur depression, hip height; the trackway is merely described as wide enough 'for lateral undulations'); Jagnandan, Russell & Higham 2014 *JEB* 217:3891-3897 (full text read - GRFs, sagittal joint angles, body pitch; no girdle rotation or lateral bending); Jagnandan & Higham 2018 *JEB* 221:jeb179564 (full text read - limb kinematics and EMG only). Zaaf, Van Damme, Herrel & Aerts 2001 *JEB* 204:1233-1246 (Eublepharis vs Gekko gekko): only the abstract could be read (paywalled at journals.biologists.com; curl returned a challenge page); the abstract lists stride/step length, frequency, duty factor and sprawling only, so I cannot say whether the full text quantifies lateral undulation. 2) For Gekko gekko, the Nanjing group's primary numbers (Wang et al. 2020 *Zoology* 140:125768; Wang et al. 2021 *Zoology* 145:125892 - 13 angular variables incl. trunk flexion and trunk deflection on the vertical plane; Wang et al. 2015 *J. Bionic Eng.* bending radius vs slope; Wang et al. 2018 *J. Comp. Physiol. A* 204:707-719 review) are all paywalled (ScienceDirect/ResearchGate 403; Springer abstract only; RMIT figshare record 27530589 has no file). The only numeric value recovered is the secondary 33.56/33.22 deg girdle-angle citation above; its plane and speed are unknown. Wang et al. 2014 *Chinese Sci. Bull.* 59:4568-4577 (Tokay on inclines): abstract read (Springer) reports only speed, stride length and duty factor (0.54 at 90 deg to 0.84 on the ceiling), nothing on bending. 3) Autumn et al. 2006 *JEB* 209:260-272 (Hemidactylus garnotii climbing vertically, mass 1.9 +/- 0.7 g, SVL 4.6-5.8 cm, 0.29-0.77 m/s): PDF read - lateral body bending is described only qualitatively (convex side alternates right/left with each diagonal couplet); no angle. 4) Zhuang & Higham 2016 *PLOS ONE* (Phelsuma madagascariensis) and Zaaf et al. 2001 *Belg. J. Zool.* (Gekko gekko limb joints; not accessible) contain no axial-bending variables. 5) Non-gekkotan comparison: Ritter 1992 *JEB* 173:1-10 (Cnemidophorus, Dipsosaurus, Gerrhonotus, Eumeces; other infraorders) - abstract only was accessible (standing wave at low speed, travelling wave at high speed); no angles read, so none are reported. 6) Open-access robotics papers that cite gekko data (Haomachai/Manoonpong 2021 preprint; Li et al. 2018 *Appl. Bionics Biomech.*; PMC11505542; PMC9944875; MDPI Electronics 2025 review - 403) were checked: apart from the Soft Robotics 2023 citation above, none gives a numerical animal trunk-bending value.

[lizards_general] (1) No published measurement of mid-trunk lateral bending (angle or amplitude) exists, as far as I could find, for Eublepharis macularius or any eublepharid: the E. macularius kinematic papers (Zaaf et al. 2001; Fuller et al. 2011; Jagnandan & Higham 2017, 2018) quantify limb joints, gait timing, pelvic yaw (~50 degree excursion) and tail displacement, not trunk curvature; Farley & Ko (1997) on Coleonyx is paywalled and its abstract gives only a COM-energy statement. (2) Ritter (1992) full text - the actual bending angles, SVLs, speeds and the standing-to-traveling-wave transition speeds - is behind the JEB paywall and no open copy was found (Semantic Scholar API returned HTTP 429; JEB page shows 'Available to Purchase'). Same for Ritter (1995, 1996) and Farley & Ko (1997). (3) Irschick & Jayne (1999) pelvic-rotation values for Callisaurus, Uma, Dipsosaurus, Cnemidophorus, Phrynosoma: JEB page blocked by CAPTCHA. (4) Kubo & Ozaki (2009), the cited source of the '11.7-25.8 degree' pelvic-rotation range: ScienceDirect 403. (5) Species-level 'maximal axial angle' values from Bergmann & Irschick (2010) are only in Wiley Supporting Information (403). (6) Rockenfeller et al. (2024) species-level spine-ROM values are in Supplementary S1 (Fig. D1), not accessed; their level-ground trials are varanids only and gekko trials are climbing. (7) Not opened at all: Kohlsdorf & Biewener (2006) Sceloporus malachiticus; Daan & Belterman (1968); Peterson (1984) Agama/Chamaeleo; Wang, Ji, Manoonpong et al. (2018) *J. Comp. Physiol. A* review; Jurestovsky/Astley (2019) 'Angles and waves' (cookie wall); Wang et al. (2020) *Zoology* gekko-climbing undulation paper (ScienceDirect abstract only). (8) I found no study that reports a lizard trunk bending angle as an explicit function of speed with a fitted slope; the only speed data are Reilly & Delancey (no change over 0.27-0.83 m/s in S. clarkii displacement and pelvic angle), Rockenfeller et al. (categorical: fast trials 19.4 vs slow 22.7 degrees), Bergmann & Irschick (no interspecific relation to max velocity), and Ritter's abstract (pattern changes standing-to-traveling with speed, amplitude not stated).

[standing_wave] (1) No published lateral trunk-bending amplitude, node position, or bend-vs-footfall phase for Eublepharis macularius or any eublepharid during level walking; the corpus's 'standing wave in the trunk' for this species traces to an Introduction sentence in Jagnandan & Higham 2017 citing Hamley 1990 (a bipedal-tail review), not a measurement. (2) Ritter 1992 full text: paywalled at JEB; no OA copy via OpenAlex, Semantic Scholar, CORE; Wayback fetch blocked — so its amplitudes, transition speeds and phase values remain unverified (only the abstract and second-hand statements). (3) Reilly & Delancey 1997 *JEB* 200:753 and Ritter 1995/1996 full texts (paywalled) — abstracts only. (4) Numeric values from Wang et al. 2020/2021 Gekko gekko papers (ScienceDirect paywalled; robot reviews that cite them quote no numbers). (5) Daan & Belterman 1968 (*Proc. KNAW C* 71:245): KNAW digital library covers only 1899–1950. (6) Rockenfeller 2024 gives no level-walking gekko values (geckos were climbing) and no regression of spine ROM on speed, only a categorical slow/medium/fast trend. (7) Whether Reilly & DeLancey's Table VI displacements are peak-to-peak or half-amplitude is not stated in the paper.

[phase] (1) No published measurement of trunk-bending phase relative to the FORELIMB cycle for *Eublepharis macularius* or any other gekkotan: the leopard-gecko data (Jagnandan & Higham 2017) cover pelvic-girdle yaw vs the hindlimb cycle only, and Fuller et al. 2011 (*E. macularius*, *Teratoscincus scincus*) reports hindlimb kinematics only; the two Gekko gecko papers that measured trunk flexion and girdle rotation angles (Wang et al. 2020, 2021, *Zoology*) are paywalled and their abstracts give no phase value. (2) The full texts of Ritter 1992, 1995 and 1996 (*J Exp Biol*) are pay-per-view on the Company of Biologists site and I found no legitimate open copy, so Ritter's own bending-vs-footfall timings for *Cnemidophorus/Dipsosaurus/Varanus/Iguana* could not be extracted. (3) Daan & Belterman 1968 (*Proc. K. Ned. Akad. Wet.*) was not accessible. (4) Body mass/SVL of the Cieri et al. 2020 monitors and tegus are not in that paper. (5) Nyakatura et al. 2019 main text and Supplementary Data 13 (which might contain spine-bending vs limb timing for *Iguana* and *Tiliqua*) were not retrievable. Note also that the answer above assumes a trot-like/diagonal gait when mapping hindlimb-referenced pelvic timing onto the forelimb cycle; for a lateral-sequence walk (e.g. hindlimb leading ipsilateral forelimb by ~ 0.38 cycle as in *Brachymeles*) the forelimb-referenced fraction would shift accordingly.

The six behaviour channels, measured

The animal's action selection is Prescott's basal ganglia with six channels. Before the most recent session, four of those six had a gate of exactly 0.0000 on every one of 900 autonomous steps — a six-channel brain running as a two-channel one. Two of the four were the world's fault and are now fixed: there was nothing to flee from and nothing warm to lie on. Two are the selector's own floor and are not fixable without changing a published number.

```
morphology/gecko_world_furnished_v1.xml · seed 3 · produced by tools/four_brains_demo.py
```

CHANNEL	PEAK GATE	STEPS WON	OUTCOME
hunt	1	359	released
explore	1	85	released
flee	0.9378	169	released
bask	1	1,275	released
rest	0	0	never released
groom	0	0	never released

The floor that caps this brain at four

Bisected on one channel with every other channel at zero: a salience of 0.19 gives a gate of exactly 0.0000 and 0.25 gives 0.98. The release threshold is **0.2008**. Nothing below it is released at all, however far ahead of its rivals it is.

basal ganglia release threshold salience	0.2008
flee salience ceiling scent alone	0.21
flee needs with hunger 0.30	0.2171
groom tonic salience	0.05
rest fatigue plateau	0.011

flee clears the bare threshold by 0.009 and is beaten by explore once hunger exceeds about 0.2; groom and rest are an order of magnitude below the threshold and cannot be released by any world. Nothing here was tuned.

Why flee is the interesting one, and was left alone

Its salience IS the published probability of a defensive reaction in this species — 0.21 to scent alone, 0.28 with sight, n = 42 — delivered through a nose whose inverse-square concentration tops out at 0.28 even at contact. Explore is 0.35 × hunger. So it clears the bare threshold by 0.009 and is beaten by explore as soon as hunger passes about 0.2: with hunger 0.30 present, flee needs 0.2171 and has 0.2100, short by 0.0071. The model therefore predicts that a starving gecko does not flee from something standing on it. **Nothing was tuned.** That is either a real prediction or a sign that a reaction probability is the wrong quantity to use as a salience, and either way it is not something to fix by choosing a bigger number (#334, #335).

The eight brains, and what each one actually is

#	BRAIN	STATE
1	hypothalamus — hunger, energy, fatigue, thermostat	built; hunger reproduces the published 2.33-day feeding interval unfitted
2	basal ganglia — action selection	built; reproduces the published selection table to four decimal places
3a	spinal rhythm — the leg oscillators	built; drives the walker with every gate unchanged
3b	brainstem — decision to command	built; thinnest evidence in the project, and says so
4	eye — retina, pretectum, tectum	gaze reproduces; prey-finding NOT ACCEPTED
5	smell — the channel this species is documented to gate on	a threat-distance path exists for flee ; the sense itself is absent
6	day/night clock	a fake switch — see below
7	memory	not built
8	learning	not built

Brain 6 is not connected, and now looks as if it is (#347). GeckoBrainEnv takes a circadian= flag and constructs Arousal() when it is True, and self.clock is reset alongside the other modules. But clock.step is never called anywhere in the file, and self.arousal is assigned exactly once, to the constant 1.0 at construction, and thereafter only READ — to be copied into info["arousal"] . Switching the flag on builds the object, resets it, and reports a number the object never produced. This was recorded rather than fixed, because fixing it means deciding what arousal should modulate, and nothing published says.

The walker that is accepted, and the one that is not

There are two walkers in this repository and they are easy to confuse, because the rejected one is the default in one of the environments. Every clip, every demo and every measurement in this document uses the accepted one: **gait profile 1ab , zero residual, no trained policy** — the hand-written CPG with its contact reflex, controller ledger row #71, accepted at 19 strides, 1.1892 Hz measured against 1.1888 commanded.

FOOT	ACCEPTED: , NO POLICY	REJECTED: + TRAINED RESIDUAL	PUBLISHED TARGET
front left	0.770	0.704	0.70
front right	0.764	0.601	0.70
hind left	0.787	0.435	0.765
hind right	0.790	0.367	0.765

How the wrong one got filmed, and why it survived (#338). GeckoBrainEnv defaults to `gait_profile="legacy"` with the frozen trained residual loaded on top, and the brain demo passed neither argument, so it took both defaults. The comment twenty lines above that constructor states the measurement above in full. The reason it went unnoticed for so long is in the table: the defect is in the HIND feet, the front pair looks fine, and front-foot duty was the only thing being checked. The user caught it. The environment exposes `info["accepted_walker"]`, which was sitting there unused; the demo now asserts on it before filming and exits if it is False. The trained residual was refuted on its own terms at #11: two runs of 3.01 M steps, best trained 3 of 6 gates against the hand-written base's 4 of 6.

What the accepted walker still fails

4 of 6 published gait gates. It does not reach front-foot duty 0.70 in the open-field measurement (0.447 / 0.444), and the caption on every clip now prints that target next to the value, because a number shown alone under the word ACCEPTED reads as a pass (#325).

The spine, fitted and then not adopted (#346)

The 24-agent pass in Part 23 found a same-species number: pelvic girdle yaw excursion of about 50° per stride with the intact tail (Jagnandan & Higham 2017, n = 10, SVL 104.6 mm, level trackway, hind duty 0.78). The accepted walker measures **24.3°**, peaking 0.21 of a cycle late. The spine is one tendon across three lateral joints and excursion is linear in its amplitude — 24.3 / 36.4 / 48.9 / 61.8 / 74.6° at amplitude 0.30 / 0.45 / 0.60 / 0.75 / 0.90 — so `spine_amp` 0.61 is DERIVED, and `spine_phase` 0.21 corrects the timing: measured 51.8° peaking at 0.62. Then the gates, one declared change per trial as the protocol demands: hind duty 0.636 → 0.650 → **0.675**, moving toward the required 0.73–0.83; but speed 0.0427 → 0.0427 → **0.0369 m/s** and net-path 0.823 → 0.712 → **0.697**, and the lab set was selected for > 0.04 m/s and > 0.8 net-path. The published spine wave breaks both while helping the one the contract is actually stuck on. **It is not the default.** It is a declared parameter set, `--set spine_amp=0.61 --set spine_phase=0.21`, and the trade is recorded rather than resolved by picking a smaller amplitude that matches nothing.

What is pending: the four behaviour items

This is where the work stops and where the next session picks it up. Four items, all of them about whether a behaviour the animal is supposed to have can actually be reached. None of them is fixable by choosing a bigger number, which is precisely why they are still here.

PENDING 1

rest is structurally unreachable

Its salience IS fatigue, and fatigue plateaus at **0.011** in an animal that strolls (#284). The basal ganglia releases nothing below **0.2008** (#335). It is short by a factor of eighteen. The honest options are: give the animal something that actually tires it, find a published tonic for rest in this species, or record that a strolling gecko in this world never rests and leave the channel unreachable. Tuning the fatigue curve to clear the threshold is forbidden by Rule 2.

PENDING 2

groom is structurally unreachable

A tonic salience of **0.05** against the same 0.2008 floor. Same three options, same prohibition. Note that grooming in this species has a real published trigger — it is a shedding and eye-cleaning behaviour, not a background tonic — so the most likely correct fix is that the salience is the wrong quantity, not the wrong size.

PENDING 3

flee releases, but only when the animal is not hungry

It clears the floor by **0.009**. With hunger 0.30 present it needs 0.2171 and has 0.2100. The model's standing prediction is that a starving gecko does not flee from something standing on it. That is either real biology or a sign that a published *probability of reaction* is the wrong quantity to feed a selector as a *salience*. Deciding which is a research question, not a coding one (#334, #335).

PENDING 4

Brain 6, the day/night clock, is a switch wired to nothing

`clock.step` is never called; `_arousal` is the constant 1.0 (#347). Either connect it — which means deciding what arousal modulates, and nothing published says — or delete the flag so the model stops reporting a number no module produced. Leaving a fake switch in place is the worse of the two.

And behind those four

ITEM	STATE
Brain 4 — prey-finding	gaze reproduces; finding the cricket is NOT ACCEPTED. The strike still reads a privileged range.
Brain 5 — smell	absent as a sense, though this species is documented to gate on it.
Brains 7 and 8 — memory, learning	not built. Barely measured in this animal, so the research has to come first.
The surface research	10 verified claims in Part 23 are written down and NOT applied to the skin that exists.
Front-foot duty	0.447 / 0.444 against a published 0.70. Two of six gait gates still fail.

The brief for the next session

Everything above this page is the record. This page is the instruction. It is written to be handed to a fresh session verbatim, and it is deliberately strict, because every one of this project's 284 refuted hypotheses started as something that sounded reasonable.

READ EVERYTHING FIRST. DO NOT BUILD ANYTHING YET.

1. Go and read the research. All of it. Actually open it.

Before you touch one line of this project, read every research paper, every agent finding and every document this report names. Not the summaries — the sources. Specifically:

1. `docs/FAILURE_MAP.md` — all 348 numbered hypotheses, in order, including the 284 refuted ones. The refuted rows are the map of where not to look and are the most valuable thing in the repository. Read the verdicts, not just the claims.
2. `docs/GECKO_SURFACE_ANATOMY.md` — the 42-agent pass. All 10 surviving claims, all 26 refuted ones, and all 6 could-not-verify sections. The refuted half is the half that matters.
3. `docs/LATERAL_UNDULATION.md` — the 24-agent pass. All 18 surviving numbers with their species distance, the 2 refuted, the 4 gaps.
4. `config/proxies.yaml` — all 152 parameters with their provenance tags. Know which are PUBLISHED, which are DERIVED and which are INVENTED before you reason about any of them.
5. The bibliography in Part 16 of this document — 217 works, 51 with DOIs. Open the ones your task depends on. **Do not cite a paper you have not opened.**
6. `docs/BUILD_LOG.md`, `docs/DECISIONS.md`, `docs/BLOCKED.md` and `docs/EYE_SPEC.md` — the long-form context behind the ledger rows.
7. `CLAUDE.md` — the standing rules. They are not advisory.

Understand this animal **deeply** before you act: its mass and every proportion in Parts 19 to 21, what each of its 38 joints can do, which of its 14 morphology gates are load-bearing, which of its eight brains exist, which of its six behaviours it can actually reach, and which of its numbers are inventions wearing a published number's clothes. Every single thing this gecko needs, you should be able to state from the sources, not from memory.

2. Then come back knowing it, and say where we are resuming

The work continues from **fixing the four pending behaviour items in Part 26**: `rest` unreachable at 0.011 against a 0.2008 floor; `groom` unreachable at 0.05; `flee` clearing that floor by 0.009 and losing to hunger; and brain 6's clock being a switch wired to nothing. That is the resumption point. Do not start somewhere else because something else looked easier.

3. Ask before you start

When you have finished reading, report back with what you found, then ASK THE USER FOR PERMISSION TO START. Do not begin editing, fitting, rebuilding or filming until the user has said yes. State clearly which of the four items you propose to take first and why, and what evidence would settle it.

The rules that are not negotiable while you work

- Every number is PUBLISHED, DERIVED or INVENTED and says which, in `config/proxies.yaml`. A plausible number wearing a published number's clothes is the worst defect available.
- **Never tune until the answer matches.** If the model needs its constants adjusted to reproduce a published result, the constants must come from a paper.
- The reward is what the model is told to want; the gates are what the animal does. Only the second counts.
- A failed prediction stays in the table. Adding a knob per failing joint is fitting the harness to the answer.
- Measure before claiming — including your own tests and your own solver. Five recorded failures were in the test, not the model.
- One source is not a citation.
- Do not cite a paper you have not opened. Do not fill a gap with something plausible. Mark anything you could not verify as unverified. Do not fill a gap with a value from a related species without naming the species and the distance.
- `docs/FAILURE_MAP.md` is updated before every commit, confirmed or refuted. **Nothing is ever deleted from it.** A refuted hypothesis later found true gets a second row, never an edit. Never renumber.
- Answering the user: short, plain language, jargon defined inline, and **never give a time estimate.**
- Use the accepted walker — `gait_profile="lab"`, `use_policy=False` — and assert on `info["accepted_walker"]` before filming anything.
- Rebuild the body only through `tools/rebuild_body.py`, never step by step by hand.

Glossary — every term this document uses

Oracle

Privileged information handed to the animal that a real animal could not have — here, the prey's exact position. A model that uses one has not solved the problem; it has been told the answer.

Shadow signal

The same privileged information, routed to the log file and never to the animal, so a failure can be attributed.

Efference copy

A copy of the motor command, used to predict the sensory consequences of the animal's own movement so they can be subtracted. It is why you do not see the world jump when you move your eyes.

Optic flow

The pattern of image motion caused by moving. It expands outward from the direction of travel — which is why subtracting a local average does not cancel it.

Tectum / optic tectum

The midbrain map of visual space. In mammals it is called the superior colliculus. It is where "something small is moving over there" is computed.

CPG — central pattern generator

A circuit in the spinal cord that produces a rhythm without needing rhythmic input. Here: four coupled oscillators, one per leg.

Duty factor

The fraction of a stride during which a foot is on the ground and carrying load. Two feet with very different duty factors is a limp.

Residual policy

A trained neural network whose output is ADDED to a hand-written controller. The accepted walker runs with this set to zero.

Gate

A pass/fail check against a published measurement. Distinct from a reward: the reward is what the model is told to want, the gate is what the animal does.

Path efficiency

Straight-line ground closed toward the prey, divided by ground actually walked. 1.0 is a direct approach; near 0 is a thrash that ended up in the right place.

Hazard ratio

How much a treatment multiplies the moment-to-moment chance of an event. HR 1.09 per trial means about 9 % more likely to reach the goal with each trial of practice.

τ (tau), time constant

How long an exponential decay takes to fall to about 37 % of its starting value.

Q10

How much a biological rate changes for a 10 °C change in temperature. Q10 = 2.3 means the rate roughly doubles.

Thigmothermic

Taking body heat by contact with a warm surface rather than from sunlight. It is why *bask* modelled as a light-seeking behaviour is the wrong mechanism for this animal.

Vomeronasal / Jacobson's organ

A chemical sense separate from smell, sampled by tongue-flicking. In this species it is the channel that fires the defensive display; vision alone cannot.

n

The number of animals in a study. n=42 is a real result; n=2 is a starting point; the sleep recording of this species is n=2.

Provenance

Where a number came from. PUBLISHED means measured and printed; DERIVED means computed from something published; INVENTED means chosen, and says so.